

# Program overview

19-May-2025 13:42

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|--------------|----------------------------------------------------------|
| Year         | 2024/2025                                                |
| Organization | Electrical Engineering, Mathematics and Computer Science |
| Education    | Master Sustainable Energy Technology (SET)               |

| Code                                                                     | Omschrijving                                            | ECTS | p1 | p2 | p3 | p4 | p5 |
|--------------------------------------------------------------------------|---------------------------------------------------------|------|----|----|----|----|----|
| <b>Master Sustainable Energy Technology 2024</b>                         |                                                         |      |    |    |    |    |    |
| <b>Core Courses 2024</b>                                                 |                                                         |      |    |    |    |    |    |
| SET3014                                                                  | Renewable Energy                                        | 5    |    |    |    |    |    |
| SET3055                                                                  | Economics and Regulations of Sustainable Energy Systems | 4    |    |    |    |    |    |
| SET3061                                                                  | Energy System Modelling                                 | 4    |    |    |    |    |    |
| SET3250TPM                                                               | Communication for Engineers                             | 3    |    |    |    |    |    |
| SET3815-M                                                                | Matlab Fundamentals                                     | 2    |    |    |    |    |    |
| SET3816-PR                                                               | System Integration Project                              | 6    |    |    |    |    |    |
| SET3901                                                                  | Graduation Project                                      | 45   |    |    |    |    |    |
| <b>Profile Courses SET 2024</b>                                          |                                                         |      |    |    |    |    |    |
| <b>Track Electrical Energy 2024</b>                                      |                                                         |      |    |    |    |    |    |
| <b>Solar Systems: Solar Power Storage 2024</b>                           |                                                         |      |    |    |    |    |    |
| EE4536                                                                   | DC and AC Microgrids                                    | 4    |    |    |    |    |    |
| EE4545                                                                   | Electrical Power Systems of the Future                  | 4    |    |    |    |    |    |
| ET4376                                                                   | Photovoltaic Basics                                     | 4    |    |    |    |    |    |
| ET4377                                                                   | Photovoltaic Technologies                               | 4    |    |    |    |    |    |
| ET4378                                                                   | Photovoltaic Systems                                    | 4    |    |    |    |    |    |
| SET3065                                                                  | Intelligent Electrical Power Grids                      | 4    |    |    |    |    |    |
| SET3080                                                                  | The Necessity of Storage Technology                     | 4    |    |    |    |    |    |
| SET3085                                                                  | Hydrogen Technology                                     | 4    |    |    |    |    |    |
| SET3095                                                                  | Electronic Power Conversion                             | 4    |    |    |    |    |    |
| SET3110                                                                  | Energy Storage in Batteries                             | 4    |    |    |    |    |    |
| <b>Waste-CHP &amp; Solar Systems: Waste &amp; CHP Solar Storage 2024</b> |                                                         |      |    |    |    |    |    |
| CH3061                                                                   | Multiphase Reactor Engineering                          | 4    |    |    |    |    |    |
| ET4376                                                                   | Photovoltaic Basics                                     | 4    |    |    |    |    |    |
| ET4377                                                                   | Photovoltaic Technologies                               | 4    |    |    |    |    |    |
| ET4378                                                                   | Photovoltaic Systems                                    | 4    |    |    |    |    |    |
| ME45134                                                                  | Process and Power Plant Design                          | 4    |    |    |    |    |    |
| SET3070                                                                  | Thermochemistry of Biomass Conversion                   | 4    |    |    |    |    |    |
| SET3080                                                                  | The Necessity of Storage Technology                     | 4    |    |    |    |    |    |
| SET3085                                                                  | Hydrogen Technology                                     | 4    |    |    |    |    |    |
| SET3110                                                                  | Energy Storage in Batteries                             | 4    |    |    |    |    |    |
| <b>Autonomous Systems: Wind Storage - Solar 2024</b>                     |                                                         |      |    |    |    |    |    |
| AE4W02TU                                                                 | Introduction to Wind Turbines: Physics and Technology   | 4    |    |    |    |    |    |
| AE4W09                                                                   | Wind Turbine Design                                     | 5    |    |    |    |    |    |
| AE4W13                                                                   | Site Conditions for Wind Turbine Design                 | 3    |    |    |    |    |    |
| ET4376                                                                   | Photovoltaic Basics                                     | 4    |    |    |    |    |    |
| ET4377                                                                   | Photovoltaic Technologies                               | 4    |    |    |    |    |    |
| ET4378                                                                   | Photovoltaic Systems                                    | 4    |    |    |    |    |    |
| SET3080                                                                  | The Necessity of Storage Technology                     | 4    |    |    |    |    |    |
| SET3085                                                                  | Hydrogen Technology                                     | 4    |    |    |    |    |    |
| SET3110                                                                  | Energy Storage in Batteries                             | 4    |    |    |    |    |    |
| <b>Wind &amp; Economics: Wind Power Economics 2024</b>                   |                                                         |      |    |    |    |    |    |
| AE4W02TU                                                                 | Introduction to Wind Turbines: Physics and Technology   | 4    |    |    |    |    |    |
| AE4W09                                                                   | Wind Turbine Design                                     | 5    |    |    |    |    |    |
| AE4W13                                                                   | Site Conditions for Wind Turbine Design                 | 3    |    |    |    |    |    |
| EE4536                                                                   | DC and AC Microgrids                                    | 4    |    |    |    |    |    |
| EE4545                                                                   | Electrical Power Systems of the Future                  | 4    |    |    |    |    |    |
| SET3065                                                                  | Intelligent Electrical Power Grids                      | 4    |    |    |    |    |    |
| SET3095                                                                  | Electronic Power Conversion                             | 4    |    |    |    |    |    |
| TPM0637SET                                                               | Economic Policy for Sustainable Energy                  | 4    |    |    |    |    |    |
| TPM0931SET                                                               | Sustainable Energy Innovation and Transitions           | 4    |    |    |    |    |    |
| TPM403SET                                                                | Technology Entrepreneurship and Sustainability          | 4    |    |    |    |    |    |
| <b>Solar &amp; Economics: Solar Power Economics 2024</b>                 |                                                         |      |    |    |    |    |    |
| EE4536                                                                   | DC and AC Microgrids                                    | 4    |    |    |    |    |    |
| EE4545                                                                   | Electrical Power Systems of the Future                  | 4    |    |    |    |    |    |
| ET4376                                                                   | Photovoltaic Basics                                     | 4    |    |    |    |    |    |
| ET4377                                                                   | Photovoltaic Technologies                               | 4    |    |    |    |    |    |
| ET4378                                                                   | Photovoltaic Systems                                    | 4    |    |    |    |    |    |
| SET3065                                                                  | Intelligent Electrical Power Grids                      | 4    |    |    |    |    |    |
| SET3095                                                                  | Electronic Power Conversion                             | 4    |    |    |    |    |    |
| TPM0637SET                                                               | Economic Policy for Sustainable Energy                  | 4    |    |    |    |    |    |

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|----------------------------------------------------------------------------------|------------------------------------------------------------------------|---|--|
| TPM0931SET                                                                       | Sustainable Energy Innovation and Transitions                          | 4 |  |
| TPM403SET                                                                        | Technology Entrepreneurship and Sustainability                         | 4 |  |
| <b>Data-Driven Energy Systems: DET - Power - Economics 2024</b>                  |                                                                        |   |  |
| EE4410                                                                           | Cyber Security of Power Grids                                          | 4 |  |
| EE4536                                                                           | DC and AC Microgrids                                                   | 4 |  |
| EE4545                                                                           | Electrical Power Systems of the Future                                 | 4 |  |
| SET3065                                                                          | Intelligent Electrical Power Grids                                     | 4 |  |
| SET3095                                                                          | Electronic Power Conversion                                            | 4 |  |
| SET3120                                                                          | Energy Systems Simulation and Digital Twins                            | 4 |  |
| SET3125                                                                          | Machine Learning Workflows for Digital Energy Systems                  | 4 |  |
| TPM0637SET                                                                       | Economic Policy for Sustainable Energy                                 | 4 |  |
| TPM0931SET                                                                       | Sustainable Energy Innovation and Transitions                          | 4 |  |
| TPM403SET                                                                        | Technology Entrepreneurship and Sustainability                         | 4 |  |
| <b>Electric Mobility Systems: Electric Mobility Power Storage 2024</b>           |                                                                        |   |  |
| EE4536                                                                           | DC and AC Microgrids                                                   | 4 |  |
| EE4545                                                                           | Electrical Power Systems of the Future                                 | 4 |  |
| ET4117                                                                           | Electrical Machines and Drives                                         | 4 |  |
| ET4291SET                                                                        | Digital modelling of electric powertrain                               | 4 |  |
| SET3065                                                                          | Intelligent Electrical Power Grids                                     | 4 |  |
| SET3080                                                                          | The Necessity of Storage Technology                                    | 4 |  |
| SET3085                                                                          | Hydrogen Technology                                                    | 4 |  |
| SET3095                                                                          | Electronic Power Conversion                                            | 4 |  |
| SET3100                                                                          | Electric vehicle and charging technology                               | 4 |  |
| SET3110                                                                          | Energy Storage in Batteries                                            | 4 |  |
| <b>Electric Mobility &amp; Economics: Electric Mobility Power Economics 2024</b> |                                                                        |   |  |
| EE4536                                                                           | DC and AC Microgrids                                                   | 4 |  |
| EE4545                                                                           | Electrical Power Systems of the Future                                 | 4 |  |
| ET4117                                                                           | Electrical Machines and Drives                                         | 4 |  |
| ET4291SET                                                                        | Digital modelling of electric powertrain                               | 4 |  |
| SET3065                                                                          | Intelligent Electrical Power Grids                                     | 4 |  |
| SET3095                                                                          | Electronic Power Conversion                                            | 4 |  |
| SET3100                                                                          | Electric vehicle and charging technology                               | 4 |  |
| TPM0637SET                                                                       | Economic Policy for Sustainable Energy                                 | 4 |  |
| TPM0931SET                                                                       | Sustainable Energy Innovation and Transitions                          | 4 |  |
| TPM403SET                                                                        | Technology Entrepreneurship and Sustainability                         | 4 |  |
| <b>Track Heating and Cooling 2024</b>                                            |                                                                        |   |  |
| <b>Heat Supply Systems: Sources - LS Heat - Buildings 2024</b>                   |                                                                        |   |  |
| ME45075                                                                          | Refrigeration & Heat Pumps Fundamentals                                | 4 |  |
| ME45111SET                                                                       | Heating and Cooling Demand of Buildings                                | 4 |  |
| ME45165SET                                                                       | Equipment for Heat and Mass Transfer                                   | 4 |  |
| SET3200                                                                          | Heating and Cooling Technologies from Near-Ambient Temperature Sources | 4 |  |
| SET3205                                                                          | Heat Storage                                                           | 4 |  |
| SET3210                                                                          | Heating and Cooling Grids                                              | 4 |  |
| SET3215                                                                          | Heating and Cooling Sources                                            | 4 |  |
| SET3220                                                                          | Heat Distribution in Buildings                                         | 4 |  |
| SET3995                                                                          | Direct Use of Geothermal Energy                                        | 4 |  |
| <b>Heat Grids &amp; Economics: Sources - LS Heat - Economics 2024</b>            |                                                                        |   |  |
| ME45165SET                                                                       | Equipment for Heat and Mass Transfer                                   | 4 |  |
| SET3200                                                                          | Heating and Cooling Technologies from Near-Ambient Temperature Sources | 4 |  |
| SET3205                                                                          | Heat Storage                                                           | 4 |  |
| SET3210                                                                          | Heating and Cooling Grids                                              | 4 |  |
| SET3215                                                                          | Heating and Cooling Sources                                            | 4 |  |
| SET3995                                                                          | Direct Use of Geothermal Energy                                        | 4 |  |
| TPM0637SET                                                                       | Economic Policy for Sustainable Energy                                 | 4 |  |
| TPM0931SET                                                                       | Sustainable Energy Innovation and Transitions                          | 4 |  |
| TPM403SET                                                                        | Technology Entrepreneurship and Sustainability                         | 4 |  |
| <b>Urban Heat &amp; Economics: Sources - Buildings - Economics 2024</b>          |                                                                        |   |  |
| ME45075                                                                          | Refrigeration & Heat Pumps Fundamentals                                | 4 |  |
| ME45111SET                                                                       | Heating and Cooling Demand of Buildings                                | 4 |  |
| SET3200                                                                          | Heating and Cooling Technologies from Near-Ambient Temperature Sources | 4 |  |
| SET3215                                                                          | Heating and Cooling Sources                                            | 4 |  |
| SET3220                                                                          | Heat Distribution in Buildings                                         | 4 |  |
| SET3995                                                                          | Direct Use of Geothermal Energy                                        | 4 |  |
| TPM0637SET                                                                       | Economic Policy for Sustainable Energy                                 | 4 |  |
| TPM0931SET                                                                       | Sustainable Energy Innovation and Transitions                          | 4 |  |
| TPM403SET                                                                        | Technology Entrepreneurship and Sustainability                         | 4 |  |
| <b>Heat &amp; Power Grids: Power - LS Heat - Buildings 2024</b>                  |                                                                        |   |  |
| EE4536                                                                           | DC and AC Microgrids                                                   | 4 |  |
| EE4545                                                                           | Electrical Power Systems of the Future                                 | 4 |  |
| ME45075                                                                          | Refrigeration & Heat Pumps Fundamentals                                | 4 |  |
| ME45111SET                                                                       | Heating and Cooling Demand of Buildings                                | 4 |  |
| ME45165SET                                                                       | Equipment for Heat and Mass Transfer                                   | 4 |  |
| SET3065                                                                          | Intelligent Electrical Power Grids                                     | 4 |  |

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|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----|--|
| SET3095                                                                                          | Electronic Power Conversion                                            | 4  |  |
| SET3205                                                                                          | Heat Storage                                                           | 4  |  |
| SET3210                                                                                          | Heating and Cooling Grids                                              | 4  |  |
| SET3220                                                                                          | Heat Distribution in Buildings                                         | 4  |  |
| <b>Urban Heat &amp; Electricity: Solar - Sources - Buildings 2024</b>                            |                                                                        |    |  |
| ET4376                                                                                           | Photovoltaic Basics                                                    | 4  |  |
| ET4377                                                                                           | Photovoltaic Technologies                                              | 4  |  |
| ET4378                                                                                           | Photovoltaic Systems                                                   | 4  |  |
| ME45075                                                                                          | Refrigeration & Heat Pumps Fundamentals                                | 4  |  |
| ME45111SET                                                                                       | Heating and Cooling Demand of Buildings                                | 4  |  |
| SET3200                                                                                          | Heating and Cooling Technologies from Near-Ambient Temperature Sources | 4  |  |
| SET3215                                                                                          | Heating and Cooling Sources                                            | 4  |  |
| SET3220                                                                                          | Heat Distribution in Buildings                                         | 4  |  |
| SET3995                                                                                          | Direct Use of Geothermal Energy                                        | 4  |  |
| <b>Industrial Heat &amp; Electricity Integration: Waste &amp; CHP - LS Heat - Economics 2024</b> |                                                                        |    |  |
| CH3061                                                                                           | Multiphase Reactor Engineering                                         | 4  |  |
| ME45134                                                                                          | Process and Power Plant Design                                         | 4  |  |
| ME45165SET                                                                                       | Equipment for Heat and Mass Transfer                                   | 4  |  |
| SET3070                                                                                          | Thermochemistry of Biomass Conversion                                  | 4  |  |
| SET3205                                                                                          | Heat Storage                                                           | 4  |  |
| SET3210                                                                                          | Heating and Cooling Grids                                              | 4  |  |
| TPM0637SET                                                                                       | Economic Policy for Sustainable Energy                                 | 4  |  |
| TPM0931SET                                                                                       | Sustainable Energy Innovation and Transitions                          | 4  |  |
| TPM403SET                                                                                        | Technology Entrepreneurship and Sustainability                         | 4  |  |
| <b>Smart Building &amp; Grid Integration: DET - LS Heat - Buildings 2024</b>                     |                                                                        |    |  |
| EE4410                                                                                           | Cyber Security of Power Grids                                          | 4  |  |
| ME45075                                                                                          | Refrigeration & Heat Pumps Fundamentals                                | 4  |  |
| ME45111SET                                                                                       | Heating and Cooling Demand of Buildings                                | 4  |  |
| ME45165SET                                                                                       | Equipment for Heat and Mass Transfer                                   | 4  |  |
| SET3120                                                                                          | Energy Systems Simulation and Digital Twins                            | 4  |  |
| SET3125                                                                                          | Machine Learning Workflows for Digital Energy Systems                  | 4  |  |
| SET3205                                                                                          | Heat Storage                                                           | 4  |  |
| SET3210                                                                                          | Heating and Cooling Grids                                              | 4  |  |
| SET3220                                                                                          | Heat Distribution in Buildings                                         | 4  |  |
| <b>Electives SET 2024</b>                                                                        |                                                                        |    |  |
| IFM4040                                                                                          | Joint Interdisciplinary Project                                        | 15 |  |
| ME45203                                                                                          | Electrolysers, Fuel Cells and Batteries                                | 4  |  |
| ME45204                                                                                          | Electrochemical Energy Storage                                         | 4  |  |
| SET3822                                                                                          | Internship SET                                                         | 15 |  |
| SET4399                                                                                          | Extra Project                                                          | 0  |  |
| WM0320TU                                                                                         | Ethics and Engineering                                                 | 3  |  |
| <b>Language courses 2024</b>                                                                     |                                                                        |    |  |
| TPM301C                                                                                          | Spoken English for International Communication (for Master students)   | 2  |  |
| TPM302B                                                                                          | Spoken English for Academic Purposes Advanced                          | 2  |  |
| TPM305A                                                                                          | Writing a Masters Thesis in English                                    | 2  |  |
| TPM306A                                                                                          | Writing Academic English using AI                                      | 2  |  |
| WM1115TU                                                                                         | Dutch Elementary 1                                                     | 3  |  |
| WM1116TU                                                                                         | Dutch Elementary 2                                                     | 3  |  |
| WM1117TU                                                                                         | Dutch Intermediate 1                                                   | 3  |  |
| <b>Entrepreneurship courses 2024</b>                                                             |                                                                        |    |  |
| TPM406A                                                                                          | Corporate Entrepreneurship and Start-ups                               | 6  |  |
| TPM416A                                                                                          | Turning Technology into Business                                       | 6  |  |
| TPM420A                                                                                          | Ready to Start-up                                                      | 6  |  |
| <b>Profile Electives 2024</b>                                                                    |                                                                        |    |  |
| <b>Waste &amp; Combined Heat and Power 2024</b>                                                  |                                                                        |    |  |
| AE4262                                                                                           | Combustion for propulsion and power technologies                       | 4  |  |
| CH3682A                                                                                          | Reactor & Kinetics (R&K)                                               | 4  |  |
| LM3741                                                                                           | Fermentation Technology & Environmental Biotechnology                  | 6  |  |
| ME45001                                                                                          | Advanced Heat Transfer                                                 | 4  |  |
| ME45026                                                                                          | Introduction to Multiphase Flows                                       | 3  |  |
| ME45070-24                                                                                       | Advanced Reaction and Separation Systems                               | 5  |  |
| ME45155                                                                                          | Computational Fluid Dynamics for Engineers                             | 5  |  |
| ME45160                                                                                          | Advanced Applied Thermodynamics                                        | 5  |  |
| <b>Economics &amp; Society 2024</b>                                                              |                                                                        |    |  |
| SEN1211                                                                                          | Agent-based Modelling                                                  | 5  |  |
| SEN1531                                                                                          | Design of Integrated Energy Systems                                    | 5  |  |
| TPM001B                                                                                          | Sociotechnology of Future Energy Systems                               | 5  |  |
| TPM012B                                                                                          | Environmental Ethics                                                   | 3  |  |
| TPM023C                                                                                          | Cost-Benefit Analysis: Theory and Application                          | 4  |  |
| TPM404B                                                                                          | Technology Entrepreneurship and Internationalization                   | 5  |  |
| TPM411A                                                                                          | Idea to Start-up IT & AI                                               | 5  |  |
| TPM413A                                                                                          | Idea to Start-up Energy & Sustainability                               | 5  |  |
| TPM414A                                                                                          | Idea to Start-up Deep Tech                                             | 5  |  |
| <b>Power Engineering 2024</b>                                                                    |                                                                        |    |  |

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| EE4665                               | Uncertainty modelling and risk assessment in electrical power systems                                | 4  |  |
| EE4C12                               | Machine Learning for Electrical Engineering                                                          | 5  |  |
| ET4103                               | High Voltage Technology                                                                              | 4  |  |
| OE44076                              | Offshore Renewable Energy                                                                            | 6  |  |
| <b>Solar Energy 2024</b>             |                                                                                                      |    |  |
| EE4585                               | Semiconductor Device Physics                                                                         | 5  |  |
| EE4670                               | PV Materials Processing and Characterization                                                         | 4  |  |
| EE4680                               | Photovoltaic Modelling                                                                               | 4  |  |
| ET4379                               | Photovoltaic Lab Course                                                                              | 4  |  |
| ME46065                              | Thin Film Materials                                                                                  | 3  |  |
| <b>Energy Storage 2024</b>           |                                                                                                      |    |  |
| AP3311                               | Neutrons, X-Rays and Positrons for Studying Microscopic Structures and Dynamics                      | 6  |  |
| CH3513                               | Electrochemistry for Renewable Energy                                                                | 4  |  |
| CH3531                               | Functional Ceramics                                                                                  | 3  |  |
| SET3090                              | Fossil-Free Fuel and Feedstock                                                                       | 4  |  |
| SET3135                              | Fuel Cell Systems                                                                                    | 3  |  |
| <b>Wind Energy 2024</b>              |                                                                                                      |    |  |
| AE4135                               | Rotor / wake Aerodynamics                                                                            | 4  |  |
| AE4202                               | CFD for Aerospace Engineers                                                                          | 3  |  |
| AE4205                               | MDO for Aerospace Applications                                                                       | 4  |  |
| AE4T40                               | Airborne Wind Energy                                                                                 | 3  |  |
| AE4W30                               | Wind Resource and Wind Farm Yield                                                                    | 4  |  |
| AE4W31                               | Floating Offshore Wind Energy                                                                        | 3  |  |
| <b>Electric Mobility 2024</b>        |                                                                                                      |    |  |
| EE4515                               | Advanced Power Electronics                                                                           | 4  |  |
| EE4C04                               | Control System Design                                                                                | 5  |  |
| ET4116                               | Power Electronics                                                                                    | 4  |  |
| ET4121                               | A.C. Machines                                                                                        | 4  |  |
| ME41106                              | Intelligent Vehicles 3ME                                                                             | 5  |  |
| RO47017                              | Vehicle Dynamics and Control                                                                         | 5  |  |
| <b>Heat in Buildings 2024</b>        |                                                                                                      |    |  |
| 4413INTPR                            | Integrated Project: Industrial and Urban Systems                                                     | 10 |  |
| AR0097                               | Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost | 5  |  |
| AR0132                               | Zero-Energy Design                                                                                   | 5  |  |
| AR0139                               | MEGA                                                                                                 | 15 |  |
| AR1U131                              | Sustainable Urban Engineering of Territory                                                           | 5  |  |
| RO47005                              | Planning & Decision Making                                                                           | 5  |  |
| SC42001                              | Control System Design                                                                                | 5  |  |
| SC42095                              | Digital Control                                                                                      | 3  |  |
| SEN1131                              | Institutional Economics for Designing in Socio-technical Systems                                     | 3  |  |
| TPM034A                              | Machine learning for socio-technical systems                                                         | 5  |  |
| <b>Heat Sources 2024</b>             |                                                                                                      |    |  |
| CIEM2304                             | Environmental and Energy Geotechnics                                                                 | 5  |  |
| ME45160                              | Advanced Applied Thermodynamics                                                                      | 5  |  |
| ME45225                              | Multiphysics transport in Energy Materials                                                           | 3  |  |
| SEN1522                              | Electricity and Gas: Market Design and Policy Issues                                                 | 5  |  |
| <b>Large Scale Heat Systems 2024</b> |                                                                                                      |    |  |
| CEGM2006                             | Subsurface Storage for Energy, Water and Climate Applications                                        | 10 |  |
| RO47019                              | Intelligent Control Systems                                                                          | 4  |  |
| SC42056                              | Optimization for Systems and Control                                                                 | 3  |  |
| SC42125                              | Model Predictive Control                                                                             | 4  |  |
| SC42130                              | Fault Diagnosis and Fault Tolerant Control                                                           | 4  |  |
| SC42155                              | Modelling of Dynamical Systems                                                                       | 3  |  |
| SC42165                              | Machine Learning for Systems and Control                                                             | 3  |  |

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| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

| <b>Master Sustainable Energy Technology 2024</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>Program Title</b>                             | Sustainable Energy Technology (SET)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>In association with the Faculty of</b>        | BK<br>CITG<br>EWI<br>3mE<br>LR<br>TNW<br>TBM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>In association with the University of</b>     | <p>The Masters degree programme in Sustainable Energy Technology is a 4TU MSc programme. The programme is offered at TU/e (Eindhoven University of Technology), TUD (Delft University of Technology) and UT (University of Twente). The programmes have similar learning objectives. The courses and specialisations at each university are different. Each programme has a different CROHO registration (programme registration according to the WHW).</p> <p>After a student is formally enrolled in the masters programme in Sustainable Energy Technology at one of the 3 universities he or she will also obtain a secondary enrolment (neveninschrijving) at the 2 other universities.</p> <p>Students are allowed to choose elective subjects from the lists of elective, specialisation and core courses from each of the 3 programmes, after consultation with the mentor/supervisor from the home university and after approval by the Board of Examiners of the home university.</p> |
| <b>Program Structure 1</b>                       | <p>The Masters degree programme in Sustainable Energy Technology has two tracks:</p> <ol style="list-style-type: none"> <li>1. Electrical Energy</li> <li>2. Heating and Cooling</li> </ol> <p>An individual exam programme (IEP) is composed as follows:</p> <ol style="list-style-type: none"> <li>1. a common core worth 24 credits;</li> <li>2. a profile cluster worth 36 credits;</li> <li>3. an elective space worth 15 credits and</li> <li>4. a graduation project worth 45 credits.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Program Structure 2</b>                       | <p>The core courses are:</p> <p>SET3014 Renewable Energy<br/> SET3250TPM Communication for Engineers<br/> SET3061 Energy System Modelling<br/> SET3055 Economics and Regulations of Sustainable Energy Systems<br/> SET3815-M Matlab Fundamentals<br/> SET3816-Pr System Integration Project</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Administration by the Faculty of</b>          | EWI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Core Courses 2024

| SET3014                                | Renewable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. A.H.M. Smets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>A transition to renewable energy is needed for our climate and welfare. In this course, you will learn how to assess the potential for energy reduction and the potential of renewable energy sources like hydropower, wind, solar, biomass, geothermal and other environmental sources of heat. The content is an introduction to the tracks in the Sustainable Energy Technology Masters that are organized around the energy carriers heat and electricity. You will be introduced to assessing the energy balance of a country or region and its importance to understand the opportunities and challenges for the transition to renewable energy. The basic principles of energy transition models will be discussed. The physical and chemical principles that determine the impact of our energy consumption on the earth's climate are briefly discussed, as well. This course gives an overview of the most important renewable energy sources, the challenges to incorporate those in power supply, heating and cooling systems and the need for storage. You'll learn how to integrate sources, storage and transportation of (thermal) energy in an energy system, and take an engineering approach to look for solutions and design a 100% sustainable energy system.</p>                                                                                                                                                         |   |
| <b>Study Goals</b>                     | <p>Using physical principles you will be able to:</p> <ul style="list-style-type: none"> <li>quantify, compare and evaluate the use of energy in transport, building, heating/cooling and manufacturing</li> <li>quantify, compare and evaluate the generation of Renewable Energy: how much can you generate from all the different sources? (Hydropower, Wind Energy, Solar Energy, Biomass, Tidal Energy, Wave Energy, geothermal, environmental sources of heat)</li> <li>quantify, compare and evaluate the power density and energy density of various storage technologies</li> </ul> <p>You will be able to analyse:</p> <ul style="list-style-type: none"> <li>large scale energy infrastructure using energy balances and identify corresponding climate effects of such energy infrastructure. the capacity factor, system efficiency, energy pay back time, circularity aspects of energy systems.</li> </ul> <p>In addition, after this course you will be able to make basic quantification an evaluation of the impact of planet earth's albedo, radiative forcing, greenhouse gas emissions on the climate of the planet (temperature, sea levels).</p> <p>You will be able to recognize, quantify and evaluate the technological principles and challenges for various Renewable Energy technologies</p> <p>After this course you will be able to identify and evaluate feasible options for a 100% renewable energy system.</p> |   |
| <b>Education Method</b>                | <p>The education method is the so-called flipped-class room.</p> <p>Lectures will be provided in short video lectures on Brightspace.</p> <p>The on-campus lectures are focused on problem-solving. Students will actively work on problems that cover the entire content of the course and the various engineering skills.</p> <p>Solutions will be discussed during the lectures.</p> <p>Homework for every lecture includes: watching the relevant video lectures and complimentary reading materials dedicated to the upcoming course and the remaining exercises of the previous lecture.</p> <p>Note, for students doing the MSc program Sustainable Energy Technology: the course is aligned with the course WM0201SET (Technical writing Technology) in Q2. The content of the SET3014 course will act as the basis to write a consultancy report in the course WM0201SET in a group of students in Q2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- written exam (100%)</li> </ul> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- written exam: written resit</li> </ul> <p>The exam takes 3 hours.</p> <p>The exam consists of open questions and multiple-choice questions.</p> <p>You can use only pen, paper and a calculator.</p> <p>The exam will be assessed with a mark with a resolution of 0.5.</p> <p>To pass the exam a score equal to or higher than 58.0 out of 100.0 is needed.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SET3055                                | Economics and Regulations of Sustainable Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Prof.dr.ir. L.J. de Vries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | G.O. Ndubuisi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Expected prior knowledge</b>        | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>This course is a general introduction to energy economics. In this course, you learn the essentials of how actors in the energy system make economic decisions, when markets work well and when they may fail, and what options the government has to remedy market failures. Specifically, the following topics are addressed:</p> <p>Introduction to energy (electricity) markets: perfect competition; oligopoly<br/> Rational (and irrational) market behaviour; ABM ?<br/> Investment decision making; LCOE<br/> Opportunity costing / discounting; NPV; IRR; discount rate<br/> Markets, (perfect) competition and market failure (esp. externalities &amp; lock-in)<br/> Private costs, external cost and social cost; empirical evidence (related to energy)<br/> LCOE; discounting; private discount rate; IRR and NPV<br/> Social cost-benefit analysis; social discount rate</p>                     |   |
| <b>Study Goals</b>                     | <p>After this course students are able to:</p> <p>evaluate and analyse the performance of energy (electricity) markets;<br/> identify cases of market failures and design remedial action to rectify these market failures;<br/> estimate the LCOE and calculate the NPV and IRR of an electricity investment project;<br/> contrast private and social costs of (renewable) electricity generation;<br/> assess technological lock-in in energy generation; and<br/> appraise the role of public policy in energy markets and in energy technology innovation.</p>                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | This course consists of lectures. Parallel to the course a simulation game will be played by the students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | To be announced.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Individual computer exam (80%)</li> <li>- Report on the simulation game (20%)</li> </ul> <p>Students should score at least 6 for both parts. All marks will be expressed on a scale of 1 to 10.</p> <p>Final grade calculation: <math>(0.8 * \text{computer exam}) + (0.2 * \text{report})</math></p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Computer exam: resit computer exam</li> <li>- Report on the simulation: resubmission of the report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | G.O. Ndubuisi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |



| SET3061                                | Energy System Modelling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. S.J. Pfenninger-Lee                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Prof.dr.ir. Z. Lukszo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Dr.ir. K. Bruninx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr. F. Lombardi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | This course is an introduction to mathematical optimisation modelling applied to diverse problems in planning and operating energy systems, in particular energy systems with a high share of variable (weather-dependent) wind and solar generation. We will go from a solid theoretical grounding in formulating different kinds of optimisation problems, to practical strategies to solve them under real-world conditions. To do so, we will formulate problems in Matlab using the YALMIP optimisation package.                                                                                                                                       |   |
| <b>Study Goals</b>                     | <p>After completing the course, you should be able to:</p> <ul style="list-style-type: none"> <li>- LO1: Explain the challenges in planning and operating renewable energy systems and the role of mathematical optimisation in addressing these challenges</li> <li>- LO2: Formulate linear, mixed-integer, and non-linear optimisation problems</li> <li>- LO3: Apply different methods to deal with parametric uncertainty in optimisation</li> <li>- LO4: Determine and justify the choice of an optimisation approach to formulate an energy system decision problem</li> <li>- LO5: Formulate and solve optimisation problems using Matlab</li> </ul> |   |
| <b>Education Method</b>                | Lectures and instructions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Relations</b>                | This course cannot be followed together with SEN1511.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>The final grade consists 100% of an individual digital exam with some parts solved and handed in on paper.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                              |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Co-Instructor</b>                   | Prof.dr.ir. Z. Lukszo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-Instructor</b>                   | Dr.ir. K. Bruninx                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| SET3250TPM                             | Communication for Engineers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Drs. R.J. Hoffmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | M. Wackers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Drs. P.C. Post                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 2<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>The course Communication for Engineers consists of two parts: Oral Presentations in Q2 (1EC) and Technical Writing in Q4 (2 EC).</p> <p><b>Oral Presentations</b><br/>The work of modern engineers can only be effective when it is communicated convincingly. In Oral Presentations, students learn how to prepare and deliver convincing and compelling presentations, geared towards a specific audience.</p> <p>The six lectures have the character of workshops in which students work on exercises and prepared individual and group presentations. Through recordings and the exchange of feedback, students are able to improve their presentation skills. Therefore, attendance is mandatory.</p> <p><b>Technical Writing</b><br/>Modern engineers are required to possess excellent communication skills. Not only do they need to know their field of study, but they also need to know how to communicate this to other experts, decision-makers, and implementers. They need to be able to write accessible, persuasive technical reports and proposals and to use a correct tone in professional correspondence.</p> <p>In this course, students in groups write a report on their research in the SIP project. The course focuses on writing technical reports, with an emphasis on both the quality of the text and the writing process.</p> <p>The five lectures and a teacher's consultation will be used for instruction, feedback, and practice. Since the course equals 2EC or 56 hours per student, students will have to put in a significant amount of time outside the classroom to pass this course. Students will be required to hand in several assignments while they are working on their main assignment. Attendance is mandatory.</p> |   |
| <b>Study Goals</b>                     | <p><b>Oral Presentations</b><br/>At the end of this course, students will master a professional level of presenting. They will be able to:</p> <ul style="list-style-type: none"> <li>- adjust academic and technical information to a specific target audience;</li> <li>- use content and rhetorical techniques to make presentations relevant and attractive to the audience;</li> <li>- use voice and body language to support the presentation;</li> <li>- apply a clear structure to a presentation;</li> <li>- use visualisation tools that support the content of the presentation and operate these;</li> <li>- deal professionally with questions from the audience.</li> </ul> <p><b>Technical Writing</b><br/>At the end of the course, students are able to:</p> <ul style="list-style-type: none"> <li>- identify and apply genre-specific characteristics for a technical report</li> <li>- assess whether a technical report meets these characteristics</li> <li>- efficiently write a clear report for a target audience that has little time</li> <li>- apply (visual and textual) structure markers to highlight key elements of the report</li> <li>- write clearly, concisely, and convincingly</li> <li>- write proper and appropriate language</li> <li>- refer correctly to sources, whilst synthesising information from multiple (recent) sources.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | <p>Oral presentations: workshops, exercises, presentations, peer feedback, reflection assignment.<br/>Technical Writing: workshops, peer feedback, presentation, teacher's consultation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>Oral Presentations: Brightspace TU Present.<br/>Technical Writing: Brightspace TU Write.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p><b>Oral Presentations:</b><br/>Final assignment: final presentation in groups. Resit opportunity based on the final presentation (same assignment, same presentation). The teacher decides what the resit entails for the and group and/or individual student(s) and what parts need to be improved.</p> <p>Attendance and formative assignments are mandatory.</p> <p><b>Technical Writing</b><br/>The final assignment is writing a report as a group.<br/>Repair opportunity: re-submit the final assignment, the repair grade will be capped at 6.0.</p> <p>Attendance and formative assignments are mandatory.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| SET3815-M                              | Matlab Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. A. Leki                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>"Learn the fundamentals of the MATLAB scripting language and model-based design using Simulink"</p> <p>Get acquainted with MATLAB and Simulink through a series of interactive lectures focusing on developing real-world programming skills and insight. A series of "activities" during lectures and weekly homework get you to increase your "hands-on" experience. Get to focus on common pitfalls and mistakes and how to avoid them. Get exposed to "good" programming style guidelines so that others can better understand and further work with, or modify your code and designs. You get to practice in a mid-term mini project and demonstrate your newly acquired skills in a final mini project.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>After having passed this course...</p> <ul style="list-style-type: none"> <li>- You will be able to install and run MATLAB/SIMULINK in either Windows, Mac OS, Linux, or Android operating systems.</li> <li>- You will be able to operate MATLAB/SIMULINK, at least basically, either by the graphical user interface or command-line interface.</li> <li>- You will be able to develop, write and evaluate simple programs that run in MATLAB/SIMULINK.</li> <li>- You will be able to import evaluate and export data in native formats and other common data formats.</li> <li>- You will be able to create visualization aids (figures, plots, graphs, charts) that comply with standards in engineering and science.</li> <li>- You will be able to choose and apply the right code and/or model blocks to guarantee compatibility with specific versions of MATLAB/SIMULINK.</li> <li>- You will be able to avoid, identify and correct common pitfalls and mistakes in writing and running code.</li> <li>- You will be able to follow basic design and programming guidelines that make MATLAB/SIMULINK programs easier to read and work with.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Method</b>                | <p>Interactive online lectures using Brightspace as a digital learning platform.</p> <p>Mini-projects carried out in groups form part of the assessment.</p> <p>The lectures are structured as follows:</p> <ol style="list-style-type: none"> <li>1) Theory and concepts are introduced with live programming demonstrations.</li> <li>2) After a block of theory is explained, a MATLAB/SIMULINK "activity" consisting of, e.g. a brief exercise, analyzing an example, or completing a brief questionnaire is carried on by every student, and the deliverable of the activity uploaded to its respective Brightspace folder along with possible questions or remarks on the Brightspace forum.</li> <li>3) If time permits, to finalize the activity its deliverable and posted questions are analyzed and discussed.</li> <li>4) These steps are repeated until the topics for the session are covered.</li> </ol> <p>Delivered activities by the student get a delivery notification before the beginning of the next lecture at the latest. Delivery of activities is compulsory to get a grade.</p> <p>The course load is spread into 8 lectures of two hours each (16 hours). Preparation time for the lectures and homework overhead (delivery of activities) is estimated at a total of 16 hours (approx 2 hours per week). A mid-term project and a final project are budgeted at 12 hours each (24 hours both) of homework time for a total study load of 2 ECTS.</p>                                                                                               |   |
| <b>Computer Use</b>                    | Having access to a PC/laptop with installation rights is mandatory in order to be able to take part in the activities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | <p>This course follows a PASS or FAIL grading scheme structured as follows:</p> <p>Formative assessment:</p> <ol style="list-style-type: none"> <li>1) In the form of "activities" that are introduced, analyzed, and if time permits solved and discussed during lectures. Activities can (and have to) be delivered at the latest before the beginning of the next lecture. Activities get a delivery acknowledgment. A student must have at least 80% of the activities delivered to get a final grade.</li> <li>2) In the form of a "mid-term" project carried on in groups, to be delivered at around half the course schedule. A lecture session is devoted for groups to peer-review and analyze the results of this project. Delivery of this project is compulsory to get a final grade.</li> </ol> <p>Summative assessment:</p> <p>A final project carried on in groups has to be delivered at a date announced in Brightspace. Your project deliverable gets assessed by the lecturer. It has to reflect that you (each student in your group) have fulfilled the learning objectives to get a PASS mark.</p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Project: resubmit deliverable</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Tags</b>                            | Matlab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| SET3816-PR                             | System Integration Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. R. Vismara                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Responsible Instructor</b>          | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>QUARTER 2</p> <p>A series of lectures will provide you with the background knowledge of electricity and heating systems necessary to complete the project.</p> <p>The lectures cover the following topics:</p> <ul style="list-style-type: none"> <li>- Fundamentals of power engineering.</li> <li>- Fundamentals of three-phase systems.</li> <li>- Fundamental of power conversion.</li> <li>- Distributed renewable electricity generation (solar and wind).</li> <li>- Battery energy storage.</li> <li>- Fundamentals of geothermal heat generation.</li> <li>- Fundamentals of heat storage.</li> <li>- Economic and environmental aspects of energy systems.</li> </ul> <p>QUARTER 3</p> <p>You will work in groups to design and model an integrated energy (electricity + heat) system.</p> <p>The project is divided into three phases:</p> <ol style="list-style-type: none"> <li>1. Design and modelling of the electricity and heat generation sub-systems.</li> <li>2. Design and modelling of the electricity and heat storage sub-systems.</li> <li>3. Integration of the sub-systems and design of a system control strategy.</li> </ol> <p>QUARTER 4</p> <p>You will perform an economic and environmental analysis of the system and prepare a scientific report summarising the project's main results.</p>                                                                                                                                                                                                                                                                 |   |
| <b>Study Goals</b>                     | <p>By the end of the course, you should be able to:</p> <ol style="list-style-type: none"> <li>1. Design and model an electrical system able to satisfy the electricity demand of a (small) community.</li> <li>2. Design and model a heating system able to satisfy the heat demand of a (small) community.</li> <li>3. Integrate different electrical and heating sub-systems into a working energy system.</li> <li>4. Design and implement a control strategy that maintains the energy balance of the integrated system.</li> <li>5. Analyse the economic and environmental impact of the project.</li> <li>6. Communicate the project process and results in a clear and scientific manner.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>QUARTER 2</p> <p>Lectures, an individual formative assignment, and teacher feedback.</p> <p>QUARTER 3</p> <p>Group work, feedback sessions, and intermediate formative assignments.</p> <p>QUARTER 4</p> <p>Group work and feedback sessions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>The following material will be available on Brightspace:</p> <ul style="list-style-type: none"> <li>- Lecture slides.</li> <li>- List of available resources for project data.</li> <li>- Assignment instructions and templates.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>QUARTER 2</p> <p>In Q2, you will:</p> <ul style="list-style-type: none"> <li>- Complete a series of individual formative assessments to prepare you for the system integration project.</li> <li>- Submit an individual summative assessment focused on energy storage.</li> </ul> <p>QUARTER 3</p> <p>In Q3, you will:</p> <ul style="list-style-type: none"> <li>- Submit two intermediate reports (formative assessment).</li> <li>- Peer review two other team's intermediate reports (summative assessment).</li> <li>- Deliver an oral presentation with the main project results (summative assessment).</li> </ul> <p>QUARTER 4</p> <p>In Q4, you will submit a final report including all steps and results of the project (summative assessment).</p> <p>The final grade will be calculated as follows:</p> <ul style="list-style-type: none"> <li>- 10% individual assessment.</li> <li>- 10% peer reviews.</li> <li>- 30% oral presentation.</li> <li>- 50% final report.</li> </ul> <p>Individual grades will be adjusted following an intra-team peer review (conducted with BuddyCheck) assessing each member's contributions to the group's work.</p> <p>For more details about assessment rules and methodology, we refer you to RRoBE Art 17.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| SET3901                                | Graduation Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 45 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x<br>Based on appointments with supervisor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Education Period</b>                | None (Self Study)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Exam Period</b>                     | Exam by appointment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Course Contents</b>                 | <p>The Graduation Project (SET3901) is the last study unit of the MSc SET programme. The project has a study load of 45 EC, which is equivalent to a full-time duration of 7,5 months.</p> <p>The topic of the project should fall within the chosen profile cluster, with the exception that it can fall under a fourth profile that the student has followed as part of their elective space.</p> <p>In the Graduation Project a student works independently on a research project, under guidance of one or more supervisors. Important is that system integration is addressed in the project. The topic can be defined in several ways:</p> <ul style="list-style-type: none"> <li>- A specialized topic addressing a specific component of the energy system within one profile. In that case the relation with the energy system must also be demonstrated in the report and during the defence.</li> <li>- A topic on system integration within one profile.</li> <li>- A topic on system integration across two or more profiles in a combined project. In that case one of the profiles will take the lead, but the student will have supervisors from all profiles.</li> </ul> |    |
| <b>Study Goals</b>                     | <p>After the Graduation Project the student should be able to:</p> <ul style="list-style-type: none"> <li>- design a SET research project</li> <li>- execute a research project</li> <li>- defend the research project</li> <li>- write a research report</li> <li>- present the research project</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Education Method</b>                | <p>Individual project</p> <p>The graduation project, as determined in the EEMCS graduation policy (see TER, IR Art 3, sub 2), follows certain steps:</p> <ol style="list-style-type: none"> <li>1. Orientation</li> <li>2. Kick-off meeting</li> <li>3. Midterm review</li> <li>4. Greenlight review</li> <li>5. Defense Day (defense and assessment)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
| <b>Prerequisites</b>                   | <p>Before being allowed to start the graduation project, the student needs to have an Individual Exam Programme (IEP) approved by the Board of Examiners and at least 60 credits of the Master's degree course work as stated in the IEP should be completed.</p> <p>Before starting the project, the student, the responsible supervisor, and the profile leader agree that the student will conduct a graduation project under the responsibility of the responsible supervisor, if necessary, with the support of a daily supervisor. The student and responsible (and daily) supervisor:</p> <ul style="list-style-type: none"> <li>- Agree on a starting date;</li> <li>- Agree on what is expected during the kick-off meeting.</li> </ul> <p>The agreements are written in the Thesis Proposal Form. This form needs to be signed by the student, the responsible supervisor, and the relevant profile leader, and then uploaded in MaRe. This is done just before or after the kick-off meeting, but prior to starting work on the Graduation Project.</p>                                                                                                                        |    |
| <b>Assessment</b>                      | <p>On behalf of the Board of Examiners, the thesis committee assesses the graduation project on the day of the presentation and defence of the project, based on the degree to which the study goals have been reached. The core members of the thesis committee determine the final grade.</p> <p>The justification for the marks for each of the study goals is summarized on the assessment form, which is signed by the chair of the thesis committee.</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Report: resubmit the report</li> <li>- Defense: resit presentation and defense</li> </ul> <p>Depending on the planning of the student it might not be possible to repair the report and/or the presentation within the same academic year.</p>                                                                                                                                                                                      |    |
| <b>Special Information</b>             | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>The Graduation Project can be carried out at Delft University of Technology or at an external company or institute. The project needs to be carried out according to MSc SET guidelines of Delft University of Technology. This is safeguarded by the requirement to have a responsible supervisor who is on the Green List of the MSc SET programme for the most relevant profile.</p> <p>NB: Internship SET (SET3822) can never be part of a Graduation Project (SET3901) and it is not allowed to carry out both an internship and the graduation project at the same company. Likewise, Extra Project (SET4399) cannot be used as preparation for Graduation Project (SET3901).</p> <p>The Graduation Project (SET3901) is subject to strict rules and procedures. Enrol in the Brightspace page MSc SET Thesis Project EEMCS for more information. One can find the step-by-step guide explaining all procedures, forms, and additional information e.g. about the thesis assessment.</p>                                                                    |    |
| <b>Co-Instructor</b>                   | Dr.ir. W.A.A.M. Bierbooms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| <b>Co-Instructor</b>                   | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Co-Instructor</b>                   | Prof.dr.ir. A.H.M. Smets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |
| <b>Co-Instructor</b>                   | L. Botto                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Profile Courses SET 2024**

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Track Electrical Energy 2024

Year

Organization

Education

2024/2025

Electrical Engineering, Mathematics and Computer Science

Master Sustainable Energy Technology (SET)

|                                         |
|-----------------------------------------|
| Solar Systems: Solar Power Storage 2024 |
| Introduction 1                          |



| EE4536                                 | DC and AC Microgrids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>          | Dr.ir. L.M. Ramirez Elizondo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | BSc Electrical Engineering or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | Identify the challenges introduced by distributed generation<br>Describe different microgrids' layouts and sizing<br>Describe the operation and control of power electronic converters in AC and DC microgrids<br>Apply droop control in AC and DC microgrids<br>Describe the role of energy management systems in AC and DC microgrids<br>Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiorgiou                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Written exam 70%, Assignments 30%<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                           |   |

| EE4545                                  | Electrical Power Systems of the Future                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>           | Dr. A. Leki                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>         | <p>Pre-requisites (entry levels acquired in Q1 and Q2):</p> <p>Master SET / Master EE / Master EWEM<br/>           SET3065 Intelligent Electrical Power Grids<br/>           ET4107 Power System Analyses II<br/>           Matlab and Simulink skills (self study tutorial is available)</p> <p>Optional<br/>           EE4665 Uncertainty modelling and risk assessment in electrical power systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | <p>The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system.</p> <p>Learning objectives(LOs):</p> <p>LO1: Explain the impact of high RES penetration on the resiliency of future power systems.</p> <p>LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration.</p> <p>LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply.</p> <p>LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.</p> |   |
| <b>Education Method</b>                 | <p>Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site</p> <p>Estimated study load: 112 hours (4 ECTs)<br/>           Lectures 20 hours<br/>           Practical 24 hours<br/>           Self study 68 hours</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                       | <p>Practicum assignments and digital exam/re-exam</p> <p>The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                          | <p>Company visit in combination with lectures and practicum<br/>           @ Mariëndaal Centre of Excellence, TenneT Arnhem<br/>           Dont forget identity card or passport!</p> <p>8:00 Departure from EWI, Travel by touring bus to Arnhem.<br/>           Lectures on Operational challenges by RES for system operations.<br/>           Lessons learnt from large disturbances.<br/>           Visit TenneT Control Room (including lunch).<br/>           Practical Exercises Operational challenges and large disturbances.<br/>           Travel by touring bus to Arnhem<br/>           16:30 Arrival at EWI</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>maximum aantal deelnemers</b>        | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| ET4376                          | Photovoltaic Basics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Coordinator              | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Contact Hours / Week<br>x/x/x/x | 0/4/0/0 lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Education Period                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Start Education                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Exam Period                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Contents                 | This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised. |   |
| Study Goals                     | Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Education Method                | Lectures, exercises, instructions and self-study using the book Solar Energy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Literature and Study Materials  | Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Assessment                      | During instruction sessions you will practice 'exam level' questions. These questions are intended to prepare you for the exam, but are not graded. The final grade is 100% determined by the written exam. All marks will be expressed on a scale of 0 to 10.<br><br>In case of an insufficient result for the written exam, you have the opportunity to take the resit exam in the next quarter.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| ET4377                                 | Photovoltaic Technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Dr. L. Mazzarella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr. I.M.F. Gordon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0 lectures; 0/0/2/0 instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior knowledge</b>        | Lecture course Photovoltaic Basics (ET4 376)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed.</p> <p>Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.</p>                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required.</p> <p>The learning objectives (LO) are as follows:</p> <p>LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation.<br/> LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design.<br/> LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Flipped-classroom video lectures and on-campus instructions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England.</p> <p>ISBN: 978 1 906860 32 5 (paperback);</p> <p>This book is also available as a free ebook:</p> <p>ISBN: 978 1 906860 73 8 (ePub)<br/> ISBN: 978 1 906860 75 2 (pdf)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10.</p> <p>For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account.</p> <p>The final mark, F, for this course will be computed as follows: <math>F = E + [A \times (10 - E)/100]</math>. The computation will be carried out using one decimal and subsequently rounded to halves.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| ET4378                                 | Photovoltaic Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. M.R. Vogt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr.ir. H. Ziar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2 lectures and 0/0/0/2 lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>        | - Basic knowledge of photovoltaics (e.g. PV Basics course ET4376)<br>- Good command of Matlab scripting language (or equivalent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.                                                                                                                                                                                                                                          |   |
| <b>Study Goals</b>                     | By the end of this course, you will be able to:<br>1. Explain the purpose of the components in different PV Systems using trends in their physical quantities.<br>2. Examine different power tracking methods and the importance of bypass diodes.<br>3. Calculate the energy yield of a PV system in conjunction with the following steps:<br>3a. Employ the calculation for optimal PV panel orientation from geographical survey.<br>3b. Determine the advantages and disadvantages of different PV systems topologies.<br>4. Design PV systems for different applications based on your own cost estimation and technical evaluation justified in a report                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Lectures, self study and teaming up for carrying out the design of an actual PV System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | - Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available)<br>- Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library)<br>- Handouts<br>- Scientific articles<br>- PV Systems course in "Solar Engineering MicroMasters" ( <a href="https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/">https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/</a> )<br>- PV Resources ( <a href="http://www.pvresources.com/">http://www.pvresources.com/</a> )<br>- PVGIS ( <a href="http://re.jrc.ec.europa.eu/pvgis/">http://re.jrc.ec.europa.eu/pvgis/</a> )<br>- Sandia National Laboratories PVPerformance ( <a href="https://pvpmc.sandia.gov/">https://pvpmc.sandia.gov/</a> )<br>- Dutch PV Portal 3.0 ( <a href="https://www.tudelft.nl/dutchpvportal">https://www.tudelft.nl/dutchpvportal</a> )                                                                                 |   |
| <b>Assessment</b>                      | For students of the 2024 cohort: Final project report (33% of the final mark) and written test (67% of the final mark).<br><br>For students of previous cohorts: Final project report (25% of the final mark) and written test (75% of the final mark).<br><br>Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual.<br><br>To access to the written test, the grade of the final project report must be equal to or greater than 5.75.<br><br>To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75.<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). |   |

| SET3065                                | Intelligent Electrical Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. P. Palensky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. J.L. Rueda Torres                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets.</p> <p>Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools.</p> <p>You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done.</p> <p>This course is part of the Power track of the SET master and can also be chosen by EE master students.</p> <p>Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.</p> |   |
| <b>Study Goals</b>                     | <p>After this course students are able to</p> <p>LO1: Describe the functions and topologies of cyber-physical energy systems</p> <p>LO2: Implement data driven modelling and numerical simulation of energy systems</p> <p>LO3: Calculate variable power flows based on optimization/probability theory</p> <p>LO4: Synthesize functional requirements of energy management systems</p> <p>LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized in the following units, all units have assignments (typically a simulation):</p> <ol style="list-style-type: none"> <li>1: The energy transition and future energy systems (Peter)</li> <li>2: Analysis of steady-state performance in variable operating conditions (Jose)</li> <li>3: Assessment and control of power system dynamic performance (Jose)</li> <li>4: Lab 1 Digital and data driven modelling of interconnected systems</li> <li>5: Lab 2 Simulation and impact analysis of renewables, storage, and demand</li> <li>6: Data acquisition and communication for energy automation (Peter)</li> <li>7: Artificial intelligence-based functionalities for situational awareness (Pedro &amp; Jochen)</li> <li>8: Cyber security of smart grids (Alex)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | On Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <p>Assignment reports (in time!) (50%)</p> <p>Multiple choice Computer Exam (50%)</p> <p>In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture.</p> <p>Final grade calculation: <math>(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Assignment reports (in time!): resubmit deliverable</li> <li>- Multiple choice Computer Exam: computer resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                    |   |

| SET3080                                | The Necessity of Storage Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <p>In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy.</p> <ul style="list-style-type: none"> <li>- Week 1: Future energy generation systems, the combination of sources on a large scale.</li> </ul> <p>Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel</p> <ul style="list-style-type: none"> <li>- Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc.</li> </ul> <p>Group work 2: program in Matlab typical use patterns for the different sectors</p> <ul style="list-style-type: none"> <li>- Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses.</li> </ul> <p>Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year.</p> <ul style="list-style-type: none"> <li>- Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities.</li> </ul> <p>Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency.</p> <ul style="list-style-type: none"> <li>- Week 5: Thermal storage, high T and low T. Combined heat and power possibilities.</li> </ul> <p>Group work 5: Investigate energy densities, efficiencies, capacities, exergies.</p> <ul style="list-style-type: none"> <li>- Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean.</li> </ul> <p>Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers.</p> <ul style="list-style-type: none"> <li>- Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.</li> </ul> |   |
| <b>Study Goals</b>                     | <p>At the end of the lectures and group work the students should be able to:</p> <ul style="list-style-type: none"> <li>- Reproduce and reflect on the role of storage in a renewable energy based society.</li> <li>- Explain how different timescales for energy storage are relevant for the future system.</li> <li>- List and explain the different storage solutions in terms of typical energy density, cost and efficiency.</li> <li>- Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy.</li> <li>- Reflect on the energy efficiency of intermediate storage.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time.</p> <p>The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week.</p> <p>The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | <p>Slides</p> <p>Articles, reports and reviews.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <p>Individual written exam (70% of the final grade).</p> <p>Group work (30% of the final grade, based on a report produced by the group).</p> <p>For one part may be scored 5,0 as long as the other part compensates this.</p> <p>Final grade calculation: <math>(0,7 * \text{written exam}) + (0,3 * \text{group project})</math></p> <p>Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- written exam: written resit</li> <li>- group work: resubmit report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| SET3085                                | Hydrogen Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed. |   |
| <b>Study Goals</b>                     | <p>After this course students:</p> <p>Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms.</p> <p>Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it.</p> <p>Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.</p>                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>This course consists of 14 2-hour lectures.</p> <p>These lecture also include some problem solving.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Literature and Study Materials</b>  | The exam will be based on the lecture slides and on selected research papers as provided on Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written exam (100%)</li> </ul> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                         |   |
| <b>Co-Instructor</b>                   | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| SET3095                                | Electronic Power Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | S.A. Rivera Iunissi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.                                                                                                                                                                                                                                                                 |   |
| <b>Study Goals</b>                     | To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | <p>Lectures</p> <p>Problem solving sessions</p> <p>Practicum Lectures</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written final exam (closed book) (70%)</li> <li>- Assignments (30%)</li> </ul> <p><math>T = \text{grade for written final exam} * 0,7 + \text{grade for assignments} * 0,3</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |



| SET3110                                | Energy Storage in Batteries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M. Wagemaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                      | A. Vasileiadis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Dr. E.M. Kelder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | T. Famprakis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr. X. Wang                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar.</p> <p>In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including:</p> <p>Redox Flow<br/>Na-ion<br/>next generation Li-ion batteries<br/>Solid State batteries<br/>Supercapacitors</p>                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>At the end of the battery lectures students should be able to:</p> <ul style="list-style-type: none"> <li>- List the major types of Li and other batteries and their characteristics.</li> <li>- Reproduce the major challenges in batteries.</li> <li>- Explain how a battery works and the functioning of each component.</li> <li>- List and explain the different reaction types in batteries.</li> <li>- Explain the different charge transfer processes in batteries.</li> <li>- List and explain the different mechanisms that limit the cycle life of batteries.</li> <li>- Apply Nernst Law describing redox reactions.</li> <li>- Calculate conductivities and overpotentials.</li> <li>- Apply the Tafel equation describing the charge transfer between electrode and electrolyte.</li> <li>- Calculate energy densities and power densities from observed properties.</li> <li>- Derive from a phase diagram the different phases that will occur upon lithiation.</li> <li>- Apply the Gibbs Phase rule to predict constant of variable output voltage.</li> <li>- Derive the voltage curve from the shape of the Gibbs Free energy.</li> <li>- Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials.</li> <li>- Compare different battery systems and differentiate.</li> <li>- Discuss scientific results observed in battery performance and to relate this to material properties.</li> </ul> |   |
| <b>Education Method</b>                | <p>The course contains fourteen 2-hour lectures.</p> <p>The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery.</p> <p>In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed.</p> <p>In the final lecture example exams will be discussed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | R.A. Huggins: Advanced Batteries. Available via the library as pdf's: <a href="http://www.springerlink.com/content/nq6404/?MUD=MP">http://www.springerlink.com/content/nq6404/?MUD=MP</a><br>Slides<br>Articles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <p>Written exam (100% of the final grade)</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Co-Instructor</b>                   | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>Dr. E.M. Kelder</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

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|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Waste-CHP & Solar Systems: Waste & CHP Solar Storage 2024

| CH3061                                 | Multiphase Reactor Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. J.R. van Ommen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | <p>The student is expected to:</p> <ol style="list-style-type: none"> <li>1. have followed a course on mass and energy balances (such as Molecular Transport Phenomena)</li> <li>2. have followed a basic course in reactor engineering (such as Reactors and Kinetics)</li> <li>3. is able to work with Python (or similar software; e.g., has followed a course on numerical techniques)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                 | <p>Multiphase reactor types: Fixed beds, trickle beds, fluidized beds, bubble columns, slurry reactors, microreactors, structured (monolith) reactors, electrocatalytic reactors.</p> <p>Engineering aspects: Flow regimes, mass transfer, conversion models. Hatta number, reaction enhancement. Residence time distribution. Catalytic reactors structured in time and space. Unifying concepts.</p> <p>Design aspects: reactor selection, decoupling of kinetics and transport phenomena, combination of reaction and separation, reaction coupling, separating catalytic steps, controlled energy input, circularity of materials.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Study Goals</b>                     | <ol style="list-style-type: none"> <li>1. The student is able to describe for each of the archetypes of multiphase reactors, the various embodiments with practical relevance.</li> <li>2. The student is able to use the scientific literature to find relevant information for the various reactor types.</li> <li>3. The student is able to analyse a multiphase reactor and to describe the physical and chemical process steps in words.</li> <li>4. The student is able to explain the basic measurement techniques used in studying multiphase reactors.</li> <li>5. The student is able to write down the mass, energy, and momentum balances for the overall system and the steps identified under objective 3.</li> <li>6. The student is able to implement customary models in the balances mentioned under 5, such that a system of equations describing the reactor is obtained.</li> <li>7. The student is able to simplify the system of equations based on a quantitative analysis and identification the rate-determining steps.</li> <li>8. The student is able to use Python to solve the systems of equations numerically for the basic types of reactors.</li> <li>9. The student is able to use the developed models to analyse defined cases.</li> <li>10. The student is able to present, both orally and in writing, the outcomes of modeling, calculations, and evaluation.</li> </ol> |   |
| <b>Education Method</b>                | <p>Each lecture will give a short overview of a specific multiphase reactor type (only 45 minutes). The slides available on Brightspace. You will get references to papers and/or chapters from textbooks that give more information about the reactor type. These will be made available via Brightspace.</p> <p>After the lecture, an assignment (about 8 hours work) will be available via Brightspace for the specific reactor type treated. These assignments are made in teams of four students; this number may be changed depending on the number of participants. You are free to choose which software package you want to use for making the assignments, but Python is recommended.</p> <p>After two weeks, each team hands in a detailed solution of the problem on paper the day before the discussion before the announced deadline. Moreover, each team prepares a summary of the solution for presentation in PowerPoint. Please bring this presentation on a memory stick. Each week, a few teams present their solution; the instructors choose the student that is presenting. The other teams challenge this solution.</p>                                                                                                                                                                                                                                                                  |   |
| <b>Computer Use</b>                    | In this course we use Python.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | All material is provided via Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The final grade consists of:</p> <ul style="list-style-type: none"> <li>* the solution to the assignments</li> <li>* an individual part; the nature of this part will depend on the number of participants</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| ET4376                          | Photovoltaic Basics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Coordinator              | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Contact Hours / Week<br>x/x/x/x | 0/4/0/0 lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Education Period                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Start Education                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Exam Period                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Contents                 | This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised. |   |
| Study Goals                     | Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Education Method                | Lectures, exercises, instructions and self-study using the book Solar Energy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Literature and Study Materials  | Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Assessment                      | During instruction sessions you will practice 'exam level' questions. These questions are intended to prepare you for the exam, but are not graded. The final grade is 100% determined by the written exam. All marks will be expressed on a scale of 0 to 10.<br><br>In case of an insufficient result for the written exam, you have the opportunity to take the resit exam in the next quarter.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| ET4377                                 | Photovoltaic Technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Dr. L. Mazzarella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr. I.M.F. Gordon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0 lectures; 0/0/2/0 instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior knowledge</b>        | Lecture course Photovoltaic Basics (ET4 376)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed.</p> <p>Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.</p>                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required.</p> <p>The learning objectives (LO) are as follows:</p> <p>LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation.<br/> LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design.<br/> LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Flipped-classroom video lectures and on-campus instructions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England.</p> <p>ISBN: 978 1 906860 32 5 (paperback);</p> <p>This book is also available as a free ebook:</p> <p>ISBN: 978 1 906860 73 8 (ePub)<br/> ISBN: 978 1 906860 75 2 (pdf)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10.</p> <p>For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account.</p> <p>The final mark, F, for this course will be computed as follows: <math>F = E + [A \times (10 - E)/100]</math>. The computation will be carried out using one decimal and subsequently rounded to halves.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| ET4378                                 | Photovoltaic Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. M.R. Vogt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr.ir. H. Ziar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2 lectures and 0/0/0/2 lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>        | - Basic knowledge of photovoltaics (e.g. PV Basics course ET4376)<br>- Good command of Matlab scripting language (or equivalent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.                                                                                                                                                                                                                                          |   |
| <b>Study Goals</b>                     | By the end of this course, you will be able to:<br>1. Explain the purpose of the components in different PV Systems using trends in their physical quantities.<br>2. Examine different power tracking methods and the importance of bypass diodes.<br>3. Calculate the energy yield of a PV system in conjunction with the following steps:<br>3a. Employ the calculation for optimal PV panel orientation from geographical survey.<br>3b. Determine the advantages and disadvantages of different PV systems topologies.<br>4. Design PV systems for different applications based on your own cost estimation and technical evaluation justified in a report                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Lectures, self study and teaming up for carrying out the design of an actual PV System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | - Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available)<br>- Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library)<br>- Handouts<br>- Scientific articles<br>- PV Systems course in "Solar Engineering MicroMasters" ( <a href="https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/">https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/</a> )<br>- PV Resources ( <a href="http://www.pvresources.com/">http://www.pvresources.com/</a> )<br>- PVGIS ( <a href="http://re.jrc.ec.europa.eu/pvgis/">http://re.jrc.ec.europa.eu/pvgis/</a> )<br>- Sandia National Laboratories PVPerformance ( <a href="https://pvpmc.sandia.gov/">https://pvpmc.sandia.gov/</a> )<br>- Dutch PV Portal 3.0 ( <a href="https://www.tudelft.nl/dutchpvportal">https://www.tudelft.nl/dutchpvportal</a> )                                                                                 |   |
| <b>Assessment</b>                      | For students of the 2024 cohort: Final project report (33% of the final mark) and written test (67% of the final mark).<br><br>For students of previous cohorts: Final project report (25% of the final mark) and written test (75% of the final mark).<br><br>Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual.<br><br>To access to the written test, the grade of the final project report must be equal to or greater than 5.75.<br><br>To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75.<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). |   |

| ME45134                                | Process and Power Plant Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr. L.L. Cutz IJchajchal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | Design of equipment for mass and heat transfer<br>Thermodynamics of mixtures and phases separation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>This course provides the fundamentals for conceptually designing processes of chemical and energy conversion plants, where basic unit operations such as reactions, energy conversion, heat and mass transfer, separations, etc. are designed, modeled and optimized.</p> <p>The conceptual process design involves the design of the flow sheet, creating the topology of the process by the optimizing the sequence of reactors, separation units and energy conversion systems to achieve the required product specification in a sustainable and economically feasible way.</p> <p>In the course the students learn how to select, sequence and model the different unit operations as well as the selection and rough sizing of the required equipment. In addition the students will learn how to integrate the different units into a process flow sheet, and how to model, optimize and rank of the different design alternatives.</p> <p>During the course the students get a hands-on experience with modern process design tools (a.o. Aspen Plus), which will be used in a larger assignment in which a conceptual design of a sustainable, and (if possible) economically feasible process will be setup and simulated.</p> <p>The course includes the following elements:</p> <ul style="list-style-type: none"> <li>Intro; context and scope of conceptual process design;</li> <li>Modelling the thermodynamics of mixtures</li> <li>Modelling of key unit operations</li> <li>Hierarchical approach (Douglas method) to conceptual process design <ul style="list-style-type: none"> <li>o reaction and separation systems</li> <li>o homogeneous systems</li> <li>o heterogeneous systems</li> <li>o steam power plant system</li> </ul> </li> <li>Simulation and optimization</li> <li>Superstructures</li> <li>Costing and project evaluation</li> <li>Safety, Health and Environmental impact considerations</li> <li>Flow-sheet design using Aspen</li> <li>Presentation (pre-recorded) of the case studies</li> </ul> |   |
| <b>Study Goals</b>                     | <p>The student can conceptually design and optimize chemical process plants.</p> <p>More specifically, the student must be able to:</p> <ol style="list-style-type: none"> <li>1. describe the context and scope of a conceptual design of a chemical plant,</li> <li>2. model and design the most important unit operations of chemical/energy conversion process plants,</li> <li>3. make a conceptual process design using the hierarchical Douglas approach for process design</li> <li>4. select, sequence and model the different unit operations in the flowsheet of a chemical process plant and determine the main dimensions of the required reaction/separation equipment,</li> <li>5. apply process flowsheeting tools for the design and simulation of process flowsheets for a chemical plant including the recycle streams</li> <li>6. Use the process simulation tools to evaluate, optimize and rank the different design alternatives for a process.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Method</b>                | Lectures (plenary and working), assignments, pre-recorded presentations and oral examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | <p>Lecture slides</p> <p>Book (selected chapters): Robin Smith<br/>Chemical Process Design and<br/>John Wiley &amp; Sons, U.S.A. (either first or second edition)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | The examination will be as follows: Oral exam [50%, must be 6 or higher] + one presentation of the block scheme prerecorded [10%] and one final report on the process design [32%] and a presentation about it (prerecorded) [8%]. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: exactly the same. The assessment is Covid-19 independent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| SET3070                                | Thermochemistry of Biomass Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
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| <b>Responsible Instructor</b>          | Dr. L.L. Cutz IJchajchal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Expected prior knowledge</b>        | To start this course you need to have a good knowledge of BSc. level algebra and calculus, thermodynamics, chemistry, kinetics, basic reactor modelling (batch, plug flow, continuously well stirred reactor), basic process engineering (mass, energy, momentum balances), and physical transport phenomena. In case you think you miss some of this pre-existing knowledge, please perform selfstudy of the first six chapters of the book 'Biomass as a Sustainable Energy Source for the Future', Wiley (2015), eds. De Jong and Van Ommen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Summary</b>                         | Biomass is available in abundance on a global level; it is naturally stored solar energy in a wide diversity of chemical structures. Biomass is a renewable resource for heat, electricity, liquid fuels, chemicals and materials. Among the renewable energy sources, biomass can be stored most easily, however it is contingent on the resources characteristics. Due to the heterogeneous nature of biomass, several conversion technologies have been developed in the context of energy conversion chains, with combustion being the oldest performed by mankind. This course deals with the background, design and implementation of different technologies to thermochemically convert biomass into valuable products.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | In this course you will learn about the following aspects of biomass conversion based on thermo-chemical conversion technologies.<br><br>- Biomass Characterisation (biomass types, properties, analyses)<br>- Pretreatment (harvesting, size reduction, drying, torrefaction)<br>- Thermochemical conversion (combustion, gasification and pyrolysis; reactor types)<br>- Transportation fuels / biorefinery (syngas conversion, economics)<br>- Process design (structured design approach)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                     | During this course, the students will gain knowledge of and insight into energy conversion technologies based on biomass thermo-chemical conversion processes for a wide range of applications, such as supply of electricity, heat and fuels. They will be provided with practical problems to solve and projects to perform in this field.<br><br>These main learning objectives can be broken down into several smaller learning objectives:<br><br>1) The student has knowledge on the typical biomass composition, related properties (like e.g. heating value, ash melting behaviour) and can critically identify and select fuel characterization techniques.<br>2) The student is able to identify and perform basic capacity calculations related to thermo-chemical biomass conversion systems for small and larger scale (combined) heat and power generation, as well as systems for producing transportation fuel.<br>3) The student is able to set up reaction equations (stoichiometry) for a wide range of energy supply related fuel conversion processes.<br>4) The student can set up and perform heat and mass balance calculations regarding single phase and simple multi-phase biomass thermo-chemical conversion reactors and extended (downstream) systems for power, heat and fuels generation.<br>5) The student knows how to perform basic ideal gas phase chemical reaction equilibrium calculations to calculate product compositions and extents of reactions given initial reaction mixture composition, pressure and temperature.<br>6) The student can understand and describe reaction kinetic expressions and also apply them in order to solve (design) problems related to the conversion of fuels and the formation of emission components.<br>7) The student demonstrates good knowledge of mass and heat transfer mechanisms and phenomena applied to the chemical conversion of both liquid and solid biomass (derived) fuels.<br>8) The student can derive equations for idealised model reactors (well-stirred reactor and plug flow reactor concepts) and apply these simplified reactor concepts to solve engineering problems related to the thermo-chemical conversion of biomass fuels, in particular combustion, gasification, pyrolysis and hydrothermal liquefaction.<br>9) The student is able to perform calculations concerning the conversion of small particles; important in this respect is the identification of the regime of conversion. Related dimensionless numbers such as the Thiele Modulus and the Biot number are key indicators in this respect.<br>10) The student is able to understand the basics of thermo-chemical conversion processes and to translate this into designing and evaluating such (sometimes hybrid) systems, including their economic evaluation based on characteristic parameters like NPV, POT, and ROI.<br>11) The student demonstrates good communication skills, both oral and written, concerning presentation of (applied) scientific results. |   |
| <b>Education Method</b>                | This course consists of six 2-hour lectures, also using blended learning methods. Each lecture is followed by a 2-hour work session or excursion(s)/lab visit(s) in the same week. These sessions are focused on working out minor assignments including literature surveys, problem solving and working on the major assignment project. During the week of exams, you will present the results of the major assignment project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Relations</b>                | This course is a core course in the Waste & Combined Heat and Power profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | Book: "Biomass, a sustainable energy source for the future", de Jong and van Ommen, Wiley., 1st edition, Wiley (2015).<br>Lecture slides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | The final grade of the course consists of the following components:<br>* the solution to the (up to three) minor assignments (team work) (40%)<br>* a larger final assignment (team work); (50%)<br>* evaluation of the final assignment's presentation; (10%)<br><br>Final grade calculation: $(0.4 * \text{minor group assignments}) + (0.5 * \text{larger group assignment}) + (0.1 * \text{presentation})$<br><br>Resit/ Repair opportunities:<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br>* minor assignments: submit 1 new minor assignment<br>* larger assignment: resubmit report<br>* Presentation: resit (new presentation)<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-Instructor</b>                   | Dr. L.L. Cutz IJchajchal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |



| SET3080                                | The Necessity of Storage Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 |
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| <b>Responsible Instructor</b>          | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy.</p> <ul style="list-style-type: none"> <li>- Week 1: Future energy generation systems, the combination of sources on a large scale.<br/>Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel</li> <li>- Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc.<br/>Group work 2: program in Matlab typical use patterns for the different sectors</li> <li>- Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses.<br/>Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year.</li> <li>- Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities.<br/>Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency.</li> <li>- Week 5: Thermal storage, high T and low T. Combined heat and power possibilities.<br/>Group work 5: Investigate energy densities, efficiencies, capacities, exergies.</li> <li>- Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean.<br/>Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers.</li> <li>- Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.</li> </ul> |   |
| <b>Study Goals</b>                     | <p>At the end of the lectures and group work the students should be able to:</p> <ul style="list-style-type: none"> <li>- Reproduce and reflect on the role of storage in a renewable energy based society.</li> <li>- Explain how different timescales for energy storage are relevant for the future system.</li> <li>- List and explain the different storage solutions in terms of typical energy density, cost and efficiency.</li> <li>- Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy.</li> <li>- Reflect on the energy efficiency of intermediate storage.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | <p>The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time.<br/>The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week.</p> <p>The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <p>Slides<br/>Articles, reports and reviews.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:<br/>Individual written exam (70% of the final grade).<br/>Group work (30% of the final grade, based on a report produced by the group).<br/>For one part may be scored 5,0 as long as the other part compensates this.</p> <p>Final grade calculation: <math>(0,7 * \text{written exam}) + (0,3 * \text{group project})</math></p> <p>Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/>- written exam: written resit<br/>- group work: resubmit report</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| SET3085                                | Hydrogen Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
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| <b>Responsible Instructor</b>          | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed. |   |
| <b>Study Goals</b>                     | <p>After this course students:</p> <p>Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms.</p> <p>Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it.</p> <p>Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.</p>                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>This course consists of 14 2-hour lectures.</p> <p>These lecture also include some problem solving.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Literature and Study Materials</b>  | The exam will be based on the lecture slides and on selected research papers as provided on Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written exam (100%)</li> </ul> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                         |   |
| <b>Co-Instructor</b>                   | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| SET3110                                | Energy Storage in Batteries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M. Wagemaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                      | A. Vasileiadis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Dr. E.M. Kelder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | T. Famprakis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr. X. Wang                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar.</p> <p>In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including:</p> <p>Redox Flow<br/>Na-ion<br/>next generation Li-ion batteries<br/>Solid State batteries<br/>Supercapacitors</p>                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>At the end of the battery lectures students should be able to:</p> <ul style="list-style-type: none"> <li>- List the major types of Li and other batteries and their characteristics.</li> <li>- Reproduce the major challenges in batteries.</li> <li>- Explain how a battery works and the functioning of each component.</li> <li>- List and explain the different reaction types in batteries.</li> <li>- Explain the different charge transfer processes in batteries.</li> <li>- List and explain the different mechanisms that limit the cycle life of batteries.</li> <li>- Apply Nernst Law describing redox reactions.</li> <li>- Calculate conductivities and overpotentials.</li> <li>- Apply the Tafel equation describing the charge transfer between electrode and electrolyte.</li> <li>- Calculate energy densities and power densities from observed properties.</li> <li>- Derive from a phase diagram the different phases that will occur upon lithiation.</li> <li>- Apply the Gibbs Phase rule to predict constant of variable output voltage.</li> <li>- Derive the voltage curve from the shape of the Gibbs Free energy.</li> <li>- Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials.</li> <li>- Compare different battery systems and differentiate.</li> <li>- Discuss scientific results observed in battery performance and to relate this to material properties.</li> </ul> |   |
| <b>Education Method</b>                | <p>The course contains fourteen 2-hour lectures.</p> <p>The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery.</p> <p>In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed.</p> <p>In the final lecture example exams will be discussed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | R.A. Huggins: Advanced Batteries. Available via the library as pdf's: <a href="http://www.springerlink.com/content/nq6404/?MUD=MP">http://www.springerlink.com/content/nq6404/?MUD=MP</a><br>Slides<br>Articles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <p>Written exam (100% of the final grade)</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Co-Instructor</b>                   | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>Dr. E.M. Kelder</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Autonomous Systems: Wind Storage - Solar 2024**

| AE4W02TU                               | Introduction to Wind Turbines: Physics and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M.B. Zaayer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Prof.dr. D.A. von Terzi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Required for</b>                    | AE4W09 (Wind Turbine Design).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Expected prior knowledge</b>        | A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | <p>By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to:</p> <ul style="list-style-type: none"> <li>describe the components and configurations of wind turbines</li> <li>describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine</li> <li>describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design</li> <li>explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc.</li> <li>identify the drivers of dynamic behaviour and calculate their typical values</li> <li>describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure</li> <li>describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes</li> </ul> |   |
| <b>Education Method</b>                | Self-study of online material, with interactive lectures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>  | Handouts, videos and additional course material on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | Written exam at the end of period 2, with a resit at the end of period 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Remarks</b>                         | This course is an elective course for students from various faculties (AE, AS, ME, EEMCS, CEG, TPM). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Set-up</b>                          | Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| AE4W09                                 | Wind Turbine Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. D.A. von Terzi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr.ir. M.B. Zaayer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | Dr. D. Zappalá                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                      | Dr.ir. D.A.M. De Tavernier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Expected prior knowledge</b>        | AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Parts</b>                           | <p>In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4.</p> <p>The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.</p>                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>Overview of the course and the group assignment</p> <p>System design and scaling rules</p> <p>Aerodynamic and structural rotor design and analysis</p> <p>Drive train and electrical system</p> <p>Wind turbine control</p> <p>The use of standards for load calculations</p> <p>Other optional material (tbd)</p>                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>In this course you learn various aspects of designing a wind turbine, including:</p> <ul style="list-style-type: none"> <li>Selecting approaches for design and analysis</li> <li>Collect information and data</li> <li>Use (simulation) programs, models and calculations</li> <li>Analyse and interpret results and draw conclusions about the design</li> <li>Make design decisions</li> </ul> <p>Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.</p> |   |
| <b>Education Method</b>                | lectures + software tutorials + group assignment + discussion meetings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Computer Use</b>                    | <p>Spreadsheet-type calculations are used to support design decisions.</p> <p>The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.</p>                                                                                                                                                                                                                                                               |   |
| <b>Literature and Study Materials</b>  | Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Remarks</b>                         | <p>This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master.</p> <p>Supplementary courses are (these treat similar subjects as this course, but more in-depth):</p> <ul style="list-style-type: none"> <li>AE4W13 Site conditions for wind turbine design</li> <li>AE4135 Rotor / wake aerodynamics</li> <li>AE4W21-14 Wind turbine aeroelasticity</li> <li>AE4W31 Floating offshore wind energy</li> <li>ET4117 Electrical machines and drives</li> <li>OE44135 Offshore wind support structures</li> </ul>                                                                                  |   |

| AE4W13                                 | Site Conditions for Wind Turbine Design                                                                                                                                                                                                                                                                                                                                                                                                             | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. W.A.A.M. Bierbooms                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr.ing. R. Schmehl                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 4<br>Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>        | It is assumed that the student has a basic understanding of fluid mechanics and stochastics (on Brightspace self tests are available). A basic skill of Matlab is assumed: you should be able to read from and write to a text file, write and/or modify simple functions (m-files). Furthermore, it is recommended that you already have a basic knowledge of wind turbines; e.g. AE4W02TU, Introduction to Wind Turbines: Physics and Technology. |   |
| <b>Course Contents</b>                 | Introduction<br>Meteorology<br>Atmospheric Boundary Layer<br>Fluid Dynamics<br>Turbulence<br>Wind power meteorology<br>Measurement instruments<br>Statistics<br>Methods in turbulence<br>Offshore wind climate<br>External conditions in standards<br>Wave and current climate and loading<br>Example analysis site data<br>Turbulence spectra<br>Satellite remote sensing techniques<br>Simulation of turbulent wind fields and gusts              |   |
| <b>Study Goals</b>                     | After successfully completing the course, students will be able to:<br>Discuss the relevance of particular site data for wind turbine design, operation and siting.<br>Analyse simple problems related to wind power meteorology<br>Interpret scientific dissemination by experts on main points<br>Apply analysis methods to convert basic site data to wind turbine design parameters                                                             |   |
| <b>Education Method</b>                | Lecture + assignments + practical                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Relations</b>                | Follow up courses<br>AE4W09, AE4135                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | <a href="http://www.eos.ubc.ca/books/Practical_Meteorology">www.eos.ubc.ca/books/Practical_Meteorology</a><br>Chapters 12 to 14; Pijush K. Kundu, Ira M. Cohen, Fluid Mechanics, Elsevier.<br>Lecture notes                                                                                                                                                                                                                                         |   |
| <b>Prerequisites</b>                   | AE4W02TU, Introduction to Wind Turbines: Physics and Technology.                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | Written / digital exam (scheduled in the exam period after Q4).<br>The resit is an written exam on basis of pass/fail (i.e. in case of a pass the grade 6 will be put in Osiris).                                                                                                                                                                                                                                                                   |   |
| <b>Set-up</b>                          | The course combines lectures and student assignments/practical. For several topics guest speakers will be invited.                                                                                                                                                                                                                                                                                                                                  |   |

| ET4376                          | Photovoltaic Basics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Coordinator              | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Contact Hours / Week<br>x/x/x/x | 0/4/0/0 lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Education Period                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Start Education                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Exam Period                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Contents                 | This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised. |   |
| Study Goals                     | Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Education Method                | Lectures, exercises, instructions and self-study using the book Solar Energy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Literature and Study Materials  | Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Assessment                      | During instruction sessions you will practice 'exam level' questions. These questions are intended to prepare you for the exam, but are not graded. The final grade is 100% determined by the written exam. All marks will be expressed on a scale of 0 to 10.<br><br>In case of an insufficient result for the written exam, you have the opportunity to take the resit exam in the next quarter.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |



| ET4377                                 | Photovoltaic Technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Dr. L. Mazzarella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr. I.M.F. Gordon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0 lectures; 0/0/2/0 instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior knowledge</b>        | Lecture course Photovoltaic Basics (ET4 376)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed.</p> <p>Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.</p>                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required.</p> <p>The learning objectives (LO) are as follows:</p> <p>LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation.<br/> LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design.<br/> LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Flipped-classroom video lectures and on-campus instructions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England.</p> <p>ISBN: 978 1 906860 32 5 (paperback);</p> <p>This book is also available as a free ebook:</p> <p>ISBN: 978 1 906860 73 8 (ePub)<br/> ISBN: 978 1 906860 75 2 (pdf)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10.</p> <p>For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account.</p> <p>The final mark, F, for this course will be computed as follows: <math>F = E + [A \times (10 - E)/100]</math>. The computation will be carried out using one decimal and subsequently rounded to halves.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| ET4378                                 | Photovoltaic Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. M.R. Vogt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr.ir. H. Ziar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2 lectures and 0/0/0/2 lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>        | <ul style="list-style-type: none"> <li>- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376)</li> <li>- Good command of Matlab scripting language (or equivalent)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | <p>This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.</p>                                                                                                                                                                                                                                           |   |
| <b>Study Goals</b>                     | <p>By the end of this course, you will be able to:</p> <ol style="list-style-type: none"> <li>1. Explain the purpose of the components in different PV Systems using trends in their physical quantities.</li> <li>2. Examine different power tracking methods and the importance of bypass diodes.</li> <li>3. Calculate the energy yield of a PV system in conjunction with the following steps: <ol style="list-style-type: none"> <li>3a. Employ the calculation for optimal PV panel orientation from geographical survey.</li> <li>3b. Determine the advantages and disadvantages of different PV systems topologies.</li> </ol> </li> <li>4. Design PV systems for different applications based on your own cost estimation and technical evaluation justified in a report</li> </ol>                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | Lectures, self study and teaming up for carrying out the design of an actual PV System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available)</li> <li>- Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library)</li> <li>- Handouts</li> <li>- Scientific articles</li> <li>- PV Systems course in "Solar Engineering MicroMasters" (<a href="https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/">https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/</a>)</li> <li>- PV Resources (<a href="http://www.pvresources.com/">http://www.pvresources.com/</a>)</li> <li>- PVGIS (<a href="http://re.jrc.ec.europa.eu/pvgis/">http://re.jrc.ec.europa.eu/pvgis/</a>)</li> <li>- Sandia National Laboratories PVPerformance (<a href="https://pvpmc.sandia.gov/">https://pvpmc.sandia.gov/</a>)</li> <li>- Dutch PV Portal 3.0 (<a href="https://www.tudelft.nl/dutchpvportal">https://www.tudelft.nl/dutchpvportal</a>)</li> </ul> |   |
| <b>Assessment</b>                      | <p>For students of the 2024 cohort: Final project report (33% of the final mark) and written test (67% of the final mark).</p> <p>For students of previous cohorts: Final project report (25% of the final mark) and written test (75% of the final mark).</p> <p>Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual.</p> <p>To access to the written test, the grade of the final project report must be equal to or greater than 5.75.</p> <p>To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>  |   |

| SET3080                                | The Necessity of Storage Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 |
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| <b>Responsible Instructor</b>          | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy.</p> <ul style="list-style-type: none"> <li>- Week 1: Future energy generation systems, the combination of sources on a large scale.<br/>Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel</li> <li>- Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc.<br/>Group work 2: program in Matlab typical use patterns for the different sectors</li> <li>- Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses.<br/>Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year.</li> <li>- Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities.<br/>Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency.</li> <li>- Week 5: Thermal storage, high T and low T. Combined heat and power possibilities.<br/>Group work 5: Investigate energy densities, efficiencies, capacities, exergies.</li> <li>- Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean.<br/>Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers.</li> <li>- Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.</li> </ul> |   |
| <b>Study Goals</b>                     | <p>At the end of the lectures and group work the students should be able to:</p> <ul style="list-style-type: none"> <li>- Reproduce and reflect on the role of storage in a renewable energy based society.</li> <li>- Explain how different timescales for energy storage are relevant for the future system.</li> <li>- List and explain the different storage solutions in terms of typical energy density, cost and efficiency.</li> <li>- Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy.</li> <li>- Reflect on the energy efficiency of intermediate storage.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | <p>The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time.<br/>The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week.</p> <p>The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <p>Slides<br/>Articles, reports and reviews.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:<br/>Individual written exam (70% of the final grade).<br/>Group work (30% of the final grade, based on a report produced by the group).<br/>For one part may be scored 5,0 as long as the other part compensates this.</p> <p>Final grade calculation: <math>(0,7 * \text{written exam}) + (0,3 * \text{group project})</math></p> <p>Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/>- written exam: written resit<br/>- group work: resubmit report</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| SET3085                                | Hydrogen Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed. |   |
| <b>Study Goals</b>                     | <p>After this course students:</p> <p>Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms.</p> <p>Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it.</p> <p>Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.</p>                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>This course consists of 14 2-hour lectures.</p> <p>These lecture also include some problem solving.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Literature and Study Materials</b>  | The exam will be based on the lecture slides and on selected research papers as provided on Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written exam (100%)</li> </ul> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                         |   |
| <b>Co-Instructor</b>                   | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| SET3110                                | Energy Storage in Batteries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M. Wagemaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                      | A. Vasileiadis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Dr. E.M. Kelder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | T. Famprakis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr. X. Wang                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar.</p> <p>In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including:</p> <p>Redox Flow<br/>Na-ion<br/>next generation Li-ion batteries<br/>Solid State batteries<br/>Supercapacitors</p>                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>At the end of the battery lectures students should be able to:</p> <ul style="list-style-type: none"> <li>- List the major types of Li and other batteries and their characteristics.</li> <li>- Reproduce the major challenges in batteries.</li> <li>- Explain how a battery works and the functioning of each component.</li> <li>- List and explain the different reaction types in batteries.</li> <li>- Explain the different charge transfer processes in batteries.</li> <li>- List and explain the different mechanisms that limit the cycle life of batteries.</li> <li>- Apply Nernst Law describing redox reactions.</li> <li>- Calculate conductivities and overpotentials.</li> <li>- Apply the Tafel equation describing the charge transfer between electrode and electrolyte.</li> <li>- Calculate energy densities and power densities from observed properties.</li> <li>- Derive from a phase diagram the different phases that will occur upon lithiation.</li> <li>- Apply the Gibbs Phase rule to predict constant of variable output voltage.</li> <li>- Derive the voltage curve from the shape of the Gibbs Free energy.</li> <li>- Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials.</li> <li>- Compare different battery systems and differentiate.</li> <li>- Discuss scientific results observed in battery performance and to relate this to material properties.</li> </ul> |   |
| <b>Education Method</b>                | <p>The course contains fourteen 2-hour lectures.</p> <p>The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery.</p> <p>In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed.</p> <p>In the final lecture example exams will be discussed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | R.A. Huggins: Advanced Batteries. Available via the library as pdf's: <a href="http://www.springerlink.com/content/nq6404/?MUD=MP">http://www.springerlink.com/content/nq6404/?MUD=MP</a><br>Slides<br>Articles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:<br/>Written exam (100% of the final grade)</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-Instructor</b>                   | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>Dr. E.M. Kelder</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Wind & Economics: Wind Power Economics 2024**

| AE4W02TU                               | Introduction to Wind Turbines: Physics and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M.B. Zaayer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Prof.dr. D.A. von Terzi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Required for</b>                    | AE4W09 (Wind Turbine Design).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Expected prior knowledge</b>        | A proper engineering background in mechanics (Newton's laws of motion), statics (forces, stresses and displacements in structures), dynamics (mass-spring-damper system) and electricity and magnetism (a.o. Lorentz force) is assumed. Knowledge of aerodynamics (lift and drag of aerofoils) is convenient, but not mandatory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | Wind turbine technology, aerodynamic theory, wind climate, energy production, drive train, control, dynamic modelling, Campbell diagram, strength and fatigue, wind farm aspects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | <p>By the end of this course, you will be able to understand what a turbine is and how it interacts with the environment: what it is like, why it is like that and how to calculate values for its characteristic properties. Specifically, you will be able to:</p> <ul style="list-style-type: none"> <li>describe the components and configurations of wind turbines</li> <li>describe the characteristics of wind, use models that quantify wind speed variations and estimate the energy yield of a wind turbine</li> <li>describe the aerodynamic processes at work in wind energy conversion, calculate the forces and power generated and produce a simple rotor design</li> <li>explain the working principles of drive train components and determine the operational conditions that affect power, torque, pitch angle etc.</li> <li>identify the drivers of dynamic behaviour and calculate their typical values</li> <li>describe the principles of structural analysis of wind turbines and carry out preliminary predictions of structural failure</li> <li>describe the effect of wind turbines on downstream conditions and quantify wind speed and turbulence in wakes</li> </ul> |   |
| <b>Education Method</b>                | Self-study of online material, with interactive lectures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>  | Handouts, videos and additional course material on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | Written exam at the end of period 2, with a resit at the end of period 3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Remarks</b>                         | This course is an elective course for students from various faculties (AE, AS, ME, EEMCS, CEG, TPM). It is also part of the SET MSc curriculum in several clusters. The course is therefore not tailored to a particular curriculum, but tries to accommodate students with different backgrounds and interests.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Set-up</b>                          | Students are expected to study the online course material at home, before the lecture. During the lecture, frequently asked questions will be addressed, difficult topics will be further explained and exercises will be used to test and improve your understanding and application of the course material.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| AE4W09                                 | Wind Turbine Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. D.A. von Terzi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr.ir. M.B. Zaayer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | Dr. D. Zappalá                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                      | Dr.ir. D.A.M. De Tavernier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Expected prior knowledge</b>        | AE4W02TU 'Introduction to wind turbines' or equivalent. When expected prior knowledge from 'Introduction to wind turbines' is missing, extra self-study may be needed to complete 'Wind turbine design' successfully.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Parts</b>                           | <p>In this course, students design a wind turbine in a group. This group assignment is divided in two parts. In the first part the preliminary design of the wind turbine is made and this needs to be finished in quarter 3. In the second part the controller is designed and detailed (load) analysis is performed. This part is completed in quarter 4.</p> <p>The assignment runs in parallel with the lectures. In each quarter a software tutorial is given to get familiar with the elements of the software that are relevant in that quarter.</p>                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>Overview of the course and the group assignment</p> <p>System design and scaling rules</p> <p>Aerodynamic and structural rotor design and analysis</p> <p>Drive train and electrical system</p> <p>Wind turbine control</p> <p>The use of standards for load calculations</p> <p>Other optional material (tbd)</p>                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>In this course you learn various aspects of designing a wind turbine, including:</p> <ul style="list-style-type: none"> <li>Selecting approaches for design and analysis</li> <li>Collect information and data</li> <li>Use (simulation) programs, models and calculations</li> <li>Analyse and interpret results and draw conclusions about the design</li> <li>Make design decisions</li> </ul> <p>Although the assessment is performed by a group assignment, effective teamwork skills are not an explicit learning goal of this course. There is no learning material or assessment to address such skills. Nevertheless, proper teamwork will contribute to a good outcome of the design activities.</p> |   |
| <b>Education Method</b>                | lectures + software tutorials + group assignment + discussion meetings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Computer Use</b>                    | <p>Spreadsheet-type calculations are used to support design decisions.</p> <p>The Matlab/simulink environment is used to develop the controller and to perform detailed analysis of the designed wind turbine. For this purpose Matlab/simulink is connected to 'FAST', a wind turbine simulation package. The use of FAST through the Matlab/simulink environment is explained in the tutorials. FAST only works on computers with MS Windows.</p>                                                                                                                                                                                                                                                               |   |
| <b>Literature and Study Materials</b>  | Lecture material and some additional information will be published on Brightspace. To execute the assignment, external sources need to be consulted as well.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | A group assignment to design a wind turbine has to be executed, reported and discussed with the teachers. The group work is graded and the final individual grade may deviate up or down from the group grade, based on individual contributions to the work. A minimum level is set for both group and individual performance to pass the course.                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Remarks</b>                         | <p>This is multidisciplinary course, attended by students from various faculties (AE, EEMCS, CEG, 3ME, AS), from the 3TU MSc track SET and from the European Wind Energy Master.</p> <p>Supplementary courses are (these treat similar subjects as this course, but more in-depth):</p> <ul style="list-style-type: none"> <li>AE4W13 Site conditions for wind turbine design</li> <li>AE4135 Rotor / wake aerodynamics</li> <li>AE4W21-14 Wind turbine aeroelasticity</li> <li>AE4W31 Floating offshore wind energy</li> <li>ET4117 Electrical machines and drives</li> <li>OE44135 Offshore wind support structures</li> </ul>                                                                                  |   |



| AE4W13                                 | Site Conditions for Wind Turbine Design                                                                                                                                                                                                                                                                                                                                                                                                             | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. W.A.A.M. Bierbooms                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr.ing. R. Schmehl                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 4<br>Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>        | It is assumed that the student has a basic understanding of fluid mechanics and stochastics (on Brightspace self tests are available). A basic skill of Matlab is assumed: you should be able to read from and write to a text file, write and/or modify simple functions (m-files). Furthermore, it is recommended that you already have a basic knowledge of wind turbines; e.g. AE4W02TU, Introduction to Wind Turbines: Physics and Technology. |   |
| <b>Course Contents</b>                 | Introduction<br>Meteorology<br>Atmospheric Boundary Layer<br>Fluid Dynamics<br>Turbulence<br>Wind power meteorology<br>Measurement instruments<br>Statistics<br>Methods in turbulence<br>Offshore wind climate<br>External conditions in standards<br>Wave and current climate and loading<br>Example analysis site data<br>Turbulence spectra<br>Satellite remote sensing techniques<br>Simulation of turbulent wind fields and gusts              |   |
| <b>Study Goals</b>                     | After successfully completing the course, students will be able to:<br>Discuss the relevance of particular site data for wind turbine design, operation and siting.<br>Analyse simple problems related to wind power meteorology<br>Interpret scientific dissemination by experts on main points<br>Apply analysis methods to convert basic site data to wind turbine design parameters                                                             |   |
| <b>Education Method</b>                | Lecture + assignments + practical                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Relations</b>                | Follow up courses<br>AE4W09, AE4135                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | <a href="http://www.eos.ubc.ca/books/Practical_Meteorology">www.eos.ubc.ca/books/Practical_Meteorology</a><br>Chapters 12 to 14; Pijush K. Kundu, Ira M. Cohen, Fluid Mechanics, Elsevier.<br>Lecture notes                                                                                                                                                                                                                                         |   |
| <b>Prerequisites</b>                   | AE4W02TU, Introduction to Wind Turbines: Physics and Technology.                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | Written / digital exam (scheduled in the exam period after Q4).<br>The resit is an written exam on basis of pass/fail (i.e. in case of a pass the grade 6 will be put in Osiris).                                                                                                                                                                                                                                                                   |   |
| <b>Set-up</b>                          | The course combines lectures and student assignments/practical. For several topics guest speakers will be invited.                                                                                                                                                                                                                                                                                                                                  |   |

| EE4536                                 | DC and AC Microgrids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>          | Dr.ir. L.M. Ramirez Elizondo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | BSc Electrical Engineering or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | Identify the challenges introduced by distributed generation<br>Describe different microgrids' layouts and sizing<br>Describe the operation and control of power electronic converters in AC and DC microgrids<br>Apply droop control in AC and DC microgrids<br>Describe the role of energy management systems in AC and DC microgrids<br>Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiorgiou                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Written exam 70%, Assignments 30%<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                           |   |

| EE4545                                  | Electrical Power Systems of the Future                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>           | Dr. A. Leki                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>         | <p>Pre-requisites (entry levels acquired in Q1 and Q2):</p> <p>Master SET / Master EE / Master EWEM<br/>           SET3065 Intelligent Electrical Power Grids<br/>           ET4107 Power System Analyses II<br/>           Matlab and Simulink skills (self study tutorial is available)</p> <p>Optional<br/>           EE4665 Uncertainty modelling and risk assessment in electrical power systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | <p>The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system.</p> <p>Learning objectives(LOs):</p> <p>LO1: Explain the impact of high RES penetration on the resiliency of future power systems.</p> <p>LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration.</p> <p>LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply.</p> <p>LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.</p> |   |
| <b>Education Method</b>                 | <p>Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site</p> <p>Estimated study load: 112 hours (4 ECTs)<br/>           Lectures 20 hours<br/>           Practical 24 hours<br/>           Self study 68 hours</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                       | <p>Practicum assignments and digital exam/re-exam</p> <p>The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                          | <p>Company visit in combination with lectures and practicum<br/>           @ Mariëndaal Centre of Excellence, TenneT Arnhem<br/>           Dont forget identity card or passport!</p> <p>8:00 Departure from EWI, Travel by touring bus to Arnhem.<br/>           Lectures on Operational challenges by RES for system operations.<br/>           Lessons learnt from large disturbances.<br/>           Visit TenneT Control Room (including lunch).<br/>           Practical Exercises Operational challenges and large disturbances.<br/>           Travel by touring bus to Arnhem<br/>           16:30 Arrival at EWI</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>maximum aantal deelnemers</b>        | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| SET3065                                | Intelligent Electrical Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. P. Palensky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. J.L. Rueda Torres                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets.</p> <p>Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools.</p> <p>You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done.</p> <p>This course is part of the Power track of the SET master and can also be chosen by EE master students.</p> <p>Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.</p> |   |
| <b>Study Goals</b>                     | <p>After this course students are able to</p> <p>LO1: Describe the functions and topologies of cyber-physical energy systems</p> <p>LO2: Implement data driven modelling and numerical simulation of energy systems</p> <p>LO3: Calculate variable power flows based on optimization/probability theory</p> <p>LO4: Synthesize functional requirements of energy management systems</p> <p>LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized in the following units, all units have assignments (typically a simulation):</p> <ol style="list-style-type: none"> <li>1: The energy transition and future energy systems (Peter)</li> <li>2: Analysis of steady-state performance in variable operating conditions (Jose)</li> <li>3: Assessment and control of power system dynamic performance (Jose)</li> <li>4: Lab 1 Digital and data driven modelling of interconnected systems</li> <li>5: Lab 2 Simulation and impact analysis of renewables, storage, and demand</li> <li>6: Data acquisition and communication for energy automation (Peter)</li> <li>7: Artificial intelligence-based functionalities for situational awareness (Pedro &amp; Jochen)</li> <li>8: Cyber security of smart grids (Alex)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | On Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <p>Assignment reports (in time!) (50%)</p> <p>Multiple choice Computer Exam (50%)</p> <p>In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture.</p> <p>Final grade calculation: <math>(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Assignment reports (in time!): resubmit deliverable</li> <li>- Multiple choice Computer Exam: computer resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                    |   |

| SET3095                                | Electronic Power Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | S.A. Rivera Iunnissi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures<br>Problem solving sessions<br>Practicum Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written final exam (closed book) (70%)</li> <li>- Assignments (30%)</li> </ul> <p><math>T = \text{grade for written final exam} * 0,7 + \text{grade for assignments} * 0,3</math></p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| TPM0637SET                             | Economic Policy for Sustainable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. E. Schröder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Co-responsible for assignments</b>  | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | <p>Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course.</p> <p>The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.</p>                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis?</p> <p>This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to:</p> <ul style="list-style-type: none"> <li>- consumption and production</li> <li>- efficiency and welfare</li> <li>- economic growth and sustainability</li> <li>- the choice of environmental policy targets and instruments</li> <li>- integrated assessment models</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing the course, students should be able to:</p> <ul style="list-style-type: none"> <li>- explain the main theoretical concepts used in environmental economics and how they relate to one another</li> <li>- explain how a market economy undervalues the environment</li> <li>- explain how environmental economists value the environment</li> <li>- apply economic theory to environmental problems such as climate change</li> <li>- evaluate economic policies for solving environmental problems such as climate change</li> </ul>                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Classes, quizzes, and exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>The main text is:</p> <ul style="list-style-type: none"> <li>- Harris &amp; Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor &amp; Francis.</li> </ul> <p>The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade is based exclusively on the performance in the final exam.</p> <p>In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| TPM0931SET                             | Sustainable Energy Innovation and Transitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. J.N. Quist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition.</p> <p>This course is the first course in the SET Economics &amp; Society profile. It can also be followed by other MSc students as an elective. Due to overlap, is not allowed to follow this course if you have already followed course TPM033A or 4413TRIG.</p>                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>After this course, the student:</p> <ol style="list-style-type: none"> <li>1. has knowledge of: <ul style="list-style-type: none"> <li>* the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective.</li> <li>* the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability.</li> </ul> </li> <li>2. has insight in: <ul style="list-style-type: none"> <li>* the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts.</li> </ul> </li> <li>3. has experience and basic skills in: <ul style="list-style-type: none"> <li>* exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.</li> <li>* formulating research questions.</li> <li>* writing a coherent, well-structured and well argued academic report.</li> <li>* critically reflecting on work of others and on their own work.</li> </ul> </li> </ol> |   |
| <b>Education Method</b>                | <p>This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>Scientific articles which are made available via the TU library or via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Both reports must be completed with at least grade 5.0. Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Elective</b>                        | <p>Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                            | <p>Energy<br/>Energy &amp; Industry<br/>Group work<br/>Sustainability</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| TPM403SET                              | Technology Entrepreneurship and Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | M. Kolagar Daronkola                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Responsible for assignments</b>     | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-responsible for assignments</b>  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | No specific knowledge is required prior to entering the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course.</p> <p>This course is the third course in the SET Economics &amp; Society profile.</p>                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to:</p> <ol style="list-style-type: none"> <li>1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing;</li> <li>2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems;</li> <li>3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products;</li> <li>4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones;</li> <li>5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular;</li> <li>6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.</li> </ol>                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | <p>Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted on Brightspace-site.<br/>Slides presented in class: Will be posted on Brightspace-site.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>The assessment is composed of several elements:</p> <ol style="list-style-type: none"> <li>1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade.</li> <li>2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade.</li> <li>3. Individual: Theory quizzes (30%)</li> </ol> <p>All marks will be expressed on a scale of 1-10.</p> |   |
| <b>Enrolment / Application</b>         | All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Remarks</b>                         | For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Targetgroup</b>                     | Master students from all master programs can apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |



|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Solar & Economics: Solar Power Economics 2024**

| EE4536                                 | DC and AC Microgrids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>          | Dr.ir. L.M. Ramirez Elizondo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | BSc Electrical Engineering or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | Identify the challenges introduced by distributed generation<br>Describe different microgrids' layouts and sizing<br>Describe the operation and control of power electronic converters in AC and DC microgrids<br>Apply droop control in AC and DC microgrids<br>Describe the role of energy management systems in AC and DC microgrids<br>Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiorgiou                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Written exam 70%, Assignments 30%<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                           |   |

| EE4545                                  | Electrical Power Systems of the Future                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>           | Dr. A. Leki                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>         | <p>Pre-requisites (entry levels acquired in Q1 and Q2):</p> <p>Master SET / Master EE / Master EWEM<br/>           SET3065 Intelligent Electrical Power Grids<br/>           ET4107 Power System Analyses II<br/>           Matlab and Simulink skills (self study tutorial is available)</p> <p>Optional<br/>           EE4665 Uncertainty modelling and risk assessment in electrical power systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | <p>The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system.</p> <p>Learning objectives(LOs):</p> <p>LO1: Explain the impact of high RES penetration on the resiliency of future power systems.</p> <p>LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration.</p> <p>LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply.</p> <p>LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.</p> |   |
| <b>Education Method</b>                 | <p>Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site</p> <p>Estimated study load: 112 hours (4 ECTs)<br/>           Lectures 20 hours<br/>           Practical 24 hours<br/>           Self study 68 hours</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                       | <p>Practicum assignments and digital exam/re-exam</p> <p>The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                          | <p>Company visit in combination with lectures and practicum<br/>           @ Mariëndaal Centre of Excellence, TenneT Arnhem<br/>           Dont forget identity card or passport!</p> <p>8:00 Departure from EWI, Travel by touring bus to Arnhem.<br/>           Lectures on Operational challenges by RES for system operations.<br/>           Lessons learnt from large disturbances.<br/>           Visit TenneT Control Room (including lunch).<br/>           Practical Exercises Operational challenges and large disturbances.<br/>           Travel by touring bus to Arnhem<br/>           16:30 Arrival at EWI</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>maximum aantal deelnemers</b>        | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| ET4376                          | Photovoltaic Basics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Coordinator              | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Contact Hours / Week<br>x/x/x/x | 0/4/0/0 lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Education Period                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Start Education                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Exam Period                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Contents                 | This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised. |   |
| Study Goals                     | Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Education Method                | Lectures, exercises, instructions and self-study using the book Solar Energy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Literature and Study Materials  | Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Assessment                      | During instruction sessions you will practice 'exam level' questions. These questions are intended to prepare you for the exam, but are not graded. The final grade is 100% determined by the written exam. All marks will be expressed on a scale of 0 to 10.<br><br>In case of an insufficient result for the written exam, you have the opportunity to take the resit exam in the next quarter.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| ET4377                                 | Photovoltaic Technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Dr. L. Mazzarella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr. I.M.F. Gordon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0 lectures; 0/0/2/0 instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior knowledge</b>        | Lecture course Photovoltaic Basics (ET4 376)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed.</p> <p>Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.</p>                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required.</p> <p>The learning objectives (LO) are as follows:</p> <p>LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation.<br/> LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design.<br/> LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Flipped-classroom video lectures and on-campus instructions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England.</p> <p>ISBN: 978 1 906860 32 5 (paperback);</p> <p>This book is also available as a free ebook:</p> <p>ISBN: 978 1 906860 73 8 (ePub)<br/> ISBN: 978 1 906860 75 2 (pdf)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10.</p> <p>For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account.</p> <p>The final mark, F, for this course will be computed as follows: <math>F = E + [A \times (10 - E)/100]</math>. The computation will be carried out using one decimal and subsequently rounded to halves.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| ET4378                                 | Photovoltaic Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. M.R. Vogt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr.ir. H. Ziar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2 lectures and 0/0/0/2 lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>        | - Basic knowledge of photovoltaics (e.g. PV Basics course ET4376)<br>- Good command of Matlab scripting language (or equivalent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.                                                                                                                                                                                                                                          |   |
| <b>Study Goals</b>                     | By the end of this course, you will be able to:<br>1. Explain the purpose of the components in different PV Systems using trends in their physical quantities.<br>2. Examine different power tracking methods and the importance of bypass diodes.<br>3. Calculate the energy yield of a PV system in conjunction with the following steps:<br>3a. Employ the calculation for optimal PV panel orientation from geographical survey.<br>3b. Determine the advantages and disadvantages of different PV systems topologies.<br>4. Design PV systems for different applications based on your own cost estimation and technical evaluation justified in a report                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Lectures, self study and teaming up for carrying out the design of an actual PV System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | - Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available)<br>- Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library)<br>- Handouts<br>- Scientific articles<br>- PV Systems course in "Solar Engineering MicroMasters" ( <a href="https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/">https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/</a> )<br>- PV Resources ( <a href="http://www.pvresources.com/">http://www.pvresources.com/</a> )<br>- PVGIS ( <a href="http://re.jrc.ec.europa.eu/pvgis/">http://re.jrc.ec.europa.eu/pvgis/</a> )<br>- Sandia National Laboratories PVPerformance ( <a href="https://pvpmc.sandia.gov/">https://pvpmc.sandia.gov/</a> )<br>- Dutch PV Portal 3.0 ( <a href="https://www.tudelft.nl/dutchpvportal">https://www.tudelft.nl/dutchpvportal</a> )                                                                                 |   |
| <b>Assessment</b>                      | For students of the 2024 cohort: Final project report (33% of the final mark) and written test (67% of the final mark).<br><br>For students of previous cohorts: Final project report (25% of the final mark) and written test (75% of the final mark).<br><br>Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual.<br><br>To access to the written test, the grade of the final project report must be equal to or greater than 5.75.<br><br>To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75.<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). |   |

| SET3065                                | Intelligent Electrical Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. P. Palensky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. J.L. Rueda Torres                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets.</p> <p>Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools.</p> <p>You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done.</p> <p>This course is part of the Power track of the SET master and can also be chosen by EE master students.</p> <p>Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.</p> |   |
| <b>Study Goals</b>                     | <p>After this course students are able to</p> <p>LO1: Describe the functions and topologies of cyber-physical energy systems</p> <p>LO2: Implement data driven modelling and numerical simulation of energy systems</p> <p>LO3: Calculate variable power flows based on optimization/probability theory</p> <p>LO4: Synthesize functional requirements of energy management systems</p> <p>LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized in the following units, all units have assignments (typically a simulation):</p> <ol style="list-style-type: none"> <li>1: The energy transition and future energy systems (Peter)</li> <li>2: Analysis of steady-state performance in variable operating conditions (Jose)</li> <li>3: Assessment and control of power system dynamic performance (Jose)</li> <li>4: Lab 1 Digital and data driven modelling of interconnected systems</li> <li>5: Lab 2 Simulation and impact analysis of renewables, storage, and demand</li> <li>6: Data acquisition and communication for energy automation (Peter)</li> <li>7: Artificial intelligence-based functionalities for situational awareness (Pedro &amp; Jochen)</li> <li>8: Cyber security of smart grids (Alex)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | On Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <p>Assignment reports (in time!) (50%)</p> <p>Multiple choice Computer Exam (50%)</p> <p>In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture.</p> <p>Final grade calculation: <math>(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Assignment reports (in time!): resubmit deliverable</li> <li>- Multiple choice Computer Exam: computer resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                    |   |

| SET3095                                | Electronic Power Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
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| <b>Responsible Instructor</b>          | S.A. Rivera Iunnissi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures<br>Problem solving sessions<br>Practicum Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written final exam (closed book) (70%)</li> <li>- Assignments (30%)</li> </ul> <p><math>T = \text{grade for written final exam} * 0,7 + \text{grade for assignments} * 0,3</math></p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |



| TPM0637SET                             | Economic Policy for Sustainable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. E. Schröder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Co-responsible for assignments</b>  | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | <p>Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course.</p> <p>The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.</p>                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis?</p> <p>This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to:</p> <ul style="list-style-type: none"> <li>- consumption and production</li> <li>- efficiency and welfare</li> <li>- economic growth and sustainability</li> <li>- the choice of environmental policy targets and instruments</li> <li>- integrated assessment models</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing the course, students should be able to:</p> <ul style="list-style-type: none"> <li>- explain the main theoretical concepts used in environmental economics and how they relate to one another</li> <li>- explain how a market economy undervalues the environment</li> <li>- explain how environmental economists value the environment</li> <li>- apply economic theory to environmental problems such as climate change</li> <li>- evaluate economic policies for solving environmental problems such as climate change</li> </ul>                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Classes, quizzes, and exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>The main text is:</p> <ul style="list-style-type: none"> <li>- Harris &amp; Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor &amp; Francis.</li> </ul> <p>The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade is based exclusively on the performance in the final exam.</p> <p>In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| TPM0931SET                             | Sustainable Energy Innovation and Transitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. J.N. Quist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition.</p> <p>This course is the first course in the SET Economics &amp; Society profile. It can also be followed by other MSc students as an elective. Due to overlap, is not allowed to follow this course if you have already followed course TPM033A or 4413TRIG.</p>                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>After this course, the student:</p> <ol style="list-style-type: none"> <li>1. has knowledge of: <ul style="list-style-type: none"> <li>* the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective.</li> <li>* the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability.</li> </ul> </li> <li>2. has insight in: <ul style="list-style-type: none"> <li>* the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts.</li> </ul> </li> <li>3. has experience and basic skills in: <ul style="list-style-type: none"> <li>* exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.</li> <li>* formulating research questions.</li> <li>* writing a coherent, well-structured and well argued academic report.</li> <li>* critically reflecting on work of others and on their own work.</li> </ul> </li> </ol> |   |
| <b>Education Method</b>                | <p>This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>Scientific articles which are made available via the TU library or via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Both reports must be completed with at least grade 5.0. Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Elective</b>                        | <p>Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                            | <p>Energy<br/>Energy &amp; Industry<br/>Group work<br/>Sustainability</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| TPM403SET                       | Technology Entrepreneurship and Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
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| Module Manager                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Instructor                      | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Instructor                      | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Instructor                      | M. Kolagar Daronkola                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Responsible for assignments     | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Co-responsible for assignments  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Contact Hours / Week<br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Education Period                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Start Education                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Exam Period                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Expected prior knowledge        | No specific knowledge is required prior to entering the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Course Contents                 | <p>This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course.</p> <p>This course is the third course in the SET Economics &amp; Society profile.</p>                                                                                                                                                                                                                                                                                           |   |
| Study Goals                     | <p>Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to:</p> <ol style="list-style-type: none"><li>1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing;</li><li>2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems;</li><li>3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products;</li><li>4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones;</li><li>5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular;</li><li>6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.</li></ol>                                                                                                                                                                                                                                                                                                                                                        |   |
| Education Method                | <p>The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Literature and Study Materials  | <p>Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted on Brightspace-site.<br/>Slides presented in class: Will be posted on Brightspace-site.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Assessment                      | <p>The assessment is composed of several elements:</p> <ol style="list-style-type: none"><li>1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade.</li><li>2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade.</li><li>3. Individual: Theory quizzes (30%)</li></ol> <p>All marks will be expressed on a scale of 1-10.</p> |   |
| Enrolment / Application         | <p>All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Remarks                         | <p>For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Targetgroup                     | <p>Master students from all master programs can apply.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

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|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Data-Driven Energy Systems: DET - Power - Economics 2024**

| EE4410                                 | Cyber Security of Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Summary</b>                         | <p>The energy system transition towards a sustainable, low carbon future is led by decarbonisation, decentralisation, and digitalisation. Power grids are undergoing a fast-paced process of digitalisation for enhanced monitoring and control capabilities and grid intelligence. However, the increased integration of digital technologies, such as the next generation of Operational Technologies (OTs) and digital substations, implies a new risk as Information Technology (IT) OT systems for smart grids are vulnerable to cyber attacks. Furthermore, the combination of heterogeneous, co-existing smart and legacy technologies generates significant vulnerabilities and cyber security challenges. Examples of cyber security incidents related to power grids already exist around the world. On December 23, 2015 cyber attacks were conducted on the power grid in Ukraine that resulted in power outages, which affected 225,000 customers. More sophisticated cyber attacks on the Ukrainian power grid followed on December 17, 2016 resulting in a power outage in the distribution network where 200 MW of load was unsupplied. The complexity of cyber attacks on power systems is on the rise. They can initiate cascading failures and result in a catastrophic blackout. A power outage can disrupt the entire energy chain including water supply, heating, and gas networks. Without power, hospitals and other critical services are severely affected. A disruption of service may lead to financial loss, damages, chaos, or even a loss of lives. With respect to security of supply and reliability of the future energy system provision, special attention is needed for new vulnerabilities and cyber threats that come with digitalisation. Accordingly, the cyber security aspects are critical for a further power grid digitalisation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>This course aims to assess and improve the cyber security of smart grids. It discusses the stakeholders of and complexities in critical infrastructures such as electrical power systems. It analyses the typical IT-OT system architectures for grid monitoring and control and explains the main cyber security concepts and policies. The course provides essential knowledge to analyse cyber security threats, threat vectors, and vulnerabilities of IT OT systems for power grids. It examines the kill chain stages of the cyber attack process on industrial control systems. The course analyses the impact of cyber attacks on power system operation in terms of loss of load, equipment damage, cascading effects, and risk of a blackout. The course studies the state-of-the-art security controls and best practices for threat intelligence management and incident response to defend against cyber attacks and effectively mitigate their impact on power system operation.</p> <p>This course focuses on cyber security of Information Technology (IT) Operational Technologies (OT) networks used for power system operation. It starts with an overview of the smart grid and its industrial control systems (EMS/SCADA) and introduces computer networks and cyber security. Next, the course will introduce the cyber kill chain method for cyber security investigation and intelligence-driven defence. Kill chains are used to understand, anticipate, recognise, and combat cyber attacks and advanced persistent threats. The course uses the kill chains of the 2015/2016 cyber attacks on Ukrainian power grid as real-world study cases throughout the course. The main IT-OT system vulnerabilities and cyber threats are analysed. The course covers threats such as passive and active reconnaissance, social engineering, and malware. It investigates various threat vectors on host and network security and advanced cyber attacks targeting power grids. The impact on power system operation is analysed. The course discusses the state-of-the-art security controls to defend and mitigate cyber attacks on power grids such as access controls, authentication management, cryptography, firewalls, secure communication protocols, network segmentation, and intrusion detection and prevention systems. The best practices to improve power grid resilience to cyber attacks are discussed, i.e., threat intelligence management, incident response, and wide area monitoring, protection and control. This course provides a deep dive into core cyber security concepts for power grids such as cryptography (symmetric and asymmetric encryption), hash functions, digital signatures, and cyber-physical anomaly detection using machine learning techniques. Cyber security demonstrations on power grids will be conducted for students in Control Room of the Future (CRoF) technology centre.</p> |   |
| <b>Study Goals</b>                     | <p>At the end of the course, you will be able to:</p> <p>LO1: Analyse cyber security threats, threat vectors, and vulnerabilities of Information Technology (IT) and Operational Technology (OT) systems used for power grid monitoring and control.<br/> LO2: Examine kill chain stages of the cyber attack process on cyber-physical power systems.<br/> LO3: Analyse the impact of cyber attacks on power system operation (instability, cascading failures and blackout).<br/> LO4: Develop security controls using machine learning to detect, prevent and mitigate cyber attacks and cyber secure the future power grids.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>Lectures: theory and real-world study cases (8 lectures x 2h/lecture)<br/> Practicums: hands-on computer labs (5 practicums x 2h/practicum)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <p>Chwan-Hwa Wu &amp; J. David Irwin. (2013). Introduction to computer networks and cybersecurity. CRC Press. ISBN: 9781498760133</p> <p>Nigel P. Smart. (2016). Cryptography made simple. Springer, 2nd Edition.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>The grade is formed based on the following elements:</p> <p>40% Assignment reports (five assignment reports submitted in time on Brightspace)<br/> 60% Final digital (computer) exam</p> <p>All assignments and reports are mandatory.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| EE4536                                 | DC and AC Microgrids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>          | Dr.ir. L.M. Ramirez Elizondo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | BSc Electrical Engineering or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | Identify the challenges introduced by distributed generation<br>Describe different microgrids' layouts and sizing<br>Describe the operation and control of power electronic converters in AC and DC microgrids<br>Apply droop control in AC and DC microgrids<br>Describe the role of energy management systems in AC and DC microgrids<br>Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiorgiou                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Written exam 70%, Assignments 30%<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                           |   |

| EE4545                                  | Electrical Power Systems of the Future                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>           | Dr. A. Leki                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>         | <p>Pre-requisites (entry levels acquired in Q1 and Q2):</p> <p>Master SET / Master EE / Master EWEM<br/>           SET3065 Intelligent Electrical Power Grids<br/>           ET4107 Power System Analyses II<br/>           Matlab and Simulink skills (self study tutorial is available)</p> <p>Optional<br/>           EE4665 Uncertainty modelling and risk assessment in electrical power systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | <p>The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system.</p> <p>Learning objectives(LOs):</p> <p>LO1: Explain the impact of high RES penetration on the resiliency of future power systems.</p> <p>LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration.</p> <p>LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply.</p> <p>LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.</p> |   |
| <b>Education Method</b>                 | <p>Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site</p> <p>Estimated study load: 112 hours (4 ECTs)<br/>           Lectures 20 hours<br/>           Practical 24 hours<br/>           Self study 68 hours</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                       | <p>Practicum assignments and digital exam/re-exam</p> <p>The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                          | <p>Company visit in combination with lectures and practicum<br/>           @ Mariëndaal Centre of Excellence, TenneT Arnhem<br/>           Dont forget identity card or passport!</p> <p>8:00 Departure from EWI, Travel by touring bus to Arnhem.<br/>           Lectures on Operational challenges by RES for system operations.<br/>           Lessons learnt from large disturbances.<br/>           Visit TenneT Control Room (including lunch).<br/>           Practical Exercises Operational challenges and large disturbances.<br/>           Travel by touring bus to Arnhem<br/>           16:30 Arrival at EWI</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>maximum aantal deelnemers</b>        | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| SET3065                                | Intelligent Electrical Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. P. Palensky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. J.L. Rueda Torres                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets.</p> <p>Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools.</p> <p>You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done.</p> <p>This course is part of the Power track of the SET master and can also be chosen by EE master students.</p> <p>Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.</p> |   |
| <b>Study Goals</b>                     | <p>After this course students are able to</p> <p>LO1: Describe the functions and topologies of cyber-physical energy systems</p> <p>LO2: Implement data driven modelling and numerical simulation of energy systems</p> <p>LO3: Calculate variable power flows based on optimization/probability theory</p> <p>LO4: Synthesize functional requirements of energy management systems</p> <p>LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized in the following units, all units have assignments (typically a simulation):</p> <ol style="list-style-type: none"> <li>1: The energy transition and future energy systems (Peter)</li> <li>2: Analysis of steady-state performance in variable operating conditions (Jose)</li> <li>3: Assessment and control of power system dynamic performance (Jose)</li> <li>4: Lab 1 Digital and data driven modelling of interconnected systems</li> <li>5: Lab 2 Simulation and impact analysis of renewables, storage, and demand</li> <li>6: Data acquisition and communication for energy automation (Peter)</li> <li>7: Artificial intelligence-based functionalities for situational awareness (Pedro &amp; Jochen)</li> <li>8: Cyber security of smart grids (Alex)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | On Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <p>Assignment reports (in time!) (50%)</p> <p>Multiple choice Computer Exam (50%)</p> <p>In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture.</p> <p>Final grade calculation: <math>(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Assignment reports (in time!): resubmit deliverable</li> <li>- Multiple choice Computer Exam: computer resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                    |   |



| SET3095                                | Electronic Power Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | S.A. Rivera Iunnissi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures<br>Problem solving sessions<br>Practicum Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written final exam (closed book) (70%)</li> <li>- Assignments (30%)</li> </ul> <p><math>T = \text{grade for written final exam} * 0,7 + \text{grade for assignments} * 0,3</math></p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| SET3120                                | Energy Systems Simulation and Digital Twins                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. M. Cvetkovic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. S.H. Tindemans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Expected prior knowledge</b>        | Solid background in energy and power systems, which should be acquired through the courses of the SET M.Sc. program. Basic experience in Python programming.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Parts</b>                           | <p>This is a new course. The final lecture structure and/or content may differ slightly from the description.</p> <p>Theory<br/>Topics covered: Energy systems simulation, Co-simulation, Energy systems digital twins and Stochastic simulation.</p> <p>Practicums per Lecture &amp; Course Project</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>Computer simulation is central to energy system planning and operational decision-making. This course introduces advanced concepts and notions of modeling and simulation applied to energy systems. The course is divided into four themes: energy systems simulation, co-simulation, digital twins, and stochastic simulation. The first theme presents energy systems simulation concepts, including the different time-domain simulation models (dynamic, steady-state, scenario-based). In the second theme, challenges related to integrating different simulation models are presented, including topics such as data exchange, standardization, and APIs. The third theme presents the concept of digital twins, providing a detailed architecture for its implementation. The difference between digital twins and (co)simulation models is also presented. Finally, the fourth theme presents how to represent uncertainty when modelling and an introduction to Montecarlo simulation.</p>                                                                     |   |
| <b>Study Goals</b>                     | <p>At the end of the Energy Systems Simulation and Digital Twins course, you will be able to:</p> <p>[LO1] Categorize energy systems simulations according to model and solver properties.<br/> [LO2] Selecting simulation scope (e.g., time horizon, geography) and detail in accordance with application requirements.<br/> [LO3] Design a co-simulation including synchronization sequence and interface.<br/> [LO4] Implement a digital twin architecture for an energy systems application.<br/> [LO5] Implement and analyze the effect of aleatory and parametric uncertainty on simulation outcomes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                | <p>The course consists of lectures, assignments, and a project. The lectures introduce different topics, while the project requests students to apply the material studied in the lecture. Assignments are given each week to practice the concepts from the lecture or extend them before using such concepts in the project. During the practicums, students get to work on the project each week. Project milestones are organized in order to monitor the progress. The assignments and the project are done in groups.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Supporting scientific papers, tutorials, and notes are published weekly on Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>The grade is formed based on the student's and teams performance. A half-way project presentation is organized in Week 5 (Practicum 5) to monitor progress, accounting for 20% of the final grade. The final project presentation (oral exam) in Week 9 accounts for 60% of the final grade, while the project reports account for 20% of the final grade. In order to successfully complete the course, the students need to score a passing grade on all project aspects and during the oral exam. All students must participate in the completion of milestones. The final report is a document that combines the content of all intermediate milestones. Submission of the final report is mandatory for a passing grade.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A &amp;#150; Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| SET3125                                | Machine Learning Workflows for Digital Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. S.H. Tindemans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | Very basic programming skills in any programming language, elementary knowledge of probability theory and statistics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                         | This course will teach students the fundamentals of Machine Learning and how these can be applied to sustainable energy systems with focus on digital technologies. After this introduction to Machine learning covering selected topics of fundamentals, the students can make use of this course by taking further courses within the scope of digital energy technologies, further machine learning courses and engage in thesis work that uses AI.                                                                                                                                                 |   |
| <b>Course Contents</b>                 | The course covers the following:<br>- statistical Learning (supervised/unsupervised),<br>- supervised regression,<br>- neural Networks,<br>- designing machine learning workflows<br>- physics-informed learning for energy applications<br>- learning with inductive bias in grids                                                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | After this course, students will be able to<br>1. describe machine learning concepts,<br>2. compare concepts of effective learning in the energy system domain<br>3. Apply machine learning training strategies using calculations and analysis of energy system problem.<br>4. design elements of an machine learning workflow for the energy system                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | Lectures, homework, project, and office hours.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Relations</b>                | This course is mutual exclusive with the course EE4C12. Students can not take this course if they have taken the course EE4C12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | Written exam (100% of grade). Passing the homeworks and project is a prerequisite to be eligible for taking the written exam.<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project pass/fail grade.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). |   |

| TPM0637SET                             | Economic Policy for Sustainable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. E. Schröder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Co-responsible for assignments</b>  | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | <p>Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course.</p> <p>The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.</p>                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis?</p> <p>This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to:</p> <ul style="list-style-type: none"> <li>- consumption and production</li> <li>- efficiency and welfare</li> <li>- economic growth and sustainability</li> <li>- the choice of environmental policy targets and instruments</li> <li>- integrated assessment models</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing the course, students should be able to:</p> <ul style="list-style-type: none"> <li>- explain the main theoretical concepts used in environmental economics and how they relate to one another</li> <li>- explain how a market economy undervalues the environment</li> <li>- explain how environmental economists value the environment</li> <li>- apply economic theory to environmental problems such as climate change</li> <li>- evaluate economic policies for solving environmental problems such as climate change</li> </ul>                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Classes, quizzes, and exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>The main text is:</p> <ul style="list-style-type: none"> <li>- Harris &amp; Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor &amp; Francis.</li> </ul> <p>The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade is based exclusively on the performance in the final exam.</p> <p>In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| TPM0931SET                             | Sustainable Energy Innovation and Transitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. J.N. Quist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition.</p> <p>This course is the first course in the SET Economics &amp; Society profile. It can also be followed by other MSc students as an elective. Due to overlap, is not allowed to follow this course if you have already followed course TPM033A or 4413TRIG.</p>                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>After this course, the student:</p> <ol style="list-style-type: none"> <li>1. has knowledge of: <ul style="list-style-type: none"> <li>* the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective.</li> <li>* the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability.</li> </ul> </li> <li>2. has insight in: <ul style="list-style-type: none"> <li>* the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts.</li> </ul> </li> <li>3. has experience and basic skills in: <ul style="list-style-type: none"> <li>* exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.</li> <li>* formulating research questions.</li> <li>* writing a coherent, well-structured and well argued academic report.</li> <li>* critically reflecting on work of others and on their own work.</li> </ul> </li> </ol> |   |
| <b>Education Method</b>                | <p>This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>Scientific articles which are made available via the TU library or via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Both reports must be completed with at least grade 5.0. Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Elective</b>                        | <p>Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                            | <p>Energy<br/>Energy &amp; Industry<br/>Group work<br/>Sustainability</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| TPM403SET                       | Technology Entrepreneurship and Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Module Manager                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Instructor                      | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Instructor                      | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Instructor                      | M. Kolagar Daronkola                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Responsible for assignments     | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Co-responsible for assignments  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Contact Hours / Week<br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Education Period                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Start Education                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Exam Period                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Expected prior knowledge        | No specific knowledge is required prior to entering the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Course Contents                 | <p>This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course.</p> <p>This course is the third course in the SET Economics &amp; Society profile.</p>                                                                                                                                                                                                                                                                                           |   |
| Study Goals                     | <p>Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to:</p> <ol style="list-style-type: none"><li>1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing;</li><li>2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems;</li><li>3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products;</li><li>4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones;</li><li>5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular;</li><li>6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.</li></ol>                                                                                                                                                                                                                                                                                                                                                        |   |
| Education Method                | <p>The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Literature and Study Materials  | <p>Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted on Brightspace-site.<br/>Slides presented in class: Will be posted on Brightspace-site.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Assessment                      | <p>The assessment is composed of several elements:</p> <ol style="list-style-type: none"><li>1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade.</li><li>2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade.</li><li>3. Individual: Theory quizzes (30%)</li></ol> <p>All marks will be expressed on a scale of 1-10.</p> |   |
| Enrolment / Application         | <p>All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Remarks                         | <p>For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Targetgroup                     | <p>Master students from all master programs can apply.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Electric Mobility Systems: Electric Mobility Power Storage 2024**

| EE4536                                 | DC and AC Microgrids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>          | Dr.ir. L.M. Ramirez Elizondo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | BSc Electrical Engineering or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | Identify the challenges introduced by distributed generation<br>Describe different microgrids' layouts and sizing<br>Describe the operation and control of power electronic converters in AC and DC microgrids<br>Apply droop control in AC and DC microgrids<br>Describe the role of energy management systems in AC and DC microgrids<br>Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiorgiou                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Written exam 70%, Assignments 30%<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                           |   |



| EE4545                                  | Electrical Power Systems of the Future                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>           | Dr. A. Leki                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>         | <p>Pre-requisites (entry levels acquired in Q1 and Q2):</p> <p>Master SET / Master EE / Master EWEM<br/>           SET3065 Intelligent Electrical Power Grids<br/>           ET4107 Power System Analyses II<br/>           Matlab and Simulink skills (self study tutorial is available)</p> <p>Optional<br/>           EE4665 Uncertainty modelling and risk assessment in electrical power systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | <p>The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system.</p> <p>Learning objectives(LOs):</p> <p>LO1: Explain the impact of high RES penetration on the resiliency of future power systems.</p> <p>LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration.</p> <p>LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply.</p> <p>LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.</p> |   |
| <b>Education Method</b>                 | <p>Lectures (conceptual aspects) 2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site</p> <p>Estimated study load: 112 hours (4 ECTs)<br/>           Lectures 20 hours<br/>           Practical 24 hours<br/>           Self study 68 hours</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                       | <p>Practicum assignments and digital exam/re-exam</p> <p>The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                          | <p>Company visit in combination with lectures and practicum<br/>           @ Mariëndaal Centre of Excellence, TenneT Arnhem<br/>           Dont forget identity card or passport!</p> <p>8:00 Departure from EWI, Travel by touring bus to Arnhem.<br/>           Lectures on Operational challenges by RES for system operations.<br/>           Lessons learnt from large disturbances.<br/>           Visit TenneT Control Room (including lunch).<br/>           Practical Exercises Operational challenges and large disturbances.<br/>           Travel by touring bus to Arnhem<br/>           16:30 Arrival at EWI</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>maximum aantal deelnemers</b>        | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| ET4117                                 | Electrical Machines and Drives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. G.R. Chandra Mouli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Responsible Instructor</b>          | Dr.ir. H. Vahedi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | W. Shi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Responsible for assignments</b>     | H. Dialani                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Responsible for assignments</b>     | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Required for</b>                    | AC Machines (ET4121)<br>Control of Electrical Drives (ET4291)<br>Electric vehicles & charging technology (SET3100)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Study Goals</b>                     | <p>By the end of the course, students must be able to:</p> <ul style="list-style-type: none"> <li>- Apply Maxwell's equations to magnetic circuits including permanent magnets</li> <li>- Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor</li> <li>- Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines</li> <li>- Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines</li> <li>- Verify the operation of DC, induction and synchronous machine using tests on an experimental setup</li> <li>- Choose the appropriate power electronic converter and method for speed control and starting of the machine</li> </ul> |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Classroom lectures and problem solving workshops,</li> <li>- Demonstrations using machines and animation in the classroom,</li> <li>- Online exercises with self-evaluation,</li> <li>- Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | <p>Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6</p> <p>Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2014 (7th edition) ISBN 9781259254666</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets.</li> <li>- Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course.</li> </ul> <p>Repair possibilities:<br/>Written exam: Resit in next quarter</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                         |   |
| <b>Remarks</b>                         | <p>In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups.</p> <p>If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| ET4291SET                              | Digital modelling of electric powertrain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J. Dong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | Lab 0,0,0,4 and 0,0,0,1 instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | Digital modelling of the power train including the converter, motor and the control, including the following contents:<br>- Dynamic modelling of DC and AC Machines supplied with power electronic converters.<br>- Modelling of mechanical systems and thermal effects in drives.<br>- Modelling and simulation of digital control of electrical drives.<br>- Controller design for cascaded motor controller.<br>- Scalar and vector control, field weakening control.                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                     | Students should be able to:<br>- Model the dynamic behaviour of DC and AC machines using state space representation.<br>- Integrate ideal power electronic converter models and cascade control systems in the drive model.<br>- Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive.<br>- Calculate the steady state performance of DC and AC drives.<br>- Implement scalar control system and vector control system for induction machines.<br>- Implement vector control system for synchronous machines.<br>- Implement field weakening control for DC and AC drives.<br>- Implement the dynamic simulation models in MATLAB/Simulink.<br>- Explain the practical design of electrical drives.<br>- Evaluate the performance of an electrical drive system. |   |
| <b>Education Method</b>                | Lecture / Self-study / Simulation practical / Discussion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | - Rik De Doncker, Duco W.J. Pille, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: <a href="https://www.springer.com/gp/book/9789400701816">https://www.springer.com/gp/book/9789400701816</a><br><br>- Self-study materials on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | Individual assignment and oral exam<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SET3065                                | Intelligent Electrical Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. P. Palensky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. J.L. Rueda Torres                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets.</p> <p>Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools.</p> <p>You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done.</p> <p>This course is part of the Power track of the SET master and can also be chosen by EE master students.</p> <p>Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.</p> |   |
| <b>Study Goals</b>                     | <p>After this course students are able to</p> <p>LO1: Describe the functions and topologies of cyber-physical energy systems</p> <p>LO2: Implement data driven modelling and numerical simulation of energy systems</p> <p>LO3: Calculate variable power flows based on optimization/probability theory</p> <p>LO4: Synthesize functional requirements of energy management systems</p> <p>LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized in the following units, all units have assignments (typically a simulation):</p> <ol style="list-style-type: none"> <li>1: The energy transition and future energy systems (Peter)</li> <li>2: Analysis of steady-state performance in variable operating conditions (Jose)</li> <li>3: Assessment and control of power system dynamic performance (Jose)</li> <li>4: Lab 1 Digital and data driven modelling of interconnected systems</li> <li>5: Lab 2 Simulation and impact analysis of renewables, storage, and demand</li> <li>6: Data acquisition and communication for energy automation (Peter)</li> <li>7: Artificial intelligence-based functionalities for situational awareness (Pedro &amp; Jochen)</li> <li>8: Cyber security of smart grids (Alex)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | On Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>Assignment reports (in time!) (50%)</li> <li>Multiple choice Computer Exam (50%)</li> </ul> <p>In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture.</p> <p>Final grade calculation: <math>(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Assignment reports (in time!): resubmit deliverable</li> <li>- Multiple choice Computer Exam: computer resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                       |   |

| SET3080                                | The Necessity of Storage Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on chemistry and physics, as well as knowledge on Matlab. Level as of MSc SET students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>In a renewable energy future energy storage is an additional necessary means to match the supply and demand of energy.</p> <ul style="list-style-type: none"> <li>- Week 1: Future energy generation systems, the combination of sources on a large scale.<br/>Group work 1: program in Matlab a numerical model for energy generation, with flexible input of PV, solar heat, wind, hydro, biofuel</li> <li>- Week 2: Future energy use possibilities: look into current typical use patterns of the different sectors, and look into possible changes: electrification, different sources, etc.<br/>Group work 2: program in Matlab typical use patterns for the different sectors</li> <li>- Week 3: Estimate the possibilities for 'demand response' in the different sectors and program this in Matlab. Deduce the resulting demand for short term and long term storage. Determine the demand for transport of energy and estimate possible transport losses.<br/>Group work 3: program in Matlab the different magnitudes for demand response, short and long storage throughout the year.</li> <li>- Week 4: Storage methods introduction, pumped hydro, mechanical, compressed air, deployment possibilities.<br/>Group work 4: calculate different factors for energy density, capacity, timescale of storage. Investigate approximate efficiency.</li> <li>- Week 5: Thermal storage, high T and low T. Combined heat and power possibilities.<br/>Group work 5: Investigate energy densities, efficiencies, capacities, exergies.</li> <li>- Week 6: Biomass and biofuels, future prospects for storage. Introduction to scalable chemical energy storage methods: batteries and fuels. What would electrification of the chemical industry mean.<br/>Group work: investigate biofuel resources, calculate efficiencies and compare to solar PV and electrolysis numbers.</li> <li>- Week 7: Hand in report and model. Group presentations of the model calculations (where each group has different boundary conditions). Discuss how the boundary conditions influence the outcome of the storage demand.</li> </ul> |   |
| <b>Study Goals</b>                     | <p>At the end of the lectures and group work the students should be able to:</p> <ul style="list-style-type: none"> <li>- Reproduce and reflect on the role of storage in a renewable energy based society.</li> <li>- Explain how different timescales for energy storage are relevant for the future system.</li> <li>- List and explain the different storage solutions in terms of typical energy density, cost and efficiency.</li> <li>- Tell about the different drivers for energy conversion including demand for storage, and demand for transport of energy.</li> <li>- Reflect on the energy efficiency of intermediate storage.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | <p>The course contains seven weeks with 4-hour contact time. The 4 hours will be divided in lectures, group work applying the lecture content and a question time.<br/>The goal is to come to a finished assignment after six weeks that can be presented in class the seventh week.</p> <p>The structure of the course is as follows: each week there is first a class in which a background is handed out, then there is a working group session where the supplied information is used to start modelling aspects of a future energy system. The students will be divided over groups with each group an individual model question. The lesson is concluded with a question session. This will continue 6 weeks and the 7th week the different models will be presented by the students.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <p>Slides<br/>Articles, reports and reviews.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:<br/>Individual written exam (70% of the final grade).<br/>Group work (30% of the final grade, based on a report produced by the group).<br/>For one part may be scored 5,0 as long as the other part compensates this.</p> <p>Final grade calculation: <math>(0,7 * \text{written exam}) + (0,3 * \text{group project})</math></p> <p>Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/>- written exam: written resit<br/>- group work: resubmit report</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| SET3085                                | Hydrogen Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed. |   |
| <b>Study Goals</b>                     | <p>After this course students:</p> <p>Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms.</p> <p>Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it.</p> <p>Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.</p>                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>This course consists of 14 2-hour lectures.</p> <p>These lecture also include some problem solving.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Relations</b>                | This course is a core course in the Sustainable Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Literature and Study Materials</b>  | The exam will be based on the lecture slides and on selected research papers as provided on Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written exam (100%)</li> </ul> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                         |   |
| <b>Co-Instructor</b>                   | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| SET3095                                | Electronic Power Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | S.A. Rivera Iunissi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.                                                                                                                                                                                                                                                                 |   |
| <b>Study Goals</b>                     | To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | <p>Lectures</p> <p>Problem solving sessions</p> <p>Practicum Lectures</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written final exam (closed book) (70%)</li> <li>- Assignments (30%)</li> </ul> <p><math>T = \text{grade for written final exam} * 0,7 + \text{grade for assignments} * 0,3</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| SET3100                                | Electric vehicle and charging technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. G.R. Chandra Mouli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | <p>SET3100 mainly builds upon the prior knowledge of the student on:</p> <ul style="list-style-type: none"> <li>- Understanding the various type of electric machines such as induction and permanent magnet machines, Torque-Speed characteristics of an electric machine in constant Torque and constant Power regions</li> <li>- Understanding the concept of power electronic converters such as various types (DC-DC, DC-AC, etc.), voltage ratio (step-up, step-down) and pulse width modulation</li> </ul> <p>Such prior knowledge can be built via courses such as SET3095/ET4119 Electronic Power Conversion, ET4117 Electric machines &amp; drives, ME44210 Drive &amp; Energy Systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>This course focusses on the technology of (hybrid) electric vehicles and its charging technology</p> <p>Tank to Wheel, Well to Tank, Well to Wheel efficiencies and emission estimation for conventional, hybrid and fully electric vehicles based on energy source for propulsion.</p> <p>Types, operational principles, advantages and disadvantages of vehicle architectures: internal combustion engine; micro, mild and full hybrids (series, parallel, series-parallel), plug-in hybrids, full electrics, fuel cell vehicles;</p> <p>Vehicle load forces &amp; dynamics, mechanical transmission components, power and energy consumption, drive cycles, battery sizing and electric vehicle range, MPGe, planetary gears as a power-split, regenerative braking.</p> <p>Principles of battery operation, battery types, the key performance parameters of batteries, from cell to battery pack, charge-discharge curves, battery and thermal management systems.</p> <p>Fundamentals and types of electric machine drives used in electric cars: its characteristics, efficiency, basics of motor control.</p> <p>Basics of power electronic converters used in electric cars and chargers, topologies and control.</p> <p>Charging methods of EVs: AC charging, DC charging, vehicle-to-grid technology, wireless charging, battery swap; charging time estimation, effects of fast charging on battery; Smart charging of EVs and vehicle-to-grid, communication protocols for smart charging.</p> <p>Modelling and investigating the performance of the electric vehicle drivetrain using quasi-static simulation in MATLAB/Simulink.</p> <p>Case study of EV: Toyota Prius HEV, Tesla Model S, Porsche Taycan, Audi e-tron, Hyundai Ioniq 5</p> <p>The course will have a design project where students have to form teams to design their own EV by making choices for the vehicle drivetrain and corresponding charging technology. Emphasis will be on justifying and quantifying the technological design choices including drivetrain architecture, well-to-wheel efficiency and emissions, energy storage, power electronics, type of motor and charging method.</p> |   |
| <b>Study Goals</b>                     | <p>The goal of this course is to educate students to design and evaluate various (hybrid) electric vehicles with respect to their performance, efficiency, sustainability, charging methodologies and potential for grid support.</p> <p>At the end of the course, the students should be able to:</p> <ul style="list-style-type: none"> <li>Evaluate the performance of hybrids (series, parallel, series-parallel), plug-in hybrid and fully electric drivetrain architectures based on power and energy consumption, efficiency, driving range, well-to-wheel emissions and charging time.</li> <li>Explain the design trade-offs in the energy sources (battery, fuel cell, super capacitors) and motors used in electric vehicles based on the key performance indicators like efficiency, energy and/or power density, lifetime, and control.</li> <li>Analyse various charging technologies for electric vehicles (conductive, inductive or battery swap) based on power electronics converters used, type of the charger (AC Vs DC), connector types, power levels, and modes.</li> <li>Investigate the performance of the (hybrid) electric vehicle drivetrain using quasi-static modelling and simulation in MATLAB/Simulink.</li> <li>Design a (hybrid) electric drivetrain by sizing and making the relevant choice of the electric machine, internal combustion engine, energy source (batteries, fuel cells, etc.), power electronic converters, transmissions components, and charging technology for a given vehicle specification.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Method</b>                | <p>Classroom lectures and problem solving workshops,</p> <p>Discussion and case study,</p> <p>Demonstrations using tools and animation in the classroom,</p> <p>Online exercises with self-evaluation,</p> <p>Design project</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>- Lecture slides &amp; handouts</li> <li>- EdX MOOC Electric cars</li> <li>- Primary reading - Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, Third Edition, 2018, Mehrdad Ehsani, Yimin Gao, Ali Emadi, CRC Press, ISBN 9781138330498</li> <li>- Primary reading - S. Rivera, S. Kouro, S. Vazquez, S. M. Goetz, R. Lizana and E. Romero-Cadaval, "Electric Vehicle Charging Infrastructure: From Grid to Battery," in IEEE Industrial Electronics Magazine, vol. 15, no. 2, pp. 37-51, June 2021, doi: 10.1109/MIE.2020.3039039.</li> <li>- Supplementary reading - Chapter 1 &amp; 14 of Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles, John G. Hayes, G. Abas Goodarzi, ISBN: 978-1-119-06364-3, 2018</li> <li>- Supplementary reading Electric and Hybrid Vehicles: Design Fundamentals, Second Edition, 2010, Iqbal Husain, CRC Press, ISBN 9781439811757</li> </ul> <p>The enlisted books are available via the TU Delft library (free e-books), via the ETV desk &amp; bookstore, via studystore.nl and the publisher website</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written exam (70% weightage) - Closed book, Scientific (non-graphic) calculator can be used. No formula sheets</li> <li>- Design assignment (30% weightage) - Group assignment with Written report + oral presentation. Students must register in groups for the design assignment within the first two weeks of the course.</li> </ul> <p>Final grade calculation: <math>(0.7 * \text{Written exam} + 0.3 * \text{Design assignment})</math></p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <p>Written exam: Resit in next quarter</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |



|         |                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------|
| Remarks | Assignment: Re-submit assignment report in the next quarter based on feedback received (Grade is capped)      |
|         | Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). |
|         | Follow up course: ET4291SET Digital modelling of electric powertrain/ ET4291 Control of Electrical Drives     |

| SET3110                         | Energy Storage in Batteries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | Dr.ir. M. Wagemaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| Instructor                      | A. Vasileiadis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Instructor                      | Dr. E.M. Kelder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Instructor                      | T. Famprakis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Instructor                      | Dr. X. Wang                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Contact Hours / Week<br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Education Period                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Start Education                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Exam Period                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Course Contents                 | <p>Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar.</p> <p>In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including:</p> <p>Redox Flow<br/>Na-ion<br/>next generation Li-ion batteries<br/>Solid State batteries<br/>Supercapacitors</p>                                                                                                                                                                                                                                                                                                                                                            |   |
| Study Goals                     | <p>At the end of the battery lectures students should be able to:</p> <ul style="list-style-type: none"> <li>- List the major types of Li and other batteries and their characteristics.</li> <li>- Reproduce the major challenges in batteries.</li> <li>- Explain how a battery works and the functioning of each component.</li> <li>- List and explain the different reaction types in batteries.</li> <li>- Explain the different charge transfer processes in batteries.</li> <li>- List and explain the different mechanisms that limit the cycle life of batteries.</li> <li>- Apply Nernst Law describing redox reactions.</li> <li>- Calculate conductivities and overpotentials.</li> <li>- Apply the Tafel equation describing the charge transfer between electrode and electrolyte.</li> <li>- Calculate energy densities and power densities from observed properties.</li> <li>- Derive from a phase diagram the different phases that will occur upon lithiation.</li> <li>- Apply the Gibbs Phase rule to predict constant of variable output voltage.</li> <li>- Derive the voltage curve from the shape of the Gibbs Free energy.</li> <li>- Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials.</li> <li>- Compare different battery systems and differentiate.</li> <li>- Discuss scientific results observed in battery performance and to relate this to material properties.</li> </ul> |   |
| Education Method                | <p>The course contains fourteen 2-hour lectures.</p> <p>The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery.</p> <p>In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed.</p> <p>In the final lecture example exams will be discussed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Course Relations                | This course is a core course in the Sustainable Energy Energy Storage profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Literature and Study Materials  | <p>R.A. Huggins: Advanced Batteries. Available via the library as pdf's: <a href="http://www.springerlink.com/content/nq6404/?MUD=MP">http://www.springerlink.com/content/nq6404/?MUD=MP</a></p> <p>Slides<br/>Articles</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Assessment                      | <p>The final grade of the course consists of the following components:</p> <p>Written exam (100% of the final grade)</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Co-Instructor                   | Dr. E.M. Kelder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |



|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Electric Mobility & Economics: Electric Mobility Power Economics 2024**

| EE4536                                 | DC and AC Microgrids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>          | Dr.ir. L.M. Ramirez Elizondo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | BSc Electrical Engineering or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | Identify the challenges introduced by distributed generation<br>Describe different microgrids' layouts and sizing<br>Describe the operation and control of power electronic converters in AC and DC microgrids<br>Apply droop control in AC and DC microgrids<br>Describe the role of energy management systems in AC and DC microgrids<br>Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiorgiou                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Written exam 70%, Assignments 30%<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                           |   |

| EE4545                                  | Electrical Power Systems of the Future                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>           | Dr. A. Leki                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>         | Pre-requisites (entry levels acquired in Q1 and Q2):<br><br>Master SET / Master EE / Master EWEM<br>SET3065 Intelligent Electrical Power Grids<br>ET4107 Power System Analyses II<br>Matlab and Simulink skills (self study tutorial is available)<br><br>Optional<br>EE4665 Uncertainty modelling and risk assessment in electrical power systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                  | <p>The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system.</p> <p>Learning objectives(LOs):</p> <p>LO1: Explain the impact of high RES penetration on the resiliency of future power systems.</p> <p>LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration.</p> <p>LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply.</p> <p>LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.</p> |   |
| <b>Education Method</b>                 | <p>Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site</p> <p>Estimated study load: 112 hours (4 ECTs)<br/>Lectures 20 hours<br/>Practical 24 hours<br/>Self study 68 hours</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                       | <p>Practicum assignments and digital exam/re-exam</p> <p>The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                          | <p>Company visit in combination with lectures and practicum<br/>@ Mariëndaal Centre of Excellence, TenneT Arnhem<br/>Dont forget identity card or passport!</p> <p>8:00 Departure from EWI, Travel by touring bus to Arnhem.<br/>Lectures on Operational challenges by RES for system operations.<br/>Lessons learnt from large disturbances.<br/>Visit TenneT Control Room (including lunch).<br/>Practical Exercises Operational challenges and large disturbances.<br/>Travel by touring bus to Arnhem<br/>16:30 Arrival at EWI</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>maximum aantal deelnemers</b>        | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| ET4117                                 | Electrical Machines and Drives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. G.R. Chandra Mouli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Responsible Instructor</b>          | Dr.ir. H. Vahedi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | W. Shi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Responsible for assignments</b>     | H. Dialani                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible for assignments</b>     | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Required for</b>                    | AC Machines (ET4121)<br>Control of Electrical Drives (ET4291)<br>Electric vehicles & charging technology (SET3100)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Expected prior knowledge</b>        | Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | By the end of the course, students must be able to:<br><ul style="list-style-type: none"> <li>- Apply Maxwell's equations to magnetic circuits including permanent magnets</li> <li>- Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor</li> <li>- Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines</li> <li>- Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines</li> <li>- Verify the operation of DC, induction and synchronous machine using tests on an experimental setup</li> <li>- Choose the appropriate power electronic converter and method for speed control and starting of the machine</li> </ul> |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>- Classroom lectures and problem solving workshops,</li> <li>- Demonstrations using machines and animation in the classroom,</li> <li>- Online exercises with self-evaluation,</li> <li>- Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6<br>Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2014 (7th edition) ISBN 9781259254666                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets.</li> <li>- Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course.</li> </ul> <p>Repair possibilities:<br/>Written exam: Resit in next quarter</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                     |   |
| <b>Remarks</b>                         | <p>In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups.</p> <p>If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| ET4291SET                              | Digital modelling of electric powertrain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J. Dong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | Lab 0,0,0,4 and 0,0,0,1 instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | Digital modelling of the power train including the converter, motor and the control, including the following contents:<br>- Dynamic modelling of DC and AC Machines supplied with power electronic converters.<br>- Modelling of mechanical systems and thermal effects in drives.<br>- Modelling and simulation of digital control of electrical drives.<br>- Controller design for cascaded motor controller.<br>- Scalar and vector control, field weakening control.                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                     | Students should be able to:<br>- Model the dynamic behaviour of DC and AC machines using state space representation.<br>- Integrate ideal power electronic converter models and cascade control systems in the drive model.<br>- Model simple mechanical systems (wheel, gears, pulley&belt) in the electrical drive.<br>- Calculate the steady state performance of DC and AC drives.<br>- Implement scalar control system and vector control system for induction machines.<br>- Implement vector control system for synchronous machines.<br>- Implement field weakening control for DC and AC drives.<br>- Implement the dynamic simulation models in MATLAB/Simulink.<br>- Explain the practical design of electrical drives.<br>- Evaluate the performance of an electrical drive system. |   |
| <b>Education Method</b>                | Lecture / Self-study / Simulation practical / Discussion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | - Rik De Doncker, Duco W.J. Pille, Andre Veltman, Advanced Electrical Drives Analysis, Modeling, Control, Springer, available: <a href="https://www.springer.com/gp/book/9789400701816">https://www.springer.com/gp/book/9789400701816</a><br><br>- Self-study materials on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | Individual assignment and oral exam<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SET3065                                | Intelligent Electrical Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. P. Palensky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. J.L. Rueda Torres                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets.</p> <p>Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools.</p> <p>You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done.</p> <p>This course is part of the Power track of the SET master and can also be chosen by EE master students.</p> <p>Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.</p> |   |
| <b>Study Goals</b>                     | <p>After this course students are able to</p> <p>LO1: Describe the functions and topologies of cyber-physical energy systems</p> <p>LO2: Implement data driven modelling and numerical simulation of energy systems</p> <p>LO3: Calculate variable power flows based on optimization/probability theory</p> <p>LO4: Synthesize functional requirements of energy management systems</p> <p>LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized in the following units, all units have assignments (typically a simulation):</p> <ol style="list-style-type: none"> <li>1: The energy transition and future energy systems (Peter)</li> <li>2: Analysis of steady-state performance in variable operating conditions (Jose)</li> <li>3: Assessment and control of power system dynamic performance (Jose)</li> <li>4: Lab 1 Digital and data driven modelling of interconnected systems</li> <li>5: Lab 2 Simulation and impact analysis of renewables, storage, and demand</li> <li>6: Data acquisition and communication for energy automation (Peter)</li> <li>7: Artificial intelligence-based functionalities for situational awareness (Pedro &amp; Jochen)</li> <li>8: Cyber security of smart grids (Alex)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | On Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <p>Assignment reports (in time!) (50%)</p> <p>Multiple choice Computer Exam (50%)</p> <p>In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture.</p> <p>Final grade calculation: <math>(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Assignment reports (in time!): resubmit deliverable</li> <li>- Multiple choice Computer Exam: computer resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                    |   |

| SET3095                                | Electronic Power Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | S.A. Rivera Iunnissi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures<br>Problem solving sessions<br>Practicum Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written final exam (closed book) (70%)</li> <li>- Assignments (30%)</li> </ul> <p><math>T = \text{grade for written final exam} * 0,7 + \text{grade for assignments} * 0,3</math></p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| SET3100                                | Electric vehicle and charging technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. G.R. Chandra Mouli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | <p>SET3100 mainly builds upon the prior knowledge of the student on:</p> <ul style="list-style-type: none"> <li>- Understanding the various type of electric machines such as induction and permanent magnet machines, Torque-Speed characteristics of an electric machine in constant Torque and constant Power regions</li> <li>- Understanding the concept of power electronic converters such as various types (DC-DC, DC-AC, etc.), voltage ratio (step-up, step-down) and pulse width modulation</li> </ul> <p>Such prior knowledge can be built via courses such as SET3095/ET4119 Electronic Power Conversion, ET4117 Electric machines &amp; drives, ME44210 Drive &amp; Energy Systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>This course focusses on the technology of (hybrid) electric vehicles and its charging technology</p> <p>Tank to Wheel, Well to Tank, Well to Wheel efficiencies and emission estimation for conventional, hybrid and fully electric vehicles based on energy source for propulsion.</p> <p>Types, operational principles, advantages and disadvantages of vehicle architectures: internal combustion engine; micro, mild and full hybrids (series, parallel, series-parallel), plug-in hybrids, full electrics, fuel cell vehicles;</p> <p>Vehicle load forces &amp; dynamics, mechanical transmission components, power and energy consumption, drive cycles, battery sizing and electric vehicle range, MPGe, planetary gears as a power-split, regenerative braking.</p> <p>Principles of battery operation, battery types, the key performance parameters of batteries, from cell to battery pack, charge-discharge curves, battery and thermal management systems.</p> <p>Fundamentals and types of electric machine drives used in electric cars: its characteristics, efficiency, basics of motor control.</p> <p>Basics of power electronic converters used in electric cars and chargers, topologies and control.</p> <p>Charging methods of EVs: AC charging, DC charging, vehicle-to-grid technology, wireless charging, battery swap; charging time estimation, effects of fast charging on battery; Smart charging of EVs and vehicle-to-grid, communication protocols for smart charging.</p> <p>Modelling and investigating the performance of the electric vehicle drivetrain using quasi-static simulation in MATLAB/Simulink.</p> <p>Case study of EV: Toyota Prius HEV, Tesla Model S, Porsche Taycan, Audi e-tron, Hyundai Ioniq 5</p> <p>The course will have a design project where students have to form teams to design their own EV by making choices for the vehicle drivetrain and corresponding charging technology. Emphasis will be on justifying and quantifying the technological design choices including drivetrain architecture, well-to-wheel efficiency and emissions, energy storage, power electronics, type of motor and charging method.</p> |   |
| <b>Study Goals</b>                     | <p>The goal of this course is to educate students to design and evaluate various (hybrid) electric vehicles with respect to their performance, efficiency, sustainability, charging methodologies and potential for grid support.</p> <p>At the end of the course, the students should be able to:</p> <ul style="list-style-type: none"> <li>Evaluate the performance of hybrids (series, parallel, series-parallel), plug-in hybrid and fully electric drivetrain architectures based on power and energy consumption, efficiency, driving range, well-to-wheel emissions and charging time.</li> <li>Explain the design trade-offs in the energy sources (battery, fuel cell, super capacitors) and motors used in electric vehicles based on the key performance indicators like efficiency, energy and/or power density, lifetime, and control.</li> <li>Analyse various charging technologies for electric vehicles (conductive, inductive or battery swap) based on power electronics converters used, type of the charger (AC Vs DC), connector types, power levels, and modes.</li> <li>Investigate the performance of the (hybrid) electric vehicle drivetrain using quasi-static modelling and simulation in MATLAB/Simulink.</li> <li>Design a (hybrid) electric drivetrain by sizing and making the relevant choice of the electric machine, internal combustion engine, energy source (batteries, fuel cells, etc.), power electronic converters, transmissions components, and charging technology for a given vehicle specification.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Method</b>                | <p>Classroom lectures and problem solving workshops,</p> <p>Discussion and case study,</p> <p>Demonstrations using tools and animation in the classroom,</p> <p>Online exercises with self-evaluation,</p> <p>Design project</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>- Lecture slides &amp; handouts</li> <li>- EdX MOOC Electric cars</li> <li>- Primary reading - Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, Third Edition, 2018, Mehrdad Ehsani, Yimin Gao, Ali Emadi, CRC Press, ISBN 9781138330498</li> <li>- Primary reading - S. Rivera, S. Kouro, S. Vazquez, S. M. Goetz, R. Lizana and E. Romero-Cadaval, "Electric Vehicle Charging Infrastructure: From Grid to Battery," in IEEE Industrial Electronics Magazine, vol. 15, no. 2, pp. 37-51, June 2021, doi: 10.1109/MIE.2020.3039039.</li> <li>- Supplementary reading - Chapter 1 &amp; 14 of Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles, John G. Hayes, G. Abas Goodarzi, ISBN: 978-1-119-06364-3, 2018</li> <li>- Supplementary reading Electric and Hybrid Vehicles: Design Fundamentals, Second Edition, 2010, Iqbal Husain, CRC Press, ISBN 9781439811757</li> </ul> <p>The enlisted books are available via the TU Delft library (free e-books), via the ETV desk &amp; bookstore, via studystore.nl and the publisher website</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written exam (70% weightage) - Closed book, Scientific (non-graphic) calculator can be used. No formula sheets</li> <li>- Design assignment (30% weightage) - Group assignment with Written report + oral presentation. Students must register in groups for the design assignment within the first two weeks of the course.</li> </ul> <p>Final grade calculation: <math>(0.7 * \text{Written exam} + 0.3 * \text{Design assignment})</math></p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <p>Written exam: Resit in next quarter</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |



|                |                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------|
| <b>Remarks</b> | Assignment: Re-submit assignment report in the next quarter based on feedback received (Grade is capped)      |
|                | Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). |
|                | Follow up course: ET4291SET Digital modelling of electric powertrain/ ET4291 Control of Electrical Drives     |

| <b>TPM0637SET</b>                      | <b>Economic Policy for Sustainable Energy</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>4</b> |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Module Manager</b>                  | Dr. E. Schröder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Co-responsible for assignments</b>  | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Expected prior knowledge</b>        | Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Course Contents</b>                 | <p>The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.</p> <p>Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis?</p> <p>This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to:</p> <ul style="list-style-type: none"> <li>- consumption and production</li> <li>- efficiency and welfare</li> <li>- economic growth and sustainability</li> <li>- the choice of environmental policy targets and instruments</li> <li>- integrated assessment models</li> </ul> |          |
| <b>Study Goals</b>                     | <p>After completing the course, students should be able to:</p> <ul style="list-style-type: none"> <li>- explain the main theoretical concepts used in environmental economics and how they relate to one another</li> <li>- explain how a market economy undervalues the environment</li> <li>- explain how environmental economists value the environment</li> <li>- apply economic theory to environmental problems such as climate change</li> <li>- evaluate economic policies for solving environmental problems such as climate change</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Education Method</b>                | Classes, quizzes, and exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Literature and Study Materials</b>  | <p>The main text is:</p> <ul style="list-style-type: none"> <li>- Harris &amp; Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor &amp; Francis.</li> </ul> <p>The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Assessment</b>                      | <p>The final grade is based exclusively on the performance in the final exam.</p> <p>In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |

| TPM0931SET                             | Sustainable Energy Innovation and Transitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. J.N. Quist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition.</p> <p>This course is the first course in the SET Economics &amp; Society profile. It can also be followed by other MSc students as an elective. Due to overlap, is not allowed to follow this course if you have already followed course TPM033A or 4413TRIG.</p>                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>After this course, the student:</p> <ol style="list-style-type: none"> <li>1. has knowledge of: <ul style="list-style-type: none"> <li>* the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective.</li> <li>* the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability.</li> </ul> </li> <li>2. has insight in: <ul style="list-style-type: none"> <li>* the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts.</li> </ul> </li> <li>3. has experience and basic skills in: <ul style="list-style-type: none"> <li>* exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.</li> <li>* formulating research questions.</li> <li>* writing a coherent, well-structured and well argued academic report.</li> <li>* critically reflecting on work of others and on their own work.</li> </ul> </li> </ol> |   |
| <b>Education Method</b>                | <p>This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>Scientific articles which are made available via the TU library or via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Both reports must be completed with at least grade 5.0. Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Elective</b>                        | <p>Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                            | <p>Energy<br/>Energy &amp; Industry<br/>Group work<br/>Sustainability</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| TPM403SET                              | Technology Entrepreneurship and Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | M. Kolagar Daronkola                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Responsible for assignments</b>     | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-responsible for assignments</b>  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | No specific knowledge is required prior to entering the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course.</p> <p>This course is the third course in the SET Economics &amp; Society profile.</p>                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to:</p> <ol style="list-style-type: none"> <li>1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing;</li> <li>2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems;</li> <li>3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products;</li> <li>4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones;</li> <li>5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular;</li> <li>6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.</li> </ol>                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | <p>Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted on Brightspace-site.<br/>Slides presented in class: Will be posted on Brightspace-site.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>The assessment is composed of several elements:</p> <ol style="list-style-type: none"> <li>1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade.</li> <li>2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade.</li> <li>3. Individual: Theory quizzes (30%)</li> </ol> <p>All marks will be expressed on a scale of 1-10.</p> |   |
| <b>Enrolment / Application</b>         | All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Remarks</b>                         | For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Targetgroup</b>                     | Master students from all master programs can apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Track Heating and Cooling 2024

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Heat Supply Systems: Sources - LS Heat - Buildings 2024**

| ME45075                                | Refrigeration & Heat Pumps Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Ramdin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Prof.dr.ir. T.J.H. Vlugt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on the first and second law of thermodynamics (a short introduction to this will also be provided during the lectures). Students who are missing this we explain the most important concepts again)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>The course consists of the following content:</p> <ul style="list-style-type: none"> <li>-Introduction of thermodynamic cycles for heating and cooling. The working field of refrigeration and heat pumps;</li> <li>-Overview of the most important refrigeration / heat pumps systems: Mechanically driven heat systems, thermally driven systems, solid state systems and gas cycles. Comparison of these systems;</li> <li>-Components of refrigeration and heat pump cycles (compressors, valves, condensers, evaporators, turbines etc.);</li> <li>-analysis of cooling and heating cycles using the 1st and second law of thermodynamics, exergy analysis;</li> <li>-gas liquification;</li> <li>-techno-economic analysis of refrigeration and heat pump cycles;</li> <li>-Working fluids. Refrigerants for mechanical vapour compression; refrigerating machines: limits of application, evaluation of ODP and GWP;</li> <li>-Gas cycle refrigerating machines. Gas-phase cycles;</li> <li>-Magnetic refrigeration / heat pumping;</li> </ul>                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course, the students are able to:</p> <ul style="list-style-type: none"> <li>explain, reproduce and apply thermodynamic concepts in relation to refrigeration machines and heat pumps, considering their economic and environmental impact.</li> <li>recognize and apply the working principles, specific characteristics and application range of the most relevant methods to generate cooling and heating.</li> <li>explain and apply the concepts of gas liquefaction</li> <li>perform an energy and exergy analysis of refrigeration machines and heat pumps.</li> <li>evaluate the impact of the alternative concepts for refrigerating machines and heat pumps.</li> <li>sizing of heat pump and refrigeration equipment.</li> <li>perform techno-economic analysis of heat pumps and refrigeration systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>Course sheets / relevant papers / additional material provided during the lectures and on Brightspace</p> <ul style="list-style-type: none"> <li>-Moran&amp;Shapiro, Fundamentals of Engineering Thermodynamics, 9th edition</li> <li>-Dincer, Refrigeration systems and applications (available online via the TU Delft library)</li> <li>-Kiss/Infante Ferreira, Heat pumps in chemical process industry (available online via TU Delft library)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| ME45111SET                             | Heating and Cooling Demand of Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr.ir. W.H. van der Spoel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr.ir. M.J. Tenpierik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A. Rasooli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A.I. Joseph Thaddeus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course is aimed at understanding the dynamics of buildings thermal behavior and at learning how to optimize their design and operation through thermal modeling, &amp; simulation. You will learn to solve problems like: how to shift or reduce peak heating demand to obtain a better match with a renewable energy system. What is thermally more efficient: a heavy concrete building or a light timber-frame building? How effective is night ventilation in warm periods? This course will provide you with answers to questions relating to dynamic thermal behavior in buildings. Amongst others you will learn about:</p> <ul style="list-style-type: none"> <li>- Energy chain from supply to demand and design steps</li> <li>- Basics of thermal comfort</li> <li>- Energy flows in buildings (transmission, ventilation &amp; infiltration, solar heat gains and internal heat gains)</li> <li>- Energy balance</li> <li>- Steady state simulations for peak load and energy usage estimates in pre-design (1 node energy balance model)</li> <li>- Transient simulation: The role of inertia in construction; setting up dynamic balance equations (Fourier equation), setting up a suitable RC network (multi-node energy balances); Solving the equations through finite difference and through transfer functions</li> </ul> <p>This is accompanied by many exercises to understand the optimization of building characteristics and peak shaving/shifting. The course ends up with an overview of energy conversion systems that can be used to supply the needed heating and cooling to buildings.</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>- explain how the heating and cooling needs of a building are determined by diverse building design parameters, weather, occupancy and comfort preferences of occupants.</li> <li>- translate this knowledge into design calculations (using steady state and transient balances) to optimize the energy demand towards zero energy buildings, to size the heating and cooling equipment, and to provide a good indoor thermal comfort.</li> </ul> <p>More specifically, you will be able to:</p> <ul style="list-style-type: none"> <li>- derive the mathematical equations describing the thermal performance of the various building components.</li> <li>- combine these equations to simulate the whole building system.</li> <li>- use dynamic simulations to answer questions about the optimal energy design, heating and cooling capacities, energy use, comfort.</li> <li>- translate the fuzzy term "comfort" into design requirements</li> <li>- simulate how thermostat control allows for peak shaving and shifting.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>The course is built around a case study that runs like a common thread between the different course items. The case study is a self-paced individual assignment divided into 4 parts, that are discussed during workshops.</p> <p>Type of contact: Lectures, videos, guided exercises, 3 workshops and excursion. Sometimes we flip the classroom.</p> <p>Number of contact hours/week: 4</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>Formative assessment: 4 assignments with answers for self-study. Each of the assignments is discussed during workshops sessions of 2 hours.</p> <p>Summative assessment: Written exam (50% of grade) + Midterm quiz (50% of grade). The midterm quiz relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the written exam are required to pass.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Tags</b>                            | <p>Analysis<br/>Building Engineering<br/>Group work<br/>Matlab<br/>Modelling<br/>Numeric Methods<br/>Thermodynamics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| ME45165SET                              | Equipment for Heat and Mass Transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Prof.dr. K. Hooman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-responsible for assignments</b>   | Dr. H.B. Eral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>   | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Summary</b>                          | The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                  | <p>Thermal energy can be converted or stored. Heat exchangers are needed in either case.</p> <p>This course will cover:</p> <ul style="list-style-type: none"> <li>Fundamentals of heat &amp; mass transfer,</li> <li>Design of different types of heat exchangers (shell and tube, plate, finned tubes ), regenerators/recuperators, cooling towers, and furnaces,</li> <li>Fouling of heat exchanger surfaces,</li> <li>characteristics of equipment for various separation processes</li> <li>the role of mass transfer in the sizing and design of the separation equipment.</li> <li>Cost Analysis,</li> <li>Pinch Technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>By the end of this course, you'd be able to design and rate heat &amp; mass transfer equipment. This can help you, for instance, to:</p> <ul style="list-style-type: none"> <li>- Quantify heating and cooling processes and requirements in various industrial applications such as: metallurgy, waste processes, food, agriculture, desalinization, freeze-drying.</li> <li>- Explain design requirements for different industrial heating and cooling processes.</li> <li>- Analyze existing industrial heating and cooling processes.</li> <li>- Identify decarbonization solutions for existing industrial heating and cooling processes.</li> <li>- Design suitable components/facilities/technologies for industrial heating and cooling processes.</li> <li>- Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes.</li> <li>- Comprehends the role of heat and mass transfer in the dimensioning of equipment for separation processes.</li> </ul>                                                                                             |   |
| <b>Education Method</b>                 | Lectures, assignments, self-study, team work, tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Computer Use</b>                     | Standard engineering packages including ASPEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>   | <p>Course material:</p> <p>1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738</p> <p>2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Wiley &amp; Sons, 2011.</p> <p>3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer.</p> <p>4: Course sheets / relevant papers</p> <p>5: R K Sinnott, Gavin P Towler: Chemical engineering design, Coulson &amp; Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020.</p> <p>6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984.</p> <p>7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004.</p> <p>Further reading</p> <p>Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son</p> <p>Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK</p> |   |
| <b>Assessment</b>                       | <p>Formative:</p> <p>In-class exercises/quizzes</p> <p>Summative:</p> <p>Group project report &amp; presentation (50%)</p> <p>Individual exam (50%)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Permitted Materials during Tests</b> | books, lecture notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |



| SET3200                                | Heating and Cooling Technologies from Near-Ambient Temperature Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>There are several technologies which can obtain heat or cool from the local environment, which are called ambient temperature sources. In this course, you will learn the process of design for ambient or near ambient temperature sources. Typically, air source systems are more variable and less efficient and less easy to operate, but easily accessible, while the shallow subsurface offers a more stable and efficient source, but less easy to access. Water sources are in between. In this course you will learn which pros and cons prevail under various conditions and how you can make an efficient design for either type.</p> <p>You will investigate two different technologies in depth, and encounter several key design aspects including:<br/> Field data collection: Undertake the design and go to the field to measure important quantities that inform the design (based on a location)<br/> Quantification of heat transfer processes<br/> Technology sizing and integration<br/> Identify key modes of operation<br/> Predicted performance</p> <p>Remark: Heat pump is applied. Only basic energy flows. No details on HP itself</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you are able to:<br/> design a H&amp;C system with a near ambient temperature H&amp;C source.<br/> evaluate design suitability to be a significant contributor to the energy needs and year-round energy flows: hydraulic connections and various modes of operation.<br/> This requires the integration of various sub technologies which you will also learn to design and quantify:<br/> Borehole Heat Exchanger (BHE), including non-stationary conduction and interference.<br/> Aquathermal: Utilisation of heat from urban water infrastructure: surface, sewerage and drinking water. This includes surveying of field conditions to inform the design process [including data collection in the field].<br/> Other sources of thermal energy: air, asphalt.<br/> Integration with Heat pump and storage facilities.<br/> Design sizing / system capacity.</p>                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | <p>Type of contact: lectures, workshops/work session, excursion, fieldwork<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting grouped tables<br/> Collegerama: nno<br/> Hybrid: no, fully on-campus</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | Materials provided: Reader, Lecture slides, papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Summative: 1) Aquathermal mini project in groups (25%)<br/> 2) Duo assignment with report describing two alternative designs with source(s) of ambient temperature (75%).</p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> - Assignment: resubmit deliverable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Tags</b>                            | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>Energy<br/> Sustainability<br/> Thermodynamics<br/> exergy</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| SET3205                                | Heat Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>Due to temporal mismatch in availability and demand for heat, heat storage is needed. In this course you will learn that the need for storage is determined by the variations and mismatches in demand and supply profiles. Also, Power to heat is enabled by heat storage. Depending on the required storage requirements (volume, cycle length, temperature level) you will learn to design a heat storage system and quantify heat losses. The heat storage technologies covered in this course are:</p> <ul style="list-style-type: none"> <li>-Sensible heat storage: subsurface, pits, tanks</li> <li>-Latent heat storage: Phase change materials</li> </ul> <p>Because heat storage systems enable the use of heat sources, but are no heat sources themselves, the economics of heat storage is often complex. Therefore, the integration of heat storage in business cases is also covered in this course.</p>                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>After taking this course you are able to:</p> <ul style="list-style-type: none"> <li>Assess the need for heat storage.</li> <li>Identify storage requirements for various supply-demand conditions.</li> <li>Select appropriate storage media for different conditions.</li> <li>Quantify and calculate heat convection, conduction and phase change heat transfer in various heat storage systems.</li> <li>Make a basic design and business case for different heat storage technologies, LCOE.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting<br/> Collegerama: no<br/> hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Formative<br/> Weekly exercises / small assignments related to individual storage technologies.<br/> The main weekly homework assignments on each of the storage technologies are put together as a portfolio in the appendix of the summative assignment report. The portfolio must be complete to be assessed for the summative assignment.</p> <p>Summative:<br/> Group (2 students) design assignment, assessed via report and presentation [75%].<br/> Exam [25%].</p> <p>Final grade calculation: <math>(0.25 * \text{written exam} + 0.75 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> -Written exam: resit next period<br/> -Group assignment: re-submit assignment</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. W.P. Breugem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SET3210                                | Heating and Cooling Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | D. Mehta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | <p>In this course all the concepts needed for planning, design, construction and operation of heating and cooling grids are covered.</p> <p>In this course you will learn the following topics:</p> <ul style="list-style-type: none"> <li>-The need for DHN and the classification of various types of DHNs.</li> <li>-Costs, planning and construction of DHN</li> <li>-Underground space requirements for DHN</li> <li>-Applicable standards/guidelines</li> <li>-Water quality, clogging, leak detection, heat exchange/delivery.</li> <li>-Hydraulic design, control valves and pumps, water hammer</li> <li>-Structural and thermal design, thermal expansion, ground friction,</li> <li>-Design of substations in different DHNs.</li> <li>-Insulation, Heat exchange with surroundings</li> <li>-Temperature levels and Control and operation</li> </ul>                                                                                          |   |
| <b>Study Goals</b>                     | <p>By the end of the course you are able to:</p> <ul style="list-style-type: none"> <li>Classify and discriminate between different types of DHN.</li> <li>Make a plan of DHN development and integration in both green and brown field conditions.</li> <li>Describe and quantify CAPEX/OPEX Costs of DHN.</li> <li>Make a conceptual hydraulic, structural and control design for different DHN-types.</li> <li>Model and simulate a DHN including substations/building interface.</li> <li>Analyse and optimise operation of DHN.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions</p> <p>Expected student participation:</p> <p>Number of contact hours/week: 4</p> <p>Room type preference: workshop setting</p> <p>Colleggerama: yes</p> <p>hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>Formative assessment: 2 assignments with answers for self-study. Each of the assignments is discussed during workshop sessions</p> <p>Summative assessment:</p> <ul style="list-style-type: none"> <li>-Written exam (50% of grade)</li> <li>-Group assignment with report showing a design and simulation of DHN (50% of grade).</li> </ul> <p>Final grade calculation: <math>(0.5 * \text{written exam} + 0.5 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>-Written exam: resit Q5</li> <li>-Group assignment: re-submit assignment</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. I.W.M. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-Instructor</b>                   | Dr. M.A. Mosteiro Romero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| SET3215                                | Heating and Cooling Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. D.F. Bruhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>The course provides an overview of sources for heating and cooling, their characteristics, and their role in the energy mix. You will learn to quantify available heating and cooling capacities based on local conditions. The following sources are covered:</p> <ul style="list-style-type: none"> <li>· Solar thermal</li> <li>· Borehole Heat Exchangers</li> <li>· Geothermal</li> <li>· Waste Heat: data centres, industry, cooling demand)</li> <li>· Aquathermal and other environmental sources</li> </ul>                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>· classify &amp; contrast available H&amp;C sources</li> <li>· calculate available capacities of heat sources depending on local conditions</li> <li>· establish (heating/cooling) supply profile</li> <li>· build a conceptual, connected model of heating and cooling sources</li> <li>· quantify the heat transport process in H&amp;C sources</li> </ul>                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Lecture (hybrid)<br/>Group work: case study</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | <p>Formative: quizzes on Classify, Contrast and Connect H&amp;C sources<br/>Case study presentation and report</p> <p>Summative:<br/>Written exam: conceptualize heat sources, quantification of heat availability profile, quantify heat transfer processes (100% weight)<br/>Scale: 1-10</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/>written exam: Written exam or oral exam (depending on the amount of students)</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| SET3220                                | Heat Distribution in Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | ME45111SET or equivalent (knowledge about Heating and Cooling demand of Buildings)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>In this course you will learn how to decide on efficient combinations of energy conversion systems accounting for available heat sources, and how to distribute and control air and energy flows in buildings whilst minimizing carbon emissions and optimizing thermal comfort. To do this, you will learn about:</p> <ul style="list-style-type: none"> <li>- Rational no regret use of heat sources in relation to acceptable temperature levels in buildings &amp; neighbourhoods, depending on buildings insulation standard</li> <li>- Relationship between building systems and large-scale outdoor systems (like ATES or geothermal)</li> <li>- Air handling systems and properties of Humid air. Humidification/de-humidification, and related energy use</li> <li>- Distribution of heat/cold by air systems and low/high temperature</li> <li>- Distribution of heat/cold by hydronic systems and low/high temperature</li> <li>- Distribution of heat/cold by mixed systems</li> <li>- Room emitters and thermal comfort</li> <li>- Basics of control in HVAC systems</li> <li>- Thermal buffers and zoning</li> <li>- Designing a complete renewable-based energy systems for a a building (energy demand, energy supply, distribution)</li> <li>- Example of a complex building including heat pump, ATES, solar collectors, PV, free air cooling and distribution to multiple rooms</li> </ul> |   |
| <b>Study Goals</b>                     | <p>By the end of the course you will be able to:</p> <ul style="list-style-type: none"> <li>- Make substantiated no-regret choices of heat sources to be used in cities and buildings, depending on geographic availability (supply), acceptable temperature levels in the buildings under consideration (demand), and principles relating to rational use of energy sources.</li> <li>- Describe the different ways to distribute heat and cold in buildings, either through air systems or through hydronic (water) systems and their advantages and disadvantages.</li> <li>- Create zones with similar suitable temperature levels compatible with heat sources, distribution and room emitters, accounting for health &amp; comfort and use of renewables.</li> <li>- Engineer required capacity for, and distribution to the zones present in building.</li> <li>- Make a basic design for control engineering and thermal buffering in HVAC systems (Heating, Ventilation and Air Conditioning systems).</li> <li>- Make a complete renewable-based energy design of a building, including an ATES system, at the level of preliminary design</li> </ul>                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | Lectures, workshops, exercises, 2 weeks group work on integration of systems, excursion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>Formative assessment: a series of short assignments with answers for self-study.</p> <p>Summative assessment: Midterm online exam (75% of grade) and groupwork on integrated design, assessed by short report and presentation (25% of grade). The midterm exam relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the midterm quiz are required to pass.</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Online midterm quiz: written resit</li> <li>- Group project: resubmit report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| SET3995                                 | Direct Use of Geothermal Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-responsible for assignments</b>   | C. Wallmeier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                  | <p>Scope: Due to the growing interest in sustainable energy production, this course presents the fundamentals of geothermal resource exploitation. Students will gain a general understanding of what a geothermal resource is and of the basic steps in their economic exploitation, including lifetime analysis, characterization of a geothermal reservoir, reservoir engineering and system design, basics of the operation and maintenance of geothermal systems, and geothermal field management, as well as some environmental aspects.</p> <p>Contents:</p> <ul style="list-style-type: none"> <li>- Introduction to geothermal energy</li> <li>- Geology and geomechanics</li> <li>- Reservoir characterization</li> <li>- Design of a geothermal system</li> <li>- Geothermal reservoir engineering</li> <li>- Numerical modeling of geothermal doublets</li> <li>- Techno-economic assessment of geothermal systems</li> </ul>                                                                                               |   |
| <b>Study Goals</b>                      | <p>After successfully completing this course you will be able to:</p> <ul style="list-style-type: none"> <li>Classify geothermal systems based on their different geological settings</li> <li>Describe and apply geomechanical principles related to geothermal systems</li> <li>Describe solute and heat transfer in porous media by mathematical equations and explain their role in different geothermal systems (Darcy flow, conductive and convective heat transfer, diffusion and advection equations)</li> <li>Develop a 2D and 3D Finite Element (FE) model in COMSOL describing a direct use geothermal system by implementing real world data</li> <li>Design a prototype geothermal system: <ul style="list-style-type: none"> <li>-Choose the appropriate location and well configuration</li> <li>-Assess the system lifetime, power, generated energy and stochastic economic output</li> <li>-Justify your geothermal development choices from an energy, engineering and economic point of view</li> </ul> </li> </ul> |   |
| <b>Education Method</b>                 | <ul style="list-style-type: none"> <li>- Lectures</li> <li>- Computer practicals</li> <li>- own study</li> <li>- group work</li> <li>- excursion (half day) (depending on class size and operator availability)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                       | <ul style="list-style-type: none"> <li>- Homework (formative)</li> <li>- Written exam (summative, 25% of final grade)</li> <li>- Group work (75% of final grade): <ul style="list-style-type: none"> <li>- Group presentation (summative 25% of final grade)</li> <li>- Group report (summative 50% of final grade)</li> </ul> </li> </ul> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for all assessments:</p> <ul style="list-style-type: none"> <li>- Written exam: resit exam</li> <li>- Group presentation: group presentation</li> <li>- Group report: group report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                       |   |
| <b>Permitted Materials during Tests</b> | Basic calculator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-Instructor</b>                    | Prof.dr. D.F. Bruhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Heat Grids & Economics: Sources - LS Heat - Economics 2024**

| ME45165SET                              | Equipment for Heat and Mass Transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Prof.dr. K. Hooman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-responsible for assignments</b>   | Dr. H.B. Eral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>   | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Summary</b>                          | The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                  | <p>Thermal energy can be converted or stored. Heat exchangers are needed in either case.</p> <p>This course will cover:</p> <ul style="list-style-type: none"> <li>Fundamentals of heat &amp; mass transfer,</li> <li>Design of different types of heat exchangers (shell and tube, plate, finned tubes ), regenerators/recuperators, cooling towers, and furnaces,</li> <li>Fouling of heat exchanger surfaces,</li> <li>characteristics of equipment for various separation processes</li> <li>the role of mass transfer in the sizing and design of the separation equipment.</li> <li>Cost Analysis,</li> <li>Pinch Technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>By the end of this course, you'd be able to design and rate heat &amp; mass transfer equipment. This can help you, for instance, to:</p> <ul style="list-style-type: none"> <li>- Quantify heating and cooling processes and requirements in various industrial applications such as: metallurgy, waste processes, food, agriculture, desalinization, freeze-drying.</li> <li>- Explain design requirements for different industrial heating and cooling processes.</li> <li>- Analyze existing industrial heating and cooling processes.</li> <li>- Identify decarbonization solutions for existing industrial heating and cooling processes.</li> <li>- Design suitable components/facilities/technologies for industrial heating and cooling processes.</li> <li>- Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes.</li> <li>- Comprehends the role of heat and mass transfer in the dimensioning of equipment for separation processes.</li> </ul>                                                                                             |   |
| <b>Education Method</b>                 | Lectures, assignments, self-study, team work, tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Computer Use</b>                     | Standard engineering packages including ASPEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>   | <p>Course material:</p> <p>1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738</p> <p>2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Wiley &amp; Sons, 2011.</p> <p>3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer.</p> <p>4: Course sheets / relevant papers</p> <p>5: R K Sinnott, Gavin P Towler: Chemical engineering design, Coulson &amp; Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020.</p> <p>6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984.</p> <p>7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004.</p> <p>Further reading</p> <p>Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son</p> <p>Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK</p> |   |
| <b>Assessment</b>                       | <p>Formative:</p> <p>In-class exercises/quizzes</p> <p>Summative:</p> <p>Group project report &amp; presentation (50%)</p> <p>Individual exam (50%)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Permitted Materials during Tests</b> | books, lecture notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |



| SET3200                                | Heating and Cooling Technologies from Near-Ambient Temperature Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>There are several technologies which can obtain heat or cool from the local environment, which are called ambient temperature sources. In this course, you will learn the process of design for ambient or near ambient temperature sources. Typically, air source systems are more variable and less efficient and less easy to operate, but easily accessible, while the shallow subsurface offers a more stable and efficient source, but less easy to access. Water sources are in between. In this course you will learn which pros and cons prevail under various conditions and how you can make an efficient design for either type.</p> <p>You will investigate two different technologies in depth, and encounter several key design aspects including:<br/> Field data collection: Undertake the design and go to the field to measure important quantities that inform the design (based on a location)<br/> Quantification of heat transfer processes<br/> Technology sizing and integration<br/> Identify key modes of operation<br/> Predicted performance</p> <p>Remark: Heat pump is applied. Only basic energy flows. No details on HP itself</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you are able to:<br/> design a H&amp;C system with a near ambient temperature H&amp;C source.<br/> evaluate design suitability to be a significant contributor to the energy needs and year-round energy flows: hydraulic connections and various modes of operation.<br/> This requires the integration of various sub technologies which you will also learn to design and quantify:<br/> Borehole Heat Exchanger (BHE), including non-stationary conduction and interference.<br/> Aquathermal: Utilisation of heat from urban water infrastructure: surface, sewerage and drinking water. This includes surveying of field conditions to inform the design process [including data collection in the field].<br/> Other sources of thermal energy: air, asphalt.<br/> Integration with Heat pump and storage facilities.<br/> Design sizing / system capacity.</p>                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | <p>Type of contact: lectures, workshops/work session, excursion, fieldwork<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting grouped tables<br/> Collegerama: nno<br/> Hybrid: no, fully on-campus</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | Materials provided: Reader, Lecture slides, papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Summative: 1) Aquathermal mini project in groups (25%)<br/> 2) Duo assignment with report describing two alternative designs with source(s) of ambient temperature (75%).</p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> - Assignment: resubmit deliverable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Tags</b>                            | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>Energy<br/> Sustainability<br/> Thermodynamics<br/> exergy</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| SET3205                                | Heat Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>Due to temporal mismatch in availability and demand for heat, heat storage is needed. In this course you will learn that the need for storage is determined by the variations and mismatches in demand and supply profiles. Also, Power to heat is enabled by heat storage. Depending on the required storage requirements (volume, cycle length, temperature level) you will learn to design a heat storage system and quantify heat losses. The heat storage technologies covered in this course are:</p> <ul style="list-style-type: none"> <li>-Sensible heat storage: subsurface, pits, tanks</li> <li>-Latent heat storage: Phase change materials</li> </ul> <p>Because heat storage systems enable the use of heat sources, but are no heat sources themselves, the economics of heat storage is often complex. Therefore, the integration of heat storage in business cases is also covered in this course.</p>                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>After taking this course you are able to:</p> <ul style="list-style-type: none"> <li>Assess the need for heat storage.</li> <li>Identify storage requirements for various supply-demand conditions.</li> <li>Select appropriate storage media for different conditions.</li> <li>Quantify and calculate heat convection, conduction and phase change heat transfer in various heat storage systems.</li> <li>Make a basic design and business case for different heat storage technologies, LCOE.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting<br/> Collegerama: no<br/> hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Formative<br/> Weekly exercises / small assignments related to individual storage technologies.<br/> The main weekly homework assignments on each of the storage technologies are put together as a portfolio in the appendix of the summative assignment report. The portfolio must be complete to be assessed for the summative assignment.</p> <p>Summative:<br/> Group (2 students) design assignment, assessed via report and presentation [75%].<br/> Exam [25%].</p> <p>Final grade calculation: <math>(0.25 * \text{written exam} + 0.75 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> -Written exam: resit next period<br/> -Group assignment: re-submit assignment</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. W.P. Breugem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SET3210                                | Heating and Cooling Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | D. Mehta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | <p>In this course all the concepts needed for planning, design, construction and operation of heating and cooling grids are covered.</p> <p>In this course you will learn the following topics:</p> <ul style="list-style-type: none"> <li>-The need for DHN and the classification of various types of DHNs.</li> <li>-Costs, planning and construction of DHN</li> <li>-Underground space requirements for DHN</li> <li>-Applicable standards/guidelines</li> <li>-Water quality, clogging, leak detection, heat exchange/delivery.</li> <li>-Hydraulic design, control valves and pumps, water hammer</li> <li>-Structural and thermal design, thermal expansion, ground friction,</li> <li>-Design of substations in different DHNs.</li> <li>-Insulation, Heat exchange with surroundings</li> <li>-Temperature levels and Control and operation</li> </ul>                                                                                          |   |
| <b>Study Goals</b>                     | <p>By the end of the course you are able to:</p> <ul style="list-style-type: none"> <li>Classify and discriminate between different types of DHN.</li> <li>Make a plan of DHN development and integration in both green and brown field conditions.</li> <li>Describe and quantify CAPEX/OPEX Costs of DHN.</li> <li>Make a conceptual hydraulic, structural and control design for different DHN-types.</li> <li>Model and simulate a DHN including substations/building interface.</li> <li>Analyse and optimise operation of DHN.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions</p> <p>Expected student participation:</p> <p>Number of contact hours/week: 4</p> <p>Room type preference: workshop setting</p> <p>Colleggerama: yes</p> <p>hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>Formative assessment: 2 assignments with answers for self-study. Each of the assignments is discussed during workshop sessions</p> <p>Summative assessment:</p> <ul style="list-style-type: none"> <li>-Written exam (50% of grade)</li> <li>-Group assignment with report showing a design and simulation of DHN (50% of grade).</li> </ul> <p>Final grade calculation: <math>(0.5 * \text{written exam} + 0.5 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>-Written exam: resit Q5</li> <li>-Group assignment: re-submit assignment</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. I.W.M. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-Instructor</b>                   | Dr. M.A. Mosteiro Romero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| SET3215                                | Heating and Cooling Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. D.F. Bruhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>The course provides an overview of sources for heating and cooling, their characteristics, and their role in the energy mix. You will learn to quantify available heating and cooling capacities based on local conditions. The following sources are covered:</p> <ul style="list-style-type: none"> <li>· Solar thermal</li> <li>· Borehole Heat Exchangers</li> <li>· Geothermal</li> <li>· Waste Heat: data centres, industry, cooling demand)</li> <li>· Aquathermal and other environmental sources</li> </ul>                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>· classify &amp; contrast available H&amp;C sources</li> <li>· calculate available capacities of heat sources depending on local conditions</li> <li>· establish (heating/cooling) supply profile</li> <li>· build a conceptual, connected model of heating and cooling sources</li> <li>· quantify the heat transport process in H&amp;C sources</li> </ul>                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Lecture (hybrid)<br/>Group work: case study</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | <p>Formative: quizzes on Classify, Contrast and Connect H&amp;C sources<br/>Case study presentation and report</p> <p>Summative:<br/>Written exam: conceptualize heat sources, quantification of heat availability profile, quantify heat transfer processes (100% weight)<br/>Scale: 1-10</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/>written exam: Written exam or oral exam (depending on the amount of students)</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| SET3995                                 | Direct Use of Geothermal Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-responsible for assignments</b>   | C. Wallmeier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                  | <p>Scope: Due to the growing interest in sustainable energy production, this course presents the fundamentals of geothermal resource exploitation. Students will gain a general understanding of what a geothermal resource is and of the basic steps in their economic exploitation, including lifetime analysis, characterization of a geothermal reservoir, reservoir engineering and system design, basics of the operation and maintenance of geothermal systems, and geothermal field management, as well as some environmental aspects.</p> <p>Contents:</p> <ul style="list-style-type: none"> <li>- Introduction to geothermal energy</li> <li>- Geology and geomechanics</li> <li>- Reservoir characterization</li> <li>- Design of a geothermal system</li> <li>- Geothermal reservoir engineering</li> <li>- Numerical modeling of geothermal doublets</li> <li>- Techno-economic assessment of geothermal systems</li> </ul>                                                                                               |   |
| <b>Study Goals</b>                      | <p>After successfully completing this course you will be able to:</p> <ul style="list-style-type: none"> <li>Classify geothermal systems based on their different geological settings</li> <li>Describe and apply geomechanical principles related to geothermal systems</li> <li>Describe solute and heat transfer in porous media by mathematical equations and explain their role in different geothermal systems (Darcy flow, conductive and convective heat transfer, diffusion and advection equations)</li> <li>Develop a 2D and 3D Finite Element (FE) model in COMSOL describing a direct use geothermal system by implementing real world data</li> <li>Design a prototype geothermal system: <ul style="list-style-type: none"> <li>-Choose the appropriate location and well configuration</li> <li>-Assess the system lifetime, power, generated energy and stochastic economic output</li> <li>-Justify your geothermal development choices from an energy, engineering and economic point of view</li> </ul> </li> </ul> |   |
| <b>Education Method</b>                 | <ul style="list-style-type: none"> <li>- Lectures</li> <li>- Computer practicals</li> <li>- own study</li> <li>- group work</li> <li>- excursion (half day) (depending on class size and operator availability)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                       | <ul style="list-style-type: none"> <li>- Homework (formative)</li> <li>- Written exam (summative, 25% of final grade)</li> <li>- Group work (75% of final grade): <ul style="list-style-type: none"> <li>- Group presentation (summative 25% of final grade)</li> <li>- Group report (summative 50% of final grade)</li> </ul> </li> </ul> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for all assessments:</p> <ul style="list-style-type: none"> <li>- Written exam: resit exam</li> <li>- Group presentation: group presentation</li> <li>- Group report: group report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                       |   |
| <b>Permitted Materials during Tests</b> | Basic calculator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-Instructor</b>                    | Prof.dr. D.F. Bruhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

| TPM0637SET                             | Economic Policy for Sustainable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 |
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| <b>Module Manager</b>                  | Dr. E. Schröder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Co-responsible for assignments</b>  | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | <p>Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course.</p> <p>The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.</p>                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis?</p> <p>This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to:</p> <ul style="list-style-type: none"> <li>- consumption and production</li> <li>- efficiency and welfare</li> <li>- economic growth and sustainability</li> <li>- the choice of environmental policy targets and instruments</li> <li>- integrated assessment models</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing the course, students should be able to:</p> <ul style="list-style-type: none"> <li>- explain the main theoretical concepts used in environmental economics and how they relate to one another</li> <li>- explain how a market economy undervalues the environment</li> <li>- explain how environmental economists value the environment</li> <li>- apply economic theory to environmental problems such as climate change</li> <li>- evaluate economic policies for solving environmental problems such as climate change</li> </ul>                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Classes, quizzes, and exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>The main text is:</p> <ul style="list-style-type: none"> <li>- Harris &amp; Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor &amp; Francis.</li> </ul> <p>The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade is based exclusively on the performance in the final exam.</p> <p>In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| TPM0931SET                             | Sustainable Energy Innovation and Transitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. J.N. Quist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition.</p> <p>This course is the first course in the SET Economics &amp; Society profile. It can also be followed by other MSc students as an elective. Due to overlap, is not allowed to follow this course if you have already followed course TPM033A or 4413TRIG.</p>                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>After this course, the student:</p> <ol style="list-style-type: none"> <li>1. has knowledge of: <ul style="list-style-type: none"> <li>* the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective.</li> <li>* the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability.</li> </ul> </li> <li>2. has insight in: <ul style="list-style-type: none"> <li>* the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts.</li> </ul> </li> <li>3. has experience and basic skills in: <ul style="list-style-type: none"> <li>* exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.</li> <li>* formulating research questions.</li> <li>* writing a coherent, well-structured and well argued academic report.</li> <li>* critically reflecting on work of others and on their own work.</li> </ul> </li> </ol> |   |
| <b>Education Method</b>                | <p>This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>Scientific articles which are made available via the TU library or via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Both reports must be completed with at least grade 5.0. Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Elective</b>                        | <p>Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                            | <p>Energy<br/>Energy &amp; Industry<br/>Group work<br/>Sustainability</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| TPM403SET                       | Technology Entrepreneurship and Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Module Manager                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Instructor                      | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Instructor                      | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Instructor                      | M. Kolagar Daronkola                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Responsible for assignments     | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Co-responsible for assignments  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Contact Hours / Week<br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Education Period                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Start Education                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Exam Period                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Expected prior knowledge        | No specific knowledge is required prior to entering the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Course Contents                 | <p>This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course.</p> <p>This course is the third course in the SET Economics &amp; Society profile.</p>                                                                                                                                                                                                                                                                                           |   |
| Study Goals                     | <p>Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to:</p> <ol style="list-style-type: none"><li>1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing;</li><li>2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems;</li><li>3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products;</li><li>4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones;</li><li>5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular;</li><li>6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.</li></ol>                                                                                                                                                                                                                                                                                                                                                        |   |
| Education Method                | <p>The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Literature and Study Materials  | <p>Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted on Brightspace-site.<br/>Slides presented in class: Will be posted on Brightspace-site.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Assessment                      | <p>The assessment is composed of several elements:</p> <ol style="list-style-type: none"><li>1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade.</li><li>2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade.</li><li>3. Individual: Theory quizzes (30%)</li></ol> <p>All marks will be expressed on a scale of 1-10.</p> |   |
| Enrolment / Application         | <p>All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Remarks                         | <p>For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Targetgroup                     | <p>Master students from all master programs can apply.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |



|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Urban Heat & Economics: Sources - Buildings - Economics 2024**

| ME45075                                | Refrigeration & Heat Pumps Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Ramdin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Prof.dr.ir. T.J.H. Vlugt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on the first and second law of thermodynamics (a short introduction to this will also be provided during the lectures). Students who are missing this we explain the most important concepts again)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>The course consists of the following content:</p> <ul style="list-style-type: none"> <li>-Introduction of thermodynamic cycles for heating and cooling. The working field of refrigeration and heat pumps;</li> <li>-Overview of the most important refrigeration / heat pumps systems: Mechanically driven heat systems, thermally driven systems, solid state systems and gas cycles. Comparison of these systems;</li> <li>-Components of refrigeration and heat pump cycles (compressors, valves, condensers, evaporators, turbines etc.);</li> <li>-analysis of cooling and heating cycles using the 1st and second law of thermodynamics, exergy analysis;</li> <li>-gas liquification;</li> <li>-techno-economic analysis of refrigeration and heat pump cycles;</li> <li>-Working fluids. Refrigerants for mechanical vapour compression; refrigerating machines: limits of application, evaluation of ODP and GWP;</li> <li>-Gas cycle refrigerating machines. Gas-phase cycles;</li> <li>-Magnetic refrigeration / heat pumping;</li> </ul>                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course, the students are able to:</p> <ul style="list-style-type: none"> <li>explain, reproduce and apply thermodynamic concepts in relation to refrigeration machines and heat pumps, considering their economic and environmental impact.</li> <li>recognize and apply the working principles, specific characteristics and application range of the most relevant methods to generate cooling and heating.</li> <li>explain and apply the concepts of gas liquefaction</li> <li>perform an energy and exergy analysis of refrigeration machines and heat pumps.</li> <li>evaluate the impact of the alternative concepts for refrigerating machines and heat pumps.</li> <li>sizing of heat pump and refrigeration equipment.</li> <li>perform techno-economic analysis of heat pumps and refrigeration systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>Course sheets / relevant papers / additional material provided during the lectures and on Brightspace</p> <ul style="list-style-type: none"> <li>-Moran&amp;Shapiro, Fundamentals of Engineering Thermodynamics, 9th edition</li> <li>-Dincer, Refrigeration systems and applications (available online via the TU Delft library)</li> <li>-Kiss/Infante Ferreira, Heat pumps in chemical process industry (available online via TU Delft library)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| ME45111SET                             | Heating and Cooling Demand of Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr.ir. W.H. van der Spoel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr.ir. M.J. Tenpierik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A. Rasooli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A.I. Joseph Thaddeus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course is aimed at understanding the dynamics of buildings thermal behavior and at learning how to optimize their design and operation through thermal modeling, &amp; simulation. You will learn to solve problems like: how to shift or reduce peak heating demand to obtain a better match with a renewable energy system. What is thermally more efficient: a heavy concrete building or a light timber-frame building? How effective is night ventilation in warm periods? This course will provide you with answers to questions relating to dynamic thermal behavior in buildings. Amongst others you will learn about:</p> <ul style="list-style-type: none"> <li>- Energy chain from supply to demand and design steps</li> <li>- Basics of thermal comfort</li> <li>- Energy flows in buildings (transmission, ventilation &amp; infiltration, solar heat gains and internal heat gains)</li> <li>- Energy balance</li> <li>- Steady state simulations for peak load and energy usage estimates in pre-design (1 node energy balance model)</li> <li>- Transient simulation: The role of inertia in construction; setting up dynamic balance equations (Fourier equation), setting up a suitable RC network (multi-node energy balances); Solving the equations through finite difference and through transfer functions</li> </ul> <p>This is accompanied by many exercises to understand the optimization of building characteristics and peak shaving/shifting. The course ends up with an overview of energy conversion systems that can be used to supply the needed heating and cooling to buildings.</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>- explain how the heating and cooling needs of a building are determined by diverse building design parameters, weather, occupancy and comfort preferences of occupants.</li> <li>- translate this knowledge into design calculations (using steady state and transient balances) to optimize the energy demand towards zero energy buildings, to size the heating and cooling equipment, and to provide a good indoor thermal comfort.</li> </ul> <p>More specifically, you will be able to:</p> <ul style="list-style-type: none"> <li>- derive the mathematical equations describing the thermal performance of the various building components.</li> <li>- combine these equations to simulate the whole building system.</li> <li>- use dynamic simulations to answer questions about the optimal energy design, heating and cooling capacities, energy use, comfort.</li> <li>- translate the fuzzy term "comfort" into design requirements</li> <li>- simulate how thermostat control allows for peak shaving and shifting.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>The course is built around a case study that runs like a common thread between the different course items. The case study is a self-paced individual assignment divided into 4 parts, that are discussed during workshops.</p> <p>Type of contact: Lectures, videos, guided exercises, 3 workshops and excursion. Sometimes we flip the classroom.</p> <p>Number of contact hours/week: 4</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>Formative assessment: 4 assignments with answers for self-study. Each of the assignments is discussed during workshops sessions of 2 hours.</p> <p>Summative assessment: Written exam (50% of grade) + Midterm quiz (50% of grade). The midterm quiz relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the written exam are required to pass.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Tags</b>                            | <p>Analysis<br/>Building Engineering<br/>Group work<br/>Matlab<br/>Modelling<br/>Numeric Methods<br/>Thermodynamics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| SET3200                                | Heating and Cooling Technologies from Near-Ambient Temperature Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>There are several technologies which can obtain heat or cool from the local environment, which are called ambient temperature sources. In this course, you will learn the process of design for ambient or near ambient temperature sources. Typically, air source systems are more variable and less efficient and less easy to operate, but easily accessible, while the shallow subsurface offers a more stable and efficient source, but less easy to access. Water sources are in between. In this course you will learn which pros and cons prevail under various conditions and how you can make an efficient design for either type.</p> <p>You will investigate two different technologies in depth, and encounter several key design aspects including:<br/> Field data collection: Undertake the design and go to the field to measure important quantities that inform the design (based on a location)<br/> Quantification of heat transfer processes<br/> Technology sizing and integration<br/> Identify key modes of operation<br/> Predicted performance</p> <p>Remark: Heat pump is applied. Only basic energy flows. No details on HP itself</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you are able to:<br/> design a H&amp;C system with a near ambient temperature H&amp;C source.<br/> evaluate design suitability to be a significant contributor to the energy needs and year-round energy flows: hydraulic connections and various modes of operation.<br/> This requires the integration of various sub technologies which you will also learn to design and quantify:<br/> Borehole Heat Exchanger (BHE), including non-stationary conduction and interference.<br/> Aquathermal: Utilisation of heat from urban water infrastructure: surface, sewerage and drinking water. This includes surveying of field conditions to inform the design process [including data collection in the field].<br/> Other sources of thermal energy: air, asphalt.<br/> Integration with Heat pump and storage facilities.<br/> Design sizing / system capacity.</p>                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | <p>Type of contact: lectures, workshops/work session, excursion, fieldwork<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting grouped tables<br/> Collegerama: nno<br/> Hybrid: no, fully on-campus</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | Materials provided: Reader, Lecture slides, papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Summative: 1) Aquathermal mini project in groups (25%)<br/> 2) Duo assignment with report describing two alternative designs with source(s) of ambient temperature (75%).</p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> - Assignment: resubmit deliverable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Tags</b>                            | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>Energy<br/> Sustainability<br/> Thermodynamics<br/> exergy</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| SET3215                                | Heating and Cooling Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. D.F. Bruhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>The course provides an overview of sources for heating and cooling, their characteristics, and their role in the energy mix. You will learn to quantify available heating and cooling capacities based on local conditions. The following sources are covered:</p> <ul style="list-style-type: none"> <li>· Solar thermal</li> <li>· Borehole Heat Exchangers</li> <li>· Geothermal</li> <li>· Waste Heat: data centres, industry, cooling demand)</li> <li>· Aquathermal and other environmental sources</li> </ul>                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>· classify &amp; contrast available H&amp;C sources</li> <li>· calculate available capacities of heat sources depending on local conditions</li> <li>· establish (heating/cooling) supply profile</li> <li>· build a conceptual, connected model of heating and cooling sources</li> <li>· quantify the heat transport process in H&amp;C sources</li> </ul>                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Lecture (hybrid)<br/>Group work: case study</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | <p>Formative: quizzes on Classify, Contrast and Connect H&amp;C sources<br/>Case study presentation and report</p> <p>Summative:<br/>Written exam: conceptualize heat sources, quantification of heat availability profile, quantify heat transfer processes (100% weight)<br/>Scale: 1-10</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/>written exam: Written exam or oral exam (depending on the amount of students)</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| SET3220                                | Heat Distribution in Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | ME45111SET or equivalent (knowledge about Heating and Cooling demand of Buildings)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>In this course you will learn how to decide on efficient combinations of energy conversion systems accounting for available heat sources, and how to distribute and control air and energy flows in buildings whilst minimizing carbon emissions and optimizing thermal comfort. To do this, you will learn about:</p> <ul style="list-style-type: none"> <li>- Rational no regret use of heat sources in relation to acceptable temperature levels in buildings &amp; neighbourhoods, depending on buildings insulation standard</li> <li>- Relationship between building systems and large-scale outdoor systems (like ATES or geothermal)</li> <li>- Air handling systems and properties of Humid air. Humidification/de-humidification, and related energy use</li> <li>- Distribution of heat/cold by air systems and low/high temperature</li> <li>- Distribution of heat/cold by hydronic systems and low/high temperature</li> <li>- Distribution of heat/cold by mixed systems</li> <li>- Room emitters and thermal comfort</li> <li>- Basics of control in HVAC systems</li> <li>- Thermal buffers and zoning</li> <li>- Designing a complete renewable-based energy systems for a a building (energy demand, energy supply, distribution)</li> <li>- Example of a complex building including heat pump, ATES, solar collectors, PV, free air cooling and distribution to multiple rooms</li> </ul> |   |
| <b>Study Goals</b>                     | <p>By the end of the course you will be able to:</p> <ul style="list-style-type: none"> <li>- Make substantiated no-regret choices of heat sources to be used in cities and buildings, depending on geographic availability (supply), acceptable temperature levels in the buildings under consideration (demand), and principles relating to rational use of energy sources.</li> <li>- Describe the different ways to distribute heat and cold in buildings, either through air systems or through hydronic (water) systems and their advantages and disadvantages.</li> <li>- Create zones with similar suitable temperature levels compatible with heat sources, distribution and room emitters, accounting for health &amp; comfort and use of renewables.</li> <li>- Engineer required capacity for, and distribution to the zones present in building.</li> <li>- Make a basic design for control engineering and thermal buffering in HVAC systems (Heating, Ventilation and Air Conditioning systems).</li> <li>- Make a complete renewable-based energy design of a building, including an ATES system, at the level of preliminary design</li> </ul>                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | Lectures, workshops, exercises, 2 weeks group work on integration of systems, excursion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>Formative assessment: a series of short assignments with answers for self-study.</p> <p>Summative assessment: Midterm online exam (75% of grade) and groupwork on integrated design, assessed by short report and presentation (25% of grade). The midterm exam relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the midterm quiz are required to pass.</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Online midterm quiz: written resit</li> <li>- Group project: resubmit report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| SET3995                                 | Direct Use of Geothermal Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-responsible for assignments</b>   | C. Wallmeier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                  | <p>Scope: Due to the growing interest in sustainable energy production, this course presents the fundamentals of geothermal resource exploitation. Students will gain a general understanding of what a geothermal resource is and of the basic steps in their economic exploitation, including lifetime analysis, characterization of a geothermal reservoir, reservoir engineering and system design, basics of the operation and maintenance of geothermal systems, and geothermal field management, as well as some environmental aspects.</p> <p>Contents:</p> <ul style="list-style-type: none"> <li>- Introduction to geothermal energy</li> <li>- Geology and geomechanics</li> <li>- Reservoir characterization</li> <li>- Design of a geothermal system</li> <li>- Geothermal reservoir engineering</li> <li>- Numerical modeling of geothermal doublets</li> <li>- Techno-economic assessment of geothermal systems</li> </ul>                                                                                               |   |
| <b>Study Goals</b>                      | <p>After successfully completing this course you will be able to:</p> <ul style="list-style-type: none"> <li>Classify geothermal systems based on their different geological settings</li> <li>Describe and apply geomechanical principles related to geothermal systems</li> <li>Describe solute and heat transfer in porous media by mathematical equations and explain their role in different geothermal systems (Darcy flow, conductive and convective heat transfer, diffusion and advection equations)</li> <li>Develop a 2D and 3D Finite Element (FE) model in COMSOL describing a direct use geothermal system by implementing real world data</li> <li>Design a prototype geothermal system: <ul style="list-style-type: none"> <li>-Choose the appropriate location and well configuration</li> <li>-Assess the system lifetime, power, generated energy and stochastic economic output</li> <li>-Justify your geothermal development choices from an energy, engineering and economic point of view</li> </ul> </li> </ul> |   |
| <b>Education Method</b>                 | <ul style="list-style-type: none"> <li>- Lectures</li> <li>- Computer practicals</li> <li>- own study</li> <li>- group work</li> <li>- excursion (half day) (depending on class size and operator availability)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                       | <ul style="list-style-type: none"> <li>- Homework (formative)</li> <li>- Written exam (summative, 25% of final grade)</li> <li>- Group work (75% of final grade): <ul style="list-style-type: none"> <li>- Group presentation (summative 25% of final grade)</li> <li>- Group report (summative 50% of final grade)</li> </ul> </li> </ul> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for all assessments:</p> <ul style="list-style-type: none"> <li>- Written exam: resit exam</li> <li>- Group presentation: group presentation</li> <li>- Group report: group report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                       |   |
| <b>Permitted Materials during Tests</b> | Basic calculator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-Instructor</b>                    | Prof.dr. D.F. Bruhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

| TPM0637SET                             | Economic Policy for Sustainable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. E. Schröder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Co-responsible for assignments</b>  | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | <p>Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course.</p> <p>The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.</p>                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis?</p> <p>This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to:</p> <ul style="list-style-type: none"> <li>- consumption and production</li> <li>- efficiency and welfare</li> <li>- economic growth and sustainability</li> <li>- the choice of environmental policy targets and instruments</li> <li>- integrated assessment models</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing the course, students should be able to:</p> <ul style="list-style-type: none"> <li>- explain the main theoretical concepts used in environmental economics and how they relate to one another</li> <li>- explain how a market economy undervalues the environment</li> <li>- explain how environmental economists value the environment</li> <li>- apply economic theory to environmental problems such as climate change</li> <li>- evaluate economic policies for solving environmental problems such as climate change</li> </ul>                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Classes, quizzes, and exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>The main text is:</p> <ul style="list-style-type: none"> <li>- Harris &amp; Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor &amp; Francis.</li> </ul> <p>The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade is based exclusively on the performance in the final exam.</p> <p>In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |



| TPM0931SET                             | Sustainable Energy Innovation and Transitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. J.N. Quist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition.</p> <p>This course is the first course in the SET Economics &amp; Society profile. It can also be followed by other MSc students as an elective. Due to overlap, is not allowed to follow this course if you have already followed course TPM033A or 4413TRIG.</p>                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>After this course, the student:</p> <ol style="list-style-type: none"> <li>1. has knowledge of: <ul style="list-style-type: none"> <li>* the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective.</li> <li>* the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability.</li> </ul> </li> <li>2. has insight in: <ul style="list-style-type: none"> <li>* the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts.</li> </ul> </li> <li>3. has experience and basic skills in: <ul style="list-style-type: none"> <li>* exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.</li> <li>* formulating research questions.</li> <li>* writing a coherent, well-structured and well argued academic report.</li> <li>* critically reflecting on work of others and on their own work.</li> </ul> </li> </ol> |   |
| <b>Education Method</b>                | <p>This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>Scientific articles which are made available via the TU library or via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Both reports must be completed with at least grade 5.0. Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Elective</b>                        | <p>Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                            | <p>Energy<br/>Energy &amp; Industry<br/>Group work<br/>Sustainability</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| TPM403SET                              | Technology Entrepreneurship and Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | M. Kolagar Daronkola                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Responsible for assignments</b>     | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-responsible for assignments</b>  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | No specific knowledge is required prior to entering the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course.</p> <p>This course is the third course in the SET Economics &amp; Society profile.</p>                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to:</p> <ol style="list-style-type: none"> <li>1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing;</li> <li>2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems;</li> <li>3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products;</li> <li>4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones;</li> <li>5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular;</li> <li>6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.</li> </ol>                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | <p>Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted on Brightspace-site.<br/>Slides presented in class: Will be posted on Brightspace-site.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>The assessment is composed of several elements:</p> <ol style="list-style-type: none"> <li>1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade.</li> <li>2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade.</li> <li>3. Individual: Theory quizzes (30%)</li> </ol> <p>All marks will be expressed on a scale of 1-10.</p> |   |
| <b>Enrolment / Application</b>         | All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Remarks</b>                         | For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Targetgroup</b>                     | Master students from all master programs can apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Heat & Power Grids: Power - LS Heat - Buildings 2024**

| EE4536                                 | DC and AC Microgrids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>          | Dr.ir. L.M. Ramirez Elizondo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | BSc Electrical Engineering or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | This course focuses on AC and DC microgrids. The design and layout of AC and DC microgrids will be defined, including the sizing of different components. Furthermore, attention will be given to the converters, control, and protection challenges in these systems.                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | Identify the challenges introduced by distributed generation<br>Describe different microgrids' layouts and sizing<br>Describe the operation and control of power electronic converters in AC and DC microgrids<br>Apply droop control in AC and DC microgrids<br>Describe the role of energy management systems in AC and DC microgrids<br>Identify different potential ancillary services that could be provided by AC and DC microgrids and describe the application of demand side management in AC and DC microgrids |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | Microgrids: Architectures and Control (Wiley - IEEE) 1st Edition, by Nikos Hatzigiorgiou                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | Written exam 70%, Assignments 30%<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                           |   |

| EE4545                                  | Electrical Power Systems of the Future                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>           | Dr. A. Leki                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>         | <p>Pre-requisites (entry levels acquired in Q1 and Q2):</p> <p>Master SET / Master EE / Master EWEM<br/>           SET3065 Intelligent Electrical Power Grids<br/>           ET4107 Power System Analyses II<br/>           Matlab and Simulink skills (self study tutorial is available)</p> <p>Optional<br/>           EE4665 Uncertainty modelling and risk assessment in electrical power systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | <p>The course deals with the development of the electrical power infrastructure as an integrated system consisting of power grid, electricity production and load. The impact of large scale integration of sustainable energy sources (such as wind and solar power) on system reliability and security of supply will be explained. Advanced concepts (MT-HVDC, FACTS, interaction between DC and AC, WAMPAC, interaction transmission and distribution) are treated. The main principles of markets and services (system services, transmission services, ancillary services) will be highlighted. The course aims to provide the student with a thorough understanding of the reliability issues involved in power system design and operation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>Designing and operating an electrical power system in a reliable, sustainable and affordable way will be a great challenge in the future. After following the course the student understands the planning and operation of the onshore and offshore power system as a whole and the impact of integrating sustainable energy sources at a large scale on system operation and on the transmission and distribution of electrical energy. The student has gained insight in different power transmission and control technologies, balancing, congestion management, interaction between transmission and distribution systems. The student is able to make power system reliability analyses in relation to planning and operation of the power system.</p> <p>Learning objectives(LOs):</p> <p>LO1: Explain the impact of high RES penetration on the resiliency of future power systems.</p> <p>LO2: Analyse the dynamic performance of future power systems with hybrid HVDC/HVAC configurations under high RES penetration.</p> <p>LO3: Design solutions from an integrated planning, operation and control perspective for integrating high RES volumes whilst maintaining the security of supply.</p> <p>LO4: Write a weekly group report to synthesize the analysis and findings from different case studies.</p> |   |
| <b>Education Method</b>                 | <p>Lectures (conceptual aspects)2-4 hours/week, practicum (solution by exercises) 4 hours/week, group interaction, self study, company visit with lecture on site</p> <p>Estimated study load: 112 hours (4 ECTs)<br/>           Lectures 20 hours<br/>           Practical 24 hours<br/>           Self study 68 hours</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                       | <p>Practicum assignments and digital exam/re-exam</p> <p>The final grade comprises: i) 2.4 points achieved through the 8 practicum assignments (0.3 each); ii) 7.6 grade of the final exam (or re-exam). The maximum achievable final grade is 10. At least 6/10 is needed to pass the course. In case of obtaining a grade lower than 6/10, all parts of the course (exam/re-exam and practicum) need to be retaken.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | Book, course notes, calculators, and formula sheet are not allowed. All electronic devices (smartphone, tablet, notebook, USB sticks) must be turned off and never be used during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Remarks</b>                          | <p>Company visit in combination with lectures and practicum<br/>           @ Mariëndaal Centre of Excellence, TenneT Arnhem<br/>           Dont forget identity card or passport!</p> <p>8:00 Departure from EWI, Travel by touring bus to Arnhem.<br/>           Lectures on Operational challenges by RES for system operations.<br/>           Lessons learnt from large disturbances.<br/>           Visit TenneT Control Room (including lunch).<br/>           Practical Exercises Operational challenges and large disturbances.<br/>           Travel by touring bus to Arnhem<br/>           16:30 Arrival at EWI</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>maximum aantal deelnemers</b>        | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| ME45075                                | Refrigeration & Heat Pumps Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Ramdin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Prof.dr.ir. T.J.H. Vlugt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on the first and second law of thermodynamics (a short introduction to this will also be provided during the lectures). Students who are missing this we explain the most important concepts again)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>The course consists of the following content:</p> <ul style="list-style-type: none"> <li>-Introduction of thermodynamic cycles for heating and cooling. The working field of refrigeration and heat pumps;</li> <li>-Overview of the most important refrigeration / heat pumps systems: Mechanically driven heat systems, thermally driven systems, solid state systems and gas cycles. Comparison of these systems;</li> <li>-Components of refrigeration and heat pump cycles (compressors, valves, condensers, evaporators, turbines etc.);</li> <li>-analysis of cooling and heating cycles using the 1st and second law of thermodynamics, exergy analysis;</li> <li>-gas liquification;</li> <li>-techno-economic analysis of refrigeration and heat pump cycles;</li> <li>-Working fluids. Refrigerants for mechanical vapour compression; refrigerating machines: limits of application, evaluation of ODP and GWP;</li> <li>-Gas cycle refrigerating machines. Gas-phase cycles;</li> <li>-Magnetic refrigeration / heat pumping;</li> </ul>                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course, the students are able to:</p> <ul style="list-style-type: none"> <li>explain, reproduce and apply thermodynamic concepts in relation to refrigeration machines and heat pumps, considering their economic and environmental impact.</li> <li>recognize and apply the working principles, specific characteristics and application range of the most relevant methods to generate cooling and heating.</li> <li>explain and apply the concepts of gas liquefaction</li> <li>perform an energy and exergy analysis of refrigeration machines and heat pumps.</li> <li>evaluate the impact of the alternative concepts for refrigerating machines and heat pumps.</li> <li>sizing of heat pump and refrigeration equipment.</li> <li>perform techno-economic analysis of heat pumps and refrigeration systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>Course sheets / relevant papers / additional material provided during the lectures and on Brightspace</p> <ul style="list-style-type: none"> <li>-Moran&amp;Shapiro, Fundamentals of Engineering Thermodynamics, 9th edition</li> <li>-Dincer, Refrigeration systems and applications (available online via the TU Delft library)</li> <li>-Kiss/Infante Ferreira, Heat pumps in chemical process industry (available online via TU Delft library)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| ME45111SET                             | Heating and Cooling Demand of Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr.ir. W.H. van der Spoel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr.ir. M.J. Tenpierik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A. Rasooli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A.I. Joseph Thaddeus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course is aimed at understanding the dynamics of buildings thermal behavior and at learning how to optimize their design and operation through thermal modeling, &amp; simulation. You will learn to solve problems like: how to shift or reduce peak heating demand to obtain a better match with a renewable energy system. What is thermally more efficient: a heavy concrete building or a light timber-frame building? How effective is night ventilation in warm periods? This course will provide you with answers to questions relating to dynamic thermal behavior in buildings. Amongst others you will learn about:</p> <ul style="list-style-type: none"> <li>- Energy chain from supply to demand and design steps</li> <li>- Basics of thermal comfort</li> <li>- Energy flows in buildings (transmission, ventilation &amp; infiltration, solar heat gains and internal heat gains)</li> <li>- Energy balance</li> <li>- Steady state simulations for peak load and energy usage estimates in pre-design (1 node energy balance model)</li> <li>- Transient simulation: The role of inertia in construction; setting up dynamic balance equations (Fourier equation), setting up a suitable RC network (multi-node energy balances); Solving the equations through finite difference and through transfer functions</li> </ul> <p>This is accompanied by many exercises to understand the optimization of building characteristics and peak shaving/shifting. The course ends up with an overview of energy conversion systems that can be used to supply the needed heating and cooling to buildings.</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>- explain how the heating and cooling needs of a building are determined by diverse building design parameters, weather, occupancy and comfort preferences of occupants.</li> <li>- translate this knowledge into design calculations (using steady state and transient balances) to optimize the energy demand towards zero energy buildings, to size the heating and cooling equipment, and to provide a good indoor thermal comfort.</li> </ul> <p>More specifically, you will be able to:</p> <ul style="list-style-type: none"> <li>- derive the mathematical equations describing the thermal performance of the various building components.</li> <li>- combine these equations to simulate the whole building system.</li> <li>- use dynamic simulations to answer questions about the optimal energy design, heating and cooling capacities, energy use, comfort.</li> <li>- translate the fuzzy term "comfort" into design requirements</li> <li>- simulate how thermostat control allows for peak shaving and shifting.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>The course is built around a case study that runs like a common thread between the different course items. The case study is a self-paced individual assignment divided into 4 parts, that are discussed during workshops.</p> <p>Type of contact: Lectures, videos, guided exercises, 3 workshops and excursion. Sometimes we flip the classroom.</p> <p>Number of contact hours/week: 4</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>Formative assessment: 4 assignments with answers for self-study. Each of the assignments is discussed during workshops sessions of 2 hours.</p> <p>Summative assessment: Written exam (50% of grade) + Midterm quiz (50% of grade). The midterm quiz relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the written exam are required to pass.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Tags</b>                            | <p>Analysis<br/>Building Engineering<br/>Group work<br/>Matlab<br/>Modelling<br/>Numeric Methods<br/>Thermodynamics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| ME45165SET                              | Equipment for Heat and Mass Transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Prof.dr. K. Hooman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-responsible for assignments</b>   | Dr. H.B. Eral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>   | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Summary</b>                          | The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                  | <p>Thermal energy can be converted or stored. Heat exchangers are needed in either case.</p> <p>This course will cover:</p> <ul style="list-style-type: none"> <li>Fundamentals of heat &amp; mass transfer,</li> <li>Design of different types of heat exchangers (shell and tube, plate, finned tubes ), regenerators/recuperators, cooling towers, and furnaces,</li> <li>Fouling of heat exchanger surfaces,</li> <li>characteristics of equipment for various separation processes</li> <li>the role of mass transfer in the sizing and design of the separation equipment.</li> <li>Cost Analysis,</li> <li>Pinch Technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>By the end of this course, you'd be able to design and rate heat &amp; mass transfer equipment. This can help you, for instance, to:</p> <ul style="list-style-type: none"> <li>- Quantify heating and cooling processes and requirements in various industrial applications such as: metallurgy, waste processes, food, agriculture, desalinization, freeze-drying.</li> <li>- Explain design requirements for different industrial heating and cooling processes.</li> <li>- Analyze existing industrial heating and cooling processes.</li> <li>- Identify decarbonization solutions for existing industrial heating and cooling processes.</li> <li>- Design suitable components/facilities/technologies for industrial heating and cooling processes.</li> <li>- Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes.</li> <li>- Comprehends the role of heat and mass transfer in the dimensioning of equipment for separation processes.</li> </ul>                                                                                             |   |
| <b>Education Method</b>                 | Lectures, assignments, self-study, team work, tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Computer Use</b>                     | Standard engineering packages including ASPEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>   | <p>Course material:</p> <p>1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738</p> <p>2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Wiley &amp; Sons, 2011.</p> <p>3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer.</p> <p>4: Course sheets / relevant papers</p> <p>5: R K Sinnott, Gavin P Towler: Chemical engineering design, Coulson &amp; Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020.</p> <p>6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984.</p> <p>7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004.</p> <p>Further reading</p> <p>Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son</p> <p>Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK</p> |   |
| <b>Assessment</b>                       | <p>Formative:</p> <p>In-class exercises/quizzes</p> <p>Summative:</p> <p>Group project report &amp; presentation (50%)</p> <p>Individual exam (50%)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Permitted Materials during Tests</b> | books, lecture notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |



| SET3065                                | Intelligent Electrical Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. P. Palensky                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. J.L. Rueda Torres                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Intelligent Electrical Power Grids - a.k.a. Smart Grids - are the "smartified" version of our electricity infrastructure. By adding controls, communication, and other digital elements these grids are expected to be more flexible, robust, efficient, and capable in order to satisfy the requirements of the future: hosting more renewable sources, catering for an active demand side, or implementing flexible peer-to-peer energy markets.</p> <p>Working with Smart Grids requires knowledge in a variety of disciplines out of electrical engineering, computer science, and mathematics. Of great importance is modeling and simulation, since Smart Grids show a level of complexity and heterogeneity that often can not be covered by analytical methods. In this course you will learn how to describe and assess a Smart Grid, mainly by using numerical tools.</p> <p>You will use various modeling and simulation tools to investigate the properties of smart power grids. Each practical results in a (very) short report. In the end, an electronic exam will be done.</p> <p>This course is part of the Power track of the SET master and can also be chosen by EE master students.</p> <p>Understanding of power flow is expected. Slides and exercises to check if you are up for it are provide on Brightspace.</p> |   |
| <b>Study Goals</b>                     | <p>After this course students are able to</p> <p>LO1: Describe the functions and topologies of cyber-physical energy systems</p> <p>LO2: Implement data driven modelling and numerical simulation of energy systems</p> <p>LO3: Calculate variable power flows based on optimization/probability theory</p> <p>LO4: Synthesize functional requirements of energy management systems</p> <p>LO5: Analyse cyber-security threats, vulnerabilities of ITOT systems, and attack vectors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized in the following units, all units have assignments (typically a simulation):</p> <ol style="list-style-type: none"> <li>1: The energy transition and future energy systems (Peter)</li> <li>2: Analysis of steady-state performance in variable operating conditions (Jose)</li> <li>3: Assessment and control of power system dynamic performance (Jose)</li> <li>4: Lab 1 Digital and data driven modelling of interconnected systems</li> <li>5: Lab 2 Simulation and impact analysis of renewables, storage, and demand</li> <li>6: Data acquisition and communication for energy automation (Peter)</li> <li>7: Artificial intelligence-based functionalities for situational awareness (Pedro &amp; Jochen)</li> <li>8: Cyber security of smart grids (Alex)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | On Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>Assignment reports (in time!) (50%)</li> <li>Multiple choice Computer Exam (50%)</li> </ul> <p>In order to successfully complete the course you need to score 60% out of 100% in the final grade. All assignments and the reports are mandatory. Practical reports are due until 24h before the next lecture.</p> <p>Final grade calculation: <math>(0.5 * \text{individual assignment} + 0.5 * \text{computer exam})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Assignment reports (in time!): resubmit deliverable</li> <li>- Multiple choice Computer Exam: computer resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                       |   |

| SET3095                                | Electronic Power Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | S.A. Rivera Iunnissi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | H.J.M. Olsthoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | This course covers the principles of power conversion with switch-mode circuits with sustainable energy applications in mind. The basic circuits of power electronics are explained, including ac-dc converters (diode rectifiers), dc-dc converters (non-isolated) and dc-ac converters (inverters). Issues specific to power electronics including pulse width modulation, methods of analysis, voltage distortion, power quality and commonly used power semiconductor switches are treated.                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | To get acquainted with applications of power electronics; to understand the principles of power electronics; to get an overview of power electronic circuits; to sketch and explain circuit diagram and their specific waveforms and functionalities; To be able to select appropriate circuits for specific power electronics applications.                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures<br>Problem solving sessions<br>Practicum Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | <p>The final grade of the course consists of the following components:</p> <ul style="list-style-type: none"> <li>- Written final exam (closed book) (70%)</li> <li>- Assignments (30%)</li> </ul> <p><math>T = \text{grade for written final exam} * 0,7 + \text{grade for assignments} * 0,3</math></p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Written exam: written resit</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| SET3205                                | Heat Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>Due to temporal mismatch in availability and demand for heat, heat storage is needed. In this course you will learn that the need for storage is determined by the variations and mismatches in demand and supply profiles. Also, Power to heat is enabled by heat storage. Depending on the required storage requirements (volume, cycle length, temperature level) you will learn to design a heat storage system and quantify heat losses. The heat storage technologies covered in this course are:</p> <ul style="list-style-type: none"> <li>-Sensible heat storage: subsurface, pits, tanks</li> <li>-Latent heat storage: Phase change materials</li> </ul> <p>Because heat storage systems enable the use of heat sources, but are no heat sources themselves, the economics of heat storage is often complex. Therefore, the integration of heat storage in business cases is also covered in this course.</p>                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>After taking this course you are able to:</p> <ul style="list-style-type: none"> <li>Assess the need for heat storage.</li> <li>Identify storage requirements for various supply-demand conditions.</li> <li>Select appropriate storage media for different conditions.</li> <li>Quantify and calculate heat convection, conduction and phase change heat transfer in various heat storage systems.</li> <li>Make a basic design and business case for different heat storage technologies, LCOE.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting<br/> Collegerama: no<br/> hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Formative<br/> Weekly exercises / small assignments related to individual storage technologies.<br/> The main weekly homework assignments on each of the storage technologies are put together as a portfolio in the appendix of the summative assignment report. The portfolio must be complete to be assessed for the summative assignment.</p> <p>Summative:<br/> Group (2 students) design assignment, assessed via report and presentation [75%].<br/> Exam [25%].</p> <p>Final grade calculation: <math>(0.25 * \text{written exam} + 0.75 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> -Written exam: resit next period<br/> -Group assignment: re-submit assignment</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. W.P. Breugem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SET3210                                | Heating and Cooling Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | D. Mehta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | <p>In this course all the concepts needed for planning, design, construction and operation of heating and cooling grids are covered.</p> <p>In this course you will learn the following topics:</p> <ul style="list-style-type: none"> <li>-The need for DHN and the classification of various types of DHNs.</li> <li>-Costs, planning and construction of DHN</li> <li>-Underground space requirements for DHN</li> <li>-Applicable standards/guidelines</li> <li>-Water quality, clogging, leak detection, heat exchange/delivery.</li> <li>-Hydraulic design, control valves and pumps, water hammer</li> <li>-Structural and thermal design, thermal expansion, ground friction,</li> <li>-Design of substations in different DHNs.</li> <li>-Insulation, Heat exchange with surroundings</li> <li>-Temperature levels and Control and operation</li> </ul>                                                                                          |   |
| <b>Study Goals</b>                     | <p>By the end of the course you are able to:</p> <ul style="list-style-type: none"> <li>Classify and discriminate between different types of DHN.</li> <li>Make a plan of DHN development and integration in both green and brown field conditions.</li> <li>Describe and quantify CAPEX/OPEX Costs of DHN.</li> <li>Make a conceptual hydraulic, structural and control design for different DHN-types.</li> <li>Model and simulate a DHN including substations/building interface.</li> <li>Analyse and optimise operation of DHN.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions</p> <p>Expected student participation:</p> <p>Number of contact hours/week: 4</p> <p>Room type preference: workshop setting</p> <p>Collegerama: yes</p> <p>hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Formative assessment: 2 assignments with answers for self-study. Each of the assignments is discussed during workshop sessions</p> <p>Summative assessment:</p> <ul style="list-style-type: none"> <li>-Written exam (50% of grade)</li> <li>-Group assignment with report showing a design and simulation of DHN (50% of grade).</li> </ul> <p>Final grade calculation: <math>(0.5 * \text{written exam} + 0.5 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>-Written exam: resit Q5</li> <li>-Group assignment: re-submit assignment</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. I.W.M. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-Instructor</b>                   | Dr. M.A. Mosteiro Romero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| SET3220                                | Heat Distribution in Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | ME45111SET or equivalent (knowledge about Heating and Cooling demand of Buildings)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>In this course you will learn how to decide on efficient combinations of energy conversion systems accounting for available heat sources, and how to distribute and control air and energy flows in buildings whilst minimizing carbon emissions and optimizing thermal comfort. To do this, you will learn about:</p> <ul style="list-style-type: none"> <li>- Rational no regret use of heat sources in relation to acceptable temperature levels in buildings &amp; neighbourhoods, depending on buildings insulation standard</li> <li>- Relationship between building systems and large-scale outdoor systems (like ATES or geothermal)</li> <li>- Air handling systems and properties of Humid air. Humidification/de-humidification, and related energy use</li> <li>- Distribution of heat/cold by air systems and low/high temperature</li> <li>- Distribution of heat/cold by hydronic systems and low/high temperature</li> <li>- Distribution of heat/cold by mixed systems</li> <li>- Room emitters and thermal comfort</li> <li>- Basics of control in HVAC systems</li> <li>- Thermal buffers and zoning</li> <li>- Designing a complete renewable-based energy systems for a a building (energy demand, energy supply, distribution)</li> <li>- Example of a complex building including heat pump, ATES, solar collectors, PV, free air cooling and distribution to multiple rooms</li> </ul> |   |
| <b>Study Goals</b>                     | <p>By the end of the course you will be able to:</p> <ul style="list-style-type: none"> <li>- Make substantiated no-regret choices of heat sources to be used in cities and buildings, depending on geographic availability (supply), acceptable temperature levels in the buildings under consideration (demand), and principles relating to rational use of energy sources.</li> <li>- Describe the different ways to distribute heat and cold in buildings, either through air systems or through hydronic (water) systems and their advantages and disadvantages.</li> <li>- Create zones with similar suitable temperature levels compatible with heat sources, distribution and room emitters, accounting for health &amp; comfort and use of renewables.</li> <li>- Engineer required capacity for, and distribution to the zones present in building.</li> <li>- Make a basic design for control engineering and thermal buffering in HVAC systems (Heating, Ventilation and Air Conditioning systems).</li> <li>- Make a complete renewable-based energy design of a building, including an ATES system, at the level of preliminary design</li> </ul>                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | Lectures, workshops, exercises, 2 weeks group work on integration of systems, excursion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>Formative assessment: a series of short assignments with answers for self-study.</p> <p>Summative assessment: Midterm online exam (75% of grade) and groupwork on integrated design, assessed by short report and presentation (25% of grade). The midterm exam relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the midterm quiz are required to pass.</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Online midterm quiz: written resit</li> <li>- Group project: resubmit report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Urban Heat & Electricity: Solar - Sources - Buildings 2024**

| ET4376                          | Photovoltaic Basics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Coordinator              | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Contact Hours / Week<br>x/x/x/x | 0/4/0/0 lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Education Period                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Start Education                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| Exam Period                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Course Contents                 | This course teaches basic knowledge about direct energy conversion of light into electricity that takes place in advanced semiconductor devices called solar cells. Another term that denotes the field of direct solar energy conversion into electricity is Photovoltaics (PV). The course starts with explaining the role of PV technology in energy transition towards sustainable energy. It continues with a short overview of solar cell technologies based on different semiconductor materials, PV markets and applications. The main parameters of radiation spectra, and the solar spectrum in particular, are explained. Suitable solar-cell materials and their optical and electronic properties that are relevant to PV energy conversion, such as absorption of light, transport of charge carries, and their recombination-generation, are discussed. The course focuses on solar-cell operating principles that enable illuminated solar cells to deliver electricity. Dark and illuminated "current-voltage (I-V)" characteristics, solar cell external parameters and their measurement are explained. The course treats the conversion efficiency limits of solar cells. The effect of illumination level, temperature, and parasitic resistances on the performance of solar cells is discussed using the equivalent circuit of a solar cell. The optimal design of solar cells is highlighted based on the analysis and minimization of optical and recombination losses in solar cells. A complete PV system is introduced, the individual components and their characteristics are explained. Practical design and sizing rules of PV systems are exercised. |   |
| Study Goals                     | Students learn about renewable energy sources, their potential and role in the energy transition towards sustainable energy. The generation of electricity directly from solar energy using solar cells is the focus of the course. This requires understanding the principles of the photovoltaic energy conversion based on the knowledge of material properties, solar cell structures, and processes of absorption of light and transport of charge carriers in the solar cells. Students learn to evaluate the performance limits of different types of solar cells and can design solar cell structures with minimal optical and collection losses in the energy conversion process. The students know the different components of a PV system and understand the design rules of a PV system for various applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Education Method                | Lectures, exercises, instructions and self-study using the book Solar Energy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Literature and Study Materials  | Book Solar Energy: The physics and engineering of photovolataic conversion technologies and systems (Publisher UIT Cambridge 2016), lecture sheets, short instructive videos placed on brightspace, old exams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Assessment                      | During instruction sessions you will practice 'exam level' questions. These questions are intended to prepare you for the exam, but are not graded. The final grade is 100% determined by the written exam. All marks will be expressed on a scale of 0 to 10.<br><br>In case of an insufficient result for the written exam, you have the opportunity to take the resit exam in the next quarter.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| ET4377                                 | Photovoltaic Technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Dr. L. Mazzarella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr. I.M.F. Gordon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/0 lectures; 0/0/2/0 instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior knowledge</b>        | Lecture course Photovoltaic Basics (ET4 376)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>This course will focus on several solar-cell technologies: silicon solar cells based on (multi)-crystalline wafers, thin-film solar cells based on thin silicon films, cadmium telluride (CdTe) and copper-indium-(gallium)-(di)selenide (CIS to CIGS), III-V solar cells and concentrators, and organic and perovskite solar cells. For these various technologies the device concepts, the specific properties of the materials required, and the manufacturing and processing methods will be discussed.</p> <p>Before these technologies will be discussed, an in-depth treatment of semiconductor device physics for solar cells and light-management concepts will be given that are applicable to all technologies. Issues included are recombination at interfaces, passivation, heterojunctions, various principles of light-scattering, anti-reflection coatings, and textured interfaces.</p>                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>This course aims at an understanding of the different solar-cell technologies. Students are expected to be able to explain the operation of these devices based on the design, processing and material properties. Therefore an understanding of the physics of these technologies is required.</p> <p>The learning objectives (LO) are as follows:</p> <p>LO1: Understand semiconductor processes and light management concepts relevant for solar-cell operation.<br/> LO2: Apply the semiconductor processes and light management concepts to existing state-of-the-art and emerging photovoltaic technologies with the objective to understand the solar-cell design.<br/> LO3: Analyze the origin of various loss mechanisms in solar cells and the impact of different design solutions to achieve optimal performance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Flipped-classroom video lectures and on-campus instructions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Arno Smets, Klaus Jäger, Olindo Isabella, René van Swaaij, and Miro Zeman: Solar Energy - The Physics and Engineering of Photovoltaic Conversion Technologies and Systems; UIT Cambridge, England.</p> <p>ISBN: 978 1 906860 32 5 (paperback);</p> <p>This book is also available as a free ebook:</p> <p>ISBN: 978 1 906860 73 8 (ePub)<br/> ISBN: 978 1 906860 75 2 (pdf)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The course will be assessed by a combination of a written final examination and three graded non-mandatory assignments. The final mark for this course, F, will be based on the final examination, E, and the average results for the non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10.</p> <p>For calculating A, the average of the three assignment results is used. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments will be valid only within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account.</p> <p>The final mark, F, for this course will be computed as follows: <math>F = E + [A \times (10 - E)/100]</math>. The computation will be carried out using one decimal and subsequently rounded to halves.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |



| ET4378                                 | Photovoltaic Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. M.R. Vogt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr.ir. H. Ziar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2 lectures and 0/0/0/2 lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>        | <ul style="list-style-type: none"> <li>- Basic knowledge of photovoltaics (e.g. PV Basics course ET4376)</li> <li>- Good command of Matlab scripting language (or equivalent)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | <p>This course will focus on understanding complete photovoltaic systems. It will start with general introduction to PV systems, overview of system components and their specifications. Several components will be explained in more detail such as power electronic converters and storage components. The different types of inverters, methods of maximal power point tracking and the overall efficiency of energy conversion of the system will be discussed. The applications of PV systems, such as off-grid, grid-connected, microgrids, PV systems with energy storage, will be presented. The course will focus on design of PV systems for different applications. The issues of connecting PV systems to the grid and the monitoring of PV system performance will be discussed.</p>                                                                                                                                                                                                                                           |   |
| <b>Study Goals</b>                     | <p>By the end of this course, you will be able to:</p> <ol style="list-style-type: none"> <li>1. Explain the purpose of the components in different PV Systems using trends in their physical quantities.</li> <li>2. Examine different power tracking methods and the importance of bypass diodes.</li> <li>3. Calculate the energy yield of a PV system in conjunction with the following steps: <ol style="list-style-type: none"> <li>3a. Employ the calculation for optimal PV panel orientation from geographical survey.</li> <li>3b. Determine the advantages and disadvantages of different PV systems topologies.</li> </ol> </li> <li>4. Design PV systems for different applications based on your own cost estimation and technical evaluation justified in a report</li> </ol>                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | Lectures, self study and teaming up for carrying out the design of an actual PV System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>- Solar Energy book by Smets, Jäger, Isabella, van Swaaij, Zeman (also free E-book available)</li> <li>- Practical Handbook of Photovoltaics - Fundamental and Applications by McEvoy, Markvart, Castañer (Second Edition, available TUDelft Library)</li> <li>- Handouts</li> <li>- Scientific articles</li> <li>- PV Systems course in "Solar Engineering MicroMasters" (<a href="https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/">https://online-learning.tudelft.nl/courses/solar-energy-pv-systems/</a>)</li> <li>- PV Resources (<a href="http://www.pvresources.com/">http://www.pvresources.com/</a>)</li> <li>- PVGIS (<a href="http://re.jrc.ec.europa.eu/pvgis/">http://re.jrc.ec.europa.eu/pvgis/</a>)</li> <li>- Sandia National Laboratories PVPerformance (<a href="https://pvpmc.sandia.gov/">https://pvpmc.sandia.gov/</a>)</li> <li>- Dutch PV Portal 3.0 (<a href="https://www.tudelft.nl/dutchpvportal">https://www.tudelft.nl/dutchpvportal</a>)</li> </ul> |   |
| <b>Assessment</b>                      | <p>For students of the 2024 cohort: Final project report (33% of the final mark) and written test (67% of the final mark).</p> <p>For students of previous cohorts: Final project report (25% of the final mark) and written test (75% of the final mark).</p> <p>Two-or-three-person teams will be formed at the first instruction. The grade of the final project report is per team, the grade of the written test is individual.</p> <p>To access to the written test, the grade of the final project report must be equal to or greater than 5.75.</p> <p>To pass the course, you need a grade of at least 5.75 in both parts (report and written exam) AND a final weighted grade of at least 5.75.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>  |   |

| ME45075                                | Refrigeration & Heat Pumps Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Ramdin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Prof.dr.ir. T.J.H. Vlugt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on the first and second law of thermodynamics (a short introduction to this will also be provided during the lectures). Students who are missing this we explain the most important concepts again)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>The course consists of the following content:</p> <ul style="list-style-type: none"> <li>-Introduction of thermodynamic cycles for heating and cooling. The working field of refrigeration and heat pumps;</li> <li>-Overview of the most important refrigeration / heat pumps systems: Mechanically driven heat systems, thermally driven systems, solid state systems and gas cycles. Comparison of these systems;</li> <li>-Components of refrigeration and heat pump cycles (compressors, valves, condensers, evaporators, turbines etc.);</li> <li>-analysis of cooling and heating cycles using the 1st and second law of thermodynamics, exergy analysis;</li> <li>-gas liquification;</li> <li>-techno-economic analysis of refrigeration and heat pump cycles;</li> <li>-Working fluids. Refrigerants for mechanical vapour compression; refrigerating machines: limits of application, evaluation of ODP and GWP;</li> <li>-Gas cycle refrigerating machines. Gas-phase cycles;</li> <li>-Magnetic refrigeration / heat pumping;</li> </ul>                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course, the students are able to:</p> <ul style="list-style-type: none"> <li>explain, reproduce and apply thermodynamic concepts in relation to refrigeration machines and heat pumps, considering their economic and environmental impact.</li> <li>recognize and apply the working principles, specific characteristics and application range of the most relevant methods to generate cooling and heating.</li> <li>explain and apply the concepts of gas liquefaction</li> <li>perform an energy and exergy analysis of refrigeration machines and heat pumps.</li> <li>evaluate the impact of the alternative concepts for refrigerating machines and heat pumps.</li> <li>sizing of heat pump and refrigeration equipment.</li> <li>perform techno-economic analysis of heat pumps and refrigeration systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>Course sheets / relevant papers / additional material provided during the lectures and on Brightspace</p> <ul style="list-style-type: none"> <li>-Moran&amp;Shapiro, Fundamentals of Engineering Thermodynamics, 9th edition</li> <li>-Dincer, Refrigeration systems and applications (available online via the TU Delft library)</li> <li>-Kiss/Infante Ferreira, Heat pumps in chemical process industry (available online via TU Delft library)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| ME45111SET                             | Heating and Cooling Demand of Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 |
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| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr.ir. W.H. van der Spoel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr.ir. M.J. Tenpierik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A. Rasooli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A.I. Joseph Thaddeus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course is aimed at understanding the dynamics of buildings thermal behavior and at learning how to optimize their design and operation through thermal modeling, &amp; simulation. You will learn to solve problems like: how to shift or reduce peak heating demand to obtain a better match with a renewable energy system. What is thermally more efficient: a heavy concrete building or a light timber-frame building? How effective is night ventilation in warm periods? This course will provide you with answers to questions relating to dynamic thermal behavior in buildings. Amongst others you will learn about:</p> <ul style="list-style-type: none"> <li>- Energy chain from supply to demand and design steps</li> <li>- Basics of thermal comfort</li> <li>- Energy flows in buildings (transmission, ventilation &amp; infiltration, solar heat gains and internal heat gains)</li> <li>- Energy balance</li> <li>- Steady state simulations for peak load and energy usage estimates in pre-design (1 node energy balance model)</li> <li>- Transient simulation: The role of inertia in construction; setting up dynamic balance equations (Fourier equation), setting up a suitable RC network (multi-node energy balances); Solving the equations through finite difference and through transfer functions</li> </ul> <p>This is accompanied by many exercises to understand the optimization of building characteristics and peak shaving/shifting. The course ends up with an overview of energy conversion systems that can be used to supply the needed heating and cooling to buildings.</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>- explain how the heating and cooling needs of a building are determined by diverse building design parameters, weather, occupancy and comfort preferences of occupants.</li> <li>- translate this knowledge into design calculations (using steady state and transient balances) to optimize the energy demand towards zero energy buildings, to size the heating and cooling equipment, and to provide a good indoor thermal comfort.</li> </ul> <p>More specifically, you will be able to:</p> <ul style="list-style-type: none"> <li>- derive the mathematical equations describing the thermal performance of the various building components.</li> <li>- combine these equations to simulate the whole building system.</li> <li>- use dynamic simulations to answer questions about the optimal energy design, heating and cooling capacities, energy use, comfort.</li> <li>- translate the fuzzy term "comfort" into design requirements</li> <li>- simulate how thermostat control allows for peak shaving and shifting.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>The course is built around a case study that runs like a common thread between the different course items. The case study is a self-paced individual assignment divided into 4 parts, that are discussed during workshops.</p> <p>Type of contact: Lectures, videos, guided exercises, 3 workshops and excursion. Sometimes we flip the classroom.</p> <p>Number of contact hours/week: 4</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>Formative assessment: 4 assignments with answers for self-study. Each of the assignments is discussed during workshops sessions of 2 hours.</p> <p>Summative assessment: Written exam (50% of grade) + Midterm quiz (50% of grade). The midterm quiz relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the written exam are required to pass.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Tags</b>                            | <p>Analysis<br/>Building Engineering<br/>Group work<br/>Matlab<br/>Modelling<br/>Numeric Methods<br/>Thermodynamics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| SET3200                                | Heating and Cooling Technologies from Near-Ambient Temperature Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
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| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>There are several technologies which can obtain heat or cool from the local environment, which are called ambient temperature sources. In this course, you will learn the process of design for ambient or near ambient temperature sources. Typically, air source systems are more variable and less efficient and less easy to operate, but easily accessible, while the shallow subsurface offers a more stable and efficient source, but less easy to access. Water sources are in between. In this course you will learn which pros and cons prevail under various conditions and how you can make an efficient design for either type.</p> <p>You will investigate two different technologies in depth, and encounter several key design aspects including:<br/> Field data collection: Undertake the design and go to the field to measure important quantities that inform the design (based on a location)<br/> Quantification of heat transfer processes<br/> Technology sizing and integration<br/> Identify key modes of operation<br/> Predicted performance</p> <p>Remark: Heat pump is applied. Only basic energy flows. No details on HP itself</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you are able to:<br/> design a H&amp;C system with a near ambient temperature H&amp;C source.<br/> evaluate design suitability to be a significant contributor to the energy needs and year-round energy flows: hydraulic connections and various modes of operation.<br/> This requires the integration of various sub technologies which you will also learn to design and quantify:<br/> Borehole Heat Exchanger (BHE), including non-stationary conduction and interference.<br/> Aquathermal: Utilisation of heat from urban water infrastructure: surface, sewerage and drinking water. This includes surveying of field conditions to inform the design process [including data collection in the field].<br/> Other sources of thermal energy: air, asphalt.<br/> Integration with Heat pump and storage facilities.<br/> Design sizing / system capacity.</p>                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | <p>Type of contact: lectures, workshops/work session, excursion, fieldwork<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting grouped tables<br/> Collegerama: nno<br/> Hybrid: no, fully on-campus</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | Materials provided: Reader, Lecture slides, papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Summative: 1) Aquathermal mini project in groups (25%)<br/> 2) Duo assignment with report describing two alternative designs with source(s) of ambient temperature (75%).</p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> - Assignment: resubmit deliverable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Tags</b>                            | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>Energy<br/> Sustainability<br/> Thermodynamics<br/> exergy</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| SET3215                                | Heating and Cooling Sources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. D.F. Bruhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>The course provides an overview of sources for heating and cooling, their characteristics, and their role in the energy mix. You will learn to quantify available heating and cooling capacities based on local conditions. The following sources are covered:</p> <ul style="list-style-type: none"> <li>· Solar thermal</li> <li>· Borehole Heat Exchangers</li> <li>· Geothermal</li> <li>· Waste Heat: data centres, industry, cooling demand)</li> <li>· Aquathermal and other environmental sources</li> </ul>                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>· classify &amp; contrast available H&amp;C sources</li> <li>· calculate available capacities of heat sources depending on local conditions</li> <li>· establish (heating/cooling) supply profile</li> <li>· build a conceptual, connected model of heating and cooling sources</li> <li>· quantify the heat transport process in H&amp;C sources</li> </ul>                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Lecture (hybrid)<br/>Group work: case study</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | <p>Formative: quizzes on Classify, Contrast and Connect H&amp;C sources<br/>Case study presentation and report</p> <p>Summative:<br/>Written exam: conceptualize heat sources, quantification of heat availability profile, quantify heat transfer processes (100% weight)<br/>Scale: 1-10</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/>written exam: Written exam or oral exam (depending on the amount of students)</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| SET3220                                | Heat Distribution in Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | ME45111SET or equivalent (knowledge about Heating and Cooling demand of Buildings)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>In this course you will learn how to decide on efficient combinations of energy conversion systems accounting for available heat sources, and how to distribute and control air and energy flows in buildings whilst minimizing carbon emissions and optimizing thermal comfort. To do this, you will learn about:</p> <ul style="list-style-type: none"> <li>- Rational no regret use of heat sources in relation to acceptable temperature levels in buildings &amp; neighbourhoods, depending on buildings insulation standard</li> <li>- Relationship between building systems and large-scale outdoor systems (like ATES or geothermal)</li> <li>- Air handling systems and properties of Humid air. Humidification/de-humidification, and related energy use</li> <li>- Distribution of heat/cold by air systems and low/high temperature</li> <li>- Distribution of heat/cold by hydronic systems and low/high temperature</li> <li>- Distribution of heat/cold by mixed systems</li> <li>- Room emitters and thermal comfort</li> <li>- Basics of control in HVAC systems</li> <li>- Thermal buffers and zoning</li> <li>- Designing a complete renewable-based energy systems for a a building (energy demand, energy supply, distribution)</li> <li>- Example of a complex building including heat pump, ATES, solar collectors, PV, free air cooling and distribution to multiple rooms</li> </ul> |   |
| <b>Study Goals</b>                     | <p>By the end of the course you will be able to:</p> <ul style="list-style-type: none"> <li>- Make substantiated no-regret choices of heat sources to be used in cities and buildings, depending on geographic availability (supply), acceptable temperature levels in the buildings under consideration (demand), and principles relating to rational use of energy sources.</li> <li>- Describe the different ways to distribute heat and cold in buildings, either through air systems or through hydronic (water) systems and their advantages and disadvantages.</li> <li>- Create zones with similar suitable temperature levels compatible with heat sources, distribution and room emitters, accounting for health &amp; comfort and use of renewables.</li> <li>- Engineer required capacity for, and distribution to the zones present in building.</li> <li>- Make a basic design for control engineering and thermal buffering in HVAC systems (Heating, Ventilation and Air Conditioning systems).</li> <li>- Make a complete renewable-based energy design of a building, including an ATES system, at the level of preliminary design</li> </ul>                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | Lectures, workshops, exercises, 2 weeks group work on integration of systems, excursion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>Formative assessment: a series of short assignments with answers for self-study.</p> <p>Summative assessment: Midterm online exam (75% of grade) and groupwork on integrated design, assessed by short report and presentation (25% of grade). The midterm exam relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the midterm quiz are required to pass.</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Online midterm quiz: written resit</li> <li>- Group project: resubmit report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| SET3995                                 | Direct Use of Geothermal Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-responsible for assignments</b>   | C. Wallmeier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                  | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                      | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                  | <p>Scope: Due to the growing interest in sustainable energy production, this course presents the fundamentals of geothermal resource exploitation. Students will gain a general understanding of what a geothermal resource is and of the basic steps in their economic exploitation, including lifetime analysis, characterization of a geothermal reservoir, reservoir engineering and system design, basics of the operation and maintenance of geothermal systems, and geothermal field management, as well as some environmental aspects.</p> <p>Contents:</p> <ul style="list-style-type: none"> <li>- Introduction to geothermal energy</li> <li>- Geology and geomechanics</li> <li>- Reservoir characterization</li> <li>- Design of a geothermal system</li> <li>- Geothermal reservoir engineering</li> <li>- Numerical modeling of geothermal doublets</li> <li>- Techno-economic assessment of geothermal systems</li> </ul>                                                                                               |   |
| <b>Study Goals</b>                      | <p>After successfully completing this course you will be able to:</p> <ul style="list-style-type: none"> <li>Classify geothermal systems based on their different geological settings</li> <li>Describe and apply geomechanical principles related to geothermal systems</li> <li>Describe solute and heat transfer in porous media by mathematical equations and explain their role in different geothermal systems (Darcy flow, conductive and convective heat transfer, diffusion and advection equations)</li> <li>Develop a 2D and 3D Finite Element (FE) model in COMSOL describing a direct use geothermal system by implementing real world data</li> <li>Design a prototype geothermal system: <ul style="list-style-type: none"> <li>-Choose the appropriate location and well configuration</li> <li>-Assess the system lifetime, power, generated energy and stochastic economic output</li> <li>-Justify your geothermal development choices from an energy, engineering and economic point of view</li> </ul> </li> </ul> |   |
| <b>Education Method</b>                 | <ul style="list-style-type: none"> <li>- Lectures</li> <li>- Computer practicals</li> <li>- own study</li> <li>- group work</li> <li>- excursion (half day) (depending on class size and operator availability)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                       | <ul style="list-style-type: none"> <li>- Homework (formative)</li> <li>- Written exam (summative, 25% of final grade)</li> <li>- Group work (75% of final grade): <ul style="list-style-type: none"> <li>- Group presentation (summative 25% of final grade)</li> <li>- Group report (summative 50% of final grade)</li> </ul> </li> </ul> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for all assessments:</p> <ul style="list-style-type: none"> <li>- Written exam: resit exam</li> <li>- Group presentation: group presentation</li> <li>- Group report: group report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                       |   |
| <b>Permitted Materials during Tests</b> | Basic calculator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-Instructor</b>                    | Prof.dr. D.F. Bruhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Industrial Heat & Electricity Integration: Waste & CHP - LS Heat - Economics 2024**



| CH3061                                 | Multiphase Reactor Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. J.R. van Ommen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | <p>The student is expected to:</p> <ol style="list-style-type: none"> <li>1. have followed a course on mass and energy balances (such as Molecular Transport Phenomena)</li> <li>2. have followed a basic course in reactor engineering (such as Reactors and Kinetics)</li> <li>3. is able to work with Python (or similar software; e.g., has followed a course on numerical techniques)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                 | <p>Multiphase reactor types: Fixed beds, trickle beds, fluidized beds, bubble columns, slurry reactors, microreactors, structured (monolith) reactors, electrocatalytic reactors.</p> <p>Engineering aspects: Flow regimes, mass transfer, conversion models. Hatta number, reaction enhancement. Residence time distribution. Catalytic reactors structured in time and space. Unifying concepts.</p> <p>Design aspects: reactor selection, decoupling of kinetics and transport phenomena, combination of reaction and separation, reaction coupling, separating catalytic steps, controlled energy input, circularity of materials.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Study Goals</b>                     | <ol style="list-style-type: none"> <li>1. The student is able to describe for each of the archetypes of multiphase reactors, the various embodiments with practical relevance.</li> <li>2. The student is able to use the scientific literature to find relevant information for the various reactor types.</li> <li>3. The student is able to analyse a multiphase reactor and to describe the physical and chemical process steps in words.</li> <li>4. The student is able to explain the basic measurement techniques used in studying multiphase reactors.</li> <li>5. The student is able to write down the mass, energy, and momentum balances for the overall system and the steps identified under objective 3.</li> <li>6. The student is able to implement customary models in the balances mentioned under 5, such that a system of equations describing the reactor is obtained.</li> <li>7. The student is able to simplify the system of equations based on a quantitative analysis and identification the rate-determining steps.</li> <li>8. The student is able to use Python to solve the systems of equations numerically for the basic types of reactors.</li> <li>9. The student is able to use the developed models to analyse defined cases.</li> <li>10. The student is able to present, both orally and in writing, the outcomes of modeling, calculations, and evaluation.</li> </ol> |   |
| <b>Education Method</b>                | <p>Each lecture will give a short overview of a specific multiphase reactor type (only 45 minutes). The slides available on Brightspace. You will get references to papers and/or chapters from textbooks that give more information about the reactor type. These will be made available via Brightspace.</p> <p>After the lecture, an assignment (about 8 hours work) will be available via Brightspace for the specific reactor type treated. These assignments are made in teams of four students; this number may be changed depending on the number of participants. You are free to choose which software package you want to use for making the assignments, but Python is recommended.</p> <p>After two weeks, each team hands in a detailed solution of the problem on paper the day before the discussion before the announced deadline. Moreover, each team prepares a summary of the solution for presentation in PowerPoint. Please bring this presentation on a memory stick. Each week, a few teams present their solution; the instructors choose the student that is presenting. The other teams challenge this solution.</p>                                                                                                                                                                                                                                                                  |   |
| <b>Computer Use</b>                    | In this course we use Python.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | All material is provided via Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The final grade consists of:</p> <ul style="list-style-type: none"> <li>* the solution to the assignments</li> <li>* an individual part; the nature of this part will depend on the number of participants</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| ME45134                                | Process and Power Plant Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr. L.L. Cutz IJchajchal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | Design of equipment for mass and heat transfer<br>Thermodynamics of mixtures and phases separation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>This course provides the fundamentals for conceptually designing processes of chemical and energy conversion plants, where basic unit operations such as reactions, energy conversion, heat and mass transfer, separations, etc. are designed, modeled and optimized.</p> <p>The conceptual process design involves the design of the flow sheet, creating the topology of the process by the optimizing the sequence of reactors, separation units and energy conversion systems to achieve the required product specification in a sustainable and economically feasible way.</p> <p>In the course the students learn how to select, sequence and model the different unit operations as well as the selection and rough sizing of the required equipment. In addition the students will learn how to integrate the different units into a process flow sheet, and how to model, optimize and rank of the different design alternatives.</p> <p>During the course the students get a hands-on experience with modern process design tools (a.o. Aspen Plus), which will be used in a larger assignment in which a conceptual design of a sustainable, and (if possible) economically feasible process will be setup and simulated.</p> <p>The course includes the following elements:</p> <ul style="list-style-type: none"> <li>Intro; context and scope of conceptual process design;</li> <li>Modelling the thermodynamics of mixtures</li> <li>Modelling of key unit operations</li> <li>Hierarchical approach (Douglas method) to conceptual process design <ul style="list-style-type: none"> <li>o reaction and separation systems</li> <li>o homogeneous systems</li> <li>o heterogeneous systems</li> <li>o steam power plant system</li> </ul> </li> <li>Simulation and optimization</li> <li>Superstructures</li> <li>Costing and project evaluation</li> <li>Safety, Health and Environmental impact considerations</li> <li>Flow-sheet design using Aspen</li> <li>Presentation (pre-recorded) of the case studies</li> </ul> |   |
| <b>Study Goals</b>                     | <p>The student can conceptually design and optimize chemical process plants.</p> <p>More specifically, the student must be able to:</p> <ol style="list-style-type: none"> <li>1. describe the context and scope of a conceptual design of a chemical plant,</li> <li>2. model and design the most important unit operations of chemical/energy conversion process plants,</li> <li>3. make a conceptual process design using the hierarchical Douglas approach for process design</li> <li>4. select, sequence and model the different unit operations in the flowsheet of a chemical process plant and determine the main dimensions of the required reaction/separation equipment,</li> <li>5. apply process flowsheeting tools for the design and simulation of process flowsheets for a chemical plant including the recycle streams</li> <li>6. Use the process simulation tools to evaluate, optimize and rank the different design alternatives for a process.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Method</b>                | Lectures (plenary and working), assignments, pre-recorded presentations and oral examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | <p>Lecture slides</p> <p>Book (selected chapters): Robin Smith<br/>Chemical Process Design and<br/>John Wiley &amp; Sons, U.S.A. (either first or second edition)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | The examination will be as follows: Oral exam [50%, must be 6 or higher] + one presentation of the block scheme prerecorded [10%] and one final report on the process design [32%] and a presentation about it (prerecorded) [8%]. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: exactly the same. The assessment is Covid-19 independent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| ME45165SET                              | Equipment for Heat and Mass Transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
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| <b>Responsible Instructor</b>           | Prof.dr. K. Hooman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-responsible for assignments</b>   | Dr. H.B. Eral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>   | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Summary</b>                          | The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                  | <p>Thermal energy can be converted or stored. Heat exchangers are needed in either case.</p> <p>This course will cover:</p> <ul style="list-style-type: none"> <li>Fundamentals of heat &amp; mass transfer,</li> <li>Design of different types of heat exchangers (shell and tube, plate, finned tubes ), regenerators/recuperators, cooling towers, and furnaces,</li> <li>Fouling of heat exchanger surfaces,</li> <li>characteristics of equipment for various separation processes</li> <li>the role of mass transfer in the sizing and design of the separation equipment.</li> <li>Cost Analysis,</li> <li>Pinch Technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>By the end of this course, youd be able to design and rate heat &amp; mass transfer equipment. This can help you, for instance, to:</p> <ul style="list-style-type: none"> <li>- Quantify heating and cooling processes and requirements in various industrial applications such as: metallurgy, waste processes, food, agriculture, desalinization, freeze-drying.</li> <li>- Explain design requirements for different industrial heating and cooling processes.</li> <li>- Analyze existing industrial heating and cooling processes.</li> <li>- Identify decarbonization solutions for existing industrial heating and cooling processes.</li> <li>- Design suitable components/facilities/technologies for industrial heating and cooling processes.</li> <li>- Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes.</li> <li>- Comprehends the role of heat and mass transfer in the dimensioning of equipment for separation processes.</li> </ul>                                                                                              |   |
| <b>Education Method</b>                 | Lectures, assignments, self-study, team work, tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Computer Use</b>                     | Standard engineering packages including ASPEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>   | <p>Course material:</p> <p>1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738</p> <p>2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Wiley &amp; Sons, 2011.</p> <p>3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer.</p> <p>4: Course sheets / relevant papers</p> <p>5: R K Sinnott, Gavin P Towler: Chemical engineering design, Coulson &amp; Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020.</p> <p>6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984.</p> <p>7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004.</p> <p>Further reading</p> <p>Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son</p> <p>Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK</p> |   |
| <b>Assessment</b>                       | <p>Formative:</p> <p>In-class exercises/quizzes</p> <p>Summative:</p> <p>Group project report &amp; presentation (50%)</p> <p>Individual exam (50%)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Permitted Materials during Tests</b> | books, lecture notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| SET3070                                | Thermochemistry of Biomass Conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. L.L. Cutz IJchajchal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Expected prior knowledge</b>        | To start this course you need to have a good knowledge of BSc. level algebra and calculus, thermodynamics, chemistry, kinetics, basic reactor modelling (batch, plug flow, continuously well stirred reactor), basic process engineering (mass, energy, momentum balances), and physical transport phenomena. In case you think you miss some of this pre-existing knowledge, please perform selfstudy of the first six chapters of the book 'Biomass as a Sustainable Energy Source for the Future', Wiley (2015), eds. De Jong and Van Ommen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Summary</b>                         | Biomass is available in abundance on a global level; it is naturally stored solar energy in a wide diversity of chemical structures. Biomass is a renewable resource for heat, electricity, liquid fuels, chemicals and materials. Among the renewable energy sources, biomass can be stored most easily, however it is contingent on the resources characteristics. Due to the heterogeneous nature of biomass, several conversion technologies have been developed in the context of energy conversion chains, with combustion being the oldest performed by mankind. This course deals with the background, design and implementation of different technologies to thermochemically convert biomass into valuable products.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                 | In this course you will learn about the following aspects of biomass conversion based on thermo-chemical conversion technologies.<br><br><ul style="list-style-type: none"> <li>- Biomass Characterisation (biomass types, properties, analyses)</li> <li>- Pretreatment (harvesting, size reduction, drying, torrefaction)</li> <li>- Thermochemical conversion (combustion, gasification and pyrolysis; reactor types)</li> <li>- Transportation fuels / biorefinery (syngas conversion, economics)</li> <li>- Process design (structured design approach)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | During this course, the students will gain knowledge of and insight into energy conversion technologies based on biomass thermo-chemical conversion processes for a wide range of applications, such as supply of electricity, heat and fuels. They will be provided with practical problems to solve and projects to perform in this field.<br><br>These main learning objectives can be broken down into several smaller learning objectives: <ol style="list-style-type: none"> <li>1) The student has knowledge on the typical biomass composition, related properties (like e.g. heating value, ash melting behaviour) and can critically identify and select fuel characterization techniques.</li> <li>2) The student is able to identify and perform basic capacity calculations related to thermo-chemical biomass conversion systems for small and larger scale (combined) heat and power generation, as well as systems for producing transportation fuel.</li> <li>3) The student is able to set up reaction equations (stoichiometry) for a wide range of energy supply related fuel conversion processes.</li> <li>4) The student can set up and perform heat and mass balance calculations regarding single phase and simple multi-phase biomass thermo-chemical conversion reactors and extended (downstream) systems for power, heat and fuels generation.</li> <li>5) The student knows how to perform basic ideal gas phase chemical reaction equilibrium calculations to calculate product compositions and extents of reactions given initial reaction mixture composition, pressure and temperature.</li> <li>6) The student can understand and describe reaction kinetic expressions and also apply them in order to solve (design) problems related to the conversion of fuels and the formation of emission components.</li> <li>7) The student demonstrates good knowledge of mass and heat transfer mechanisms and phenomena applied to the chemical conversion of both liquid and solid biomass (derived) fuels.</li> <li>8) The student can derive equations for idealised model reactors (well-stirred reactor and plug flow reactor concepts) and apply these simplified reactor concepts to solve engineering problems related to the thermo-chemical conversion of biomass fuels, in particular combustion, gasification, pyrolysis and hydrothermal liquefaction.</li> <li>9) The student is able to perform calculations concerning the conversion of small particles; important in this respect is the identification of the regime of conversion. Related dimensionless numbers such as the Thiele Modulus and the Biot number are key indicators in this respect.</li> <li>10) The student is able to understand the basics of thermo-chemical conversion processes and to translate this into designing and evaluating such (sometimes hybrid) systems, including their economic evaluation based on characteristic parameters like NPV, POT, and ROI.</li> <li>11) The student demonstrates good communication skills, both oral and written, concerning presentation of (applied) scientific results.</li> </ol> |   |
| <b>Education Method</b>                | This course consists of six 2-hour lectures, also using blended learning methods. Each lecture is followed by a 2-hour work session or excursion(s)/lab visit(s) in the same week. These sessions are focused on working out minor assignments including literature surveys, problem solving and working on the major assignment project. During the week of exams, you will present the results of the major assignment project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Relations</b>                | This course is a core course in the Waste & Combined Heat and Power profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | Book: "Biomass, a sustainable energy source for the future", de Jong and van Ommen, Wiley., 1st edition, Wiley (2015).<br>Lecture slides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | The final grade of the course consists of the following components:<br>* the solution to the (up to three) minor assignments (team work) (40%)<br>* a larger final assignment (team work); (50%)<br>* evaluation of the final assignment's presentation; (10%)<br><br>Final grade calculation: $(0.4 * \text{minor group assignments}) + (0.5 * \text{larger group assignment}) + (0.1 * \text{presentation})$<br><br>Resit/ Repair opportunities:<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br>* minor assignments: submit 1 new minor assignment<br>* larger assignment: resubmit report<br>* Presentation: resit (new presentation)<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Co-Instructor</b>                   | Dr. L.L. Cutz IJchajchal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| SET3205                                | Heat Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
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| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>Due to temporal mismatch in availability and demand for heat, heat storage is needed. In this course you will learn that the need for storage is determined by the variations and mismatches in demand and supply profiles. Also, Power to heat is enabled by heat storage. Depending on the required storage requirements (volume, cycle length, temperature level) you will learn to design a heat storage system and quantify heat losses. The heat storage technologies covered in this course are:</p> <ul style="list-style-type: none"> <li>-Sensible heat storage: subsurface, pits, tanks</li> <li>-Latent heat storage: Phase change materials</li> </ul> <p>Because heat storage systems enable the use of heat sources, but are no heat sources themselves, the economics of heat storage is often complex. Therefore, the integration of heat storage in business cases is also covered in this course.</p>                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>After taking this course you are able to:</p> <ul style="list-style-type: none"> <li>Assess the need for heat storage.</li> <li>Identify storage requirements for various supply-demand conditions.</li> <li>Select appropriate storage media for different conditions.</li> <li>Quantify and calculate heat convection, conduction and phase change heat transfer in various heat storage systems.</li> <li>Make a basic design and business case for different heat storage technologies, LCOE.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting<br/> Collegerama: no<br/> hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Formative<br/> Weekly exercises / small assignments related to individual storage technologies.<br/> The main weekly homework assignments on each of the storage technologies are put together as a portfolio in the appendix of the summative assignment report. The portfolio must be complete to be assessed for the summative assignment.</p> <p>Summative:<br/> Group (2 students) design assignment, assessed via report and presentation [75%].<br/> Exam [25%].</p> <p>Final grade calculation: <math>(0.25 * \text{written exam} + 0.75 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> -Written exam: resit next period<br/> -Group assignment: re-submit assignment</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. W.P. Breugem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SET3210                                | Heating and Cooling Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | D. Mehta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | <p>In this course all the concepts needed for planning, design, construction and operation of heating and cooling grids are covered.</p> <p>In this course you will learn the following topics:</p> <ul style="list-style-type: none"> <li>-The need for DHN and the classification of various types of DHNs.</li> <li>-Costs, planning and construction of DHN</li> <li>-Underground space requirements for DHN</li> <li>-Applicable standards/guidelines</li> <li>-Water quality, clogging, leak detection, heat exchange/delivery.</li> <li>-Hydraulic design, control valves and pumps, water hammer</li> <li>-Structural and thermal design, thermal expansion, ground friction,</li> <li>-Design of substations in different DHNs.</li> <li>-Insulation, Heat exchange with surroundings</li> <li>-Temperature levels and Control and operation</li> </ul>                                                                                          |   |
| <b>Study Goals</b>                     | <p>By the end of the course you are able to:</p> <ul style="list-style-type: none"> <li>Classify and discriminate between different types of DHN.</li> <li>Make a plan of DHN development and integration in both green and brown field conditions.</li> <li>Describe and quantify CAPEX/OPEX Costs of DHN.</li> <li>Make a conceptual hydraulic, structural and control design for different DHN-types.</li> <li>Model and simulate a DHN including substations/building interface.</li> <li>Analyse and optimise operation of DHN.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions</p> <p>Expected student participation:</p> <p>Number of contact hours/week: 4</p> <p>Room type preference: workshop setting</p> <p>Colleggerama: yes</p> <p>hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>Formative assessment: 2 assignments with answers for self-study. Each of the assignments is discussed during workshop sessions</p> <p>Summative assessment:</p> <ul style="list-style-type: none"> <li>-Written exam (50% of grade)</li> <li>-Group assignment with report showing a design and simulation of DHN (50% of grade).</li> </ul> <p>Final grade calculation: <math>(0.5 * \text{written exam} + 0.5 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>-Written exam: resit Q5</li> <li>-Group assignment: re-submit assignment</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. I.W.M. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-Instructor</b>                   | Dr. M.A. Mosteiro Romero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| TPM0637SET                             | Economic Policy for Sustainable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. E. Schröder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Co-responsible for assignments</b>  | Dr. S.T.H. Storm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | <p>Prior knowledge of algebra and calculus is required. In general, the typical master student in a STEM subject has mathematical competencies that exceed those required for this course.</p> <p>The course builds on some of the ideas covered in the SET course Economics and Regulations of Sustainable Energy Systems in Q1, especially the parts taught by Gideon Ndubuisi and Servaas Storm. We will review some of these ideas, but in general non-SET students will have to catch up.</p>                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>Why do markets create environmental problems? How and when can markets be harnessed to solve environmental problems? Are there environmental limits to economic growth? What should governments do to speed up the energy transition and fight the global climate crisis?</p> <p>This is a course in environmental and ecological economics designed for master students with little background in economics. It introduces students to key economic concepts and theories with respect to:</p> <ul style="list-style-type: none"> <li>- consumption and production</li> <li>- efficiency and welfare</li> <li>- economic growth and sustainability</li> <li>- the choice of environmental policy targets and instruments</li> <li>- integrated assessment models</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing the course, students should be able to:</p> <ul style="list-style-type: none"> <li>- explain the main theoretical concepts used in environmental economics and how they relate to one another</li> <li>- explain how a market economy undervalues the environment</li> <li>- explain how environmental economists value the environment</li> <li>- apply economic theory to environmental problems such as climate change</li> <li>- evaluate economic policies for solving environmental problems such as climate change</li> </ul>                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Classes, quizzes, and exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>The main text is:</p> <ul style="list-style-type: none"> <li>- Harris &amp; Roach (2021) Environmental and Natural Resource Economics: A Contemporary Approach (5th ed.), Taylor &amp; Francis.</li> </ul> <p>The course goes beyond the textbook and covers selected topics in more depth, on the presumption that master students have learned how to learn and thus they can learn fast. Additional materials will be posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The final grade is based exclusively on the performance in the final exam.</p> <p>In normal circumstances the assessment is based on a closed-book computer exam on campus, which includes multiple-choice questions. In case of unforeseen circumstances or due to measures resulting from COVID-19, the assessment will be based on an open-book exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| TPM0931SET                             | Sustainable Energy Innovation and Transitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. L.M. Kamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. J.N. Quist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course deals with transitions towards sustainability and with theoretical frameworks and strategies for analysing and influencing such transitions. The theoretical frameworks and strategies taught in this course are Strategic Niche Management (SNM), Functions of Innovation Systems (FIS), the Multilevel Perspective, Backcasting and Transition Management. The course focuses on niches around sustainable innovations at the sectoral level as part of larger transitions at the (inter)national level. Since the course is part of the SET program, the lectures and papers focus in particular on sustainable energy innovations and the sustainable energy transition.</p> <p>This course is the first course in the SET Economics &amp; Society profile. It can also be followed by other MSc students as an elective. Due to overlap, is not allowed to follow this course if you have already followed course TPM033A or 4413TRIG.</p>                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>After this course, the student:</p> <ol style="list-style-type: none"> <li>1. has knowledge of: <ul style="list-style-type: none"> <li>* the most relevant innovation theories to analyse transitions towards sustainability: functions of innovation systems, strategic niche management, transition theory and the multilevel perspective.</li> <li>* the key characteristics of the strategies strategic niche management, backcasting and transition management, which can be applied to explore and facilitate transitions towards sustainability.</li> </ul> </li> <li>2. has insight in: <ul style="list-style-type: none"> <li>* the opportunities, limitations, strengths and weaknesses of the intervention strategies and the innovation theories and concepts.</li> </ul> </li> <li>3. has experience and basic skills in: <ul style="list-style-type: none"> <li>* exploring and evaluating niche developments and innovation processes which aim to contribute to the sustainable energy transition.</li> <li>* formulating research questions.</li> <li>* writing a coherent, well-structured and well argued academic report.</li> <li>* critically reflecting on work of others and on their own work.</li> </ul> </li> </ol> |   |
| <b>Education Method</b>                | <p>This course consists of lectures, homework assignments and a project assignment consisting of group work and individual work. In the project assignment student groups analyse a self-chosen sustainable energy niche. The technology in the niche must have the potential to radically change the energy sector and must be in the phase of first introduction the niche phase. The aim of the analysis is to assess the current status of the niche in question, identify bottlenecks and opportunities, and propose a strategy for enabling further growth of the niche.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>Scientific articles which are made available via the TU library or via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Students are evaluated based on the group report (75% of the final grade) and on the individual assignment report (25% of the final grade). Both reports must be completed with at least grade 5.0. Furthermore, students have to deliver weekly homework assignments and participate actively in the lectures. All marks will be expressed on a scale of 1 to 10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Elective</b>                        | <p>Yes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                            | <p>Energy<br/>Energy &amp; Industry<br/>Group work<br/>Sustainability</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |



| TPM403SET                              | Technology Entrepreneurship and Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | M. Kolagar Daronkola                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Responsible for assignments</b>     | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-responsible for assignments</b>  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | No specific knowledge is required prior to entering the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course.</p> <p>This course is the third course in the SET Economics &amp; Society profile.</p>                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to:</p> <ol style="list-style-type: none"> <li>1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing;</li> <li>2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems;</li> <li>3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products;</li> <li>4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones;</li> <li>5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular;</li> <li>6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.</li> </ol>                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | <p>Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted on Brightspace-site.<br/>Slides presented in class: Will be posted on Brightspace-site.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>The assessment is composed of several elements:</p> <ol style="list-style-type: none"> <li>1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade.</li> <li>2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade.</li> <li>3. Individual: Theory quizzes (30%)</li> </ol> <p>All marks will be expressed on a scale of 1-10.</p> |   |
| <b>Enrolment / Application</b>         | All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Remarks</b>                         | For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Targetgroup</b>                     | Master students from all master programs can apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

**Smart Building & Grid Integration: DET - LS Heat - Buildings 2024**

| EE4410                                 | Cyber Security of Power Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. A.I. Stefanov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Summary</b>                         | <p>The energy system transition towards a sustainable, low carbon future is led by decarbonisation, decentralisation, and digitalisation. Power grids are undergoing a fast-paced process of digitalisation for enhanced monitoring and control capabilities and grid intelligence. However, the increased integration of digital technologies, such as the next generation of Operational Technologies (OTs) and digital substations, implies a new risk as Information Technology (IT) OT systems for smart grids are vulnerable to cyber attacks. Furthermore, the combination of heterogeneous, co-existing smart and legacy technologies generates significant vulnerabilities and cyber security challenges. Examples of cyber security incidents related to power grids already exist around the world. On December 23, 2015 cyber attacks were conducted on the power grid in Ukraine that resulted in power outages, which affected 225,000 customers. More sophisticated cyber attacks on the Ukrainian power grid followed on December 17, 2016 resulting in a power outage in the distribution network where 200 MW of load was unsupplied. The complexity of cyber attacks on power systems is on the rise. They can initiate cascading failures and result in a catastrophic blackout. A power outage can disrupt the entire energy chain including water supply, heating, and gas networks. Without power, hospitals and other critical services are severely affected. A disruption of service may lead to financial loss, damages, chaos, or even a loss of lives. With respect to security of supply and reliability of the future energy system provision, special attention is needed for new vulnerabilities and cyber threats that come with digitalisation. Accordingly, the cyber security aspects are critical for a further power grid digitalisation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>This course aims to assess and improve the cyber security of smart grids. It discusses the stakeholders of and complexities in critical infrastructures such as electrical power systems. It analyses the typical IT-OT system architectures for grid monitoring and control and explains the main cyber security concepts and policies. The course provides essential knowledge to analyse cyber security threats, threat vectors, and vulnerabilities of IT OT systems for power grids. It examines the kill chain stages of the cyber attack process on industrial control systems. The course analyses the impact of cyber attacks on power system operation in terms of loss of load, equipment damage, cascading effects, and risk of a blackout. The course studies the state-of-the-art security controls and best practices for threat intelligence management and incident response to defend against cyber attacks and effectively mitigate their impact on power system operation.</p> <p>This course focuses on cyber security of Information Technology (IT) Operational Technologies (OT) networks used for power system operation. It starts with an overview of the smart grid and its industrial control systems (EMS/SCADA) and introduces computer networks and cyber security. Next, the course will introduce the cyber kill chain method for cyber security investigation and intelligence-driven defence. Kill chains are used to understand, anticipate, recognise, and combat cyber attacks and advanced persistent threats. The course uses the kill chains of the 2015/2016 cyber attacks on Ukrainian power grid as real-world study cases throughout the course. The main IT-OT system vulnerabilities and cyber threats are analysed. The course covers threats such as passive and active reconnaissance, social engineering, and malware. It investigates various threat vectors on host and network security and advanced cyber attacks targeting power grids. The impact on power system operation is analysed. The course discusses the state-of-the-art security controls to defend and mitigate cyber attacks on power grids such as access controls, authentication management, cryptography, firewalls, secure communication protocols, network segmentation, and intrusion detection and prevention systems. The best practices to improve power grid resilience to cyber attacks are discussed, i.e., threat intelligence management, incident response, and wide area monitoring, protection and control. This course provides a deep dive into core cyber security concepts for power grids such as cryptography (symmetric and asymmetric encryption), hash functions, digital signatures, and cyber-physical anomaly detection using machine learning techniques. Cyber security demonstrations on power grids will be conducted for students in Control Room of the Future (CRoF) technology centre.</p> |   |
| <b>Study Goals</b>                     | <p>At the end of the course, you will be able to:</p> <p>LO1: Analyse cyber security threats, threat vectors, and vulnerabilities of Information Technology (IT) and Operational Technology (OT) systems used for power grid monitoring and control.<br/> LO2: Examine kill chain stages of the cyber attack process on cyber-physical power systems.<br/> LO3: Analyse the impact of cyber attacks on power system operation (instability, cascading failures and blackout).<br/> LO4: Develop security controls using machine learning to detect, prevent and mitigate cyber attacks and cyber secure the future power grids.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>Lectures: theory and real-world study cases (8 lectures x 2h/lecture)<br/> Practicums: hands-on computer labs (5 practicums x 2h/practicum)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <p>Chwan-Hwa Wu &amp; J. David Irwin. (2013). Introduction to computer networks and cybersecurity. CRC Press. ISBN: 9781498760133</p> <p>Nigel P. Smart. (2016). Cryptography made simple. Springer, 2nd Edition.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>The grade is formed based on the following elements:</p> <p>40% Assignment reports (five assignment reports submitted in time on Brightspace)<br/> 60% Final digital (computer) exam</p> <p>All assignments and reports are mandatory.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| ME45075                                | Refrigeration & Heat Pumps Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Ramdin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Prof.dr.ir. T.J.H. Vlugt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Basic knowledge on the first and second law of thermodynamics (a short introduction to this will also be provided during the lectures). Students who are missing this we explain the most important concepts again)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>The course consists of the following content:</p> <ul style="list-style-type: none"> <li>-Introduction of thermodynamic cycles for heating and cooling. The working field of refrigeration and heat pumps;</li> <li>-Overview of the most important refrigeration / heat pumps systems: Mechanically driven heat systems, thermally driven systems, solid state systems and gas cycles. Comparison of these systems;</li> <li>-Components of refrigeration and heat pump cycles (compressors, valves, condensers, evaporators, turbines etc.);</li> <li>-analysis of cooling and heating cycles using the 1st and second law of thermodynamics, exergy analysis;</li> <li>-gas liquification;</li> <li>-techno-economic analysis of refrigeration and heat pump cycles;</li> <li>-Working fluids. Refrigerants for mechanical vapour compression; refrigerating machines: limits of application, evaluation of ODP and GWP;</li> <li>-Gas cycle refrigerating machines. Gas-phase cycles;</li> <li>-Magnetic refrigeration / heat pumping;</li> </ul>                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>By the end of this course, the students are able to:</p> <ul style="list-style-type: none"> <li>explain, reproduce and apply thermodynamic concepts in relation to refrigeration machines and heat pumps, considering their economic and environmental impact.</li> <li>recognize and apply the working principles, specific characteristics and application range of the most relevant methods to generate cooling and heating.</li> <li>explain and apply the concepts of gas liquefaction</li> <li>perform an energy and exergy analysis of refrigeration machines and heat pumps.</li> <li>evaluate the impact of the alternative concepts for refrigerating machines and heat pumps.</li> <li>sizing of heat pump and refrigeration equipment.</li> <li>perform techno-economic analysis of heat pumps and refrigeration systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>Course sheets / relevant papers / additional material provided during the lectures and on Brightspace</p> <ul style="list-style-type: none"> <li>-Moran&amp;Shapiro, Fundamentals of Engineering Thermodynamics, 9th edition</li> <li>-Dincer, Refrigeration systems and applications (available online via the TU Delft library)</li> <li>-Kiss/Infante Ferreira, Heat pumps in chemical process industry (available online via TU Delft library)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| ME45111SET                             | Heating and Cooling Demand of Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr.ir. W.H. van der Spoel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Prof.dr.ir. M.J. Tenpierik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A. Rasooli                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assistant</b>                       | A.I. Joseph Thaddeus                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>This course is aimed at understanding the dynamics of buildings thermal behavior and at learning how to optimize their design and operation through thermal modeling, &amp; simulation. You will learn to solve problems like: how to shift or reduce peak heating demand to obtain a better match with a renewable energy system. What is thermally more efficient: a heavy concrete building or a light timber-frame building? How effective is night ventilation in warm periods? This course will provide you with answers to questions relating to dynamic thermal behavior in buildings. Amongst others you will learn about:</p> <ul style="list-style-type: none"> <li>- Energy chain from supply to demand and design steps</li> <li>- Basics of thermal comfort</li> <li>- Energy flows in buildings (transmission, ventilation &amp; infiltration, solar heat gains and internal heat gains)</li> <li>- Energy balance</li> <li>- Steady state simulations for peak load and energy usage estimates in pre-design (1 node energy balance model)</li> <li>- Transient simulation: The role of inertia in construction; setting up dynamic balance equations (Fourier equation), setting up a suitable RC network (multi-node energy balances); Solving the equations through finite difference and through transfer functions</li> </ul> <p>This is accompanied by many exercises to understand the optimization of building characteristics and peak shaving/shifting. The course ends up with an overview of energy conversion systems that can be used to supply the needed heating and cooling to buildings.</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you will be able to:</p> <ul style="list-style-type: none"> <li>- explain how the heating and cooling needs of a building are determined by diverse building design parameters, weather, occupancy and comfort preferences of occupants.</li> <li>- translate this knowledge into design calculations (using steady state and transient balances) to optimize the energy demand towards zero energy buildings, to size the heating and cooling equipment, and to provide a good indoor thermal comfort.</li> </ul> <p>More specifically, you will be able to:</p> <ul style="list-style-type: none"> <li>- derive the mathematical equations describing the thermal performance of the various building components.</li> <li>- combine these equations to simulate the whole building system.</li> <li>- use dynamic simulations to answer questions about the optimal energy design, heating and cooling capacities, energy use, comfort.</li> <li>- translate the fuzzy term "comfort" into design requirements</li> <li>- simulate how thermostat control allows for peak shaving and shifting.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>The course is built around a case study that runs like a common thread between the different course items. The case study is a self-paced individual assignment divided into 4 parts, that are discussed during workshops.</p> <p>Type of contact: Lectures, videos, guided exercises, 3 workshops and excursion. Sometimes we flip the classroom.</p> <p>Number of contact hours/week: 4</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>Formative assessment: 4 assignments with answers for self-study. Each of the assignments is discussed during workshops sessions of 2 hours.</p> <p>Summative assessment: Written exam (50% of grade) + Midterm quiz (50% of grade). The midterm quiz relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the written exam are required to pass.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Tags</b>                            | <p>Analysis<br/>Building Engineering<br/>Group work<br/>Matlab<br/>Modelling<br/>Numeric Methods<br/>Thermodynamics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| ME45165SET                              | Equipment for Heat and Mass Transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
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| <b>Responsible Instructor</b>           | Prof.dr. K. Hooman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Co-responsible for assignments</b>   | Dr. H.B. Eral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>   | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Summary</b>                          | The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                  | <p>Thermal energy can be converted or stored. Heat exchangers are needed in either case.</p> <p>This course will cover:</p> <ul style="list-style-type: none"> <li>Fundamentals of heat &amp; mass transfer,</li> <li>Design of different types of heat exchangers (shell and tube, plate, finned tubes ), regenerators/recuperators, cooling towers, and furnaces,</li> <li>Fouling of heat exchanger surfaces,</li> <li>characteristics of equipment for various separation processes</li> <li>the role of mass transfer in the sizing and design of the separation equipment.</li> <li>Cost Analysis,</li> <li>Pinch Technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                      | <p>By the end of this course, youd be able to design and rate heat &amp; mass transfer equipment. This can help you, for instance, to:</p> <ul style="list-style-type: none"> <li>- Quantify heating and cooling processes and requirements in various industrial applications such as: metallurgy, waste processes, food, agriculture, desalinization, freeze-drying.</li> <li>- Explain design requirements for different industrial heating and cooling processes.</li> <li>- Analyze existing industrial heating and cooling processes.</li> <li>- Identify decarbonization solutions for existing industrial heating and cooling processes.</li> <li>- Design suitable components/facilities/technologies for industrial heating and cooling processes.</li> <li>- Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes.</li> <li>- Comprehends the role of heat and mass transfer in the dimensioning of equipment for separation processes.</li> </ul>                                                                                              |   |
| <b>Education Method</b>                 | Lectures, assignments, self-study, team work, tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Computer Use</b>                     | Standard engineering packages including ASPEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>   | <p>Course material:</p> <p>1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738</p> <p>2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Wiley &amp; Sons, 2011.</p> <p>3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer.</p> <p>4: Course sheets / relevant papers</p> <p>5: R K Sinnott, Gavin P Towler: Chemical engineering design, Coulson &amp; Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020.</p> <p>6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984.</p> <p>7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004.</p> <p>Further reading</p> <p>Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son</p> <p>Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK</p> |   |
| <b>Assessment</b>                       | <p>Formative:</p> <p>In-class exercises/quizzes</p> <p>Summative:</p> <p>Group project report &amp; presentation (50%)</p> <p>Individual exam (50%)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Permitted Materials during Tests</b> | books, lecture notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| SET3120                                | Energy Systems Simulation and Digital Twins                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. M. Cvetkovic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. S.H. Tindemans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Expected prior knowledge</b>        | Solid background in energy and power systems, which should be acquired through the courses of the SET M.Sc. program. Basic experience in Python programming.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Parts</b>                           | <p>This is a new course. The final lecture structure and/or content may differ slightly from the description.</p> <p>Theory<br/>Topics covered: Energy systems simulation, Co-simulation, Energy systems digital twins and Stochastic simulation.</p> <p>Practicums per Lecture &amp; Course Project</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>Computer simulation is central to energy system planning and operational decision-making. This course introduces advanced concepts and notions of modeling and simulation applied to energy systems. The course is divided into four themes: energy systems simulation, co-simulation, digital twins, and stochastic simulation. The first theme presents energy systems simulation concepts, including the different time-domain simulation models (dynamic, steady-state, scenario-based). In the second theme, challenges related to integrating different simulation models are presented, including topics such as data exchange, standardization, and APIs. The third theme presents the concept of digital twins, providing a detailed architecture for its implementation. The difference between digital twins and (co)simulation models is also presented. Finally, the fourth theme presents how to represent uncertainty when modelling and an introduction to Montecarlo simulation.</p>                                                                     |   |
| <b>Study Goals</b>                     | <p>At the end of the Energy Systems Simulation and Digital Twins course, you will be able to:</p> <p>[LO1] Categorize energy systems simulations according to model and solver properties.<br/> [LO2] Selecting simulation scope (e.g., time horizon, geography) and detail in accordance with application requirements.<br/> [LO3] Design a co-simulation including synchronization sequence and interface.<br/> [LO4] Implement a digital twin architecture for an energy systems application.<br/> [LO5] Implement and analyze the effect of aleatory and parametric uncertainty on simulation outcomes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                | <p>The course consists of lectures, assignments, and a project. The lectures introduce different topics, while the project requests students to apply the material studied in the lecture. Assignments are given each week to practice the concepts from the lecture or extend them before using such concepts in the project. During the practicums, students get to work on the project each week. Project milestones are organized in order to monitor the progress. The assignments and the project are done in groups.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Supporting scientific papers, tutorials, and notes are published weekly on Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>The grade is formed based on the student's and teams performance. A half-way project presentation is organized in Week 5 (Practicum 5) to monitor progress, accounting for 20% of the final grade. The final project presentation (oral exam) in Week 9 accounts for 60% of the final grade, while the project reports account for 20% of the final grade. In order to successfully complete the course, the students need to score a passing grade on all project aspects and during the oral exam. All students must participate in the completion of milestones. The final report is a document that combines the content of all intermediate milestones. Submission of the final report is mandatory for a passing grade.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A &amp;#150; Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| SET3125                                | Machine Learning Workflows for Digital Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr. S.H. Tindemans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | Very basic programming skills in any programming language, elementary knowledge of probability theory and statistics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                         | This course will teach students the fundamentals of Machine Learning and how these can be applied to sustainable energy systems with focus on digital technologies. After this introduction to Machine learning covering selected topics of fundamentals, the students can make use of this course by taking further courses within the scope of digital energy technologies, further machine learning courses and engage in thesis work that uses AI.                                                                                                                                                 |   |
| <b>Course Contents</b>                 | The course covers the following:<br>- statistical Learning (supervised/unsupervised),<br>- supervised regression,<br>- neural Networks,<br>- designing machine learning workflows<br>- physics-informed learning for energy applications<br>- learning with inductive bias in grids                                                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | After this course, students will be able to<br>1. describe machine learning concepts,<br>2. compare concepts of effective learning in the energy system domain<br>3. Apply machine learning training strategies using calculations and analysis of energy system problem.<br>4. design elements of an machine learning workflow for the energy system                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | Lectures, homework, project, and office hours.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Relations</b>                | This course is mutual exclusive with the course EE4C12. Students can not take this course if they have taken the course EE4C12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | Written exam (100% of grade). Passing the homeworks and project is a prerequisite to be eligible for taking the written exam.<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project pass/fail grade.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5). |   |



| SET3205                                | Heat Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>Due to temporal mismatch in availability and demand for heat, heat storage is needed. In this course you will learn that the need for storage is determined by the variations and mismatches in demand and supply profiles. Also, Power to heat is enabled by heat storage. Depending on the required storage requirements (volume, cycle length, temperature level) you will learn to design a heat storage system and quantify heat losses. The heat storage technologies covered in this course are:</p> <ul style="list-style-type: none"> <li>-Sensible heat storage: subsurface, pits, tanks</li> <li>-Latent heat storage: Phase change materials</li> </ul> <p>Because heat storage systems enable the use of heat sources, but are no heat sources themselves, the economics of heat storage is often complex. Therefore, the integration of heat storage in business cases is also covered in this course.</p>                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>After taking this course you are able to:</p> <ul style="list-style-type: none"> <li>Assess the need for heat storage.</li> <li>Identify storage requirements for various supply-demand conditions.</li> <li>Select appropriate storage media for different conditions.</li> <li>Quantify and calculate heat convection, conduction and phase change heat transfer in various heat storage systems.</li> <li>Make a basic design and business case for different heat storage technologies, LCOE.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions<br/> Expected student participation:<br/> Number of contact hours/week: 4<br/> Room type preference: workshop setting<br/> Collegerama: no<br/> hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | Course slides, models and papers provided via Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Formative<br/> Weekly exercises / small assignments related to individual storage technologies.<br/> The main weekly homework assignments on each of the storage technologies are put together as a portfolio in the appendix of the summative assignment report. The portfolio must be complete to be assessed for the summative assignment.</p> <p>Summative:<br/> Group (2 students) design assignment, assessed via report and presentation [75%].<br/> Exam [25%].</p> <p>Final grade calculation: <math>(0.25 * \text{written exam} + 0.75 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:<br/> In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/> -Written exam: resit next period<br/> -Group assignment: re-submit assignment</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. W.P. Breugem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Co-Instructor</b>                   | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Co-Instructor</b>                   | Dr. A. Daniilidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SET3210                                | Heating and Cooling Grids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | D. Mehta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | <p>In this course all the concepts needed for planning, design, construction and operation of heating and cooling grids are covered.</p> <p>In this course you will learn the following topics:</p> <ul style="list-style-type: none"> <li>-The need for DHN and the classification of various types of DHNs.</li> <li>-Costs, planning and construction of DHN</li> <li>-Underground space requirements for DHN</li> <li>-Applicable standards/guidelines</li> <li>-Water quality, clogging, leak detection, heat exchange/delivery.</li> <li>-Hydraulic design, control valves and pumps, water hammer</li> <li>-Structural and thermal design, thermal expansion, ground friction,</li> <li>-Design of substations in different DHNs.</li> <li>-Insulation, Heat exchange with surroundings</li> <li>-Temperature levels and Control and operation</li> </ul>                                                                                          |   |
| <b>Study Goals</b>                     | <p>By the end of the course you are able to:</p> <ul style="list-style-type: none"> <li>Classify and discriminate between different types of DHN.</li> <li>Make a plan of DHN development and integration in both green and brown field conditions.</li> <li>Describe and quantify CAPEX/OPEX Costs of DHN.</li> <li>Make a conceptual hydraulic, structural and control design for different DHN-types.</li> <li>Model and simulate a DHN including substations/building interface.</li> <li>Analyse and optimise operation of DHN.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Method</b>                | <p>Type of contact: Lectures, work sessions</p> <p>Expected student participation:</p> <p>Number of contact hours/week: 4</p> <p>Room type preference: workshop setting</p> <p>Collegerama: yes</p> <p>hybrid: no</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Formative assessment: 2 assignments with answers for self-study. Each of the assignments is discussed during workshop sessions</p> <p>Summative assessment:</p> <ul style="list-style-type: none"> <li>-Written exam (50% of grade)</li> <li>-Group assignment with report showing a design and simulation of DHN (50% of grade).</li> </ul> <p>Final grade calculation: <math>(0.5 * \text{written exam} + 0.5 * \text{group assignment})</math></p> <p>Resit/ Repair opportunities:</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>-Written exam: resit Q5</li> <li>-Group assignment: re-submit assignment</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Co-Instructor</b>                   | Dr.ir. I.W.M. Pothof                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-Instructor</b>                   | Dr. M.A. Mosteiro Romero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| SET3220                                | Heat Distribution in Buildings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.M.J. Bokel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Prof.dr.ing. A.C. Boerstra                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | ME45111SET or equivalent (knowledge about Heating and Cooling demand of Buildings)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>In this course you will learn how to decide on efficient combinations of energy conversion systems accounting for available heat sources, and how to distribute and control air and energy flows in buildings whilst minimizing carbon emissions and optimizing thermal comfort. To do this, you will learn about:</p> <ul style="list-style-type: none"> <li>- Rational no regret use of heat sources in relation to acceptable temperature levels in buildings &amp; neighbourhoods, depending on buildings insulation standard</li> <li>- Relationship between building systems and large-scale outdoor systems (like ATES or geothermal)</li> <li>- Air handling systems and properties of Humid air. Humidification/de-humidification, and related energy use</li> <li>- Distribution of heat/cold by air systems and low/high temperature</li> <li>- Distribution of heat/cold by hydronic systems and low/high temperature</li> <li>- Distribution of heat/cold by mixed systems</li> <li>- Room emitters and thermal comfort</li> <li>- Basics of control in HVAC systems</li> <li>- Thermal buffers and zoning</li> <li>- Designing a complete renewable-based energy systems for a a building (energy demand, energy supply, distribution)</li> <li>- Example of a complex building including heat pump, ATES, solar collectors, PV, free air cooling and distribution to multiple rooms</li> </ul> |   |
| <b>Study Goals</b>                     | <p>By the end of the course you will be able to:</p> <ul style="list-style-type: none"> <li>- Make substantiated no-regret choices of heat sources to be used in cities and buildings, depending on geographic availability (supply), acceptable temperature levels in the buildings under consideration (demand), and principles relating to rational use of energy sources.</li> <li>- Describe the different ways to distribute heat and cold in buildings, either through air systems or through hydronic (water) systems and their advantages and disadvantages.</li> <li>- Create zones with similar suitable temperature levels compatible with heat sources, distribution and room emitters, accounting for health &amp; comfort and use of renewables.</li> <li>- Engineer required capacity for, and distribution to the zones present in building.</li> <li>- Make a basic design for control engineering and thermal buffering in HVAC systems (Heating, Ventilation and Air Conditioning systems).</li> <li>- Make a complete renewable-based energy design of a building, including an ATES system, at the level of preliminary design</li> </ul>                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | Lectures, workshops, exercises, 2 weeks group work on integration of systems, excursion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>Formative assessment: a series of short assignments with answers for self-study.</p> <p>Summative assessment: Midterm online exam (75% of grade) and groupwork on integrated design, assessed by short report and presentation (25% of grade). The midterm exam relates closely to the formative assignments.</p> <p>An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the midterm quiz are required to pass.</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Online midterm quiz: written resit</li> <li>- Group project: resubmit report</li> </ul> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

|              |                                                          |
|--------------|----------------------------------------------------------|
| Year         | 2024/2025                                                |
| Organization | Electrical Engineering, Mathematics and Computer Science |
| Education    | Master Sustainable Energy Technology (SET)               |

| Electives SET 2024  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Program Structure 1 | <p>For more information about the List of Electives, see the Brightspace Organisation of the Master Sustainable Energy Technology.</p> <p>The elective space should be used for one of more of the following purposes:</p> <ul style="list-style-type: none"> <li>To gain work experience (SET3822 Internship MSc SET, 15 credits),. Please note: The internship has a maximum fulltime duration of three months, this includes writing the final report. The internship cannot be part of a graduation project and it is not allowed to carry out both your internship and your thesis at the same company.</li> <li>To gain research experience (SET4399 Extra Project, max. 15 credits),</li> <li>To broaden their knowledge of the SET field,</li> <li>To deepen knowledge of the SET profiles (profile electives),</li> <li>To gain entrepreneurship skills (max. 6 credits),</li> <li>To further develop language skills (max. 3 credits).</li> </ul> <p>See the most up to date list of automatically approved elective courses on the Brightspace page of MSc SET. Other courses will be taken into consideration if and when submitted to the Board of Examiners.</p> |

| IFM4040                                | Joint Interdisciplinary Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 15 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | Prof.dr.ir. J. Hellendoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Instructor</b>                      | Ir. B.J.E. de Bruin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| <b>Education Period</b>                | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Start Education</b>                 | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Exam Period</b>                     | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| <b>Expected prior knowledge</b>        | <p>To enter JIP, a student should</p> <ul style="list-style-type: none"> <li>be a 2nd-year master student;</li> <li>have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft.</li> <li>be full time available (40 hours per week) in Q5;</li> <li>meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV</li> </ul> <p>Students will be selected by the JIP core staff based on motivation, study progress and recommendation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Course Contents</b>                 | <p>The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&amp;D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman, VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Study Goals</b>                     | <p>Meta-cognitive abilities attributable to interdisciplinary learning</p> <p>LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (ILO:K)</p> <p>Scientific and intellectual development</p> <p>LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I)</p> <p>Research and design capabilities</p> <p>LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J)</p> <p>Collaboration and communication in an interdisciplinary team</p> <p>LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K)</p> <p>Self-adjustment and reflection capabilities</p> <p>LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Education Method</b>                | <p>Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |
| <b>Assessment</b>                      | <p>Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions:</p> <p>Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills.</p> <p>Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level.</p> <p>In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback)</p> <p>Summative assessment on module level</p> <p>Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es).</p> <p>Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.</p> |    |

The minimum grade for the summative assessment is  $\geq 5.8$ .

Process grade is 20%

The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

| ME45203                                | Electrolysers, Fuel Cells and Batteries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | J.W. Haverkort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0.0.0.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Familiarity with transport phenomena (e.g. a course on heat transport), basic fluid dynamics, elementary differential equations, and introductory electrochemistry. Ideal preparation include the courses ME45201, CH3513, or SET3110. A previous course on fuel cells, batteries, or similar, or additional self-study may also be sufficient.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>Electrochemistry plays a pivotal role in the transition towards a sustainable energy future. Renewable electrical energy can be stored in batteries or used to synthesise chemicals or fuels in electrochemical reactors, like an electrolyser producing hydrogen. This course will introduce to a student, the knowledge and skills required to understand, model and optimize such electrochemical devices.</p> <p>Content overview:</p> <p>Porous electrode theory, multiphase flow, and transport models will be used to derive analytical current-voltage relations of various electrochemical devices including fuel cells, batteries, redox flow batteries, and electrolysers.</p> <p>Some of the topics that will be discussed in this course include:</p> <ul style="list-style-type: none"> <li>- The Nernst-Planck equations for ionic transport in binary electrolytes.</li> <li>- Porous electrode theory using the concept of an electrochemical Thiele modulus.</li> <li>- The flooded agglomerate and relative permeability models for gas diffusion electrodes.</li> <li>- The single particle and Newman's P2D battery models.</li> <li>- A vertical current and gas distribution model for electrolysers.</li> </ul>                           |   |
| <b>Study Goals</b>                     | <p>After successful completion of this course, the student is able to independently:</p> <ol style="list-style-type: none"> <li>1. explain and apply porous electrode theory.</li> <li>2. derive 0D, 1D, and 2D mathematical models to describe the current-voltage relationship of various electrochemical cells including flow, multiple phases, and porous electrodes.</li> <li>3. optimize an electrochemical cell design with respect to, for example, energy efficiency.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Method</b>                | Two lecture hours + two exercise hours per week.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Relations</b>                | This course is a natural sequel to ME45201, CH3513, and SET3110 for students interested in analytical modelling of electrochemical systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Books</b>                           | Electrolysers, Fuel Cells and Batteries: Analytical Modelling. (J.W. Haverkort) <a href="https://doi.org/10.59490/tb.93">https://doi.org/10.59490/tb.93</a> (open access)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | Written exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |

| ME45204                               | Electrochemical Energy Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4 |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>         | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                     | J.W. Haverkort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>               | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                    | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior knowledge</b>       | Basic thermodynamics. Basic chemistry and knowledge of kinetics is not required, but will be helpfull                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                | One of the main challenges that mankind faces in the 21st century is the energy transition from fossil fuels and feedstocks to renewable sources. Most renewable energy sources, such as solar and wind energy, are intermittent. Therefore, large scale energy storage is required to successfully increase the share of these renewable energy sources. One way of achieving this is by storing energy using electrochemical systems and devices. This course introduces the principals of electrochemical energy storage and conversion to students new to field. Examples of topics discussed during this course are the thermodynamics and kinetics of electrochemical systems, transport phenomena, the electrochemical double layer, electrocatalysis and an introduction to electrochemical methods and measurements. We will discuss these topics in the light of electrochemical applications such as batteries, electrolyzers and fuel cells.                                   |   |
| <b>Study Goals</b>                    | <p>After successful completion of this course, the student is able to independently:</p> <ol style="list-style-type: none"> <li>1. Apply electrochemical nomenclature to describe the electrochemical cell and electrochemical measurements;</li> <li>2. Explain the concept of the electrochemical double layer and mass transport in the electrolyte;</li> <li>3. Apply basic principles of electrochemical thermodynamics, such as the Nernst equation, overpotentials and (standard) equilibrium potentials;</li> <li>4. Apply basic principles of electrochemical kinetics, such as the Tafel equation and the Buttlar-Volmer model;</li> <li>5. Explain the principles that determine the electrocatalytic performance of materials;</li> <li>6. Interpret scientific literature reports on electrochemical energy storage and conversion systems and devices.</li> <li>7. Explain the working principles of different fuel cell, electrolyzers and battery technologies.</li> </ol> |   |
| <b>Education Method</b>               | Combination of two to four lecture hours per week and self-study. Homework assignments will be given after every lecture and will be discussed in class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Relations</b>               | A suitable follow-up course may be ME45203: Electrolyzers, Fuel Cells and Batteries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b> | Study materials will be the lecture slides, homework exercises and practice exams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Books</b>                          | Part of the lectures have large overlap with the book 'Electrochemical Engineering' by Fuller and Harb, which is recommended reading but not required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                     | The method of assessment will be a written exam. If campus examination is not possible the prescribed assessment will be as follows: oral exam and presentation on a literature report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| SET3822                                | Internship SET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 15 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |
| <b>Exam Period</b>                     | Exam by appointment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| <b>Course Contents</b>                 | <p>The Internship SET (SET3822) is a project on a sustainable energy engineering topic performed at a company or institute, either in the Netherlands or abroad. The internship has a study load of 15 EC, which is equivalent to a maximum full-time duration of 3 months, this includes writing the final report.</p> <p>General guidelines are that it should concern a design, model, system integration or methodology implementation, or an experimental study relevant for the company involved. The topic of the project should fall within at least one of the profiles of the chosen profile cluster. Literature study can only be a minor part of the internship.</p> <p>When carrying out Internship SET (SET3822) at a company or institute, the following conditions apply:</p> <ul style="list-style-type: none"> <li>- The student joins in with the company or institute culture in general, including sharing the company-established balance between on-site and online work.</li> <li>- The student joins in with the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and access to and use of the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...).</li> </ul> |    |
| <b>Study Goals</b>                     | <p>There are five main learning objectives (LO) for Internship SET (SET3822):</p> <p>LO1: The student is able to contribute to research and development in a company within the field of sustainable energy technology.</p> <p>LO2: The student is able to report the results to the company supervisor and the TU Delft supervisor.</p> <p>LO3: The student is able to communicate and collaborate in a professional environment.</p> <p>LO4: The student is able to describe the effect of sustainable energy innovations on the society.</p> <p>LO5: The student is able to reflect on their own role and the added value of the internship to their study programme.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| <b>Education Method</b>                | <p>Project</p> <p>The Internship SET (SET3822) follows certain steps:</p> <ol style="list-style-type: none"> <li>1. Orientation</li> <li>2. Kick-off meeting</li> <li>3. Delivering the final report</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Prerequisites</b>                   | <p>In order to ensure that students start the internship with a solid skill set, the student can apply for Internship SET after the third quarter of the first year of the MSc SET programme. Before starting the internship, the student needs to have an Individual Exam Programme (IEP) approved by the Board of Examiners and the courses of the first three quarters of the first year of the MSc SET programme must have been completed. A student can also start when at least 45 credits of the Master's degree course work as stated in the IEP have been completed.</p> <p>Before starting the project, the student, the responsible supervisor, the company supervisor and the profile leader agree that the student will conduct an internship project under the responsibility of the responsible supervisor, with the support of the company supervisor. The student and responsible and company supervisor:</p> <ul style="list-style-type: none"> <li>- Agree on a starting date;</li> <li>- Agree on what is expected during the kick-off meeting.</li> </ul> <p>The agreements are written in the Internship Application Form. This form needs to be signed by the student, the responsible supervisor, and the relevant profile leader, and then submitted to the Internship Office. This is done just before the kick-off meeting.</p>                                                                                                                                                  |    |
| <b>Assessment</b>                      | <p>The student needs to write a technical report during the internship and send it to the company supervisor and the TU Delft supervisor on the final day of the internship.</p> <p>The student needs to write a reflection note and a testimonial and send these documents, together with the technical report, to the internship office, on the final day of the internship.</p> <p>The company supervisor needs to send the evaluation within two weeks after the end of the internship to the TU Delft supervisor.</p> <p>The TU Delft supervisor needs to send the TU evaluation and the company evaluation to the internship office, within two weeks after receiving the evaluation from the company supervisor.</p> <p>When the internship office has received the two reports and the two assessment forms, the grade will be processed within a week.</p> <p>The Internship Grading Scheme will be used as a guideline to determine the mark. Please, consult the Brightspace page Internship SET for the detailed grading scheme.</p> <p>The TU Delft supervisor is responsible for the final grading. The internship will be graded with a pass or a fail (not numerically).</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:<br/>- Report: resubmit deliverable</p>                               |    |
| <b>Special Information</b>             | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>The project needs to be carried out according to MSc SET guidelines of Delft University of Technology. This is safeguarded by the requirement to have a responsible supervisor who is on the SET green list for the most relevant profile.</p> <p>NB: Internship SET (SET3822) can never be part of a Graduation Project (SET3901) and it is not allowed to carry out both an internship and the graduation project at the same company.</p> <p>The Internship SET (SET3822) is subject to strict rules and procedures. Enrol in the Brightspace page Internship SET for more information. One can find the step-by-step guide explaining all procedures, forms, and additional information e.g. about grants and funding for internships abroad.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
| <b>Co-Instructor</b>                   | <p>L. Botto</p> <p>Dr. L.M. Kamp</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |



|                      |                           |
|----------------------|---------------------------|
| <b>Co-Instructor</b> |                           |
| <b>Co-Instructor</b> | Prof.dr.ir. A.H.M. Smets  |
| <b>Co-Instructor</b> | Prof.dr. P.J. Vardon      |
| <b>Co-Instructor</b> | Dr.ir. W.A.A.M. Bierbooms |

| SET4399                                | Extra Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x Based on appointments with supervisor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | Exam by appointment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | <p>Extra Project (SET4399) is a small research project on a sustainable energy engineering topic carried out within a research group at Delft University of Technology. The project has a maximum study load of 15 EC, with an equivalent maximum full-time duration of 3 months, this includes writing the final report.</p> <p>General guidelines are that it should concern a design, model, system integration or methodology implementation, or an experimental study relevant for the research group involved. The topic should fall within at least one of the profiles of the chosen Profile Cluster. Literature study can only be a minor part of the project.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                     | <p>There are five main learning objectives (LO) for Extra Project (SET4399):</p> <p>LO1: The student is able to contribute to research and development within the field of sustainable energy technology in a TU Delft research group.</p> <p>LO2: The student is able to report the results in a comprehensive manner to the TU Delft supervisor.</p> <p>LO3: The student is able to communicate to and collaborate with members of the research group.</p> <p>LO4: The student is able to describe the effect of the research and its relationship with sustainable energy innovations on society.</p> <p>LO5: The student is able to reflect on their own role and the added value of the project to their study programme.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                | <p>Project</p> <p>The Extra Project (SET4399) follows certain steps:</p> <ol style="list-style-type: none"> <li>1. Orientation</li> <li>2. Kick-off meeting</li> <li>3. Delivering the final report</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Prerequisites</b>                   | <p>In order to ensure that students start the project with a solid skill set, the student can apply for the Extra Project after the third quarter of the first year of the MSc SET programme. Before the application for Extra Project (SET4399), the student needs to have an Individual Exam Programme (IEP) approved by the Board of Examiners and the courses of the first three quarters of the first year of the MSc SET programme must have been completed. A student can also start when at least 45 credits of the Master's degree course work as stated in the IEP has been completed.</p> <p>Before starting the project, the student, the responsible supervisor and the profile leader agree that the student will conduct a research project under the responsibility of the responsible supervisor. The student and responsible supervisor:</p> <ul style="list-style-type: none"> <li>- Agree on the starting date;</li> <li>- Agree on what is expected during the kick-off meeting.</li> </ul> <p>The agreements are written in the Extra Project Application Form. This form needs to be signed by the student, the responsible supervisor and the Director of Studies (for students within the Electrical Energy track) or the master coordinator (for students within the Heating &amp; Cooling track). This is done just before the kick-off meeting.</p> |   |
| <b>Assessment</b>                      | <ul style="list-style-type: none"> <li>- The student needs to write a scientific report during the Extra Project and send it to the TU Delft supervisor on the final day of the project.</li> <li>- The student needs to write a reflection note and a testimonial and send these documents, together with the scientific report, to the responsible instructor, on the final day of the project.</li> <li>- The TU Delft supervisor needs to send the assessment form to the responsible instructor, within two weeks after receiving the documents from the student.</li> <li>- When the responsible instructor has received the two reports and the assessment form, your grade will be processed within a week.</li> </ul> <p>Please, consult the Brightspace page SET4399 Extra Project for the detailed grading scheme.</p> <p>The Extra Project will be graded with a pass or a fail (not numerically).</p> <p>Resit/ Repair opportunities:<br/>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:</p> <ul style="list-style-type: none"> <li>- Report: resubmit deliverable</li> </ul>                                                                                                                               |   |
| <b>Special Information</b>             | <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> <p>The project needs to be carried out according to MSc SET guidelines of Delft University of Technology. This is safeguarded by the requirement to have a responsible supervisor who is on the SET green list for the most relevant profile.</p> <p>NB: Extra Project (SET4399) can never be used as preparation for a Graduation Project (SET3901).</p> <p>The Extra Project is subject to strict rules and procedures. Enrol in the Brightspace page SET4399 Extra Project for more information. One can find the step-by-step guide explaining all procedures, forms, and additional information e.g. about the assessment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-Instructor</b>                   | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| WM0320TU                               | Ethics and Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | M. Sand                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | M. Sand                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | A.R. Gammon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | A.R. Gammon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                 | You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.                                                                                                                  |   |
| <b>Study Goals</b>                     | <p>After having completed the course you:</p> <ul style="list-style-type: none"> <li>can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology.</li> <li>have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds.</li> <li>are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers.</li> <li>are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.</li> </ul> |   |
| <b>Education Method</b>                | A series of 6 lectures and 5 interactive work sessions with partial assessment, concluded with a written test.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>Written exam (60%, individual grade), presentation and active participation during the working group sessions (40%, same grade for all members of a presentation group).</p> <p>In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75.</p> <p>It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade.</p> <p>These partial grades are only valid for 1 academic year, meaning that if one or both are not passed during resit, both assessments need to be retaken the following academic year.</p>                                                                            |   |
| <b>Enrolment / Application</b>         | Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Remarks</b>                         | The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course WM0329TU (6 ects).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Category</b>                        | MSc niveau                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Language courses 2024

| TPM301C                               | Spoken English for International Communication (for Master students)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2 |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                 | Drs. M.A. Swennen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                     | Drs. D.W. Laponder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                     | Drs. K.I. van der Linden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                     | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Co-responsible for assignments</b> | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week x/x/x/x</b>   | 0/3/0/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>               | 2<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                | 2<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                    | 2<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                | This speaking course focuses on the language and skills students need to participate successfully at the university. Students will engage in a number of activities both inside and out of the classroom - to practise and improve their English. The vocabulary focus will be on general academic language, so that after the course students can interact with more confidence in an academic setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                    | The aim of the course is to develop language skills needed for the university environment. You will learn phrases and words for a range of skills needed here, for example participating in discussions; rephrasing and explaining; giving an opinion, (dis)agreeing, making suggestions; describing and presenting a concept.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>               | Group and pair work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b> | To be announced in Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Practical Guide</b>                | You can only take this course when you have taken our Placement Test, and received a Course Recommendation for this course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                     | <p>We offer this Placement Test regularly. The Course Recommendation that you receive is valid for a year. This means, for example, that you can take the test in October, and sign up for a course that starts in April.</p> <p>Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session.</p> <p>The assignments must be submitted each week before the deadline set by the instructor.</p> <p>You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.</p>                                                                  |   |
| <b>Enrolment / Application</b>        | <p>Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English.</p> <p>For more information on this 'test' please visit our website: <a href="https://www.tudelft.nl/tbm/itav/onderwijs/english-unit/placement-test">https://www.tudelft.nl/tbm/itav/onderwijs/english-unit/placement-test</a></p> <p>Enrolment for the course (and the Placement Test) is via Brightspace. Type 'TPM301C'(or 'Placement Test') in Course search (Catalog). Make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.</p> <p>You do not need to register for this course in Osiris.</p> <p>Please note that you must also enrol in a group to reserve a place.</p> |   |
| <b>Remarks</b>                        | <p>You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.</p> <p>The course will only be offered if there is a sufficient number of participants.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Elective</b>                       | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Tags</b>                           | English proficiency<br>Group work<br>Small groups                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Category</b>                       | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| TPM302B                                | Spoken English for Academic Purposes Advanced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Drs. K.I. van der Linden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Module Manager</b>                  | Drs. M.A. Swennen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Drs. D.W. Laponder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Drs. K.I. van der Linden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Co-responsible for assignments</b>  | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 3/0/3/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context.</p> <p>A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles.</p> <p>The sessions revolve around student interaction.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Study Goals</b>                     | <p>The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information.</p> <p>Another study course goal is to expand and improve the students' general vocabulary.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Method</b>                | Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>  | To be announced in Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Practical Guide</b>                 | This is a course for both BSc and MSc students who would like to expand and improve their English language skills.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session.</p> <p>The assignments must be submitted each week before the deadline set by the instructor.</p> <p>You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Enrolment / Application</b>         | <p>Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. When you have taken our Placement Test you will receive a course recommendation. With this recommendation you can sign up for this course. You can only register for courses that are recommended to you, or do not require a Placement Test.</p> <p>For more information on this 'test' please visit our website: <a href="http://www.itav.tudelft.nl">www.itav.tudelft.nl</a></p> <p>Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris.</p> <p>Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.</p> |   |
| <b>Remarks</b>                         | <p>The course will only be offered if there is a sufficient number of participants.</p> <p>Please note that it is not a presentation course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Tags</b>                            | <p>English proficiency</p> <p>Group work</p> <p>Small groups</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Category</b>                        | BSc and MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

| TPM305A                                | Writing a Masters Thesis in English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Drs. D.W. Laponder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Drs. M.A. Swennen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Drs. D.W. Laponder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Drs. K.I. van der Linden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                      | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Responsible for assignments</b>     | Drs. D.W. Laponder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Co-responsible for assignments</b>  | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                     | The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure your thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.                                                                                                                                                                                          |   |
| <b>Education Method</b>                | Learning by doing and learning from lectures, corrections, peer review and tutor feedback.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Books</b>                           | Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English.</p> <p>You may not submit texts that were previously submitted for any other course. Students who submit their weekly assignments late, repeatedly miss deadlines or lack commitment may be asked to leave the course.</p> <p>An 85% attendance record is required. This means students may miss no more than one session (preferably none).</p> <p>Students will receive their course credits when their work shows sufficient progress and weekly deadlines and all other conditions are met.</p> |   |
| <b>Special Information</b>             | Enrolment is through Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Remarks</b>                         | The course will only be offered when there is sufficient demand. If there are not enough enrolments, the course may be cancelled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Tags</b>                            | English proficiency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| TPM306A                                | Writing Academic English using AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Drs. K.I. van der Linden                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Drs. M.A. Swennen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Co-responsible for assignments</b>  | S.F. Johnson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/x/0/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 2<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 2<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 2<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | The Writing Academic English using AI course is designed to elevate students' writing proficiency through the strategic use of AI tools. Students learn the conventions of academic writing, such as text structure, paragraph structure, sentence structure, grammatical variation, and formal-vs-informal words. They will practise editing and proofreading with AI, focusing on improving grammar, style, and coherence. They also learn to critically evaluate the AI-generated responses. Additionally, the course will cover ethical considerations in the use of AI in writing, ensuring students understand issues related to plagiarism and data privacy.                                                                  |   |
| <b>Study Goals</b>                     | This course aims to familiarise students with academic writing conventions and show them how to use large language models to improve their writing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                | This is a 7-week course with one lesson a week.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>  | Course material will be announced in Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Books</b>                           | Course material will be announced in Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Prerequisites</b>                   | <p>Before enrolling in this course you must first take our Placement Test. When you have taken our Placement Test you will receive a course recommendation. With this recommendation you can sign up for this course. You can only register for courses that are recommended to you, or do not require a Placement Test.</p> <p>Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris.</p> <p>Please note that you must also choose a group to reserve a place in the class. Type 'TPM306A' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.</p> |   |
| <b>Assessment</b>                      | Continuous assessment. Participants submit various writing assignments in which they demonstrate their writing skills and their ability to critically evaluate the output given by AI tools.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Remarks</b>                         | The course will only be offered if there is a sufficient number of participants.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Tags</b>                            | English proficiency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

| WM1115TU                                | Dutch Elementary 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                   | Drs. G. Hoezen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Responsible for assignments</b>      | Drs. G. Hoezen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Co-responsible for assignments</b>   | Dr. S.J. van Boxtel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 3/3/3/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                 | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                  | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                      | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                  | Dutch<br>English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Required for</b>                     | After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>         | No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a> .                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Parts</b>                            | Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                  | Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                      | The goal of this course is to acquire basic Dutch language skills on CEF level A1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                 | Interactive group sessions, individual work through online course materials.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>   | Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                       | The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade.<br>If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | No materials admitted during tests.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Enrolment / Application</b>          | --- for MSc/BSc students ---<br>There is a two-step-enrolment procedure.<br>1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will self-study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1.<br>2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU)<br>You can enrol for one of the groups.<br><br>--- for others (PhDs, employees) ---<br>Go to the website <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a> for registration. |   |
| <b>Special Information</b>              | Brightspace, <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a> or <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Remarks</b>                          | Regular courses take place at the campus in groups of max. 15 participants.<br>There is an option to take this course as a self-study programme with a final exam + a speaking test. Contact the secretaries office for registration. <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a><br>More information: <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a>                                                                                                                                                                                                                                                                                                    |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Tags</b>                             | Broad<br>Diverse<br>Small groups                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Targetgroup</b>                      | Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch.<br>People who already know some Dutch (approx. 1000 words or more) can contact <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a> for a level test.                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Self Test</b>                        | If you want to do self-study, you can register for a written test + speaking test by sending an email to <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Category</b>                         | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |



| WM1116TU                                | Dutch Elementary 2                                                                                                                                                                                                                                                                                                                                                                                         | 3 |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                   | Drs. G. Hoezen                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Responsible for assignments</b>      | Drs. G. Hoezen                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Co-responsible for assignments</b>   | Dr. S.J. van Boxtel                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 3/3/3/3                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                 | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                  | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                      | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                  | Dutch                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Required for</b>                     | After this course, participants can continue learning in Dutch Intermediate 1 (course code WM1117TU)                                                                                                                                                                                                                                                                                                       |   |
| <b>Expected prior knowledge</b>         | A completed course Elementary 1 (WM1115TU). It is also possible to enrol if you have a similar level: CEF A2. In that case you have to sit a level test. Contact <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a> .                                                                                                                                                                |   |
| <b>Parts</b>                            | Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                  | Dutch Elementary 2 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.                                                           |   |
| <b>Study Goals</b>                      | The goal of this course is to acquire basic Dutch language skills on CEF level A2.                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                 | Interactive group sessions, individual work through online course materials.                                                                                                                                                                                                                                                                                                                               |   |
| <b>Literature and Study Materials</b>   | Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                       | The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade.<br>If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period. |   |
| <b>Permitted Materials during Tests</b> | No materials admitted during tests.                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Enrolment / Application</b>          | --- for MSc/BSc students ---<br><br>Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU)<br><br>--- for others (PhDs, employees) ---<br>Go to the website <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a> for registration.                                                                                                                                                  |   |
| <b>Special Information</b>              | Brightspace, <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a> or <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a>                                                                                                                                                                                                                                                          |   |
| <b>Remarks</b>                          | Regular courses take place at the campus in groups of max. 15 participants.<br>There is an option to take this course as a self-study programme with a final test + a speaking test.. Contact the secretaries office for registration. <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a><br>More information: <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a>              |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Tags</b>                             | Broad<br>Diverse                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Targetgroup</b>                      | Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a> for a level test.                                                                                                                                                              |   |
| <b>Category</b>                         | MSc level                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| WM1117TU                                | Dutch Intermediate 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3 |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                   | Drs. G. Hoezen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Responsible for assignments</b>      | Drs. G. Hoezen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-responsible for assignments</b>   | Dr. S.J. van Boxtel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 3/3/3/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                 | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                  | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                      | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                  | Dutch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Required for</b>                     | After this course, participants can continue learning in Dutch Intermediate 2. This is a paid course, also for students and will not give you ECTS credits.                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>         | A completed course Elementary 2 (WM1116TU). It is also possible to enrol if you have a similar level: CEF A2+. In that case you have to sit a level test. Contact <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a> .                                                                                                                                                                                                                                                                                                         |   |
| <b>Parts</b>                            | Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests and gap filling tests.                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                  | Dutch Intermediate 1 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                      | The goal of this course is to acquire basic Dutch language skills on CEF level A2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Method</b>                 | Interactive group sessions, individual work through online course materials.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>   | Wesdijk en Blom Sciarone e.a., "Tweede Ronde, herziene editie. Nederlands voor buitenlanders. De Delftse Methode". Text book.                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                       | The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: listening tests and written gap filling tests. Attendance also contributes to the final grade.<br>If this final grade is not sufficient, students sit a written exam, a speaking test or a combination of both. These tests take place after the regular course period.                                                                                                                                           |   |
| <b>Permitted Materials during Tests</b> | No materials admitted during tests.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Enrolment / Application</b>          | --- for MSc/BSc students ---<br><br>Enrolment through Brightspace in the course Dutch Intermediate 1 (WM1117TU)<br><br>--- for others (PhDs, employees) ---<br>Go to the website <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a> for registration.                                                                                                                                                                                                                                                                                          |   |
| <b>Special Information</b>              | Brightspace, <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a> or <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a>                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Remarks</b>                          | Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation.<br>There is an option to take this course as a self-study programme with a final test + speaking test. Contact the secretary's office for registration. <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a><br>More information: <a href="http://www.dm.tudelft.nl">www.dm.tudelft.nl</a> |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Tags</b>                             | Broad<br>Diverse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Targetgroup</b>                      | Students and others who want to improve their Dutch knowledge. If you are not sure whether this course is on the right level for you, please contact <a href="mailto:delftsemethode@tudelft.nl">delftsemethode@tudelft.nl</a> for a level test.                                                                                                                                                                                                                                                                                                      |   |
| <b>Category</b>                         | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

#### Entrepreneurship courses 2024

| TPM406A                                | Corporate Entrepreneurship and Start-ups                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Responsible for assignments</b>     | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Co-responsible for assignments</b>  | Dr. L. Hartmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/x/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior knowledge</b>        | No prior knowledge is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Summary</b>                         | <p>This course examines how large corporations engage startups to tap into new markets, technologies and business models.</p> <p>You will develop knowledge and skills to analyse strategic situations of existing corporates and how they innovate through engaging startups. The course provides an eye on corporate startup engagement from both the corporate and startup perspective.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>Large international organizations often find themselves in a luxurious position compared to small start-ups. They have more financial capital, access to markets, and more technological knowledge. However, large international organizations are confronted with increasing external turbulence, technological advancement and globalization, which puts their luxurious position in jeopardy. At the same time the luxurious position constrains them to depart from present activity and focus on radical new innovations. Hence, corporates try to renew themselves through internal entrepreneurial behaviour, but also by initiating start-ups and acquiring start-ups to experiment with new technology. Many times, these start-ups originate in academic environments. Therefore we pay specific attention to the emergence of academic start-ups and how corporate organizations provide incubation facilities and venture capital support to engage startups to renew themselves.</p> <p>This course examines the various forms of corporate entrepreneurship and the relationship with corporate strategy in particular. You will develop knowledge and skills to analyze strategic situations and design the most appropriate solutions in terms of corporate entrepreneurship strategy and organization.</p> |   |
| <b>Study Goals</b>                     | <p>Understand the concept of corporate entrepreneurship; the contexts, the forms, and the relationship with strategic management</p> <p>Analyze strategic situations and design appropriate corporate entrepreneurship strategies and organizations</p> <p>Analyse the contribution of internal and external ventures in the renewal of large corporates</p> <p>Understand the emergence of academic spin-offs and how they relate to the technology development of large established organisations</p> <p>Identify the various key actors that facilitate and support the emergence of new business ventures and academic spin-offs</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Method</b>                | <p>Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Each student will be expected to be present during lectures (two times a week), actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Literature and Study Materials</b>  | <p>Journal articles:<br/>Will be posted in Brightspace</p> <p>Slides presented in class:<br/>Will be posted in Brightspace</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Practical Guide</b>                 | <p>Given the interactive nature of this course, attendance at classes is mandatory. Only very serious grounds for absence (such as illness accompanied with doctors note) are acceptable. Holidays and (paid or voluntary) work obligations do not count as acceptable reasons for absence. This is a full-time course with scheduled holidays. Unjustified non-attendance will result in expulsion from this course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>The assessment includes three components.</p> <p>1) The first component is a team level online presentation of a case analyses of an existing company. Each team will analyse a practical company case provided to you and based on literature of a specific lecture session you will present your findings online in class (40%)</p> <p>2) The second component is a written report of a large firm in which you will analyse with your team the startup engagement activities this firm has put in place to renew itself. The report is an analyses of the company activities w.r.t. corporate venturing and an reflection of these activities by making use of literature. The report accounts for 40% of the final grade.</p> <p>3) The third component is an individual analyses of a startup that is involved in the corporate startup engagement activities of your team case company. It reflects on the motivation for the startup engagement from both startup and corporate perspective. It accounts for 20% of the final grade.</p>                                                                                                                                                                                                                                                                             |   |
| <b>Enrolment / Application</b>         | <p>All students across the various faculties of the Delft University of Technology can take this course.</p> <p>==&gt; You can directly enroll in Brightspace.</p> <p>==&gt; If you need to use the mytudelft app, you may see that you cannot register to this course. BUT that information is for the exam, which does not apply to this course, since there is no exam. Hence, you can register and enroll in Brightspace.</p> <p>==&gt; You may need approval from your exam committee, when you take this course within you elective space. But please register also in Brightspace and we can plan the course.</p> <p>==&gt; You can always take this course as an extra course, not part of your elective list, and it will be included in grade list.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Special Information</b>             | <p>Each student team will present during the lectures, actively participate during the lectures, complete assigned readings, read posted lectures, listen to thought leader presentations and submit written assignments when due.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Remarks</b>                         | <p>For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Targetgroup</b>                     | <p>Master students from all master programs can apply.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Category</b>                        | <p>MSc level</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| TPM416A                         | Turning Technology into Business                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Module Manager                  | Dr. L. Hartmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Responsible for assignments     | Dr. L. Hartmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Co-responsible for assignments  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Contact Hours / Week<br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Education Period                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| Start Education                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| Exam Period                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Course Contents                 | Turning Technology into Business (TTiB) aims to equip participants with a strong conceptual foundation to actively understand the dynamic process of technology-based entrepreneurship. Participants learn how business strategies are best formulated and how (through entrepreneurship) technology can create value. TTiB consists of seven lectures, introducing the theoretical backgrounds of technological, market and business analyses. A unique aspect of TTiB is that existing technologies (developed and patented by the TU Delft) are used as case subjects. The patent project is the focal point of the curriculum. Each group of 4-5 students will be assigned an original patent, and is expected to evaluate the commercial potential of this technology. This includes choosing promising applications (products) for the technology, and making recommendations for the most suitable business model to commercialize the technology. The patent project should provide a coherent and structured answer to the central question: which strategy is most likely to generate business from this patent? In addition to equipping participants with a thorough knowledge of the course subject, the section Technology, Strategy & Entrepreneurship (TSE) hopes that this course will be the starting point for a variety of university spin-offs in which the students will participate. To accommodate this, TSE offers the course TPM420A- Ready to Startup! |   |
| Study Goals                     | This course aims to equip students with a strong conceptual foundation to an active understanding of two domains:<br>(1) The dynamic process of technological innovation through concepts such as technology life-cycles, dominant design, disruptive technologies, Schumpeterian competition and the diffusion of innovations<br>(2) How business strategies are formulated and, through entrepreneurship, technology can create value. This multi-faceted process of technology commercialization process is addressed in terms of assessing technology position, discovering market opportunities, competitive analysis, appropriability and the various modes of entrepreneurship.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Education Method                | 7 interactive lectures, participant-centered case studies, homework assignments, classroom assignments, individual group coaching.<br>Final Presentations in February.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Literature and Study Materials  | Will be provided during the course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Assessment                      | Each group (of 4 or 5 students) writes a report on their findings, analysis and recommendations for the technology (patent) they used as a case for this course. On the Final Day, the group has to give a 10 minute oral presentation.<br>The report makes up 80% of the final grade. The final grade is summative.<br>This team grade will be modified according to the individual student's class contributions and his/her performance within the group (peer review).<br>There is no resit possibility.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Enrolment / Application         | MAXIMUM CAPACITY is limited to 75 participants. Register on BrightSpace from October 1.<br>Because of the huge popularity of this course, there is a selection procedure which consists of pre-course assignments. One of these assignments is a letter of motivation: why do you want to participate in this course? Based on the motivation and the other pre-course assignments, the final selection of 75 participants will be made.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| Remarks                         | For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Targetgroup                     | Advanced Master students, PDEng and PhD students, and staff researchers from all faculties within the TU Delft.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| Category                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| TPM420A                                | Ready to Start-up                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | P. Harish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | P. Harish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Responsible for assignments</b>     | P. Harish                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Co-responsible for assignments</b>  | Dr. L. Hartmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/x/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 1<br>2<br>3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Summary</b>                         | <p>This course is offered twice in a year; First starting week 3/4 of Q1 and ending in Q2 and second starting week 3/4 of Q3 until end Q4</p> <p>Ready to Startup (RtS) is an elective master course that has been the crown jewel of entrepreneurial education at TU Delft for over two decades. Bringing together business experts and students eager to start companies, it has one simple goal: turn engineers into successful founders.</p> <p>RtS focuses predominantly on Business Strategy: gaining market traction, prototyping, growing the team, devising a financial plan and fundraising strategy, protecting IP, and foundational activities (setting up a company, sharing ownership, etc.)</p> <p>The course has the prerequisite that teams should be formed and have a business ideas that they want to further analyse and validate. It is meant for students who plan to start a company.</p> <p>After RtS, teams must be at the level required for YES!Delft incubation and Delft Enterprises investments.</p> |   |
| <b>Course Contents</b>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Study Goals</b>                     | <p>At the end of the course, you will be able to</p> <ul style="list-style-type: none"> <li>- Reflect on your current business model and identify further necessary validation activities. Analyse market entry strategies and consider alternative models.</li> <li>- Learn how to negotiate, attract first users, and seek endorsement of stakeholders (such as investors and advisors).</li> <li>- Plan your core operational activities such as company foundation, IP protection, product development strategy and team development</li> <li>- Devise financial plans and appropriate fundraising strategies</li> <li>- Convey a convincing narrative of your business and be able to provide constructive feedback to other startup plans.</li> <li>- Demonstrate effective entrepreneurial behaviour by making swift decisions based on newly gathered information and adjusting your model accordingly.</li> </ul>                                                                                                          |   |
| <b>Education Method</b>                | <p>The course is spread out over 12 in-person sessions on Wednesday afternoon (13h30-17h30), where the core elements of a startup plan are presented. You work on assignments that are aimed at specific elements of the core startup plan, and you will by apply them to your startup project and present your findings during the course.</p> <p>In this way you make evident progress with your startup. The course combines state of the art business methods, translated to the needs of starting entrepreneurs, with consults from practitioners. Topics are discussed using case studies. You will present your progress, provide peer-feedback as well as receive advice from mentor experts. The course ends with a Final Day, consisting of a jury pitch and handing in your business plan.</p>                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>  | Handouts and literature are available trough in Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Prerequisites</b>                   | Enthusiasm, entrepreneurial spirit and a good business idea.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The course has 4 grading components.</p> <ul style="list-style-type: none"> <li>- Business plan: final report (40%)</li> <li>- Financial plan (10%)</li> <li>- Final Day performance (30%)</li> <li>- Overall progress: homework and attendance (20%)</li> </ul> <p>In general, you are only allowed to miss one lecture (with a valid reason submitted beforehand). Furthermore, we will monitor your progress during the course, with an official assessment halfway through the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Enrolment / Application</b>         | <p>Application process you need to register in advance to the course.</p> <p>Teams of students with clear ideas about a business they want to start supported with evidence that it addresses a need, will have priority to enter.</p> <p>==&gt; You can directly enroll in Brightspace.</p> <p>==&gt; If you need to use the mytudelft app, you may see that you cannot register to this course. BUT that information is for the exam, which does not apply to this course, since there is no exam. Hence, you can register and enroll in Brightspace.</p> <p>==&gt; You may need approval from your exam committee, when you take this course within your elective space. But please register also in Brightspace and we can plan the course.</p> <p>==&gt; You can always take this course as an extra course, not part of your elective list, and it will be included in grade list.</p>                                                                                                                                        |   |
| <b>Remarks</b>                         | <p>For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Targetgroup</b>                     | Master students from all master programs can apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Profile Electives 2024

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Waste & Combined Heat and Power 2024



| AE4262                                 | Combustion for propulsion and power technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4 |
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| <b>Responsible Instructor</b>          | Prof.dr. A. Gangoli Rao                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Responsible Instructor</b>          | F. De Domenico                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Responsible Instructor</b>          | Dr. I. Langella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | G. Kelesidis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4+2lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Expected prior knowledge</b>        | WI1421LR-I Calculus I, WI1402LR Calculus II, AE1240-I Thermodynamics, AE4238 Aero Engine Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Summary</b>                         | This course is designed to provide the student with a fundamental understanding of combustion processes and their application within advanced propulsion and power technologies. The goal is to reach a knowledge on how to improve the efficiency and mitigate pollutant formation of practical combustion systems. Since currently about 90% of the worldwide energy usage (e.g. transportation, electricity generation, heating) is provided by combustion, it is of the utmost importance to continue developing highly efficient combustion technologies with the lowest possible level of emissions. This is especially true for aero engines, whereby the use of combustion engines will continue in the foreseeable future.                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <ol style="list-style-type: none"> <li>1. Thermodynamics of combustion</li> <li>2. Chemical kinetics</li> <li>3. Transport processes</li> <li>4. Premixed and diffusion flames</li> <li>5. Laminar and turbulent reactive flows</li> <li>6. Combustion experiments and diagnostics</li> <li>7. Combustion modelling</li> <li>8. Laboratory Tutorial &amp; Exercise</li> <li>9. Practical combustion systems for aerospace applications</li> <li>10. Pollutants and emissions</li> <li>11. Hydrogen combustion physics and technology</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Study Goals</b>                     | <p>Write chemical equations and perform stoichiometry calculations.<br/>         Compute heating values of fuels, enthalpies of formation and adiabatic flame temperature.<br/>         Apply principles of thermodynamics to analyse reacting mixtures at chemical equilibrium.<br/>         Describe the fundamentals of the physics involved in premixed- and diffusion flames, laminar and turbulent.<br/>         Discuss the physics related to turbulent reacting flows and chemistry interaction.<br/>         Explain experimental measurement techniques and their role in understanding combustion.<br/>         Discuss critical mechanisms for improving the efficiency and reducing the emissions of air-breathing combustion systems.<br/>         Provide phenomenological descriptions of pollutant formation of conventional combustion systems.<br/>         Elaborate on the potential and challenges related to the use of alternative fuels in combustion systems.</p> |   |
| <b>Education Method</b>                | The course consists of weekly lectures and exercise sessions (practicals). In addition, the students are required to complete a project assignment and perform a laboratory exercise. The project assignment is relevant to practical combustion systems for aerospace applications. The laboratory exercise deals with the investigation of methane or hydrogen laminar flames. The students will gain hands-on experience in designing the experimental matrix, acquiring data, performing data analysis and writing a technical report. Both laboratory exercise and assignment are mandatory to be admitted to the oral exam.                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Relations</b>                | Aero Engine Technology<br>Hydrogen Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | Slides, lecture notes and selected material from several textbooks are made available (e.g., J. Warnatz, U. Maas, and R.W. Dibble COMBUSTION;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Books</b>                           | J. Warnatz · U. Maas · R.W. Dibble "Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>To pass the course, the students must</p> <p>Complete laboratory exercise and project assignment; and<br/>         Pass the oral exam with a minimum grade of 5.5 out of 10</p> <p>The project assignment consists of several parts, including analysis on the data collected during the laboratory exercise and further theoretical and numerical analyses of combustion phenomena. Note attendance to the laboratory exercise and submission of the assignment are both compulsory for the admission to the oral exam.<br/>         The work on the assignment should be summarized in a report, to be written by a single student or in a group of maximum three.<br/>         The final grade is the result of the oral examination (60%) and the grade obtained in the assignment (40%).</p>                                                                                                                                                                                         |   |

| CH3682A                                 | Reactor & Kinetics (R&K)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Prof.dr.ir. A. Urakawa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                       | J. Palomo Jiménez                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                       | Prof.dr.ir. A. Urakawa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/6/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                      | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>         | BSc level Chemical Reaction Engineering; Numerical Methods, Python.<br>Note: it is absolutely required to have a good working understanding of BSc (undergrad) level chemical reaction engineering, thermodynamics and transport phenomena.                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                  | Kinetics<br>- Homogeneous reactions: Equilibrium and kinetics<br>- Heterogeneous reactions: Adsorption, surface reactions, kinetics<br>- Catalysis: Transport limitations, effectiveness, heat effects, catalyst deactivation<br><br>Reactors<br>- Brief review of conversion and selectivity in ideal reactor types<br>- Energy balance and isothermal/non-isothermal reactors<br>- Non-ideal reactors and residence time distribution<br>- Fixed-bed reactors: Transport limitations<br>- Advanced reactors such as fluidised-bed reactors    |   |
| <b>Study Goals</b>                      | This course deals with experimental and theoretical aspects of kinetics and reactor theory. After the course, the student will be able to:<br>- formulate kinetic models for mechanisms of complex reactions<br>- interpret kinetic studies and use reactor measurements to validate such models<br>- describe and model complex reactors with diffusion, dispersion and heat effects<br>- propose reactor designs based on kinetic insights                                                                                                    |   |
| <b>Education Method</b>                 | Lectures, homework (theoretical and computer based, often use of Python).                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Computer Use</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>   | Chemical Reactor Analysis and Design Fundamentals, James B. Rawlings & John G. Ekerdt, among others<br>+ Course Handouts                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                       | The final grade is based on:<br>- Mid-term computer exam (35%);<br>- Final written exam (50%) that is based on theory (no coding/programming);<br>- Results of multiple exercises/homeworks that involve analytical problems and programming (15%).<br><br>Re-examination is possible for both the mid-term computer exam and final written exam. The weight within the final grade will be the same as in the first exam, 35% and 50 %, respectively. Only the exam with the highest grade will be considered for the final grade calculation. |   |
| <b>Permitted Materials during Tests</b> | During mid-term computer exam, course slides will be provided as supporting material.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| LM3741                                 | Fermentation Technology & Environmental Biotechnology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. R. Kleerebezem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Prof.dr.ir. H.J. Noorman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr.ir. R. Kleerebezem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/5/2/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>        | Microbial diversity, Black box kinetics of growth/ product formation, mass balances, transport phenomena.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>The FTEB course is divided into two sections: Microorganism characterization (i), and Bioreactor design (ii). Microorganism characterization deals with stoichiometry, kinetics, and thermodynamics of microbial processes in the field of FT and EB. Specific attention is paid to identification of parameters and parameter sensitivity. The bioreactor design section includes batch, fed-batch and continuous bioreactor operation. Specific attention is paid to mixing, gas-liquid mass transfer, biofilm and other biomass retention based bioprocesses, as well as integration of different microorganisms in a single process configuration. All lectures are supplemented with practical examples.</p> <p>Students will elaborate in groups of 3-4 students a case study that should result eventually in the design of a bioprocess. As in the lecture series, students will first elaborate the properties of the microorganisms required and subsequently on the implementation in the bioreactor. Both parts of the case study will be discussed during poster presentations.</p> |   |
| <b>Study Goals</b>                     | <p>After this course the students are able to analyse and design fermentation and environmental bioprocesses.</p> <p>Most relevant transferable skills:</p> <ul style="list-style-type: none"> <li>- Problem solving</li> <li>- Teamwork</li> <li>- Communication</li> <li>- Responsibility</li> <li>- Time management</li> <li>- Presentation skills</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | Lectures and a case study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>  | Lecture notes, lecture slides and relevant scientific papers, which will be provided via Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | Case study (group work) and exam (individual). The scores for the case study and the exam determine for 50% the final grade. The minimum exam score to pass is 5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Enrolment / Application</b>         | On Brightspace. Not later than one week before the first lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Schedule</b>                        | <p>Part 1: Microorganism</p> <ul style="list-style-type: none"> <li>- Introduction</li> <li>- Stoichiometry</li> <li>- Thermodynamics</li> <li>- Basic kinetics</li> <li>- Including product formation</li> <li>- Determining parameters from experimental data</li> <li>- Fed-batch, batch, chemostat</li> <li>- Case study part 1: microorganisms</li> </ul> <p>Part 2: Bioreactor</p> <ul style="list-style-type: none"> <li>- Mass transfer modelling</li> <li>- Design transfer processes in bioreactors</li> <li>- Biofilms structure,function, flux analysis, and process design</li> <li>- Integrating mass transfer, biofilms, and biomass retention</li> <li>- FT-examples</li> <li>- EB-examples</li> <li>- Case study part 2: Bioreactor</li> </ul>                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Location</b>                        | Delft                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| ME45001                                 | Advanced Heat Transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
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| <b>Responsible Instructor</b>           | Dr.ir. J.W.R. Peeters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                       | Dr. R. Delfos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 6/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Expected prior knowledge</b>         | A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena.<br>Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in ME or equivalent; basic understanding of Fourier series.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                  | <p>In this course the concepts &amp; mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures.</p> <p>During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems.</p> <p>Subjects in order of appearance:</p> <ul style="list-style-type: none"> <li>- A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy.</li> <li>- A refresher on Conduction, Convection and Radiation.</li> <li>- Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms.</li> <li>- Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques &amp; numerical techniques; relaxation.</li> <li>- Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number &amp; Stefan condition.</li> <li>- Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques &amp; numerical techniques; stability criteria.</li> <li>- Forced &amp; Free convection: Nusselt, Stanton, Prandlt &amp; Peclet numbers; Analysis &amp; the physics behind empirical correlations. The role of boundary conditions.</li> <li>- Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.</li> </ul> |   |
| <b>Study Goals</b>                      | <p>More specifically: The student is able to</p> <ol style="list-style-type: none"> <li>1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner.</li> <li>2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches.</li> <li>3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction.</li> <li>4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions.</li> <li>5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries.</li> <li>6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations.</li> <li>7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications.</li> </ol> <p>Indicate implications when more detailed distributions of convective heat transfer are involved.</p>                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                 | <ul style="list-style-type: none"> <li>- Lectures (2x2 hours per week) including experiments &amp; computer demonstrations.</li> <li>- After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided.</li> <li>- Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment.</li> <li>- Total number of contact hours: 6 hours per week.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Computer Use</b>                     | Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>   | <p>Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used:<br/>A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'.</p> <p>The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical.</p> <p>Alternative books on heat transfer from your Bachelor's course (including Cengel &amp; Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Instruction if it's not standard.</p> <p>Lecture slides, short videos, Matlab scripts &amp; other study materials required will be provided via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                       | The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Permitted Materials during Tests</b> | Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted.<br>One (1) equation sheet of 2 sides A4 can be taken and/or is provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Enrolment / Application</b>          | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Department</b>                       | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| ME45026                          | Introduction to Multiphase Flows                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3 |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor           | Dr.ir. W.J. Hogendoorn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Instructor                       | Prof.dr.ir. C. Poelma                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Contact Hours / Week<br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Education Period                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Start Education                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Exam Period                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Course Language                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| Course Contents                  | A multiphase flow is the simultaneous flow of two or more immiscible phases of matter like gasses, liquids and/or solids. Such flows are important in many environmental and industrial processes and equipment. Examples are transport of food and production of pharmaceuticals, dredging, steel making, refineries and chemical- and nuclear power plants, sediment transport in rivers, cavitation, etc. This course, inspired by these processes, gives a broad introduction to the fundamental fluid mechanics of such flows. Topics include: classification of multiphase flows into dispersed and separated multiphase flows, relevant dimensionless numbers, stability of two-phase flows, modelling approaches, flow pattern maps, physics of droplets, bubbles, and particles; dynamics of particle-laden flow, effects of turbulence, experimental techniques and computational methods for multiphase flows. Examples will be given of the use of multiphase flow theory in industrial applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Study Goals                      | <p>At the end of this course the student is able to:</p> <p>1.a classify multiphase flows into separated and dispersed flows, including solid particles, gas bubbles and/or liquid droplets.<br/>1.b describe similarities and differences between the dynamics of solid particles, gas bubbles and liquid droplets.<br/>1.c formulate appropriate interfacial conditions for a fluid/fluid and a fluid/solid interface.<br/>1.d formulate relevant dimensionless numbers for multiphase flows.<br/>1.e explain the concept of dynamic similarity of multiphase flows.</p> <p>2.a understand the stability mechanisms of two-phase flows such as for instance a liquid jet in a gas (Rayleigh-Plateau instability) and two horizontal fluid layers in relative motion with each other (Kelvin-Helmholtz instability).<br/>2.b classify gas/liquid pipe flows into various flow patterns, dependent on the superficial gas and liquid velocities. And explain physical mechanisms for the transition between those patterns.</p> <p>3.a understand averaging methods to describe macroscopic behavior of multiphase flows.<br/>3.b apply simplified 1D (cross-section averaged) models.</p> <p>4.a describe the settling of a solid sphere under gravity in a stagnant fluid in terms of the Archimedes number and the fluid/solid mass density ratio.<br/>4.b describe possible interactions between solid particles and turbulence of the carrier phase.<br/>4.c describe the dynamics of characteristic particle-laden flows such as sediment transport through a pipe.</p> <p>5.a explain changes in the shape of a rising gas bubble in a stagnant fluid dependent on the Reynolds number, Eotvos number and Morton number.<br/>5.b estimate the terminal rise velocity of a bubble in a stagnant liquid from the equation of motion and an appropriate model for the drag.<br/>5.c give physical mechanisms for the breakup of bubbles/droplets in turbulence.<br/>5.d Explain the dynamics and mechanisms of cavitating flows</p> <p>6. describe the merits and shortcomings of different experimental methods for multiphase flows.</p> <p>7. describe the merits and shortcomings of different numerical methods for simulating multiphase flows.</p> |   |
| Education Method                 | Oral lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Assessment                       | Paper presentation (20%)<br>Written exam (80%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Permitted Materials during Tests | Course books and lecture notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| ME45070-24                    | Advanced Reaction and Separation Systems | 5 |
|-------------------------------|------------------------------------------|---|
| <b>Responsible Instructor</b> | Prof.dr.ir. A.A. Kiss                    |   |

| ME45155                                | Computational Fluid Dynamics for Engineers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. M.J.B.M. Pourquie                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Prof.dr. R. Pecnik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr. P. Simões Costa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Expected prior knowledge</b>        | WB2542 (thermofluids), ME45042 or ME45043 (advanced fluid dynamics), WBMT2049 T2 and T3 or similar (basic numerical analysis), some elementary programming skill (python, matlab or any other)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>This is a course on computational fluid dynamics related to process and energy systems. Some elementary knowledge of numerical methods (discretisation of a derivative, numerical solution of ordinary differential equations) is assumed. See expected prior knowledge.</p> <p>In general the course will cover:</p> <ol style="list-style-type: none"> <li>(1) general conservation equations for mass, heat and momentum transport (Navier-Stokes equations)</li> <li>(2) numerical solution of multi-dimensional, stationary and time-dependent problems as encountered in low-speed CFD applications; <ol style="list-style-type: none"> <li>(2.1) numerical schemes for temporal discretization;</li> <li>(2.2) numerical schemes for spatial discretization;</li> <li>(2.3) imposition boundary conditions</li> </ol> </li> <li>(3) turbulence modeling</li> <li>(4) wave equation</li> <li>(5) numerical methods for high-speed (compressible) flows</li> <li>(6) high performance computing</li> </ol>                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>After learning the content of the course the student will have the following capabilities:</p> <ol style="list-style-type: none"> <li>(1) Describe the two most popular methods in commercial CFD, finite differences and finite volumes</li> <li>(2) solve simple demonstrative problems in fluid flow and heat transfer by programming them in python or matlab, using finite differences and finite volumes</li> <li>(3) recognize the effects of numerical methods on the solution, such as numerical diffusion and numerical dispersion and to explain how to make these effects smaller</li> <li>(4) recognize numerical instability, to list several ways to avoid it and to analyze stability of simple methods analytically</li> <li>(5) solve fluid flow and heat transfer problems with the commercial CFD package Fluent or the open source CFD package OpenFOAM, which includes the following: set up a calculation, generate a simple grid, set appropriate boundary conditions, choose suitable discretisation, interpret and validate the results. Alternatively, in some cases, instead of using a commercial package, students can decide to write and validate their own code which they will use for calculating a fluid mechanics problem.</li> </ol> |   |
| <b>Education Method</b>                | Lectures, practical exercises (2x2 hours per week)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>  | <p>Course material:</p> <ol style="list-style-type: none"> <li>(1) Sheets/handouts</li> <li>(2) J.H. Ferziger and M. Peric, Computational methods for Fluid Dynamics, Springer Verlag.</li> <li>(3) P. Moin, Fundamentals of Engineering Numerical Analysis, Cambridge University Press, 2001.</li> </ol> <p>References from literature:</p> <ol style="list-style-type: none"> <li>(1) C. Hirsch, Numerical computation of internal and external flows, Volume I Fundamentals of numerical discretization, Volume II Computational methods for inviscid and viscous flows, Chichester, Wiley &amp; Sons, 1988, 1990</li> <li>(2) C.A.J. Fletcher, Computational techniques for Fluid Dynamics, Volume I Fundamental and general techniques, Volume II Specific techniques for different flow categories, Berlin, Springer, 2-nd ed. 1991.</li> <li>(3) H.K. Versteegh, W. Malalasekara, An introduction of computational fluid dynamics. The finite volume method. Second edition. Pearson Education.</li> </ol>                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | Written exam + assignment with report; written exam may be online type depending on circumstances                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>          |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| ME45160                                | Advanced Applied Thermodynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. B.P. Tighe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Dr. O. Moulτος                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Required for</b>                    | EFPT track                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>        | Thermodynamics at the level of a BSc in Mechanical Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <ul style="list-style-type: none"> <li>- Review of fundamentals of thermodynamics, including thermodynamic potentials, Maxwell relations, and the 1st and 2nd Law</li> <li>- Equilibrium in one-component systems: equations of state, chemical potential, fugacity, Clausius-Clapeyron, etc.</li> <li>- Mixtures: ideal gas mixtures, partial molar quantities, Gibbs-Duhem, vapor- liquid equilibrium, activity coefficient models, and osmosis</li> <li>- Model equations of state</li> <li>- First introduction to statistical mechanics</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                     | After this course, you will be able to calculate phase and material properties in thermodynamic equilibrium for multi-phase and/or multi-component mixtures under ideal and non-ideal process conditions. You will be able to use equations of state and activity coefficient models, and to calibrate and validate those models using thermodynamic data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | Lectures, optional homework, review session                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Computer Use</b>                    | homework assignments require scientific computing tools such as python, Matlab (or comparable) and basic programming methods such as for loops                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Books</b>                           | Holyst Poniewierski                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | Written final exam; open book                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Economics & Society 2024



| SEN1211                                | Agent-based Modelling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. C.N. van der Wal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr.ir. I. Nikolic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                      | Dr. O. Kammouh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | Dr. C.N. van der Wal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. I. Nikolic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Co-responsible for assignments</b>  | Dr. O. Kammouh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>This course has two focus points: 1. agent-based modelling and 2. complex adaptive systems theory for ABM.</p> <p>Firstly, you will develop agent-based modelling skills by creating an agent-based model in a small group. This course is not meant as an introduction to basic programming, or a thorough introduction to Python/Mesa, but will rather focus on understanding the concepts of agent-based modelling and how model design choices affect the model created and how it can be used. Additional reading materials are provided for self-study in getting up to speed in basic programming, Python, and Mesa, as well as refresher exercises in the practical sessions. There will be a weekly lecture focused on agent-based modelling theory. In addition, there will be a weekly practical session focused on further developing your agent-based modelling skills.</p> <p>Secondly, there will be weekly lectures focusing on complex adaptive systems theory for agent-based models. Different characteristics of these systems, theoretical concepts and their applications will be discussed. Further reading will be offered to broaden your view and deepen your understanding.</p> |   |
| <b>Study Goals</b>                     | <p>After following this course, the student can:</p> <ol style="list-style-type: none"> <li>1. understand, modify and create agent-based simulations in Python with the Mesa library.</li> <li>2. identify relevant properties of a system and determine whether the system can be classified as a complex adaptive system or not.</li> <li>3. apply the definitions of complex adaptive system properties, as listed in the course topics, to different types of socio-technical systems and agent-based models</li> <li>4. reflect on the traditional engineering systems thinking from the complex adaptive systems-perspective and understand the implications of changing the traditional perspective.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>This course consists of a series of lectures and a modeling project.</p> <p><b>Lectures</b><br/>Lectures introduce the theoretical concepts and their application for agent-based models and complex adaptive systems. All the lectures form the basis for the practical sessions, as well as the knowledge for the theoretical exam.</p> <p><b>Modelling project</b><br/>You will be creating an agent-based model in a small group. The project deliverable consists of the simulation code and a report documenting the problem statement, model design, performed experiments and data analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Computer Use</b>                    | The course involves developing and experimenting with simulations in Python with the Mesa library.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>The final grade of the course is based 50% on the written theory exam and 50% on the modelling project (simulation + report).</p> <p>Both parts must be completed with a passing grade (at least 5.75) in order to pass the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| SEN1531                                | Design of Integrated Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. F. Lombardi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Instructor</b>                      | Dr.ir. P.W. Heijnen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Responsible for assignments</b>     | Dr. F. Lombardi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Co-responsible for assignments</b>  | Dr.ir. E.J.L. Chappin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/x/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | The SEN Energy Track modules taught in Q1 and Q2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>Students are challenged to tackle an integrated energy system design problem in the form of a modelling project. Students are expected to come up with a creative (set of) design(s). Afterwards, they evaluate the expected performance of one or more of these designs as a future system or improvement to existing energy systems.</p> <p>The design challenge is complex and can only be met: 1) by taking into account dynamics on the short and long term, 2) by tapping into relevant knowledge on technological developments, 3) by using various quantitative analyses, 4) by considering various parts of the value chain, in terms of energy, mass, financial and information flows and 6) by relating and addressing the multi-actor and multi-level complexities.</p> <p>The quantitative analyses are to be executed with a set of quantitative tools, some used in earlier courses in the Energy Track, some developed by research institutes and companies, and others used in ongoing academic research at the faculty, made available for this course.</p> |   |
| <b>Study Goals</b>                     | <p>By the end of this course you should be able to:</p> <ul style="list-style-type: none"> <li>Apply the main steps needed for the design of integrated energy systems</li> <li>Discuss the design space of integrated energy systems, consisting of multiple carriers, multiple relevant geographical and temporal scales, and multiple sectors</li> <li>Conceptualize key processes that affect the performance of an integrated energy system that play out on short, medium and long term</li> <li>Select and use quantitative simulation models for designing integrated energy systems</li> <li>Reflect on aspects that are placed outside the design space, i.e. the relationship to food, water, built environment, etc.</li> </ul>                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | <p>The core task within this module is to carry out a group project in which an integrated energy system is designed and evaluated by using various models. Groups of students use various models and interact with model experts: teachers, researchers but also various companies are involved.</p> <p>There are lectures and visiting hours for debating modelling choices and details, but also for reflection and debate on modelling and simulation. Students do peer review and present their results for each other and for model experts. This enables academic reflection as well as direct interaction with the potential work field.</p>                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>A reader is provided with the approach applied and descriptions of all the models available.</p> <p>Various energy transition models and simulations are made available for use in the projects. Some were introduced in earlier courses, some are extracted from ongoing energy research at the TU Delft, and others come from various companies in the field that do their own energy system modelling and analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>An individual grade will be assigned on the basis of the following inputs:</p> <ul style="list-style-type: none"> <li>- The group project report that describes the integrated energy systems design including an assessment of its expected performance</li> <li>- An individual appendix with a personal reflection and description of contribution to the project</li> <li>- The contribution to peers, through peer review and discussions</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| TPM001B                                | Sociotechnology of Future Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5 |
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| <b>Module Manager</b>                  | Dr.ir. I. Bouwmans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr. A.F. Correljé                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Co-responsible for assignments</b>  | Dr. A.F. Correljé                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/x/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>        | <p>PLEASE NOTE: As of academic year 2025-2026 this course will be DISCONTINUED. The (identical) course SEN154a in Q4 will be open as elective.</p> <p>Students need to have sufficient prior knowledge in the fields of energy technology, economics and policy. They should be familiar with concepts such as intermittency, Carnot efficiency, BEVs, energy vs power, network externalities, sunk costs, fixed and variable costs, ROI and NPV.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>This module addresses the changing technological and societal characteristics of future energy systems.</p> <p>Contemporary energy systems are confronted with technical and socio-economic developments, such as the climate crisis, renewable technologies, ICT-based control innovations, energy scarcity and environmental concerns. These changes will affect operational roles and responsibilities, market structures and actors interests, and business models for energy services. Moreover, they may require new forms of governance to ensure reliable operations that meet socio-economic performance criteria. The course discusses the technical and socio-economic aspects of future energy systems, dealing with issues like the integration of new forms of distributive solar PV or large-scale off-shore wind energy into existing networks, various forms of sustainable energy sources, alternative energy carriers, storage methods, and the dynamics of transition processes.</p> |   |
| <b>Study Goals</b>                     | <p>At the end of this module the students know:</p> <ul style="list-style-type: none"> <li>- Which technical and societal developments can be expected in the energy sector.</li> <li>- How new energy technologies and societal developments may reshape operational roles and responsibilities, energy markets and business models, and governance arrangements.</li> <li>- How to compare and analyze future energy systems, develop plausible transition paths towards them, and estimate the potential of new energy technologies within various scenarios.</li> </ul> <p>At the end of this module the students can:</p> <ul style="list-style-type: none"> <li>- Formulate descriptions of the technical and societal characteristics of future energy systems.</li> <li>- Make a conceptual design of an energy system, given certain demands.</li> <li>- Formulate a balanced advice on issues in the energy sector.</li> </ul>                                                                    |   |
| <b>Education Method</b>                | Interactive sessions, self-study, online & class exercises, class discussion. Reading material: scientific papers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | Individual written exam (on the computer, on campus) consisting of essay questions. Any reference material on paper and/or in pdf format on memory stick is allowed; no internet access other than to the examination site is allowed. Sufficient participation in the online exercises (one exercise may be missed) is a prerequisite for taking the exam, but not for the resit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Targetgroup</b>                     | Master students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| TPM012B                               | Environmental Ethics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3 |
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| <b>Module Manager</b>                 | A.R. Gammon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Co-responsible for assignments</b> | Dr. J.B. van Grunsven                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week x/x/x/x</b>   | 0/0/0/2 + 2 tutorials (werkcollege) in week 4.4 & 4.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>               | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                    | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Expected prior knowledge</b>       | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                | <p>Environmental Ethics is an elective course that TU Delft students, or study abroad/exchange students, from any degree program are invited to take. No prior knowledge or experience with ethics or philosophy is assumed.</p> <p>This course introduces key ideas, arguments, and concepts from Environmental Ethics, the area of philosophy concerned with the nonhuman world. Through the weekly meetings, readings, and course assignments, we'll consider questions like:</p> <ul style="list-style-type: none"> <li>--- Do nonhuman entities (plants, animals, ecosystems, etc.) have moral value?</li> <li>--- Is there really such a thing as 'nature,' existing separately from humans?</li> <li>--- What is biodiversity, and how should we conserve it?</li> <li>--- Who benefits when environments are protected, and who is harmed?</li> <li>--- What makes climate change a moral issue, and what should we do to address it?</li> <li>--- Are we living in the Anthropocene, and if so, what does that mean about how we should live?</li> </ul> <p>We'll also think about what environmental ethics means for engineers specifically, asking questions like:</p> <ul style="list-style-type: none"> <li>--- How can thinking environmentally attune us to others in space and time, and what responsibilities do engineers have to do this?</li> <li>--- How do other traditions of thought (feminism, indigenous philosophies) consider human/nonhuman relations, and what might these, alongside environmental ethics, contribute to how engineers and engineering practices think about the nonhuman world?</li> </ul> <p>Students are encouraged to pursue their own personal and/or professional interests throughout the course. Students choose the topic of their final assignment.</p> <p>This course can be taken for 3ECTS, as described below, or for 5ECTS, with the addition of a course paper. (For the 5ECTS option, please see course TBM013B)</p> |   |
| <b>Study Goals</b>                    | <ol style="list-style-type: none"> <li>1. Describe key concepts in environmental ethics</li> <li>2. Apply environmental ethical concepts and ideas to new problems, topics, media</li> <li>3. Understand and analyze the social and ethical dimensions of climate change and related issues of mitigation, adaptation, and climate engineering</li> <li>4. Work in groups to develop, feedback, and present a small project (in tutorials)</li> <li>5. Critically reflect about your relation to the environment (as an engineer and person)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Method</b>               | <p>This course is held in person.</p> <p>The course meets weekly (weeks 4.1-4.7) with two required tutorials in weeks 4.4 &amp; 4.6.</p> <p>All meetings are a mix of lecture, discussion, and in-class activity. Most weeks reading or viewing is required ahead of the meeting.</p> <p>Attendance of the weekly meetings is strongly encouraged: quizzes are given most weeks, and participation is also graded as part of the course (see assessment for more information).</p> <p>all course materials will be provided via Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Prerequisites</b>                  | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                     | <p>Assessment:</p> <p>The 3ECTS course is assessed through several parts/assignments, described below. There is no exam.</p> <ul style="list-style-type: none"> <li>25% - weekly quizzes (in class, no option to retake missed quizzes. Best 5 out of 6, graded individually)</li> <li>25% - written responses (individual)</li> <li>25% - tutorial activity (designed and presented in small groups in course tutorials, graded per group)</li> <li>10% - stakeholder feedback</li> <li>15% - participation (engagement in class meetings and/or in Brightspace discussions, feedback on other groups' work in tutorial, graded individually)</li> </ul> <p>Final Grades:</p> <p>The final grade is calculated according to the weights indicated above. To pass the course, the final grade has to be 5.75 or higher. All partial grades have to be 5.0 or higher.</p> <p>There is no resit because there is no exam. However, partial grades are valid for all parts and can be carried from one year to the next if students do not pass or complete all parts of the course. This is only allowed for insufficient or incomplete grades, not to improve passing grades.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Hours</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| TPM023C                                | Cost-Benefit Analysis: Theory and Application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
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| <b>Module Manager</b>                  | Dr. E. Schröder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | N. Mouter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | R. Stöllinger                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Co-responsible for assignments</b>  | N. Mouter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>Cost-benefit analysis (CBA) has become a widely used (often mandatory) tool for public policy decision making on infrastructural projects or environmental and climate policies. Through CBA, the potential current and future gains and losses of proposed projects, programmes and policies are identified, and then converted into monetary units in order to make them comparable with alternatives. In essence, certain decision rules of CBA help determine whether the project or programme under consideration is desirable from a societal perspective.</p> <p>CBA comprises not only the private costs and benefits, which individual actors or entities (e.g. project, firm, industry) take into account in their cost calculations and decisions but also the wider social costs and benefits. Many of the social costs and benefits are of a non-financial nature. For instance, in the construction of a new highway, besides the financial aspects such as toll revenues or construction costs, it is imperative to also assess the non-financial benefits such as increased safety (lives saved, accidents avoided) and non-financial costs such as the environmental damage (noise, soil sealing, loss of natural habitat).</p> <p>Because CBA evaluates policy and/or project proposals from the point of view of the public interest, the market prices used in CBA are corrected for possible market distortions or failure (for example, unpriced environmental damage). Nevertheless, the application of CBA faces difficulties and limitations that need to be taken into account. What discount factors are to be applied to determine the present and future costs and benefits of a policy proposal? Why and how do the private benefits and costs of a policy proposal differ from the social benefits and costs? How, if at all, can environmental damage, or human health and life risks be expressed in monetary units? How can market prices be adjusted to account for external effects? How can alternative measures of welfare be used to identify the benefits and costs of a project?</p>                                                                                                                                                                                                                                                                                                                         |   |
| <b>Study Goals</b>                     | <p>At the end of the course, students are expected to acquire the ability to:</p> <ul style="list-style-type: none"> <li>- recognise contexts where CBA is a useful methodology to assess a project, along with its limitations.</li> <li>- understand the micro-economic foundations of CBA.</li> <li>- criticise and defend the CBA methodology from an ethical perspective.</li> <li>- recognise the different perspectives of CBA.</li> <li>- apply all the steps necessary to conduct a simple analytical CBA with given data.</li> <li>- interpret the results of a cost-benefit analysis.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                | Lectures and in-class exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Relations</b>                | This master-level course is part of the TPM Faculty's Economics & Finance specialization/elective package, but it can also be selected as a standalone elective course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | <p>There are two mandatory types of the assessment and a formative assessment:</p> <ol style="list-style-type: none"> <li>1. An individual essay (30%) mandatory</li> <li>2. An individual written exam (60%) mandatory</li> <li>3. Two mini assignments (group work) (10%) formative</li> </ol> <p>Rules for the individual essay: The essay is mandatory. The minimum grade of the essay required for passing the course is 5.8. There is one resit attempt. The maximum grade students can achieve in the second attempt for the essay is a 6.0. Note that the second attempt for the essay is only meant for students who do not pass and NOT intended for students who have passed nor new students who would like to enrol in the course after the first attempt. Further details about the individual essay will be provided later in Brightspace.</p> <p>Rules for the exam: The final exam is mandatory and takes the form of a written exam. The minimum grade of the exam required for passing the course is 5.8. There is one resit attempt.</p> <p>Rules for mini assignments: Each of the two mini assignments is worth 5% of the grade. The mini assignments consist of a set of questions to be solved and submitted by students in groups at the specified deadline. Submissions of group works after the deadline are not accepted. For the purpose of the mini assignments, students are assigned to groups of approximately 5 students. The mini assignments are intended as a formative assessment. For this reason there is NO resit attempt. There is no minimum requirement for the mini assignments either. (If a group decides not to hand in any of the assignments the maximum number of points they can achieve is 90%.)</p> <p>Rules for the final grade: The final grade required for passing the course is 5.8. The final grade is established as the weighted average of the individual assessments calculated as:<br/> <math>\text{GRADE}_{\text{essay}} * 0.3 + \text{GRADE}_{\text{exam}} * 0.6 + \text{GRADE}_{\text{miniassignment}} * 0.1</math>.</p> <p>Only the final grade will be entered in OSIRIS. Please register for the exam in OSIRIS on time (see: <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-examinations">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-examinations</a>).</p> |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| TPM404B                                | Technology Entrepreneurship and Internationalization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 5 |
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| <b>Module Manager</b>                  | H. Khodaei                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | J. Schmutzler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Co-responsible for assignments</b>  | Dr.ing. V.E. Scholten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <p>This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship.</p> <p>Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.</p> |   |
| <b>Study Goals</b>                     | <p>Upon finishing the course you will be able to:</p> <p>LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms.<br/> LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas.<br/> LO 3 Identify and analysis international opportunities across different markets.<br/> LO 4 Describe and apply the born-global start-up growth process.<br/> LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process.<br/> LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time.<br/> LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies.<br/> LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.</p>                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course.</p> <p>Book:<br/> Trott, P., Hartmann, D., Scholten, V., van der Duin, P., &amp; Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA</p> <p>Journal articles: Will be posted in Brightspace<br/> Slides presented in class: Will be posted in Brightspace</p>                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | <p>The assessment is as follows:</p> <ul style="list-style-type: none"> <li>- Team: Case analyses (30%) and a final report and presentation (40%)</li> <li>- Individual: theory quizzes (30%)</li> </ul> <p>All marks will be expressed on a scale of 1-10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

| TPM411A                                | Idea to Start-up IT & AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | E. Çelik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | Dr. A. Giga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | E. Çelik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Responsible for assignments</b>     | E. Çelik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Co-responsible for assignments</b>  | Dr. A. Giga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Expected prior knowledge</b>        | No prior knowledge is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Summary</b>                         | <p>In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity using emerging digital technologies. Applications can range from mobility, cyber security to predictive maintenance etc. You will test its validity, and build a viable business model a roadmap from an idea to an innovation-driven company. On this journey, you will be coached by lecturers and learn practical insights from guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | <p>You will focus on information and computing technologies (e.g AI, big data, blockchain) that are transforming traditional industries and societies at large.</p> <p>To analyse the opportunities for new startups we focus on entrepreneurial tools such as the opportunity canvas; value proposition development and customer empathy mapping, market segmentation and value chain analyses and the initial operations to develop a minimum viable product.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Study Goals</b>                     | <p>Develop a business Idea by identifying a relevant economic problem or unmet need that is addressed by digital solutions, and advanced information and computing technologies.</p> <p>Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment</p> <p>Analyse the competitive landscape and devise your products competitive advantage, given specific challenges of protecting IP in the digital world. Create a blueprint of your product.</p> <p>Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. Understand how these strategies adjust when trying to implement your solution in well-established systems of traditional industries.</p> <p>Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence.</p> <p>Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model.</p> <p>Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing</p> |   |
| <b>Education Method</b>                | <p>You will work in teams which are formed in the first sessions of the course. One aspect of a startup will be discussed each week and you will apply into your own projece immediately. You will present your work, provide peer-feedback as well and receive advise from coaches. The course ends with a jury pitch and a 4-pager of the business proposition.</p> <p>Team coaching will help you validate the idea and we provide individual coaching for personal entrepreneurial development.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Books</b>                           | <p>The primary textbook for this course is an organic handbook by Bill Aulet: Disciplined Entrepreneurship: 24 Steps to a Successful Startup. Wiley, 2023.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>The assessment is based on your project.</p> <p>It includes class assignments and participation (40%), final business model report (30%), online pitch presentation (15%), and team evaluation (15%). Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Enrolment / Application</b>         | <p>All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Remarks</b>                         | <p>For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Targetgroup</b>                     | Master students from all master programs can apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| TPM413A                                | Idea to Start-up Energy & Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                  | Dr. A. Giga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | Dr. A. Giga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | E. Çelik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Responsible for assignments</b>     | Dr. A. Giga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Co-responsible for assignments</b>  | E. Çelik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 1<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Summary</b>                         | <p>-----</p> <p>This course is NOT connected to the CleanTech Challenge.<br/>(A previous course Sustainable Business Study was linked to that competition, but this course is not in any way connected.)</p> <p>-----</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>A sustainable business idea is a product, a process or a service that enables sustainability, where sustainability is defined as the capacity for the biosphere and human civilization to co-exist indefinitely.<br/>The course stimulates and assists students in creating a sustainable business idea and assessing the feasibility and the market potential thereof by testing it against prospective customers and key players in the specific industry of application.<br/>Students work together in teams which will be self-formed or if needed formed by the instructor with the aim to create optimal diversity.<br/>Each team needs to come up with an original sustainable startup idea, or a sustainable business idea connected to pressing sustainability challenges of our day.<br/>Each sustainable business idea must be technology-based we do not accept apps and definitely no consulting of any kind.</p> <p>The course consists of two interconnected parts:<br/>1. Students attend lectures and workshops.<br/>2. Teams carry out a feasibility study for their sustainable business idea.</p> |   |
| <b>Study Goals</b>                     | <ul style="list-style-type: none"> <li>- Students learn how to assess the market potential of a sustainable product, process, or service.</li> <li>- Students gain basic knowledge on the commercialization of a sustainable business idea.</li> <li>- Students learn about different business models and gain insight in how to choose a market opportunity in the value chain.</li> <li>- Students gain practical knowledge about starting up a business.</li> <li>- Students experience real business practice by feedback from both lecturers, external advisers and experienced entrepreneurs in particular.</li> <li>- Students learn how to give a convincing business presentation.</li> <li>- Students learn to be critical about their own business case and those of their fellow students.</li> <li>- Students learn how to make a proper evaluation of the feasibility of their idea and business plan.</li> </ul>                                                                                                                                                                                          |   |
| <b>Education Method</b>                | <p>We offer a mix of lectures, case studies, and coaching sessions.<br/>The class is fast-paced and interactive.<br/>Many of the classes are in the form of coaching sessions and team exercises, therefore class attendance is mandatory.<br/>Lectures and online content will provide students with general guidelines for each business model milestone.<br/>Depending on availability, we will bring guest speakers, usually entrepreneurs, with experience in the energy and sustainability industry.<br/>Team coaching will help students to validate their ideas and develop their entrepreneurial skills.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | <p>Mandatory books:</p> <p>-----</p> <p>Aulet Disciplined Entrepreneurship: 24 Steps to a Successful Startup<br/>MacKay Sustainable Energy - Without the Hot Air</p> <p>Recommended books:</p> <p>-----</p> <p>Scott Berkun The Myths of Innovation<br/>Goldstein Planetary Improvement</p> <p>More literature and reading suggestions will be provided during the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Prerequisites</b>                   | A BSc is required. Admission to Master students only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | The grade is based on: final group report, final presentation, midterm presentation and short progress report, peer evaluation and active participation in the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Enrolment / Application</b>         | Enrol for the course on Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Tags</b>                            | Business<br>Economics<br>Energy<br>Project<br>Sustainability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Category</b>                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |



| TPM414A                         | Idea to Start-up Deep Tech                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Module Manager                  | Dr. A. Giga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Responsible for assignments     | Dr. A. Giga                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Co-responsible for assignments  | E. Çelik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| Contact Hours / Week<br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Education Period                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Start Education                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Exam Period                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Expected prior knowledge        | No prior knowledge is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Course Contents                 | <p>In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity, test its validity, and build a viable business model a roadmap from an idea to a sustainable innovation-driven company. On this journey, you will be helped by guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology venture, whether you want to build or join one. While the starting point is tackling the societal and market problems and unmet needs, the solution ideas must be based on substantial scientific advances and cutting-edge engineering that can be used to build competitive advantage in the market (via patent for example). Feel free to bring your own ideas and technologies, but you do not need one to enjoy this course. All you need is curiosity about building a business on solid technological foundations.</p> <p>Fast-paced and interactive, the class consists of mostly coaching sessions and team exercises, therefore class attendance is very important. Online content and short in-class lectures provide you with general guidelines for each business model milestone. There will be weekly assignments, building blocks to a full-fledged business model. Final deliverables include: a logbook of primary market research (yes, youll do interviews!), business model report, and a pitch presentation in front of a jury of startup professionals.</p> <p>You start the course with forming teams around a market problem or unmet need. Then, you gradually build a business model, mainly by answering questions such as: Who are my customers and what value do I create for them? How big is this market? How should my minimum viable product look like? How do I build a competitive advantage in the market (e.g. by patenting)? How will I market and deliver this product to the customer? Can this be a profitable business? You seek answers through customer, stakeholder, and expert interviews, by developing step-by-step interview protocols and executing them.</p> |   |
| Study Goals                     | <ul style="list-style-type: none"><li>- Develop a business idea based on a relevant economic problem or unmet need.</li><li>- Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment</li><li>- Develop a blueprint of your product and map its position in a competitive landscape.</li><li>- Identify relevant stakeholders in your startups ecosystem.</li><li>- Formulate a business model and a go-to-market strategy.</li><li>- Perform initial assessment of the profitability of the business idea.</li><li>- Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence.</li><li>- Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by modifying the business model.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Education Method                | <p>The course offers a mix of online lectures, case studies, and coaching sessions. Depending on availability, we will bring guest speakers with relevant industry and entrepreneurial experience.</p> <p>Team coaching with industry experts will help you validate the idea and we provide individual coaching for personal entrepreneurial development.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Assessment                      | <p>The grade is based on: final group report, final presentation, midterm presentation and short progress report, peer evaluation and active participation in the course. Also, to pass the course, participation and regular submission of class assignments is necessary. Active participation and peer evaluation are assessed for each student, the rest is evaluated on a group level.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Enrolment / Application         | <p>All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| Remarks                         | <p>For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check <a href="https://www.tudelft.nl/dce/education">https://www.tudelft.nl/dce/education</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| Elective                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| Tags                            | Business<br>Economics<br>Project<br>Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Targetgroup                     | Master students from all master programs can apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| Category                        | MSc level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Power Engineering 2024

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>EE4665</b>                         | <b>Uncertainty modelling and risk assessment in electrical power systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>4</b> |
| <b>Responsible Instructor</b>         | Dr. S.H. Tindemans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Instructor</b>                     | Dr.ir. J.L. Rueda Torres                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Co-responsible for assignments</b> | K. Bölät                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Contact Hours / Week</b>           | 0/0/4/0<br>x/x/x/x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Education Period</b>               | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Start Education</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Exam Period</b>                    | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Course Language</b>                | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>Expected prior knowledge</b>       | <p>Elementary knowledge of probability theory and statistics (discrete and continuous probability distributions, expectation/variance/standard deviation, quantiles, least squares regression).</p> <p>Basic programming skills (examples code samples will be provided in Python, but other languages may be used for hand-in exercises).</p> <p>Working knowledge of power system structure, components and actors.</p> <p>This course is aimed at MSc EE students in the EPE track and MSc SET students with the Power profile.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Course Contents</b>                | <p>This course considers the challenges faced by operators when operating and planning electrical power systems in the face of uncertainty balancing cost, reliability and environmental objectives. The course specifically focuses on quantitative modelling of uncertainty and performing quantitative risk assessments to support decision making.</p> <p>The course starts with a description of uncertainties that are important to power system operation and planning, and common metrics for system performance. Case studies in power system reliability and system capacity planning are introduced. In this context, various approaches to modelling uncertainties are introduced. Historical data is used to create models for component reliability. Students will learn how to compute system level reliability measures from bottom-up reliability models, using enumeration approaches and Monte Carlo sampling. Special attention is given to the estimation sample variation using the normal assumption and bootstrap approaches. Finally, the impact of risk metrics and decision criteria on decision making is discussed, thereby giving students insight into the whole chain from measurements to models to model-based decision making.</p>                                                                                          |          |
| <b>Study Goals</b>                    | <p>After completing this course, students will be able to:</p> <ol style="list-style-type: none"> <li>1) Calculate commonly used reliability measures and interpret their values in the context of Western European power systems.</li> <li>2) Construct and use Markov models for component reliability.</li> <li>3) Follow prescribed steps to construct reliability models from failure and repair data.</li> <li>4) Calculate the reliability of model systems on the basis of constituent component models.</li> <li>5) Implement Monte Carlo sampling and bootstrapping methods to estimate risk measures and the accuracy of such estimates.</li> <li>6) Compare model representations of uncertainty, and illustrate these in the context of power system reliability and planning.</li> <li>7) Evaluate different grid development options using quantitative risk assessment and discuss how a decision for one option depends on the choice of risk measure and decision criterion.</li> </ol>                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Education Method</b>               | Lectures (8x 2 hours); Computer labs (8x 2 hours); additional exercises for self study.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>Literature and Study Materials</b> | <ol style="list-style-type: none"> <li>1) Book, available as e-book through TU Delft (no additional cost) or physical copy: Bart W. Tuinema, José L. Rueda Torres, Alexandru I. Stefanov, Francisco M. Gonzalez-Longatt, and Mart A. M. van der Meijden, "Probabilistic Reliability Analysis of Power Systems," Springer Nature Switzerland AG 2020, May 2020. DOI: <a href="https://doi.org/10.1007/978-3-030-43498-4">https://doi.org/10.1007/978-3-030-43498-4</a>.</li> <li>2) Supplementary notes, videos, selected papers, reports and book chapters, available through Brightspace.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Assessment</b>                     | <p>Completion with 'pass' mark of a small group assignment and computer lab exercise (with basic report) is required. The final grade is determined by a written exam (80%) and computer exercise with accompanying report (20%). A minimum mark of 5.0 must be attained for each component to pass. Component marks are given on a scale 0-10, rounded to the nearest 0.1 increment; the final grade is rounded to the nearest 0.5 increment.</p> <p>One resit opportunity is provided for the written exam. One repair opportunity is offered for the pass/fail assignments (two weeks after the initial deadline, unless agreed otherwise). A repair opportunity will also be offered for the graded exercise if the initial mark was between 4.0 and 5.7 (out of 10), with a deadline 3 weeks after the initial grade is announced. A maximum mark of 6.0 out of 10 can be obtained for the exercise through this route. If a student passes only part of the course in this academic year, all parts need to be retaken.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |          |
| <b>Elective</b>                       | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Tags</b>                           | Energy<br>Modelling<br>Programming<br>Stochastics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Maximum number of participants</b> | 30.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |

| EE4C12                                 | Machine Learning for Electrical Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. P.P. Vergara Barrios                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Responsible Instructor</b>          | Dr. J.L. Cremer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Responsible Instructor</b>          | Dr. J.H.G. Dauwels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Instructor</b>                      | Dr.ing. R.K. Bishnoi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr. S.H. Tindemans                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | Basic programming skills in any programming language, elementary knowledge of probability theory and statistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Summary</b>                         | This course will teach students the fundamentals of Machine Learning and how these can be applied to electrical engineering applications. After this introduction to Machine learning covering the fundamentals, the students can make use of this course by either taking advanced ML courses or by applying their learnings to detailed track courses.                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <ol style="list-style-type: none"> <li>1. Introduction to data analytics</li> <li>2. Regression</li> <li>3. Classification</li> <li>4. Feature engineering, selection</li> <li>5. Neural Networks</li> <li>6. Develop an ML workflow</li> <li>7. Geometric Deep Learning</li> <li>8. Tree-based methods</li> <li>9. Reinforcement learning</li> <li>10. Hardware in ML</li> </ol>                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>After this course, students will be able to</p> <ol style="list-style-type: none"> <li>1. Draw conclusions by analysing data in the context of electrical engineering</li> <li>2. Compare machine learning concepts</li> <li>3. Use scikit-learn package in Python</li> <li>4. Apply training strategies for machine learning models to electrical engineering problems</li> <li>5. Design a workflow on an electrical engineering example</li> </ol>                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Interactive lectures, intuition-based,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Relations</b>                | This course is mutual exclusive with the course SET3125. Students can not take this course if they have taken the course SET3125.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Literature and Study Materials</b>  | Machine Learning Refined, Foundations, Algorithms, and Applications by Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, Cambridge University Press; 2nd edition; 2020; ISBN: 9781108480727.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>Written exam (70%), project (20%), homework (10%). In case of insufficient results for the homework or project, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is to resubmit the individual project or homework within 10 days after announcing the homework or project grade.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |

| ET4103                                 | High Voltage Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.ir. P.T.M. Vaessen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Responsible Instructor</b>          | Dr. M. Ghaffarian Niasar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Practical Coordinator</b>           | I.M. Splinter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | This course presents the basic principles of High Voltage technology and practical application of this knowledge in the power industry. Proper design of High Voltage components requires knowledge and insight of electric fields and breakdown mechanisms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | By the end of the course students will be able to: <ul style="list-style-type: none"> <li>- Demonstrate insight in the relation between charge, electric field and voltage</li> <li>- Identify and calculate capacitive voltage distributions</li> <li>- Calculate and analyze electric fields using analytical and numerical methods</li> <li>- Describe the breakdown mechanisms in different insulation materials and the phenomena leading to this</li> <li>- Explain methods of field grading and field control</li> <li>- Describe impulse breakdown phenomena (for example lightning)</li> <li>- Evaluate the effect of combinations of materials and interfaces on electric fields</li> <li>- Remember maximum permissible field strengths in different materials and configurations</li> </ul> |   |
| <b>Education Method</b>                | Lectures, demonstrations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | Kreuger, F.H., Industrial High Voltage. Vol.1, 2, 3, ISBN 90-6275-561-5 (available for purchase at the course), hand-outs (available at the course)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Prerequisites</b>                   | Electromagnetics (EE3P11) or equivalent courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                      | Written exam (70%) and project (30%)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
|                                        | In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|                                        | Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| OE44076                                | Offshore Renewable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 6 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. A. Jarquin Laguna                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Ir. A.C.M. van der Stap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/6/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | IMPORTANT: as of the academic year 2024/2025 this course has been replaced by OE44170 - 'Offshore Renewables Technologies' (5 EC points)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Study Goals</b>                     | This course will introduce the students to the technical aspects and detailed understanding of ocean energy technologies (including floating wind, wave, and tidal current). The course combines lectures on offshore renewable energy technologies, CO2 and energy storage. The students will participate in a realistic project teamwork with young industry coaches and working on actual cases with stakeholder and risk management assessment, modelling and energy extraction analysis from the offshore environment, including an overview of the project development of safe and reliable offshore energy technologies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
|                                        | by successfully completing the course you should be able to: <ul style="list-style-type: none"> <li>Explain and describe the physical principles of energy conversion for wave energy and tidal stream .</li> <li>Compare different marine renewable energy sources based on calculations of their energy content. A few examples of such sources include offshore wind, ocean waves, tidal currents</li> <li>Identify and evaluate the design considerations and technical challenges of offshore wind, wave and tidal energy technologies.</li> <li>Generate future-proof solutions for offshore energy projects with a focus on minimization of the CO2 footprint.</li> <li>Define technical, political, contractual, economic and operational constraints imposed on industrial projects throughout its entire life cycle.</li> <li>Explain the crucial importance of uncertainties and economics in the decision process in the course of developing a project.</li> <li>Apply basic project analysis and management techniques.</li> <li>Accommodate conflicting requirements in an offshore design environment.</li> </ul>                                                                                                                        |   |
| <b>Education Method</b>                | Lectures and project teamwork with industry coaches.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | Grading based on a group assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Enrolment / Application</b>         | Registration:<br>To participate in this course, registration via MyTUDelft is not needed.<br>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a> .<br><br>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a> .<br><br>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a> . |   |
| <b>Department</b>                      | ME Department Maritime & Transport Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Solar Energy 2024

| EE4585                                 | Semiconductor Device Physics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 5 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R.A.C.M.M. van Swaaij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0 lectures; 0/2/0/0 instructions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Expected prior knowledge</b>        | Some background in solid-state physics is helpful, for instance the lecture courses on Solid-State Physics (EE4705).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>In this lecture first the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion is discussed. This knowledge is applied to p-n junctions. The current through this junction can be described well by the drift-diffusion model. For a metal-semiconductor junction the model describing the current-voltage relation ship is based the thermionic emission.</p> <p>The p-n junction is the base for the bipolar transistor. In the bipolar junction transistor (BJT) the reverse current of one p-n junction (the base-collector junction) is controlled by the injection of charge carriers from a forward biased p-n junction (the emitter-base junction). Some characteristics, like the Early effect and current crowding, are discussed.</p> <p>Also another transistor, the MOSFET, is discussed. This device is based on a MOS capacitor to which we will pay attention first. In this transistor the current is led via the source contact through a conductive channel in a semiconductor to the drain contact. The conductivity of the channel can be controlled by changing the bias on the gate contact. Characteristic for this device is that the current-voltage relationship is more linear in contrast to the bipolar transistor where it is exponential.</p> <p>Following the discussion on ideal MOSFETs, we will then focus on non-ideal characteristics of MOSFETs, like channel-length modulation and short-channel effects. Additionally, we will also pay attention to threshold voltage modification by varying the dopant concentration and MOS scaling.</p> <p>Finally, we will pay attention to power devices, like power MOSFETs, thyristors, and insulated-gate bipolar transistors (IGBT).</p> |   |
| <b>Study Goals</b>                     | <p>This course aims at a thorough understanding of the physics of semiconductor devices. Students are expected to be able to explain the operation of these devices based on their design and on the material properties. In addition, they should be able to derive mathematically the most important device characteristics, based on the physics of the device.</p> <p>The above description leads to the following learning objectives (LO) for this course:</p> <p>LO1: Describe processes in the semiconductor devices (p-n junction, BJT, metal-semiconductor junction, MOSFET, and power devices).</p> <p>LO2: Derive the current-voltage characteristics of these semiconductor devices, and understand and analyze these characteristics in relation to the device design.</p> <p>LO3: Being able to draw band diagrams, and understand the relations between these band diagrams on one hand and charge-carrier concentrations and device characteristics on the other hand.</p> <p>LO4: Understand and be able to construct the equivalent diagram of the above-mentioned semiconductor devices.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Lectures and instructions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Literature and Study Materials</b>  | Donald A. Neamen: Semiconductor Physics and Devices, basic principles, McGraw Hill, 2012, 4th edition, ISBN 978-007-108902-9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                      | <p>The assessment is based on a written examination (100%).</p> <p>Exam admission conditions:<br/>Access to the examination is only possible if sufficient instruction questions have been prepared, as will be explained below. This procedure, called student-led tutoring, works as follows:</p> <p>*Before each of the 7 instruction sessions students are asked to indicate which of the 4 questions they will prepare by putting a tick in relevant section of Brightspace environment for this course.</p> <p>* At the instruction sessions students will be randomly picked to explain one of the questions they ticked (and consciously prepared) to the other students and the moderator.</p> <p>* At the end of each session, all exercises that have been ticked by at least one student have been discussed and the correct answers have been presented.</p> <p>* Students will need to have prepared at least 16 out of 28 questions (i.e., 57% of all questions; this is equivalent to 4 full instruction sessions) in order to get access to the written exam and, if needed, the resit exam.</p> <p>Repair option:<br/>If students have not managed to prepare at least 16 questions, a repair option is available in Week 8 of the quarter in which an extra instruction session with 4 questions will be organized. The latter implies that students who have ticked less than 12 questions (i.e., less than 43% of all questions) will not be admitted to the exams.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                     |   |

| EE4670                                 | PV Materials Processing and Characterization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. A.W. Weeber                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. L. Mazzarella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 1<br>2<br>Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>The course is a follow-up of PV Technologies. Details of processing and characterization techniques of PV absorber materials and the solar cells functional coating will be discussed. Examples of lecture topics are advanced light management structures, novel surface passivating layers and contacting schemes. Background of deposition technologies such as Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD) and Atomic Layer Deposition (ALD) will be given. Furthermore, working principles of special coatings and its functionality when applied to different solar cell concepts, including tandem devices, will be presented.</p> <p>This course is an elective one following the solar profile. PV Basics and PV Technologies should have been concluded successfully.</p>                              |   |
| <b>Study Goals</b>                     | <p>After this course students are able to understand processing mechanisms, material properties, and functionality and working principle of the novel functional coating applied to different solar cell concepts. Students are expected to be able to explain these working principles and have a solid background on different deposition technologies and characterization techniques, and will be able to describe the effect on the solar cells output. Students will be able to analyse and evaluate recent technology developments and the background of it, and summarize them during an oral presentation for the entire group.</p> <p>As we can see, these learning objectives are specific about the content, and they are phrased in terms of observable behaviour: understand, explain, evaluate, present and describe.</p> |   |
| <b>Education Method</b>                | <p>This course consists of eight 2-hour lectures. In the same week a 2 hour work session / instruction will be organized for a deeper understanding of the technologies discussed in the lectures. During the whole course the students will work on assignments which need to be submitted as reports. All assignments/reports should be assessed as sufficiently good (no specific mark). In week 9 (final schedule will be given at the beginning of the course) you will present results of a literature study on a special topic covered by the course for the entire group.</p>                                                                                                                                                                                                                                                    |   |
| <b>Literature and Study Materials</b>  | <p>Book (not obligatory, expensive): Photovoltaic Solar Energy: from fundamentals to applications by Angèle Reinders et al.</p> <p>Material from online course: PVCDROM, PV-manufacturing.org</p> <p>Slides during lectures</p> <p>Articles (referred to in lectures)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | <p>The final grade consists of the following parts:<br/> presentation (team work) (50% of final grade);<br/> written exam (individual) (50%);<br/> Both marks will be expressed on a scale of 1 to 10 and should be 5.80 or higher.<br/> Reports on the assignments shall be completed as well, and assessed as sufficiently good (no specific mark).</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                   |   |
| <b>maximum aantal deelnemers</b>       | <p>Expected number of student: Maximum of 30. Use Brightspace and inform our secretary Mitch Bloem (m.a.bloem@tudelft.nl) for attending the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |



| EE4680                                 | Photovoltaic Modelling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. R. Santbergen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Instructor</b>                      | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Responsible for assignments</b>     | Dipl.-Ing. Y. Blom                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Co-responsible for assignments</b>  | F.S. Saitta                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>Modelling is an important tool for gaining deeper insight in the working principles and loss mechanisms of solar cells and photovoltaic systems. Researchers use this insight to develop more efficient solar cells and to maximize the energy yield of photovoltaic systems.</p> <p>In this course we will cover the entire photovoltaic modelling chain: modelling of photovoltaic materials, devices (solar cells), modules and systems. The focus will be on understanding the working principles of these models, and using them to analyse the effects of design parameters on solar cell and photovoltaic system performance.</p> <p>The online course material consists of two parts, each consisting of several phases:</p> <ol style="list-style-type: none"> <li>1) PV materials and device modelling <ol style="list-style-type: none"> <li>1.1) PV materials and device characterization</li> <li>1.2) optical modelling of PV devices</li> <li>1.3) electrical modelling of PV devices</li> </ol> </li> <li>2) PV systems modelling <ol style="list-style-type: none"> <li>2.1) modelling of irradiance</li> <li>2.2) modelling of energy yield</li> </ol> </li> </ol> <p>In addition there will be software demos, given by expert users, allowing you to familiarize yourself with the advanced modelling software commonly used by PV researchers.</p> <p>The prior knowledge from the courses PV basics and PV systems is required. This course PV modelling will prepare you for an MSc thesis project that involves modelling of solar cells or photovoltaic systems, as well as a future career in photovoltaics research.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Study Goals</b>                     | <p>The online course material consists of two parts: 1) PV materials and device modelling, 2) PV systems modelling. Each consist of several phases:</p> <p>At the end of phase 1.1 (characterization of PV materials and devices) you should:</p> <ul style="list-style-type: none"> <li>Be familiar with the main techniques used in the characterization of PV materials and devices</li> <li>Understand the importance of a correct characterization for the modelling of PV</li> <li>Be able to extract useful and important information (trends and properties) from the characterization of both PV materials and devices</li> </ul> <p>At the end of phase 1.2 (optical modelling of PV devices) you will;</p> <ul style="list-style-type: none"> <li>have a complete overview of the modelling approaches commonly used for photovoltaic optical modelling, and which are their strength and limitations</li> <li>get a thorough understanding of different optical modelling methods and which structures they are best suited to model</li> <li>improve and practice your skills on optical modelling, by carrying out activities with our own GenPro4 software</li> </ul> <p>At the end of phase 1.3 (electrical modelling of PV devices) you should be able to:</p> <ul style="list-style-type: none"> <li>have a complete overview of the modelling approaches commonly for photovoltaics, both optical and electrical</li> <li>get a thorough understanding of different electrical modelling methods</li> <li>improve and practice your skills on optical and electrical modelling, by carrying out activities with our own advanced semiconductor analysis (ASA) software.</li> </ul> <p>At the end of phase 2.1 (modelling of irradiance) you should be able to:</p> <ul style="list-style-type: none"> <li>Gather and analyse meteorological data, including the utilization of geographic information systems,</li> <li>Obtain information on horizon and shading at a specific location, and derive the relevant location parameters,</li> <li>Combine meteorological data and location properties to calculate the irradiance on the angle of incidence with different levels of detail, and</li> <li>Optimize the orientation of a module to maximize the irradiance on the angle of incidence.</li> </ul> <p>At the end of phase 2.2 (modelling of the energy yield) you should be able to:</p> <ul style="list-style-type: none"> <li>Model the operating temperature of modules,</li> <li>Obtain information from module datasheet,</li> <li>Combine module properties with temperature and irradiance modelling to calculate DC power (and yield),</li> <li>Model the performance of inverters (DC to AC) and batteries,</li> <li>Analyse load profiles, and use them to size a PV system (module and batteries), and</li> <li>Calculate the AC power and energy yield of an entire system, and analyse its performance from an economic point of view.</li> </ul> <p>After the software demos you will be able to explain the input, working principles and output of the following commonly used software packages:</p> <ul style="list-style-type: none"> <li>Complete EASE (extracting optical constants from spectroscopic ellipsometry measurement of PV materials)</li> <li>HFSS (rigorous solving of Maxwell equation for EM-wave (light) propagation in PV devices)</li> <li>Sentaurus (modelling charge transport in PV devices by solving semiconductor equations)</li> <li>Radiance (PV module plane of array irradiance calculation using backward ray-tracing)</li> <li>PV Syst (PV system energy yield calculation)</li> </ul> |   |
| <b>Education Method</b>                | <p>This course consists of recorded video lectures, interactive lectures and instructions. The video lectures provide you with the basic information. The interactive lectures give you the opportunity to ask questions to the teacher and discuss selected topics in depth. In addition you will work on modeling assignments. During the instruction sessions the modeling assignments will be explained and feedback will be given on your (preliminary) results.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | <p>Every week you make and submit modeling assignments, which are graded. The weighted average counts for 70% of the final grade. In addition, every week you write a short report about the software demo, which is graded as well. The average of this counts for the remaining 30% of the final grade.</p> <p>In case of an insufficient final result, or in case you missed a week (e.g. due to illness), you will be offered a repair option, consisting of a resubmission of the assignments for one of the weeks.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

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|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Remarks</b> | <p>The on-campus Q&amp;A session is not mandatory. Because the course material (video lectures, live demos) is given online, this course can also be taken by students who cannot physically come to the campus.</p> <p>Because the assessment is done through weekly assignments, without an exam, the final grade (and credits) can be obtained already before the exam period (end of October). This can be relevant for those who have not yet obtained sufficient credits to start their graduation project.</p> |
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| ET4379                                 | Photovoltaic Lab Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. O. Isabella                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Responsible Instructor</b>          | Dr.ir. R. Vismara                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0 Q1=Q5! Also possible online in Q6, Q7, Q8!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | <p>The aim is to offer you an advanced practical course on solar cells and photovoltaic systems. The following areas of practical training will be part of the course:</p> <ol style="list-style-type: none"> <li>(1) Characteristics of artificial and natural light</li> <li>(2) Characterization of the performance of lab-scale solar cells</li> <li>(3) Behavior of photovoltaic modules in different temperature and shading conditions</li> <li>(4) Charge and discharge properties of batteries</li> <li>(5) Performance of power electronic components (charge controllers, inverters, maximum power point trackers)</li> <li>(6) Operation of entire photovoltaic systems</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>By the end of the course, you should be able to:</p> <ol style="list-style-type: none"> <li>(1) Experiment with photovoltaic systems and all of their components under lab and real-life conditions.</li> <li>(2) Describe the working principle behind each component of a photovoltaic system.</li> <li>(3) Analyze the operation of all the elements of photovoltaic systems and of the system as a whole.</li> <li>(4) Compare the performance of different types of components of photovoltaic systems.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | Laboratory sessions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | Manuals and templates are available on BrightSpace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Prerequisites</b>                   | This course is meant for students who have already passed at least the PV Basics course (ET4376) and preferably also PV Technologies (ET4377) and PV Systems (ET4378).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>The assessment is based on reporting results from the individual tasks using an Excel template provided on Brightspace. Two-person teams will be formed at the plenary session, the attendance of which is compulsory. The final grade is per team. The final grade is the average of the six technical reports. A grade <math>\geq 5.0</math> in all six technical reports is required to complete the course successfully. The final grade is per team and is calculated as the average of the six technical reports. In case of failure of a technical report, you can use buffer sessions on weeks 1.7 and 1.8 to repeat the experiments.</p> <p>---</p> <p>A virtual version of this course is made available for students to attend in Q2, Q3, and Q4. Each of the six tasks of the on-campus course will run in the internet browser of your choice in the form of a web application accessible via Brightspace. The assessment of the online version is entirely automatic, using numerical exercises and multiple-choice questions. A grade <math>\geq 5.0</math> in all six tasks is necessary to pass the course. The final grade is the average of the six grades. A maximum of one retry session is allowed. Attending one of these instances of the course, no teams will be formed, and the final grade is individual.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p> |   |
| <b>Enrolment / Application</b>         | <p>The on-campus course can accommodate up to 24 students. Anyone interested in attending this course must apply as soon as possible.</p> <p>In case of applications exceeding the capacity, a selection process will take place and, before the beginning of the instance in Q5, the line-up of students will be announced.</p> <p>Student(s) who failed on-campus in Q5 can re-take the virtual version of the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Remarks</b>                         | <p>This practical course will be given on campus only in Q1.</p> <p>Conversely, a virtual version will be available for students to attend in Q2, Q3, and Q4. The enrolment is open on Brightspace from Q1.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

| ME46065                                | Thin Film Materials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J.G. Buijnsters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Summary</b>                         | <p>This is a mixed lectures and project-based course. The course consists of at least 8 plenary lectures and a group project:</p> <ul style="list-style-type: none"> <li>- The lectures will cover the basics of thin film technology: deposition techniques, vacuum technology, film growth, microstructure and application domains. The lectures have the purpose to provide the necessary background and technological information to fuel the group projects.</li> <li>- The project consists of carrying out in a group (typically ~3 students) a literature research on a specific thin-film materials topic, which can be chosen from a shortlist of topics or is self-chosen. The students will present their joint group research in oral and written form.</li> </ul> <p>Plenary progress presentations will be given to the class. By the end of the course, each group will hand in a final paper and will give a final presentation to conclude their work.</p>                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | <p>The course consists of 8 plenary lectures and a student project.</p> <p>The topics of the lectures are</p> <ol style="list-style-type: none"> <li>1. Introduction, liquid deposition techniques</li> <li>2. Physical vapor deposition (evaporation &amp; sputtering)</li> <li>3. Chemical vapor deposition</li> <li>4. Atomic layer deposition</li> <li>5. Plasma surface treatment</li> <li>6. Nucleation &amp; Growth modes</li> <li>7. CVD diamond</li> </ol> <p>The project consists in carrying out in a group of few students a literature research on a specific thin film topic, chosen out of the following:</p> <ol style="list-style-type: none"> <li>1. hard coatings</li> <li>2. self-lubricating coatings</li> <li>3. optical coatings</li> <li>4. decorative coatings</li> <li>5. thin films in micro/nano devices (MEMS/NEMS)</li> <li>6. thin-film based batteries</li> <li>7. thin-film photovoltaics</li> <li>8. coatings for implants</li> </ol> <p>The students can also propose a topic of their own interest.</p> <p>The students will prepare a literature review paper on their topic and will present their progress in several plenary presentations. The class will challenge them with critical questions. At the end of the course a final presentation will be given by all groups.</p> |   |
| <b>Study Goals</b>                     | <p>At the end of the course, the student is able to:</p> <ol style="list-style-type: none"> <li>1. describe the various liquid and vacuum deposition technologies</li> <li>2. discuss the pros and cons of each of these techniques</li> <li>3. identify deposition-microstructure-properties relationships for a given deposition method</li> <li>4. explain the main mechanisms of thin film growth</li> <li>5. find, extract and order relevant information from open literature</li> <li>6. explore new trends in thin films and their applications</li> <li>7. professionally present information in written and oral form</li> <li>8. critically assess the information found</li> <li>9. critically assess the information presented by the other groups</li> <li>10. cooperate with groupmates such that everybody is responsible for the content.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | <p>Lectures and assignments (literature study in group, presentations, discussions, paper, and peer review). Lectures and presentations will be given on TU Delft campus.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature and Study Materials</b>  | <p>Course material will be shared by the teacher and posted on Brightspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | <p>The grade will be composed as follows:</p> <ol style="list-style-type: none"> <li>1. Final paper 50%</li> <li>2. Final presentation 30%</li> <li>3. Mid-term presentations 20%</li> </ol> <p>A facultative, individual multiple-choice quiz is held towards the end of the lecture series. In addition, a peer evaluation form will be used to assess whether all team members have equally contributed to the work.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                            |   |
| <b>Department</b>                      | ME Department Precision & Microsystems Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Energy Storage 2024

| AP3311                                 | Neutrons, X-Rays and Positrons for Studying Microscopic Structures and Dynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 6 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. W.G. Bouwman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                      | Dr.ir. N.H. van Dijk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr. H. Schut                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior knowledge</b>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | <p>The microscopic structure and dynamics of condensed matter in physics, chemistry and biology is obtained by a wide variety of scattering techniques. Structures are determined on length scales ranging from 10<sup>-11</sup> to 10<sup>-5</sup> m, dynamics on time scales from 10<sup>-14</sup> to 10<sup>-6</sup> s. In this course we discuss the different neutron, X-ray and positron techniques, illustrated by a broad range of scientific and technological applications, for which recent scientific articles will be studied.</p> <p>After a general introduction an overview is given of the scattering theory, both for neutrons and X-rays. Then the different techniques will be discussed in more detail. Diffraction (either by X-rays or neutrons) yields information about the structure of single crystals, powders, and liquids. Small-angle scattering (SAXS and SANS) reveals the shape and arrangement of colloidal systems, micelles or polymers in a solvent. Reflectometry is used to find the composition of layered structures like proteins adsorbed at surfaces or magnetic layers in recording materials. Polarized neutrons are used in two different ways. First, for determining magnetic structures and dynamics. Second, as a special technique (spin-echo) to study the larger length scales and time scales (polymer dynamics). Inelastic scattering measures phonons in crystals, diffusion in liquids and the vibrational spectrum of molecules. Neutron sources (fission and spallation), X-ray sources (from X-ray tube, via synchrotron to XFEL) and instrumentation will be an integral part of this course.</p> <p>An introduction to positron annihilation (PA) will be given. Positron annihilation lifetime, Doppler Broadening techniques are particularly suited for the detection and characterisation of (open volume) defects such as vacancies, vacancy clusters and voids in e.g. metals, alloys, semiconductors and polymers. The main application of the Angular Correlation of Annihilation Radiation (ACAR) technique is the study of the electronic structure of solids, and that of defects, precipitates and nanocrystals.</p> <p>A visit to the neutron and positron experimental facilities at the Reactor Institute Delft will be part of the course.</p> |   |
| <b>Study Goals</b>                     | <p>Being able to explain the theoretical background of neutron and X-ray scattering and positron annihilation techniques;<br/> Being able to talk about the applications of these techniques;<br/> Being able to perform basic calculations and interpretations of the techniques;<br/> Being able to read and discuss recent articles on the application of the techniques;<br/> Being able to talk in depth about one of the topics to be chosen by the candidate in combination with a recent research article.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | <p>The course consists of 14 times 2 hours contact on the following chapters:<br/> I. Theoretical background neutron and X-ray scattering<br/> II. Neutron experimental techniques<br/> III. X-ray experimental techniques<br/> IV. Diffraction from crystalline materials, theoretical, experimental, application<br/> V. Diffraction from amorphous materials and liquids, theoretical, experimental, application<br/> VI. Small-angle scattering, theoretical, experimental, application<br/> VII. Reflectometry, theoretical, experimental, application<br/> VIII. Polarized neutrons, theoretical, experimental, application<br/> IX. Imaging and tomography with neutrons and X-rays, theoretical, experimental, application<br/> X. Positron annihilation, theoretical, experimental, application</p> <p>Each 2-hours-lecture is accompanied by a set of assignments from the website <a href="https://e-learning.pan-training.eu/">https://e-learning.pan-training.eu/</a>. This has to be completed before the next class. In the class question about the previous lecture will be discussed as well as recent scientific articles. In turn the students have to give short presentations about the assignments and about the scientific articles.</p> <p>Those students who have attended at least 12 of the 14 lectures and sent in at least 11 of the 13 assignments are allowed to enter the oral exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>  | <p>Lecture notes. The accompanying material with the website <a href="https://e-learning.pan-training.eu/">https://e-learning.pan-training.eu/</a>.<br/> For mathematical background we recommend the book D.S. Sivia, Elementary Scattering Theory for X-ray and Neutron Users, Oxford University Press, ISBN 978-0-19-922868-3.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>Having done at least 11 of the 13 times homework is a requirement for taking the oral exam. The oral exam of 45 minutes determines the final grade. The student proposes an article or scientific topic connected to the course for the first 15 minutes of the exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

| CH3513                                  | Electrochemistry for Renewable Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | D.A. Vermaas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                       | Dr. A.J. Houtepen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/6/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Expected prior knowledge</b>         | Thermodynamics (e.g. 4051CHTHEY) and standard chemistry background (e.g., BSc education MST or similar)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                  | <p>Electrochemistry provides an important platform to convert (renewable) electricity into valuable products, chemicals, fuels and coatings. This course provides an overview of several electrochemical processes of societal importance, linking fundamental understanding with applied utilization. The scope of the course will cover processes from the nanoscale to the meter-scale, and methods to understand fundamental phenomena in practical systems.</p> <p>This course will include:</p> <ul style="list-style-type: none"> <li>- Electrochemical potentials and definitions (electromotive force, formal potential, Donnan potential)</li> <li>- Types of electrochemical reactions (reversibility, multi-step reactions)</li> <li>- Electrochemistry at liquid-solid, liquid-liquid, solid-solid and gas-solid interfaces</li> <li>- Electrode reaction kinetics in steady and unsteady state (Cottrell equation)</li> <li>- Microscopic theories of charge transfer and (Marcus model, energy states, tunneling)</li> <li>- Ion transport through diffusion and migration</li> <li>- Electrochemical characterization techniques (cyclic voltammetry, chronopotentiometry, impedance, etc.)</li> <li>- Electric double layers (Helmholtz model, Gouy-Chapman model)</li> <li>- Examples of water electrolysis, CO<sub>2</sub> reduction</li> <li>- Industrial electrochemical systems (large scale electrolyzers, batteries, etc.)</li> <li>- Electrochemical equipment (potentiostats, reference electrodes, electrodes, electrolytes, membranes)</li> <li>- Experimental setup (including practicum)</li> </ul> |   |
| <b>Study Goals</b>                      | <p>After this course, the student is able to:</p> <ul style="list-style-type: none"> <li>- Explain electrochemical responses in complex electrochemical cells (steady and non-steady state, and mono- or multi-ionic systems) to peers;</li> <li>- Demonstrate how theories for electrochemistry thermodynamics, kinetics (Marcus model, Tafel model) and electrolyte processes are applied to basic redox systems</li> <li>- Break down limiting factors (overpotentials or limiting currents) in given practical electrochemical conversions</li> <li>- Set up and perform suitable electrochemical tests for characterizing redox couples in practical electrochemical conversions</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Method</b>                 | <p>This course consists of:</p> <p>Instruction videos,<br/>Lectures and exercise sessions (combined),<br/>Practicum,<br/>Guest lectures</p> <p>Please be aware that attendance at the practicum is mandatory. The practicum will be scheduled 2x a timeslot of 2 hours in March.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>   | <p>Bard &amp; Faulkner, Electrochemical methods fundamentals and applications, 2nd edition</p> <p>Please stick to the 2nd edition of the book, and do not buy the 3rd edition! The 2nd edition is less expensive and more useful for the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                       | <p>Grading is based on:</p> <ul style="list-style-type: none"> <li>- Homework assignments (0%, counting for 0.2 point bonus)</li> <li>- Practicum report (30%)</li> <li>- Exam at the end of the course (70%)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Permitted Materials during Tests</b> | <p>Calculator (no graphical)</p> <p>Formula sheet</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Continuing Courses</b>               | <p>Possible follow-up courses are:</p> <p>CH3222: Energy Storage in Batteries (Q3)</p> <p>ME45203: Electrolyzers, Fuel Cells and Batteries (analytical modelling) (Q4)</p> <p>MSc thesis</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

| CH3531                                  | Functional Ceramics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr. E.M. Kelder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                      | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>         | BSc: general and inorganic chemistry, basic quantum chemistry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | Ceramic materials show a wide variety in chemical, optical and electrical properties. They are used in many different applications, ranging from catalysts, fuel cells, batteries and sensors to solar cells, photo-catalysts and luminescent nanoparticles. This course will treat the basic properties of functional ceramic materials. The following topics will be included: - crystal structures - bonding in inorganic solids - basic electronic structure and properties of semiconductors - defect chemistry including point defects, non-stoichiometry and doping - ionic and electronic charge transport - semiconductor junctions and space charges. |   |
| <b>Study Goals</b>                      | After this course, the student is able to:<br>- Correlate crystal structure and band structure to optical and electrical properties of ceramic materials<br>- Describe the operating principles of simple devices based on semiconductor junctions (solar cells, diodes, photo-electrodes) and ionic conduction (Nernst sensors, (Li-ion)-batteries, solid oxide fuel cells (SOFCs), and catalysts)<br>- Identify and predict relationships between point defects, dopants, space charges, and ionic/electronic conduction in both semiconductors and ionic conductors, and perform simple calculations                                                         |   |
| <b>Education Method</b>                 | Lectures, self study, instruction classes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Books</b>                            | Optional:<br>- "Physical Ceramics" by Y-M Chiang, D. P. Birnie, and W.D. Kingery, John Wiley & Sons, ISBN: 0-471-59873-9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Reader</b>                           | - Several handouts and the lecture slides will be made available on Brightspace                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                       | The exam will be a closed book exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Permitted Materials during Tests</b> | - Calculator, overview of most important equations will be handed out during exam, the reference sheets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| SET3090                       | Fossil-Free Fuel and Feedstock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b> | Dr. T.E. Burdyny                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Responsible Instructor</b> | M. Rana                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>             | Prof.dr. F.M. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>             | Dr. R. Kortlever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>             | Prof.dr.ir. E.L.V. Goetheer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>            | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>        | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>        | Lecture Topic<br>1 Course Intro / Overview Current fossil infrastructure and major processes<br>2 Routes to fossil-free fuels and feedstocks<br>3 Water Electrolysis for H <sub>2</sub> production<br>4 Water Electrolysis for H <sub>2</sub> production<br>5 Deep-dive into factors controlling electrochemical performance<br>6 Deep-dive into factors controlling electrochemical performance<br>7 H <sub>2</sub> + CO <sub>2</sub> + Thermochemistry (Sabatier, reverse water-gas shift, Fischer-Tropsch, methanol Synthesis)<br>8 H <sub>2</sub> + CO <sub>2</sub> + Thermochemistry (Sabatier, reverse water-gas shift, Fischer-Tropsch, methanol Synthesis)<br>9 H <sub>2</sub> + N <sub>2</sub> + Thermochemistry (Haber-Bosch Ammonia as an energy carrier)<br>10 H <sub>2</sub> + N <sub>2</sub> + Thermochemistry (Haber-Bosch Ammonia as an energy carrier)<br>11 Formic Acid/Organic-based Electrolysis<br>12 CO <sub>2</sub> Electrolysis<br>13 Scaling Electrolyzers (Ir calculation/recycling, sizes needed (PV area, etc.))<br>14 Course Recap |   |
| <b>Study Goals</b>            | At the end of this course, students will be able to:<br>Analyze both traditional and electrochemical processes able to produce fuels and feedstocks without the input of fossil fuels.<br>Evaluate the effect of intermittency on the operation and scaling of these processes.<br>Compare thermochemical and electrochemical conversion processes and their current technology readiness levels<br>Understand and explain the necessity of using ambient molecules (H <sub>2</sub> O, CO <sub>2</sub> , N <sub>2</sub> ) for a successful energy transition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Education Method</b>       | Standard 2-hour lectures will be used throughout the course. The lectures provide learning activities to introduce the conceptual material, and for students to test this out in example. These relate to the above learning objectives.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>             | We use Brightspace forums as an outside of class hours medium for discussion and questions.<br>The course grade will be assessed fully by a 100% Final Exam taking place in the regular exam periods.<br><br>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.<br><br>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| SET3135                                | Fuel Cell Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. A. Purushothaman Vellayani                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Responsible Instructor</b>          | Dr.ir. L. van Biert                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior knowledge</b>        | Basic thermodynamic principles. Prior knowledge of (electro)chemistry will be helpful but is not required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>Fuel cell systems convert the chemical energy in fuels directly into electricity, enabling clean and efficient power generation from renewable fuels. This course provides an overview of different fuel cell systems and discusses the various fuel cell types, working principles and characteristics. Particular attention is given to design, integration, operation and application of polymer electrolyte membrane fuel cell (PEMFC) and solid oxide fuel cell (SOFC) systems.</p> <p>Specific topics include:</p> <ul style="list-style-type: none"> <li>- Fuel cell principles and fuel cell types.</li> <li>- Electrochemical power production: open circuit voltage and reversible voltage, the Nernst equation, fuel cell voltage and efficiency, fuel and oxidant utilization.</li> <li>- Fuel cell polarization: current density, activation losses, fuel crossover and internal currents, ohmic losses, concentration losses.</li> <li>- PEMFCs: electrolyte materials and structure, electrode materials and structure, gas channels and separator plates, water management, cell cooling, fuel and air supply, system design.</li> <li>- SOFCs: electrolyte materials, electrode materials and structure, configuration, design and performance of cells, stacks and systems, internal and external reforming, combined cycles.</li> <li>- Fuel processing: desulphurisation, reforming, partial oxidation, water gas shift, purification, combustion of residual fuel, gasification and gas cleaning, heat integration.</li> <li>- System thermodynamics: exergy losses in fuel cell systems and components.</li> <li>- Applications: systems for stationary power production and mobility.</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing this course, you should understand physical and chemical processes taking place in a fuel cell and be able to create conceptual designs of fuel cell systems for different electricity generation applications.</p> <p>More specifically, you should be able to:</p> <ol style="list-style-type: none"> <li>1. Describe the physical and chemical processes taking place in different types of fuel cells.</li> <li>2. Calculate fuel cell voltage, power and fuel consumption in various operating conditions.</li> <li>3. Select appropriate the balance of plant components for a particular fuel cell and fuel type.</li> <li>4. Design and evaluate different fuel cell-based systems with using system thermodynamics.</li> <li>5. Appraise various fuel cells with regard to their characteristics and intended application.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | lectures / tutorials / self-study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Literature and Study Materials</b>  | <p>Course material:</p> <p>Fuel Cell Systems Explained. James Larminie, Andrew Dicks, John Wiley &amp; Sons, LTD, 1999, ISBN 0-471-49026-1</p> <p>Course slides</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Written Exam</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |



|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Wind Energy 2024

| AE4135                                 | Rotor / wake Aerodynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. C.J. Simao Ferreira                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Responsible Instructor</b>          | W. Yu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Gast Instructor</b>                 | Dr.ir. T. Sinnige                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/2/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior knowledge</b>        | Basic aerodynamics, level BSc Aerospace Engineering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Parts</b>                           | The lectures are given in the 3rd and 4th period (second semester), in 2 hour lectures weekly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>Introduction to rotary wing aerodynamics. Applications in aircraft, propulsion, fans and wind turbines. Conservation laws. Actuator disk/momentum theory. Limitations. Helicopter rotor vertical flight and windmill brake state. Figure of merit. Wind turbine Betz optimum. Lift and drag devices</p> <p>Blade elementmomentum method, Tip correction methods. Correction for finite nr. of blades and heavily loaded rotors.</p> <p>Aerodynamic characteristics of airfoils for rotor application. Aerodynamic properties of pitch and stall controlled wind turbine. Wind turbine rotor blade design.</p> <p>Vortex line methods. Vortex wake structure. Frozen and free wake, vortex core modelling.</p> <p>Vortex panel methods. Advanced wake models. Acceleration potential method.</p> <p>Detailed rotor near wake structure. Experimental wake velocities and wake vorticity structure.</p> <p>3D effects, Stall delay. Yawed flow and dynamic inflow. Autogiro, helicopter rotor in forward flight.</p> <p>Unsteady aerodynamics and dynamic stall effects. Theodorsens Theory. Effects of tower shadow and wind shear.</p> <p>Aeroacoustics and rotor aeroacoustics.</p> <p>Vertical axis wind turbine rotor and Voight-Schneider propeller</p> <p>Effects of inflow turbulence intensity on blade loads. Near and far wake structure</p> <p>Wind farm aerodynamics. Rotor-wake interaction. Single and multiple wakes. Effects upon loads and performance.</p> |   |
| <b>Study Goals</b>                     | <p>Provide an overview of the phenomena and models present in aerodynamics of rotors, with special emphasis in horizontal axis wind turbine rotors. Propellers, vertical axis (crossflow) wind turbine rotors and helicopter rotors will also be addressed, but with less detail.</p> <p>"Hands on" introduction to the different computational models used nowadays to analyse the aerodynamics of rotors.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | Lecture and exercises + 3 group assignment + final written (digital) or oral exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Course lectures. The students can download the course lectures, background material and assignments from the accompanying online site. Brightspace will also be used extensively for submitting exercises and response to students</p> <p>Recommended literature</p> <p>The recommended literature for each chapter will be indicated in the course lectures.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | Written assignments + oral or written (digital) final examination. The resit is an oral exam or digital exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Remarks</b>                         | Fundamental knowledge of Applied Aerodynamics is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Set-up</b>                          | <p>There will be three major group assignments that must be completed and reported during the course.</p> <p>The evaluation will consist of a written or oral examination based upon the course content and the assignment reports.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |

| AE4202                                  | CFD for Aerospace Engineers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Prof.dr.ir. S. Hickel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                       | Dr.ir. A.H. van Zuijlen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Responsible for assignments</b>      | Dr.ir. A.H. van Zuijlen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Co-responsible for assignments</b>   | Dr. D. Modesti                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                  | CFD for Aerospace Engineers sets the grounds of the essential physical, mathematical, and numerical models for turbulent fluid flows and shows the opportunities and limits of fluid flow simulations in the aerospace industry. This course includes: (1) short survey of history, basic equations and dimensionless numbers, (2) turbulence theory and models for turbulence simulations with DNS, RANS, LES, (3) generation of suitable computational grids, (4) discretization methods, boundary conditions, effect of numerical truncation (5) solvers for large linear systems, error and residuum, (6) visualization, (7) validation and verification, (8) hands-on tutorials with ANSYS CFX and OpenFOAM. |   |
| <b>Study Goals</b>                      | After completion of this course, students (1) know and understand the methods and models used in state-of-the-art CFD software, (2) are able to set-up and run CFD simulations, and (3) can critically analyze and evaluate CFD results.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Method</b>                 | Lectures, hands-on tutorials, take-home assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Books</b>                            | van Dyke: An Album of Fluid Motion; The Parabolic Press<br>Ferziger and Peric: Computational Methods for Fluid Dynamics; Springer Verlag<br>Hirsch: Numerical Computation of Internal and External Flows: Volume 1 - The Fundamentals of Computational Fluid Dynamics; Butterworth Heinemann<br>Pope: Turbulent Flows; Cambridge University Press<br>Sagaut, Deck and Terracol: Multiscale and Multiresolution Approaches in Turbulence; Imperial College Press<br>Wilcox: Turbulence Modeling for CFD; DCW Industries                                                                                                                                                                                            |   |
| <b>Reader</b>                           | Detailed lecture slides and tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                       | Written exam and report on final assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Permitted Materials during Tests</b> | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| AE4205                                 | MDO for Aerospace Applications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. G. la Rocca                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr.ir. M.F.M. Hoogreef                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Dr. F. Oliviero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | <p>Familiarity with MATLAB and basic aircraft design knowledge.</p> <p>MATLAB is used for tutorials and home assignment, thus students without MATLAB experience must gain this through self-study</p> <p>Students with no experience in aircraft design should consult textbooks like Torenbeek, Raymer and Roskam to familiarize with the fundamentals of wing sizing process and general concepts such as wing loading, aerodynamic coefficients, fuel fraction method, drag and its components, etc.</p>                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | <p>The course will introduce students to multidisciplinary design optimization (MDO), as a modern methodology to support complex product development.</p> <p>Main topics:</p> <ul style="list-style-type: none"> <li>- What and why MDO</li> <li>- Parameterization techniques</li> <li>- Numerical optimisation methods (focus on gradient based methods)</li> <li>- Decomposition and coordination of complex computational systems</li> <li>- the eXtended Design Structure Matrix (XDSM)</li> <li>- MDO architectures</li> <li>- Intro to surrogate modeling, discrete&amp;mixed integer problems, genetic algorithms and Multi Objective Optimisation</li> </ul>                                  |   |
| <b>Study Goals</b>                     | <p>The main goals are the following:</p> <ul style="list-style-type: none"> <li>- understanding of Multidisciplinary Design Optimization (MDO) methods and architectures</li> <li>- hands-on experience in setting up and solving MDO problems</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Method</b>                | <p>The course consists of classical lectures and computer tutorials using own laptop.</p> <p>During the tutorials, students are introduced to some of the tools discussed during lecture and can make hands-on experience during interactive sessions with the teachers. In particular, students will be introduced to the MATLAB optimization function fmincon and its use for gradient based optimization.</p> <p>A significant amount of time is left for self-study, where students will have to work in pairs on one take-home assignment.</p>                                                                                                                                                    |   |
| <b>Computer Use</b>                    | <p>The use of a personal computer is required</p> <p>MATLAB (including its optimization toolbox) will be used for the MDO practical sessions and assignment. Student licenses are provided by the TUD</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>- lecture sheets and some notes (only available digitally on Brightspace)</li> <li>- scientific papers</li> <li>- some software modules</li> <li>- Suggested (not compulsory) book: <ul style="list-style-type: none"> <li>- Multidisciplinary design optimization supported by knowledge based engineering. By J. Sobieszczanski-Sobieski, A. Morris and M. van Tooren, Wiley ISBN 978-1-118-49212-3</li> <li>The book is available in e-format from the TUD library (Chapter 9 is used in the course AE4204 on Knowledge Based Engineering)</li> <li>- Engineering Design Optimization, by J.R.R.A. Martins and A. Ning (free on line)</li> </ul> </li> </ul> |   |
| <b>Assessment</b>                      | <p>There is no conventional exam for this course.</p> <p>The assessment is based on the quality of one take-home assignment, which will be executed in teams of two students.</p> <p>The assignment is about setting up and solving an MDO problem using Matlab. The deliverables are the Matlab code and a report.</p> <p>Two different take-home assignments are defined for the two yearly examination sessions.</p>                                                                                                                                                                                                                                                                                |   |
| <b>Tags</b>                            | <p>Algorithmics<br/>Analysis<br/>Matlab<br/>Modelling<br/>Optimalisation<br/>Programming<br/>Project</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |

| AE4T40                                 | Airborne Wind Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ing. R. Schmehl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. D. Ragni                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | BSc. completed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Parts</b>                           | Lectures include latest material on the developments in the field of airborne wind energy, with a thematical focus on kite power generation.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                 | Lecture 1: Introduction to airborne wind energy<br>Lecture 2: History of kite applications<br>Lecture 3: Implemented AWE concepts<br>Lecture 4: Physics of tethered flight<br>Lecture 5: Theory of tethered flight in three dimensions<br>Lecture 6: Flight navigation and maneuvering<br>Lecture 7: Operation of AWE systems<br>Lecture 8: Practical use of AWE systems<br>Lecture 9: AWE in the future energy system<br>Lecture 10: Design and flight dynamics of kites<br>Lecture 11: Aerodynamics of kites<br>Lecture 12: Aeroelasticity of kites<br>Practical: AWE workshop with Jupyter Notebook |   |
| <b>Study Goals</b>                     | Give an overview of developments in airborne wind energy, developng mathematical models for kites and designing kite power systems for industrial scale generation of energy.                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Method</b>                | Lecture & workshop                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Prerequisites</b>                   | Any MSc student can take this course, given a good understanding of vector analysis and basic trigonometry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Set-up</b>                          | 12 lectures of two lecture hours each in the first semester followed by a written examination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |

| AE4W30                                 | Wind Resource and Wind Farm Yield                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr. S.J. Watson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr.ir. D.J.N. Allaerts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | This course will cover how the output of wind farm arrays is predicted based on modelling of the wind resource and combined wake interactions. The first part of the course will cover the different wind resource models which are used ranging from simple mass consistent models, to linearised solvers, computational fluid dynamics models and mesoscale models. The use of models in both onshore and offshore environments will be covered including the importance of atmospheric stability. The second part of the course will consider the application of models to assess the yield from wind farms considering wakes effects. Wake models will be reviewed including simple semi-empirical models to kinematic models and the modelling of wakes in CFD models. Once again, the importance of atmospheric stratification will be considered and the potential impact on wind farms of certain mesoscale phenomena, e.g. gravity waves offshore, will be assessed and quantified. Large wind farm phenomena will be studied including the so-called deep array effect and wind farm blockage. |   |
| <b>Study Goals</b>                     | Course Aim: to give students a firm foundation in how to predict wind farm yield from prevailing wind conditions.<br><br>Course Objectives:<br><br>1) To understand what tools are available to predict wind resource over a wide area<br>2) To apply to such models to potential wind farm sites onshore and offshore<br>3) To quantify the importance of terrain effects and atmospheric stability on the wind resource<br>4) To understand what tools are available to predict wake effects in a large wind farm<br>5) To apply wake models in a realistic wind farm situation<br>6) To appreciate the importance to wind farm yield of phenomena such as terrain complexity (onshore), proximity to the coast (offshore), atmospheric stability, mesoscale effects (e.g. gravity waves), wind farm layout and wind farm scale                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Students will learn through a combination of guided reading, group work, online resources (e.g. lectures, videos, quizzes), face-to-face lectures, coursework and exams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | The students will undertake two course assignments (resource assessment and wind farm yield prediction) worth 30% each, spread out during the course and an oral exam worth 40% at the end of the course.<br><br>Assignment 1: application of a model to assess and optimise wind resource and energy yield for an onshore site with a report to be assessed.<br>Assignment 2: application of a model to assess and optimise energy yield for an offshore site incorporating different wake models with a report to be assessed.<br>Exam: oral test to assess understanding of the principles of wind resource assessment and wind farm yield prediction.                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| AE4W31                                 | Floating Offshore Wind Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. A.C. Viré                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Dr. F. Taruffi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr.ir. D.A.M. De Tavernier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are:</p> <ul style="list-style-type: none"> <li>- Why floating offshore wind, ongoing projects and trends</li> <li>- Aerodynamics and control of floating offshore wind turbines</li> <li>- Hydrodynamics of floating offshore wind turbines</li> <li>- Small-scale and full-scale testing</li> <li>- Logistics and project development of large floating offshore wind farms</li> <li>- Alternative wind turbine concepts for floating foundations</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing the course, the students will be able to:</p> <ul style="list-style-type: none"> <li>- Explain the relevance and need for floating offshore wind energy in the current energy transition</li> <li>- Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines</li> <li>- Calculate the effect of a floating motion on the turbine performance</li> <li>- Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks</li> <li>- Explain the industrial challenges in developing a large floating offshore wind farm</li> <li>- Explain the differences between bottom-fixed and floating turbines in the overall supply chain</li> <li>- Explain pros and cons of different wind energy concepts in a floating environment</li> </ul>                                                                             |   |
| <b>Education Method</b>                | The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | The course is assessed through a written exam at the end of the course and the assignments performed during the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

Year

Organization

Education

2024/2025

Electrical Engineering, Mathematics and Computer Science

Master Sustainable Energy Technology (SET)

Electric Mobility 2024

| EE4515                                 | Advanced Power Electronics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. Z. Qin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Dr.ir. H. Vahedi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4 + lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | It is expected that the student has taken all the previous courses in the power electronics teaching line: ET4119, ET4116, ET4145                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Contents</b>                 | The focus of this course is on providing both in-depth theories and practical knowledge regarding the power converter design that are not covered in the previous courses. The topics include power converter control, switching behaviour, power loss modelling, passive components, and thermal management.                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Study Goals</b>                     | <p>The lab practicum provides a chance to test the performance of the power switches and control of power converters</p> <p>After this course, the students will be able to</p> <ul style="list-style-type: none"> <li>- explain the concerns and trade-offs in the power converter design, including the switching - behavior, the passive components, the thermal management, and control</li> <li>- use the knowledge and design a grid-tied inverter with given requirements regarding THD, efficiency and size</li> <li>- use simulation tools to evaluate the converter design</li> <li>- test with the lab setup, and analyse the test results</li> </ul> |   |
| <b>Education Method</b>                | Lectures and lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Literature and Study Materials</b>  | <p>1. Lecture slides</p> <p>2. Books:</p> <ul style="list-style-type: none"> <li>a. R.W. Erickson , D. Maksimovi, Fundamentals of Power Electronics, 2020</li> <li>b. J.G. Kassakian, D.J. Perreault, G.C. Verghese, M.F. Schlecht, Principles of Power electronics, 2nd Edition, 2023</li> <li>c. R. Teodorescu, M. Liserre, P. Rodriguez, Grid Converters for Photovoltaic and Wind Power Systems, 2011</li> </ul>                                                                                                                                                                                                                                             |   |
| <b>Assessment</b>                      | <p>Assignments (30%) and project (70%)</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                          |   |



| EE4C04                                 | Control System Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. A.J.J. van den Boom                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams<br/> Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation<br/> Dynamic response, relation to modes, the matrix exponential<br/> Realization of transfer function models by state space descriptions, coordinate changes, canonical forms<br/> Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback<br/> Application of LQ regulator<br/> Observability, detectability, unobservable modes, state-estimation observer design<br/> Output feedback synthesis and separation principle<br/> Reference signal modeling, integral action for zero steady-state error.<br/> Analysis in robust stability and robust performance.</p> <p>Basics of model predictive control (MPC). Different model-structures.<br/> Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Study Goals</b>                     | <p>By taking this course, the student</p> <ul style="list-style-type: none"> <li>- will be able to master the introduced theoretical concepts in systems theory and feedback control design and</li> <li>- will be able to practically apply these concepts to design projects and tasks</li> <li>- and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.</li> <li>- will be able to translate a predictive control problem into a standards setting and solve the predictive control problem.</li> </ul> <p>More specifically, the student will be able to:</p> <ul style="list-style-type: none"> <li>- Translate differential equation models into state-space and transfer function descriptions</li> <li>- Rationalize differences between state-space and transfer function approaches</li> <li>- Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties</li> <li>- Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state</li> <li>- Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control</li> <li>- Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers</li> <li>- Formulate the separation principle and employ it for the design of output feedback</li> <li>- Build reference models and achieve zero steady-state error using integral control.</li> <li>- set up a predictive control problem.</li> <li>- Solve the standard predictive control problem;</li> <li>- Master the main analytical details in stability proofs of predictive control schemes;</li> </ul> |   |
| <b>Education Method</b>                | Lectures 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | <p>Textbook (its use is strongly recommended):</p> <p>K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009</p> <p>Available online for download:<br/> <a href="http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf">http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf</a></p> <p>The link for downloading the lecture notes for model predictive control will be available in the first quarter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                      | <p>Written examination.</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| ET4116                                 | Power Electronics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. P. Bauer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Responsible Instructor</b>          | Dr.ir. H. Vahedi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Instructor</b>                      | Dr. Z. Qin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/3/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>        | ET4119<br>ET4145                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | Resonant converters, snubbers, Space Vector theory, Vector and scalar PWM modulation, Simulation and modelling of power electronics, High Power Converters, Applications of power electronics for: HVDC transmission, DC Grids, Electric Utility (Facts and Custom Power), Power Conditioners, UPS systems etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                     | <p>Learn objectives per topic</p> <p>1. Resonant Converters</p> <ul style="list-style-type: none"> <li>- to know why resonant converters are applied,</li> <li>- advantages/disadvantages</li> <li>- which categories of resonant converters there are</li> <li>- how they are constructed and how they work</li> <li>- when thyristors and when transistors are applied, influence on the switches,</li> <li>- to dimension the switches for a resonant converters,</li> <li>- to understand how to dimension the resonant tank</li> </ul> <p>2. PWM modulation</p> <ul style="list-style-type: none"> <li>- to understand the different modulation techniques for single phase and three phase converters</li> <li>- to know the main feature of PWM modulation for different types</li> <li>- to understand space vector (phasor)</li> <li>- to know the Clark, Park transformation and to apply it for three phase converters</li> <li>- to know the following principles:</li> </ul> <ul style="list-style-type: none"> <li>- carrier modulation-scalair</li> <li>- carrier modulation-vector</li> <li>- over modulation</li> <li>- preprogrammed modulation, harmonic elimination</li> <li>- number of pulses for harmonic elimination, advantages, disadvantages</li> <li>- hysteresis modulation</li> <li>- carrier modulation with feedback</li> <li>- scalar and vector modulation with real time optimisation, predictive current - control method</li> </ul> <p>3. Applications</p> <p>Understand the following:</p> <ul style="list-style-type: none"> <li>- HVDC Transmission</li> </ul> |   |
| <b>Education Method</b>                | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Literature and Study Materials</b>  | Mohan, N., T.M. Undeland and W.P. Robbins, 'Power Electronics', Wiley, 2nd ed. 1995 (to obtain by the ETV).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>Written and oral</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Remarks</b>                         | Consultation by appointment, actual course information available on brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

| ET4121                                 | A.C. Machines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. J. Dong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Required for</b>                    | Control of Electrical Drives (ET4291)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Expected prior knowledge</b>        | Electrical machines and drives (ET4117)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>The following subjects are discussed:</p> <ul style="list-style-type: none"> <li>- winding function and inductances;</li> <li>- time and spatial harmonics;</li> <li>- skin effect in electrical machines;</li> <li>- phasor, space vector and transformation in electrical machines;</li> <li>- the salient pole synchronous machine;</li> <li>- modeling and simulation of the dynamic performance of DC and AC machines.</li> </ul>                                                                |   |
| <b>Study Goals</b>                     | <p>After following this course the students should be able to</p> <ul style="list-style-type: none"> <li>- describe and calculate the steady-state performance of AC machines;</li> <li>- calculate spatial harmonics and inductances of given winding distributions;</li> <li>- derive the dynamic models of DC and AC machines using space vectors;</li> <li>- model and simulate the dynamic performance of DC and AC machines;</li> <li>- evaluate the limitations of the dynamic models.</li> </ul> |   |
| <b>Education Method</b>                | Lecture / Computer Simulation / Discussion                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Literature and Study Materials</b>  | <ul style="list-style-type: none"> <li>- A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill.</li> <li>- lecture notes on Brightspace</li> </ul>                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | <p>Oral exam</p> <p>In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).</p>                                                                                                                                                            |   |
| <b>Remarks</b>                         | <ul style="list-style-type: none"> <li>- Presence at the computer simulation session is obliged.</li> <li>- Submission of assignments on Brightspace is obliged.</li> </ul>                                                                                                                                                                                                                                                                                                                              |   |

| ME41106                         | Intelligent Vehicles 3ME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor          | Prof.dr. D. Gavrila                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| Practical Coordinator           | R.M. Ensing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Contact Hours / Week<br>x/x/x/x | 0.6.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Education Period                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Start Education                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| Exam Period                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| Course Language                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Course Contents                 | <p>ME41106 Intelligent Vehicles is an introduction course on the (on-board) AI technology of automated driving. Whereas the application setting is that of vehicles, most concepts apply to the broader mobile robotics setting. Students seeking a one-course overview of AI for mobile robotics are thus encouraged to attend, also from outside ME (e.g. EEMCS, Civil Engineering and Aerospace Engineering faculties).</p> <p>Introduction<br/>Course organization<br/>Intelligent Vehicles (IV) domain<br/>Motivation for automated driving<br/>SAE levels of automation<br/>Main components of IV technology<br/>Remaining challenges<br/>Sensor Overview (camera, radar, LiDAR)<br/>3D Machine vision<br/>Perspective camera model<br/>Extrinsic and intrinsic camera transformations<br/>Stereo vision<br/>Object Detection and Classification<br/>Detection vs. classification<br/>Object proposals<br/>Handcrafted features (e.g. HOG) &amp; classification (e.g. linear SVM)<br/>End-to-end learning: neural networks, deep learning<br/>Performance metrics: confusion matrices, precision vs. recall, ROC curves<br/>State Estimation<br/>Bayesian Filtering<br/>Kalman Filtering<br/>Particle Filtering<br/>Object Tracking<br/>Data Association<br/>Track Management<br/>Self-Localization &amp; Sensor Fusion<br/>Absolute vs. relative localization<br/>Ego-motion compensation (e.g. odometry)<br/>Extrinsic sensor calibration<br/>Environment representations (grids, voxels)<br/>Motion Planning<br/>Planning as graph-search (the A* algorithm)<br/>Trajectory generation and optimization<br/>The Road Ahead<br/>IV technology currently on the market<br/>Future developments</p> <p>The course has a substantial practicum component (about 65% of course work load), where learned concepts are put into practice by means of programming assignments (Python, using Jupyter notebooks).</p> |   |
| Study Goals                     | At the end of the course students will understand the main methodical components of a intelligent vehicle (mobile robot). They will be able to program basic algorithms and test these on simulated and real-world data. Students will be able to express an educated opinion on the benefits and risks of automated driving, the current developments from driver assistance to driver-less cars, and the forces driving this transformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Education Method                | <p>Lectures (2-4 hours per week)<br/>Practicum (2 hours per week)</p> <p>Formative feedback: Practica<br/>- Pairwise collaboration (alternating random groups)<br/>- Pass/Fail (counts for knock-out criteria)<br/>- No late submissions or resit (except due to motivated medical conditions)</p> <p>Students can in principle run the software locally on their own computers (laptop/PC). However, we provide no support for setting up the necessary software environment (i.e. Anaconda). If the software environment cannot be installed properly (e.g. a few processors like M1 and ARM might not work), the student will need to revert to using the computers in the designated lab rooms.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Course Relations                | <p>This course, ME41106 Intelligent Vehicles <i>*cannot*</i> be selected in combination with RO47004 Machine Perception, due to the content overlap between the courses. This means, for example, that this course ME41106 cannot be selected by students following the MS Robotics program (for which RO47004 is required).</p> <p>This course mainly addresses the on-board AI technology of an intelligent vehicle. Complementary to this course is CIE 5805 Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment which focuses more on automated vehicles as part of a larger intelligent transportation system (i.e. including smart infrastructure, connected/cooperative driving) and on associated issues at the macro level (e.g. traffic flow efficiency, fuel consumption, behavior adaptation).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Literature and Study Materials  | <p>Slides<br/>Hand-outs<br/>recommended text book: Artificial Intelligence: A Modern Approach (Russel and Norvig) 3rd edition freely available as PDF<br/>Jupyter notebooks</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| Prerequisites                   | <p>Basic linear algebra and probability theory<br/>Intermediate Python programming skills, e.g. Numpy, object oriented programming (classes, inheritance), modules, namespaces and scope.<br/>Recommended: Pattern Recognition / Machine Learning (RO47002, CSE2510, or equivalent)</p> <p>It is possible to start the course with beginner Python skills if your aim is to take advantage of this course to also improve your programming expertise (a laudable aim, given its importance for both industry and academia). Be aware, however, that this</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |

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|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Assessment</b>              | <p>catching up will likely result in an increased study load (e.g. 1-2 EC).</p> <p>Final practicum assignment (50%)</p> <ul style="list-style-type: none"> <li>- Individual assignment</li> <li>- Late submission: max. one day late for -1.0 grade point</li> <li>- Students having an initial grade of &lt; 5.8 will be offered an opportunity for a remake to obtain a (maximum) grade of 5.8 for this component in the Q3 quarter.</li> </ul> <p>Final written exam (50%)</p> <ul style="list-style-type: none"> <li>- Written exam on campus.</li> </ul> <p>Knock-out criteria</p> <ul style="list-style-type: none"> <li>- At most one of the Pass/Fail practicum assignments is Fail</li> <li>- The minimum final practicum assignment grade is 5</li> <li>- The minimum final written exam grade is 5</li> <li>- The minimum final course grade is 6</li> </ul> <p>The final practicum assignment will be evaluated on the computers in the designated lab rooms. If a student chooses to work using own computing equipment, the student needs to test the software on the designated computers before handing-in.</p> <p>Both final practicum assignment and final written exam results remain valid for one year. Each result can separately be transferred to the next academic year, upon request.</p> |
| <b>Enrolment / Application</b> | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                |
| <b>Special Information</b>     | <p>For more information about this course, see the slides of the Introduction lecture from past academic year at <a href="http://intelligent-vehicles.org/teaching/courses/">http://intelligent-vehicles.org/teaching/courses/</a> . TU Delft students might also be able to view the lectures from past academic year on Collegerama.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Department</b>              | ME Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| RO47017                                | Vehicle Dynamics and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | B. Shyrokau                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0.0.0.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <p>This course provides fundamental knowledge on vehicle motion, vehicle-road interaction and the main technical principles and aspects of automotive control systems. This course discusses vehicle modelling, controller design and testing procedures to evaluate vehicle motion and automotive control systems.</p> <p>Tire dynamics and modelling<br/>Wheel slip, slip angle, tire force / moments, influence of camber angle, pressure, temperature effect, combined slip, friction circle, transient behavior<br/>Tire modelling and wheel dynamics</p> <p>Lateral motion and vehicle handling<br/>Kinematics, understeer gradient, bicycle model, steady-state handling<br/>Transient motion, frequency response, handling testing, vehicle stability</p> <p>Antilock braking system (ABS) and vehicle dynamics control (VDC)<br/>General concepts of ABS and different control principals: rule-based, mixed slip-deceleration, advanced control techniques<br/>General concept of VDC, yaw rate based controller, body slip angle based controller, integrated vehicle control using control allocation</p> <p>Planar motion<br/>Planar vehicle modelling<br/>Static equilibrium, lateral load transfer, roll center and axis, roll stiffness, anti-roll bars<br/>Longitudinal load transfer, pitch center, anti-dive, anti-lift</p> <p>Steering &amp; assist logic<br/>Wheel kinematics, steering system modelling, column- and rack-assist<br/>Steering assist logic, steering feel and assessment</p> <p>Path-following and evasive control of automated vehicles<br/>Vehicle road model, error definition, an overview of path-following controllers, PD controller<br/>Evasive maneuver, advanced control techniques, MPC-based evasive steering assist</p> <p>Vertical motion and comfort<br/>Introduction to vibration, quarter car, 2DoF and 3DoF models, Ride comfort<br/>Suspension control: skyhook, groundhook, and mixed control</p> <p>Vehicle state estimation<br/>General concept of VSE, vehicle velocity estimation, Kalman filtering (linear and extended), wheel load reconstruction methods</p> <p>Final presentations of the group project</p> |   |
| <b>Study Goals</b>                     | <p>After successfully completing this course, you will be able to:</p> <ul style="list-style-type: none"> <li>explain the tire-road interaction and be able to implement the tire model and compare it with the established one</li> <li>describe the concept of vehicle handling and lateral stability and derive the equations of motion of a nonlinear vehicle model</li> <li>interpret the load transfer effects and implement four-wheel planar vehicle model including the wheel kinematics, steering system and steering assist logic</li> <li>explain the concepts of wheel slip control, vehicle stability control, automated lateral control and vehicle state estimation</li> <li>implement control strategies for vehicle stability control (PID with gain scheduling or LQR) and automated lateral control (MPC)</li> <li>derive the equations of vertical motion of a quarter-car vehicle model and explain the influence of vehicle and road parameters on vehicle vibration behaviour</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Method</b>                | <p>Lectures (4 hours per week)<br/>Office hours (2 hours per week)<br/>Group project &amp; presentation</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Prerequisites</b>                   | <p>Basic programming experience (e.g. Matlab / Simulink) at the BSc level<br/>Dynamics and control at the BSc level<br/>knowledge about Model Predictive Control is recommended</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>Homework assignments (25%) individual<br/>Group project (25%) collaboration in groups of 4-5 students<br/>Written exam (50%) individual</p> <p>The examination is a closed-book digital exam on campus including multi-select and open questions.</p> <p>The grade for the homework assignments will be the average grade of the 3 assignments.</p> <p>Knock-out criteria</p> <ul style="list-style-type: none"> <li>o Minimum average grade for homework assignments is a 5.0</li> <li>o Minimum average grade for group project is a 5.0</li> <li>o Minimum grade for the exam is a 5.0</li> <li>o Final grade minimum is a 5.75</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

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|                   | OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a> . |
| <b>Department</b> | ME Department Cognitive Robotics                                                                                                                                                                              |
| <b>Contact</b>    | Barys Shyrokau, <a href="mailto:b.shyrokau@tudelft.nl">b.shyrokau@tudelft.nl</a>                                                                                                                              |

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| Year         | 2024/2025                                                |
| Organization | Electrical Engineering, Mathematics and Computer Science |
| Education    | Master Sustainable Energy Technology (SET)               |

Heat in Buildings 2024



| 4413INTPR                              | Integrated Project: Industrial and Urban Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 |
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| <b>Module Manager</b>                  | Dr.ir. P. Ibarra Gonzalez                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| <b>Instructor</b>                      | Dr.ir. J.N. Quist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Instructor</b>                      | Dr.ir. P. Ibarra Gonzalez                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Required for</b>                    | This course is part of the MSc Industrial Ecology (joint degree between Leiden University and Delft University of Technology). The course builds up on the knowledge and skills taught in the first semester of the MSc Industrial Ecology. This course is part of the MSc Industrial Ecology (joint degree between Leiden University and Delft University of Technology). The course builds up on the knowledge and skills taught in the first semester of the MSc Industrial Ecology.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Course Contents</b>                 | <p>Many sustainability challenges are related to urban systems and industrial systems. The way urban and industrial environments and their infrastructures are built and transformed and how people live in them and work in them has an impact on greenhouse gas emission, resource depletion, pollution and the generation of waste.</p> <p>In this course, we approach urban environments and industrial systems from an ecosystem perspective. We focus on three important flows energy, water, construction materials, and resources. Analyzing these flows reveals the complexity of urban and industrial environments. The flow systems are related in multiple ways. An intervention to improve one flow can have a negative impact on another flow. Spatial interventions hold potentials for synergies, for example between an improved local water management and more liveable green spaces for local inhabitants, or the possibility to find the synergy between an industrial cluster and an urban area nearby.</p> <p>To understand and work with this complexity, we start with a careful analysis of a provided urban plan or a provided industrial network, that can be described as an eco-industrial park or a circular economy region. Each flow system of the plan is analyzed and modelled. The outcomes are combined into an integral sustainability assessment of the provided urban plan or provided industrial network.</p> <p>In the second part of the course, the assessment forms the base for improvements the provided urban plan or industrial network. The shape and position of the buildings and open spaces are adjusted where needed and flow related improvements are translated to spatial interventions. Or in the case of the industrial network, the outline of the process, the flowsheet, or the value chain are translated to design interventions. This can lead to conflicts between different interventions. Therefore, trade-offs are identified and negotiated while synergies are thought and strengthened. Here, aspects as spatial qualities, cluster behaviour, liveability and ecology are taken into account as well, leading to an integrated new urban plan or industrial network.</p> <p>This new plan or network is put in the institutional context of its location. Stakeholders that are relevant for a successful realization and maintenance of the plan are identified. The final outcome is an improved, integrated and implemented sustainable urban development or sustainable industrial development.</p> <p>For acting successfully in the complexity of urban environments and industrial environments negotiation, collaboration and communication are crucial skills. Therefore, this course pays extra attention to transferable skills in this field.</p> <p>This course is part of the joint degree MSc Industrial Ecology, offered by Leiden University and Delft University of Technology. More information about the course is published in the online study guide of Leiden University:<br/> <a href="https://studiegids.universiteitleiden.nl/en">https://studiegids.universiteitleiden.nl/en</a></p> |    |
| <b>Study Goals</b>                     | <p>After completing this course, you are able to</p> <ul style="list-style-type: none"> <li>- Assess the sustainability of a provided urban plan or provided industrial system by applying calculation models for key resource and waste flows.</li> <li>- Propose spatial and technical adjustments that improve the provided urban plan or provided industrial system, based on your assessment. You are able to analyze the effect of your adjustments by applying your calculation models and use the gained insight to further elaborate the plan accordingly.</li> <li>- Identify trade-offs and synergies between your proposed improvements and key qualities of urban plans or industrial systems. You are able to argue, how trade-offs should be mitigated and synergies should be strengthened. You are able to translate your argumentation into further improvements, leading to an integrated sustainable urban plan or industrial system.</li> <li>- Relate your improved urban plan or industrial system to a provided institutional context. You are able to identify key stakeholders required for a successful and synergetic implementation and maintenance of your improved and integrated sustainable urban plan or industrial system.</li> <li>- Apply several transferable skills, such as collaboration, presenting and giving feedback.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Education Method</b>                | <p>Q3: Lectures, workshops and self-study<br/> Q4: Lectures, workshops and self-study, group work with table reviews, presentations and peer feedback</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| <b>Assessment</b>                      | You are assessed on the basis of (group) reports related to the contents of the course. In addition, you are also assessed and/or asked to reflect on certain transferable skills.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
| <b>Special Information</b>             | <p>All communication will take place via Leiden University.</p> <p>Link to the study guide of the Leiden University: <a href="https://studiegids.universiteitleiden.nl/courses/104729/industrial-ecology-project">https://studiegids.universiteitleiden.nl/courses/104729/industrial-ecology-project</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Remarks</b>                         | .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Module Material</b>                 | Course material will be provided via brightspace (Leiden University).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |

| AR0097                                  | Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5 |
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| <b>Course Coordinator</b>               | Dr. A. Meijer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                       | Dr. A. Meijer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                       | Dr.ir. A. Straub                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                       | Dr. L.C.M. Itard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Responsible for assignments</b>      | Dr.ir. A. Straub                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                      | Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                 | Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>   | All literature will be handed in class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                       | Tender Game 50% of grade and Written Exam 50% of grade.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Hours</b>                       | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Permitted Materials during Tests</b> | Calculator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Enrolment / Application</b>          | <p>Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see:<br/> <a href="https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams">https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams</a><br/>           Additional enrolment information for non-BK students:</p> <p>Course enrolment via Brightspace only is not considered a valid registration<br/>           Enrolment in the BK-enrolment system BIS (<a href="https://bis.bk.tudelft.nl/">https://bis.bk.tudelft.nl/</a>) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below)<br/>           The course can be taken only after confirmation of placement in BIS by BK-enrolment</p> |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Period of Education</b>              | 10 weeks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Concept Schedule</b>                 | Monday- and Tuesday morning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Minimum number of participants</b>   | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course evaluation</b>                | For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| AR0132                                 | Zero-Energy Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 |
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| <b>Course Coordinator</b>              | Prof.dr.ir. A.A.J.F. van den Dobbelsteen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Coordinator</b>              | Ir. S. Broersma                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Responsible for assignments</b>     | Prof.dr.ir. A.A.J.F. van den Dobbelsteen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | Week 3.1-3.9<br>4 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Expected prior knowledge</b>        | A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | <p>The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered.</p> <p>The focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof.</p> <p>With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.</p> |   |
| <b>Study Goals</b>                     | <p>The student is able to:</p> <ul style="list-style-type: none"> <li>- develop an integrated energy-neutral climate design</li> <li>- make energy calculations and optimize the energy performance of a building</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | <p>Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting.</p> <p>The assignment will in principle be completed in groups of 3.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | <p>Knowledge of the theory is tested through a report and an oral presentation.</p> <p>Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.<br/>(A retake can be done in the next year)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Period of Education</b>             | Quarter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Concept Schedule</b>                | Monday afternoon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Minimum number of participants</b>  | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Maximum number of participants</b>  | 36                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course evaluation</b>               | For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| AR0139                         | MEGA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 15 |
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| Course Coordinator             | Dr. S. Brancart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |
| Course Coordinator             | Prof.dr. M. Overend                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| Course Coordinator             | Dr. A. Luna Navarro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| Course Coordinator             | Dr. M. Turrin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
| Instructor                     | Dr.ir. M.C. Lugten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| Responsible for assignments    | Dr. S. Brancart                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    |
| Education Period               | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| Start Education                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| Exam Period                    | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| Course Language                | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| Course Contents                | <p>MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations).</p> <p>By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster &amp; Partners, Stichting Hoogbouw.</p> <p>Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.</p> |    |
| Study Goals                    | <p>The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.</p> <p>Specified for this course:<br/>After successful completion of the course, the student will be able to:</p> <ul style="list-style-type: none"><li>- work in an interdisciplinary design process;</li><li>- understand and apply discipline-related knowledge in projects for large or tall buildings.</li><li>- develop design strategies to achieve high building performance;</li><li>- integrate numeric analysis and simulations data to address design choices.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| Education Method               | <p>In this course, the education methods are:</p> <ul style="list-style-type: none"><li>- Lectures</li><li>- Collaborative workshops with other students</li><li>- Exposure to external architectural practice and external experts</li><li>- Tutoring</li><li>- Presentations</li></ul> <p>The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| Literature and Study Materials | <p>Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |
| Assessment                     | <p>Assessment is twofold:</p> <ul style="list-style-type: none"><li>- Group assessment for integral group design based on presentations</li><li>- Individual assessment for discipline report</li></ul> <p>The students mark is a combination of the group assessment and individual assessment.</p> <p>Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.</p> <p>A retake can be done in the next year.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| Enrolment / Application        | <p>Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see:<br/><a href="https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams">https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams</a><br/>Additional enrolment information for non-BK students:</p> <p>Course enrolment via Brightspace only is not considered a valid registration<br/>Enrolment in the BK-enrolment system BIS (<a href="https://bis.bk.tudelft.nl/">https://bis.bk.tudelft.nl/</a>) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below)<br/>The course can be taken only after confirmation of placement in BIS by BK-enrolment</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| Period of Education            | Quarter 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| Minimum number of participants | For this course the minimum number of participants is 15.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| Maximum number of participants | For this course the maximum number of participants is 90.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| Course evaluation              | For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |

| AR1U131                                | Sustainable Urban Engineering of Territory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5 |
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| <b>Course Coordinator</b>              | Dr. F.L. Hooimeijer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Coordinator</b>              | L. Qu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Responsible for assignments</b>     | Dr. F.L. Hooimeijer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 15 hours per week starting from week 1 and ending in week 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Summary</b>                         | This quarter focusses on the making or remaking of the city and this course more specially on the city as technical construction: the natural and man-made conditions of the urban landscape, in order to integrate all interest in an urban plan. The act of integration and design requires systemic knowledge on a wide range of subjects. In this course the city is seen as a technical construction, with the central question being how to reconnect this construction to the natural systems of the territory? Therefore, this course provides vital input for the parallel studio of Q2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>In the master track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable urban development. This quarter focusses on the making or remaking of the city and this course more specially on the city as technical construction: the natural and man-made conditions of the urban landscape, in order to integrate all interest in an urban plan. The act of integration and design requires systemic knowledge on a wide range of subjects. In the course the city is seen as a technical construction, with the central question being how to reconnect this construction to the natural systems of the territory?</p> <p>In new to build areas, or green fields, there is a lot of freedom to engineer or not to engineer the natural system. One of the greatest urban challenges these days is urban renewal. This is a complex enterprise in itself because you deal with the existing urban use and fabric. On top of that we need to deal with trends like climate change and the energy transition. One dimension that is reintroduced in the urban project is the subsurface system, which plays a crucial role in water management, ecology and energy supply. The subsurface system was for a long time excluded from the urban planning and design process, because it was considered a technical aspect that was dealt with by civil engineers, not as part of the urban design. However, the subsurface sets the conditions with highest impact: it is more costly and takes more time to change a cable system then it is to build a building or construct a road. Especially the idea that the natural system has already been altered for urban use, and thus lost, is preventing innovative solutions that deal with climate change and the energy transition in urban renewal. In order to incorporate natural and technical conditions in urban plans, a better cooperation with civil engineers and a better understanding of what they do not know is crucial for urban designers. This is what the course aims at.</p> <p>To bring order in all the technical information we make use of the System Exploration Environment and Subsurface (SEES) that uses six functional layers with different dynamics, professional domains and knowledge fields: people, metabolism, occupation, public space, infrastructure and subsurface. The subsurface layer is ordered in (for the urban planner and designer) recognizable themes: water, energy, civil constructions and subsoil. Within the themes there is a logical order in the qualities of the subsurface (<a href="http://www.ruimtexmilieu.nl">www.ruimtexmilieu.nl</a>). SEES connects the subsurface information with the urban surface in order to inspire and set clear boundaries for the development of the urban surface. It is used for analysing potential problems, chances, demands and supports a creative interaction early in the process of urban planning.</p> |   |
| <b>Study Goals</b>                     | <p>At the end of this course, students are able to:</p> <ul style="list-style-type: none"> <li>Explain what for them is sustainable development and how technology is integrated.</li> <li>Evaluate the skills of an urban designer in relation to sustainability.</li> <li>Explain the city as a hybrid system of natural and human characteristics both the surface and subsurface level.</li> <li>Analyse the synergy between natural and human systems on the surface and subsurface level.</li> <li>Identify new approaches in which the conditions and chances that are given by the natural system are re-introduced.</li> <li>Demonstrate and design spatial principles or an urban ensemble with the desired performances, by combining design and engineering, including natural and human systems on the surface and subsurface level to develop ethical standards considering sustainable development.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | Interactive lectures and intensive workshops, full time three weeks starting in week 2 of the quarter. Mainly group work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>The assessment of the assignments is done using the rubric that is available in the quarter guide on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself.</p> <p>A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page.</p> <p>Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Enrolment / Application</b>         | <p>Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: <a href="https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams">https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams</a></p> <p>Additional enrolment information for non-BK students:</p> <p>Course enrolment via Brightspace only is not considered a valid registration</p> <p>Enrolment in the BK-enrolment system BIS (<a href="https://bis.bk.tudelft.nl/">https://bis.bk.tudelft.nl/</a>) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below)</p> <p>The course can be taken only after confirmation of placement in BIS by BK-enrolment</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Remarks</b>                         | The maximum marking period is 15 working days.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Period of Education</b>             | Quarter 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Concept Schedule</b>                | <p>The AR1U131 course is 15 days scheduled in the first 4 weeks of Q2. The AR1U131 course schedule is tuned with the AR1U100 course.</p> <p>The actual schedule will be available via Bright Space &gt; My Timetable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Minimum number of participants</b>  | For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

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|---------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Maximum number of participants</b> | For this course of the Master track of Urbanism the maximum number of participants is 100.   |
| <b>Course evaluation</b>              | For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment). |

| RO47005                                | Planning & Decision Making                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 5 |
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| <b>Responsible Instructor</b>          | Dr. J. Alonso-Mora                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0.6.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>This course provides an overview of motion planning and decision-making techniques and their practical application in robotics. Planning and Decision-making are critical components of autonomy in robotic systems. These components are responsible for making decisions that range from path planning and motion planning to coverage and task planning to taking actions that help robots understand the world around them better. This course studies underlying algorithmic techniques used for planning and decision-making in robotics and examines case studies in ground and aerial robots, mobile manipulation platforms and multi-robot systems. The students will learn the algorithms and implement them in a series of programming-based projects.</p> <p>In particular, we will first cover the fundamentals of planning (workspace, configuration space, representation of obstacles and robot models). We will then cover a broad set of methods, which include reactive methods and feedback control; discrete, combinatorial and probabilistic planning; planning under differential constraints; planning under uncertainty; learning in planning; and planning for multi-robot systems. Finally, we will briefly touch upon task assignment and vehicle routing.</p> <p>Course structure:<br/> Introduction<br/> Fundamentals: 3D rotations, configuration space, algorithm properties<br/> Differential constraints: recap of modeling &amp; control of manipulators, ground robots, aerial robots<br/> Graph-search fundamentals (depth/breadth first, Dijkstra, A*, heuristics)<br/> Combinatorial algorithms (cell decomposition, shortest-path roadmaps, motion primitives)<br/> Sampling-based algorithms (PRM, RRT, RRT*)<br/> Reactive algorithms: collision avoidance, potential fields, velocity obstacles<br/> Kinodynamic planning<br/> Trajectory optimization (fundamentals, global, local, Model Predictive Control)<br/> Planning under uncertainty (Markov Decision Processes, learning)<br/> Multi-robot motion planning (joint configuration, optimization-based, coverage)<br/> Task assignment (Hungarian, auction, linear program)<br/> Intro to vehicle routing</p> |   |
| <b>Study Goals</b>                     | <p>After completing this course, students will be able to:</p> <ul style="list-style-type: none"> <li>explain the role of Motion Planning and Decision-Making (PDM) in Robotics, and describe possible applications</li> <li>identify different classes of robotic systems, the associated mathematical models for their kinematics, dynamics and motion and the relevant spaces for PDM, such as the configuration and state spaces</li> <li>describe and compare algorithms for motion planning and decision-making of mobile robots and multi-robot systems, including: reactive algorithms, optimization-based algorithms, combinatorial algorithms, sampling-based algorithms and learning-based methods</li> <li>analyze a PDM problem, consider constraints, objectives and select the appropriate PDM methods to apply</li> <li>design and implement motion planners of moderate complexity to solve motion-planning tasks and navigate mobile robots</li> <li>perform PDM experiments, evaluate the results, and draw sound conclusions</li> <li>present the findings of a PDM experiment clearly in a report using graphs and scientific English</li> <li>understand PDM experiments descriptions found in scientific literature, and reproduce results</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | <p>Lecture hours/week: 4<br/> Q&amp;A sessions/week: 2<br/> Practical assignments and project (home) hours/week: 6<br/> Self-study hours/week: 4</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Books</b>                           | <p>Planning Algorithms, S. LaValle <a href="http://planning.cs.uiuc.edu/">http://planning.cs.uiuc.edu/</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Prerequisites</b>                   | <p>And additional papers/references provided by the lecturer.</p> <p>Linear Algebra<br/> Programming in Python</p> <p>Rigid body dynamics (recommended)<br/> Dynamics &amp; Control (recommended)<br/> Probability Theory (recommended)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                      | <p>Practicum assignments (10%)<br/> Project (40%) Collaboration in groups of 3 students<br/> - No resit (repeat the following year)<br/> Written Exam (50%) individual</p> <p>Knock-out criteria<br/> - Minimum grade for the project is a 5.0<br/> - Minimum grade for the exam is a 5.0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/> To participate in this course, registration via MyTUDelft is not needed.<br/> More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Department</b>                      | ME Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| SC42001                                | Control System Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5 |
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| <b>Responsible Instructor</b>          | Dr.ir. A.J.J. van den Boom                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Contents</b>                 | <p>State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation. Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design. Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance.</p> <p>Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Study Goals</b>                     | <p>By taking this course, the student</p> <ul style="list-style-type: none"> <li>- will be able to master the introduced theoretical concepts in systems theory and feedback control design</li> <li>- will be able to practically apply these concepts to design projects and tasks</li> <li>- and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses.</li> <li>- will be able to translate a predictive control problem into a standards setting and solve the predictive control problem.</li> </ul> <p>More specifically, the student will be able to:</p> <ul style="list-style-type: none"> <li>- Translate differential equation models into state-space and transfer function descriptions</li> <li>- Rationalize differences between state-space and transfer function approaches</li> <li>- Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties</li> <li>- Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state</li> <li>- Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control</li> <li>- Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers</li> <li>- Formulate the separation principle and employ it for the design of output feedback</li> <li>- Build reference models and achieve zero steady-state error using integral control.</li> </ul> <ul style="list-style-type: none"> <li>- set up a predictive control problem.</li> <li>- Solve the standard predictive control problem;</li> <li>- Master the main analytical details in stability proofs of predictive control schemes;</li> </ul> |   |
| <b>Education Method</b>                | Lectures 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Textbook (its use is strongly recommended):</p> <p>K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009</p> <p>Available online for download:<br/> <a href="http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf">http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf</a></p> <p>The link for downloading the lecture notes for model predictive control will be available in the first quarter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | Written examination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Department</b>                      | ME Department Delft Center for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |



| SC42095                                | Digital Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 |
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| <b>Responsible Instructor</b>          | A. Dabiri                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Instructor</b>                      | M.A. Sharifi Kolarijani                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | SC42015, SC42000 or similar.<br>Knowledge of classical control techniques (systematic and realistic PID design, frequency domain approaches) as well as state space theory is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Course Contents</b>                 | Computer control. Sampling of continuous-time signals. The sampling theorem. Aliasing. Discrete-time systems. State-space systems in discrete-time. The z-transform. Selection of sampling-rate. Analysis of discrete-time systems. Stability. Controllability, reachability and observability. Disturbance models. Reduction of effects of disturbances. Design methods. Approximations of continuous design. Digital PID-controller. State-space design methods. Observers. Pole-placement. Optimal design methods. Linear Quadratic control. Prediction. LQG-control. Implementation aspects of digital controllers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Study Goals</b>                     | The student must be able to:<br>1. describe the essential differences between continuous time and discrete-time control<br>2. transform a continuous time description of a system into a discrete-time description<br>3. calculate input-output responses for discrete-time systems<br>4. analyse the system characteristics of discrete-time systems<br>5. employ a pole-placement method on a discrete-time system<br>6. implement an observer to calculate the states of a discrete time system<br>7. apply optimal control on discrete-time systems<br>8. describe the functioning of the Kalman-filter as a dynamic observer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Method</b>                | Lectures and computer exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Computer Use</b>                    | Matlab/Simulink is used to carry out the exercises of this course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | Course material:<br>Lecture notes are made available on Brightspace<br><br>Textbooks:<br>K.J. Astrom, B. Wittenmark 'Computer-controlled Systems', Prentice Hall ,1997, 3rd edition.<br>L. Keviczky et. al 'Control Engineering', Springer, 2019.<br>L. Keviczky et. al 'Control Engineering: MATLAB Exercises', Springer, 2019.<br><br>References from literature:<br>B.C. Kuo 'Digital Control Systems', Tokyo, Holt-Saunders, 1980<br>G.F. Franklin, J.D. Powell 'Digital Control of Dynamic Systems', 1989, 2nd edition, Addison-Wesley                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | Project assignment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Enrolment / Application</b>         | Registration:<br>To participate in this course, registration via MyTUDelft is not needed.<br>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a> .<br><br>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a> .<br><br>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a> . |   |
| <b>Remarks</b>                         | Old course code: WB2305                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Department</b>                      | ME Department Delft Center for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |



| SEN1131                                | Institutional Economics for Designing in Socio-technical Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3 |
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| <b>Module Manager</b>                  | Dr.ir. R. van Bergem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | G. Tran                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                      | Dr.ir. R. van Bergem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Responsible for assignments</b>     | Dr.ir. R. van Bergem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Co-responsible for assignments</b>  | S. Renes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <p>This course provides fundamental concepts from the field of Institutional Economics that are instrumental for the design of institutions in socio-technical systems. Relevant frameworks and theories will be elaborated in relation to the design of socio-technical systems. The fundamentals of institutional economics will be applied to concrete design challenges, like the transition to sustainable infrastructures. It will be demonstrated how to relate Institutional Economics to interdisciplinary research challenges, including engineering, political science, law and philosophy. This provides interesting perspectives to other courses in the curriculum. Typical questions that are addressed in this module include:</p> <p>What is the specific contribution of Institutional Economics to the design of socio-technical systems?<br/> How to frame and interpret design challenges from an Institutional Economic perspective?<br/> What is an institutional design?<br/> How can institutional designs be related to the technical design of complex systems?<br/> How to operationalize concepts from institutional economics and apply them to empirical design challenges?</p> |   |
| <b>Study Goals</b>                     | <p>After the completion of the course, the student is able to:</p> <ul style="list-style-type: none"> <li>to understand and analyse complex institutional arrangements from an Institutional Economic perspective</li> <li>reflect on the possible contribution of Institutional Economics in addressing different design challenges</li> <li>apply approaches from Institutional Economics to concrete design challenges</li> <li>position and apply different approaches from Institutional Economics to interdisciplinary design challenges.</li> <li>establish a coherent institutional design given the features of the technical system(s) embedded in their socio-political environment</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Method</b>                | Lectures and workshops                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Literature and Study Materials</b>  | Reader of relevant study material, including cases to be discussed in the workshops                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | On-campus: Digital exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: online: timed take-home exam (open book)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

| TPM034A                                | Machine learning for socio-technical systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5 |
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| <b>Module Manager</b>                  | Dr.ir. S. van Cranenburgh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Instructor</b>                      | Dr. G. Marangoni                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr. A.P. Afghari                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Co-responsible for assignments</b>  | Dr. A.P. Afghari                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/6/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Students require prior knowledge of linear algebra, statistics, probability theory and data structures. Furthermore, basic Python programming knowledge is required. It is recommended (but not mandatory) to take the course EPA122A "Spatial Data Science".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | <p>Machine Learning (ML) is increasingly seen as a crucial part of the puzzle to solving the socio-technical challenges of today's networked and urbanised knowledge-driven societies. Successful adoption of ML does, however, not only require skilled computer scientists who do hard-core programming. Also, professionals are needed who have both the domain knowledge of socio-technical systems and a profound understanding of ML.</p> <p>This course aims to provide students in the socio-technical domain with a profound understanding of ML. It prepares students for the challenges and questions ML will pose them in their later careers.</p> <p>To this end, the course consists of three parts:</p> <ol style="list-style-type: none"> <li>1. Fundamentals of ML</li> <li>2. Explainability of ML</li> <li>3. Applications of ML for socio-technical challenges</li> </ol> <p>In part 1, students learn about ML fundamentals and methods. These weeks, a critical technical foundation is laid for grasping the strengths and weaknesses of ML for the analysis and design of socio-technical systems.</p> <p>Part 2 is devoted to the explainability of ML. For public decision-making as well as for decision-making in high-stake contexts, such as, e.g. autonomous vehicles, legal systems transparency and explainability of the models are of critical importance. Students learn several popular explainability techniques and discuss their value for applications in socio-technical systems.</p> <p>In part 3, a group of scholars provides exemplary applications of ML in socio-tech systems. This serves two purposes: (1) to show where and how ML is applied for analysis and application in socio-technical systems and (2) to deepen reflection on the impact of ML-based solutions and interventions on individuals, organisations, and society. Consecutively, students work in a group on a final project, building forth on one of the presented applications. This project brings together the three parts of this course: Fundamentals of ML, Explainability of ML, and application of ML for socio-technical challenges. Students need to apply ML models and techniques to real-world data in a notebook and interpret and communicate their results, taking into account the socio-technical setting through a presentation.</p> <p>The course consists of oral lectures and lab sessions. The aim of the lab sessions is to show and reinforce how the ML models, explainable ML techniques, and ML ideas presented in the oral lectures are put into practice. Also, they help students gather hands-on machine learning skills. The lab sessions involve a series of exercises in the form of Jupyter notebooks.</p> <p>The course consists of ~2 oral lectures and 1 lab session per week. Attendance of the lectures and lab sessions is highly recommended to keep up with the course but is not mandatory.</p> |   |
| <b>Study Goals</b>                     | <p>After the course, students can:</p> <ol style="list-style-type: none"> <li>1. explain fundamental concepts of machine learning (ML).</li> <li>2. explain the workings of a selected number of ML models and eXplainable AI (XAI) techniques, and apply these to empirical data.</li> <li>3. identify applications of ML and XAI techniques in real-world socio-technical systems.</li> <li>4. examine the impact of ML-based solutions and interventions on individuals, organisations, and society through XAI.</li> <li>5. conduct an in-depth analysis of a real-world socio-technical challenge, by applying ML and XAI to empirical data.</li> <li>6. reflect on the strengths and limitations of ML and XAI in real-world socio-technical systems.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Method</b>                | <p>The education methods used in this course are:</p> <ol style="list-style-type: none"> <li>1. Oral lectures with discussions</li> <li>2. Lab sessions</li> <li>3. Assignments</li> <li>4. Group work</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Computer Use</b>                    | Jupyter notebooks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | <p>The final grade is computed by taking the weighted average across the three parts:</p> <ol style="list-style-type: none"> <li>1. Individual (digital) exam, ANS, 50%, Minimum grade: 5.0</li> <li>2. Individual assignments, Jupyter notebooks, 25%, Minimum grade: 5.0</li> <li>3. Group project, 25%, Presentation and notebook, Minimum grade: 5.0</li> </ol> <p>For students to pass the course, the final grade must be equal to or larger than 5.8.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Tags</b>                            | Artificial intelligence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |

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|--------------|----------------------------------------------------------|
| Year         | 2024/2025                                                |
| Organization | Electrical Engineering, Mathematics and Computer Science |
| Education    | Master Sustainable Energy Technology (SET)               |

Heat Sources 2024

| CIEM2304                                | Environmental and Energy Geotechnics                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. A.C. Dieudonné                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Module Manager</b>                   | Dr.ir. A.C. Dieudonné                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Instructor</b>                       | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 5hrs lectures/workshops                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Contents</b>                  | 1. Multi-physics (thermal, hydraulic, mechanical) processes in environmental and energy geotechnics<br>2. Conceptual and mathematical modelling<br>3. Design of environmental and/or energy geotechnics systems<br>4. Evaluation of designs                                                                                                                                                                                                                                                                       |   |
| <b>Study Goals</b>                      | On completion of this course, the student will be able to:<br><br>[LO1]: Identify and describe processes in environmental and energy geotechnical systems<br>[LO2]: Develop a conceptual model of coupled processes occurring in environmental and energy geotechnical systems<br>[LO3]: Derive and solve a mathematical model to quantify the impact on geotechnical systems of environmental or energetic loads<br>[LO4]: Design a geotechnical project optimising the environmental and/or energetic behaviour |   |
| <b>Education Method</b>                 | Balance of lectures, supervised tutorials and self-study.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                       | Written formative assessment (based on check-list / rubric).<br><br>Written summative assessment based on 2 design assignments. The pass grade is a 5.8/10.                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Expected prior Knowledge</b>         | MUDE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Academic Skills</b>                  | Report writing, critical assessment, problem solving.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Literature &amp; Study Materials</b> | Will be provided via lecture material on Brightspace.<br><br>Fredlund and Rahardjo (1993) Soil Mechanics for Unsaturated Soils. Wiley.                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Judgement</b>                        | See Assessment<br><br>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br><a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a>                                                                                                                                          |   |
| <b>Permitted Materials during Exam</b>  | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Collegerama</b>                      | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| ME45160                                | Advanced Applied Thermodynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. B.P. Tighe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Instructor</b>                      | Dr. O. Moulτος                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Required for</b>                    | EFPT track                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Expected prior knowledge</b>        | Thermodynamics at the level of a BSc in Mechanical Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <ul style="list-style-type: none"> <li>- Review of fundamentals of thermodynamics, including thermodynamic potentials, Maxwell relations, and the 1st and 2nd Law</li> <li>- Equilibrium in one-component systems: equations of state, chemical potential, fugacity, Clausius-Clapeyron, etc.</li> <li>- Mixtures: ideal gas mixtures, partial molar quantities, Gibbs-Duhem, vapor- liquid equilibrium, activity coefficient models, and osmosis</li> <li>- Model equations of state</li> <li>- First introduction to statistical mechanics</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Study Goals</b>                     | After this course, you will be able to calculate phase and material properties in thermodynamic equilibrium for multi-phase and/or multi-component mixtures under ideal and non-ideal process conditions. You will be able to use equations of state and activity coefficient models, and to calibrate and validate those models using thermodynamic data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                | Lectures, optional homework, review session                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Computer Use</b>                    | homework assignments require scientific computing tools such as python, Matlab (or comparable) and basic programming methods such as for loops                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Books</b>                           | Holyst Poniewierski                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | Written final exam; open book                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |

| ME45225                                | Multiphysics transport in Energy Materials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3 |
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| <b>Responsible Instructor</b>          | L. Botto                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Instructor</b>                      | Dr.ir. R.M. Hartkamp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Expected prior knowledge</b>        | Elements of fluid dynamics, heat/mass transfer and thermodynamics, at the level of e.g. TUD Batchelor courses WB2542 and WB2543. Familiarity with modelling via PDE, at the level of e.g. WBMT2048.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>This is a course in modelling of transport phenomena in energy storage materials and devices. It is an excellent introduction to multiphysics problems for students interested in learning how one can simulate or model the coupling of heat, mass and electricity in the devices that drive the energy transition. Example of such devices are batteries, supercapacitors, hydrogen/ CO<sub>2</sub> storage devices, or materials for heat storage. Many courses treat transport phenomena as separate subjects, while this course unifies different transport phenomena exploiting analogies between different transport fields, with an applied emphasis on energy materials.</p> <p>The course is particularly suitable for student wishing to later specialize in numerical simulation of conservation laws (e.g. CFD courses such as Computational Fluid Dynamics for Engineers); for students with an interest in experimental, theoretical or simulation-based fluid dynamics (e.g. as a foundation for the Turbulence, Rheology or Multiphase Flow courses) and who wish to learn about the coupling of momentum and species transport ; or for students who seek a solid preparation to face confidently the study of coupled transport theories used in the modelling of electrochemical and heat transfer systems, see e.g. the EFPT Electrochemical Energy Storage course.</p> <p>Different from other transport phenomena courses within the EFPT curriculum, this course covers the modelling of diffusive and convective transport of generic species in materials that are not homogeneous, but are made of different phases in the form of compacted dry powders, solid foams, inks, slurries etc. For example, the course covers how to estimate the effective electrical or heat conductivity of materials used as battery electrodes, a key issue when modelling batteries.</p> <p>Among other topics, the courses will give a practical introduction to Comsol Multiphysics, probably one of the most widespread software for the simulation of general multiphysics problems.</p> <p>Topics include:</p> <p><b>MATHEMATICAL MODELLING:</b> Mathematical analogies between conservation laws for heat, mass, electrons, ions and electric fields; Modes of microscopic transport: diffusion, convection, reaction/adsorption and their microscale modelling; From micro to macro: upscaling microscopic fluxes to the macroscale; the science behind engineering correlations; Extract general constitute laws for multiphysics fields: the most important ideas; Averaging of linear and non-linear equations; Bounds on transport properties for statistically homogeneous, periodic or percolating fields; effective conductivities of composites</p> <p><b>MICRO/NANOPHYSICS:</b> Coupling of mass, momentum or heat transport with electric fields: electrokinetic transport in bulk and at interfaces; Upscaling adsorption in porous media: application to hydrogen and CO<sub>2</sub> adsorption; How to analyse images, e.g. Scanning Electron Microscope images, to estimate transport properties in mesoporous materials</p> <p><b>NUMERICAL SIMULATIONS:</b> How to use commercial codes (e.g. Comsol or Fluent) to compute experimentally testable transport properties; basic approaches to the discretizations of coupled multiphysics problems</p> <p><b>APPLICATIONS:</b> use of the theory in practical case studies: battery electrodes, high-surface-area capacitors, hydrogen storage media, membranes for CO<sub>2</sub> sequestration, multicomponent/multiphase heat storage materials</p> |   |
| <b>Study Goals</b>                     | <p>After completion of the course, you will be able to:</p> <ul style="list-style-type: none"> <li>- Read confidently scientific or technical literature related to multiphysics transport in energy materials</li> <li>- Analyse experimental or simulation data for multiscale mesoporous materials to extract key transport properties such as, for example, multiphase heat/mass/electron conductivities</li> <li>- Develop computational models of engineering transport in energy materials and equipment, and evaluate the accuracy of meso- and macro-scale transport correlations used in these models</li> <li>- Learn how to be the go-to-person in multi-physics modelling in an industrial or scientific organisation, by learning the unique jargon and method of analysis of this field</li> <li>- Save time in your research by learning how to develop back-of-the-envelope calculations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | The course is based on lectures on theory and practice, and in-class discussion of practical case studies. Lectures will be complemented by a few assignments analysing particular technologies. There is no single textbook: the study material is taken from book chapters, research knowledge, or scientific articles and will be given by the instructor. Some books will be suggested by the instructor for the students interested in learning certain topics more in depth.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Assessment</b>                      | Assessment is based on a written exam and an assignment on a multiphysics topic agreed with the instructor. The assignment includes research on a specific topic and a presentation in front of the class. The exam carries 50% of the final grade.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Department</b>                      | ME Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact</b>                         | For information about the course, feel free to contact the course instructor Dr. Lorenzo Botto at <a href="mailto:l.botto@tudelft.nl">l.botto@tudelft.nl</a> (the instructor will be happy with sharing lecture notes from the past year with students interested in evaluating whether or not they should take the course)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| SEN1522                                 | Electricity and Gas: Market Design and Policy Issues                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5 |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Module Manager</b>                   | Prof.dr.ir. L.J. de Vries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                       | Dr. A.F. Correljé                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Responsible for assignments</b>      | Prof.dr.ir. L.J. de Vries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-responsible for assignments</b>   | Dr. A.F. Correljé                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/6/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                      | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>         | Economics of Infrastructures SPM2230 or EPA1233) or an equivalent introduction to micro economics is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                  | <p>Introduction, models of liberalization, network regulation</p> <p>Conceptual framework of market design</p> <p>Gas system &amp; actor description</p> <p>Electricity system &amp; actor description</p> <p>Long-term development of electricity markets</p> <p>Long-term development of gas markets</p> <p>Policy options for intervention in electricity and gas markets</p> <p>Perspectives on long-term, sustainable energy market design</p> <p>Energy market modeling</p> <p>Electricity market simulation game.</p>                                                                     |   |
| <b>Study Goals</b>                      | At the end of this course, students should be able to analyze energy policy issues from a technical, economic, and institutional perspective, to make informed choices regarding market design and policy options and to develop improvements to these options. In this course, the students technical, economic, legal and policy making skills are applied to the complex reality of regulation and market design in electricity and gas sectors.                                                                                                                                              |   |
| <b>Education Method</b>                 | The course makes use of interactive lectures. The reading materials are partly developed by ourselves and consist partly of relevant, recent policy-related documents. Quantitative modeling exercises are used to learn how electricity prices come about, the influence of policy instruments and renewable energy on these prices and to learn about investment decisions. An elaborate simulation game is used to apply this knowledge and gain insights in the dynamic, long-term development of an electricity market. Finally, an excursion provides in inside view in an energy company. |   |
| <b>Computer Use</b>                     | An exercise requires the development of a simulation model. Excel may be used for this purpose. In addition, an on-line market simulation game is part of this course.                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Literature and Study Materials</b>   | Readers and other reading materials such as scientific papers will be provided through the online learning environment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                       | Digital exam (70%), modeling exercise (20%) and simulation game (10%). Each of these elements must be passed with a grade of 5,75 or higher.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Permitted Materials during Tests</b> | None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Remarks</b>                          | This course overlaps in part with SET3055 and is therefore not available to M.Sc. SET students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Elective</b>                         | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Tags</b>                             | <p>Economics</p> <p>Energy</p> <p>Policy Analysis</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Targetgroup</b>                      | CoSEM Energy Track Students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Year</b>         | <b>2024/2025</b>                                                |
| <b>Organization</b> | <b>Electrical Engineering, Mathematics and Computer Science</b> |
| <b>Education</b>    | <b>Master Sustainable Energy Technology (SET)</b>               |

Large Scale Heat Systems 2024



| CEGM2006                               | Subsurface Storage for Energy, Water and Climate Applications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 10 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | Prof.dr. H. Hajibeygi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Module Manager</b>                  | Prof.dr. H. Hajibeygi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Module Manager</b>                  | Prof.dr.ir. M. Bakker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Instructor</b>                      | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Instructor</b>                      | Dr. B.M. van Breukelen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Instructor</b>                      | Dr. D.V. Voskov                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Instructor</b>                      | Prof.dr. S. Geiger                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| <b>Instructor</b>                      | Dr.ir. C.M. van der Graaf                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
| <b>Instructor</b>                      | Dr.ir. A.C. Dieudonné                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Instructor</b>                      | Dr. A.M.H. Pluymakers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | Diverse: lectures, computer practicals and lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
| <b>Course Contents</b>                 | The subsurface is increasingly used to store fluids associated with energy and for climate change mitigation. There are many potential technologies, and each has its place. In this module, the key technologies which are available and under-development are detailed, including key aspects of flow of fluids in the subsurface. These concepts are utilised to evaluate the technologies, such that we can efficiently use the subsurface. Students will move from broad concepts and how to assess the applicability (Learning Unit 1), through theoretical aspects of how the systems work (Learning Unit 2) to evaluation and design (Learning Unit 3).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Study Goals</b>                     | <p>After completion of this module the students are able:</p> <p>CO-SS-LO1. Explain how earth underground resources can be used and sustained in an environmentally responsible manner for cyclic storage of fluids and energy (ILO: 4, 10).</p> <p>CO-SS-LO2. Design a conceptual framework for a given storage project, considering the geology and geomechanics of the host rock and physics of fluids under dynamics (ILO: 1, 3, 4, 8, 11).</p> <p>CO-SS-LO3. Evaluate the performance of storage systems, including the influence of their design parameters and nearby systems (ILO: 4, 5, 6, 7, 9).</p> <p>CO-SS-LO4. Evaluate geochemical processes in order to evaluate the environmental performance of systems (ILO: 4, 5, 6, 7, 9).</p> <p>CO-SS-LO5. Weigh the various advantages and disadvantages of subsurface storage options against each other and propose an optimal solution (ILO: 3, 6, 9, 10).</p> <p>CO-SS-LO6. Communicate their findings clearly in an appropriate comprehensive manner for various technical and non-technical stakeholders (ILO: 2).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
| <b>Education Method</b>                | <p>Teaching and learning that encourages students to take responsibility for their learning process:</p> <p>Learning Unit 1: Subsurface storage technology<br/>- Lectures. Book chapters and handouts specifically relevant to each lecture should be studied beforehand. In addition to theory explanation, students will be engaged through exercises and quizzes.</p> <p>Learning Unit 2: Heat, solute and compressible fluid transport<br/>- Lectures on the transport of dissolved substances and heat in groundwater. Small scale experimental demonstrations are given during the lectures in which students participate.<br/>- Based on the scientific contents of the lectures, practical sessions will be performed to model the fundamental concepts behind successful storage of fluids in the subsurface across scales. These are mostly in the form of weekly computer modelling exercises using, for example, MODFLOW, MT3D, SEAWAT, DARTS, DARSim, or MRST. Python scripts will be used to generate input files, read output files, and analyse the results.<br/>- Group experiments in the lab will be performed to quantify physics of the stored fluids, e.g. H<sub>2</sub> and CO<sub>2</sub>.</p> <p>Learning Unit 3: Storage design and evaluation project<br/>- Project based learning. Students will form multi-disciplinary groups. Each group will work on a specific energy scenario and conceptualise, design and evaluate a storage system based on their scenario. Groups will report findings on a weekly basis to other groups in form of oral presentations. These presentations will focus on the key messages to different stakeholders. These outputs will be formatively assessed. The final designs and evaluations will be communicated in writing (in form of report) and orally (via a presentation).</p> |    |
| <b>Practical Guide</b>                 | <p>Learning Unit 1: Subsurface storage technology (1 ECTS)<br/>Link to learning objectives of the module:<br/>CO-SS: LO 1, LO 5</p> <p>The various technologies related to subsurface storage and the evaluation of subsurface storage are detailed, including the current state of the art and future research directions. Theoretical and quantitative methods to gain insights into technical and environmental performance are taught. Evaluation methodologies, such as exergy analysis and multi-criteria analysis, are outlined.</p> <p>Topics:</p> <ul style="list-style-type: none"> <li>- Context of subsurface storage: <ul style="list-style-type: none"> <li>o global overview of Climate &amp; Energy status</li> <li>o from production to storage to mitigation.</li> </ul> </li> <li>- Overview of the technologies and advancements <ul style="list-style-type: none"> <li>o Energy storage in porous media &amp; cavities (reservoirs and salt caverns):<br/> <ul style="list-style-type: none"> <li>• Borehole thermal energy storage</li> <li>• Aquifer thermal energy storage</li> <li>• Hydrogen storage</li> <li>• Compressed air energy storage</li> <li>• Supercritical CO<sub>2</sub> energy storage</li> </ul> </li> <li>o Novel storage topics <ul style="list-style-type: none"> <li>o CO<sub>2</sub> storage</li> <li>o Water storage (managed aquifer recharge (MAR))</li> </ul> </li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |    |

- Assessment methods
  - o Exergy Analyses
  - o Multi-criteria analysis
  - o Environmental performance
  - o Geomechanics

Learning Unit 2: Heat, solute and compressible fluid transport (6 ECTS)

Link to learning objectives of the module:

CO-SS: LO 2, LO 3, LO 4

The theoretical aspects of heat, solute and compressible flow in the subsurface are presented, including the flow of heat via conduction, advection and convection, the transport of dissolved solutes and the transport of compressible fluids (e.g. CO<sub>2</sub> and H<sub>2</sub>). Reactivity and the effect on transport, leading towards changes in properties or pollution are considered. The coupled behaviour of processes is detailed, e.g. the effect of temperature on physical properties and rate constants.

Topics:

- Solute transport in groundwater
- Heat transport in the surface and groundwater
- Coupled groundwater flow and concentration/heat transport
  - o density and viscosity dependency on salinity and temperature
- Reactive transport of contaminants
  - o Effect of temperature on rate constants.
  - o Mobilization, adsorption and resorption of trace elements due to flow/concentration/temperature
- Compressible fluid flow
  - o CO<sub>2</sub>
  - o H<sub>2</sub>
  - o Laboratory experiment (geochemistry): measuring CO<sub>2</sub> dissolution, pH change
- Subsurface flow
  - o Pore scale to km scale modelling
- Simulation methods
  - o Assignments and exercises using various open-source simulators: e.g. MODFLOW, MT3D, SEAWAT, PHT3D and/or PHAST, DARTS, DARSim, MRST
- Applications include seawater intrusion, managed aquifer recharge, aquifer thermal energy storage, CO<sub>2</sub> storage, contaminant plumes.

Learning Unit 3: Storage design and evaluation project (3 ECTS)

Link to learning objectives of the module:

CO-SS: LO 2, LO 3, LO 4, LO 5, LO 6

Students are presented with an energy scenario where they must evaluate the most appropriate storage concept and present a detailed design. The design must include predictions of performance, an evaluation of environmental impact and include analysis of uncertainties. The students must demonstrate the ability to select appropriate theory, modelling concepts and data, perform numerical analysis and critically evaluate the results to form conclusions. They must clearly and precisely communicate their results to their peers.

Topics:

- Scenario evaluation
- Application of theory and tools from other two units
- Critical evaluation of results
- Presentation of project via written technical report and presentation for different stakeholders (focus on non-technical users in the final presentation)

## Assessment

Formative assessment (feedback)

Learning Units 1 and 2

During the first two units weekly tasks have to be delivered. Tasks comprise written reports based on laboratory experiments, computer exercises and theoretical knowledge. In addition, class quizzes will be given.

The tasks will be formatively assessed on completeness and quality by involved lecturers. These build towards: (i) the portfolio, and (ii) the activities in Learning Unit 3.

Learning Unit 3

Groups will give weekly presentations on their progress and considering different stakeholders. The presentations will be formatively assessed on completeness and quality by involved lecturers, and peer feedback will be solicited. In addition, formative peer feedback within the group will encourage equal contribution.

Summative assessment (grading)

Portfolio of individual learning activities at the end of Learning Units 1&2

50% of final module grade.

Assessment based on:

Individual cumulative grade. Combination of (approximately) weekly assignments defined during lectures and workshops of Units 1 and 2. The students will have received formative comments after each assignment and can upgrade the assignments for the final grade.

Integrated module assessment at the end of Learning Unit 3

50% of final module grade.

Assessment based on:

Group report

Group presentation at the end of the project

Answering questions after the presentation

A 1-2 page(s) individual report on critical evaluation of all group studies (also to stimulate reading others group reports and actively ask questions during presentations and evaluate based on capacity, efficiency, safety, exergy, public perceptions, ).

A successful participation in the module, requires a minimum evaluation of 5.8 in all examinations

## Expected prior Knowledge

For all MSc students CE, ENV, AES.

## Academic Skills

Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.

## Literature & Study Materials

Learning Unit 1: Subsurface storage technology

Books:

- o The city of the future by BNA (Branchevereniging Nederlandse Architectenbureaus), publisher Blauwdruk (2019); and Stad x Klimaat Het gebouw als watermachine by BNA and TU Delft (2020) (in Dutch). The study is interactive available at <https://www.bna.nl/kennis/bna-onderzoek/stad-x-klimaat> which can be automatically translated by Chrome Browser to English, for international students. These books comprise and review a number of practical cases concerning integral ecoengineering solutions for socio-environmental problems occurring in the built environment.
- o Sustainable energy without the hot air by David Mackay: provides an overview of alternative energy options.
- o Carbon Dioxide Capture and Storage, by Intergovernmental Panel on Climate Change (UNEP): this open-access report provides guidelines for climate-driven actions related to subsurface storage of greenhouse gases, especially CO<sub>2</sub>.

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|                                        | <p>Specific literature in form of scientific papers and book chapters on diverse subjects of subsurface storage (energy and climate) related to techno-economical, experimental, modelling, computational and analytical techniques will be provided via BrightSpace. Experimental material for the CO<sub>2</sub>/H<sub>2</sub> storage practical, as well as for the sensor workshop will be provided during the practical.</p> <p>Learning Unit 2: Heat, solute and compressible fluid transport<br/> Manuals of the open-source software and the associated packages for the modelling software used.<br/> Assignments will be developed in the form of Jupyter Notebooks so that they are fully reproducible and documented.<br/> Selected papers on the modeling of ATEs, MAR and storage of compressible fluids.</p> <p>Learning Unit 3: Storage design and evaluation project<br/> Resources from Learning Units 1 and 2.<br/> Additional specific literature in form of scientific papers and book chapters on multi-criteria analysis and scenario analysis will be provided via BrightSpace</p> |
| <b>Judgement</b>                       | <p>See Assessment</p> <p>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br/> <a href="https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations">https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Permitted Materials during Exam</b> | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Collegerama</b>                     | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| RO47019                                | Intelligent Control Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. C. Della Santina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/0/4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>        | Fundamentals of Control theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | <p>Fundamentals of Coding in Python and MatLab</p> <p>Theory and application of intelligent control system i.e., the combination of machine learning and control theory.</p> <p>Topics:</p> <ul style="list-style-type: none"> <li>* Introduction to nonlinear control and ICS</li> <li>* Learning nonlinear dynamical systems</li> <li>* Sum of Basis functions</li> <li>* Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS</li> <li>* Symbolic regression</li> <li>* Gaussian processes</li> <li>* Reinforcement learning</li> <li>* Imitation learning</li> <li>* Adaptive control</li> <li>* Iterative learning control</li> </ul> <p>We will aim for a balance of practical intuition and theoretical understanding: we will discuss examples of applications and go deeper into the theory when feasible.</p>                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Study Goals</b>                     | <p>At the end of this course, you will be able to:</p> <p>LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc.</p> <p>LO2 Implement 'intelligent control' techniques in Python.</p> <p>LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem.</p> <p>LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Method</b>                | Lectures (recorded and in class) and one assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Relations</b>                | <p>Course title until 20/21 "Knowledge Based Control Systems"</p> <p>Course code until 21/22 "SC42050"</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>  | <p>Lecture notes: R. Babuska. Knowledge-Based Control Systems.</p> <p>Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | <p>The examination will be as follows:</p> <p>(i) practical assignment and (ii) written exam, open book.</p> <p>You must score at least 5.5 on both assignments to get a pass.</p> <p>In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a remote open book with several fraud prevention measures.</p> <p>In accordance with the master program regulations:</p> <ul style="list-style-type: none"> <li>* The students will have the opportunity to resit if they fail the written examination.</li> <li>* If your practical assignment is graded between 5 and 5.75, you will have two weeks to integrate the assessors feedback. If this is done satisfactorily, you can be rewarded with a mark that is not higher than 6/10. The exact instructions for the addition will be announced after the grading of the PA is concluded.</li> </ul>                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding course registration, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) exam registration is obligatory. More information regarding exam registration, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Tags</b>                            | <p>Artificial intelligence</p> <p>Complex Mathematics</p> <p>Intensive</p> <p>Linear Algebra</p> <p>Mathematics</p> <p>Modelling</p> <p>Programming</p> <p>Project</p> <p>Proofs</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Department</b>                      | ME Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Contact</b>                         | Cosimo Della Santina, <a href="mailto:C.DellaSantina@tudelft.nl">C.DellaSantina@tudelft.nl</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| SC42056                                 | Optimization for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 |
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| <b>Responsible Instructor</b>           | Prof.dr.ir. B.H.K. De Schutter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Expected prior knowledge</b>         | basic experience with Matlab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                  | <p>Essentially, almost all engineering problems are optimization problems. If a civil engineer designs a bridge, then one of the main objectives is to obtain the cheapest design or the design that can be implemented most rapidly, where of course several specifications and constraints such as size, strength, safety, etc. have to be taken into account. When developing a new type of engine, we look for the most economical design, the cheapest design, or the design with the highest performance. A process engineer wants a production unit to deliver a final product of maximal quality, with minimal expenditure of energy or with maximal output flow. When composing a portfolio, a financial engineer tries to maximize the expected profits, subject to the given risk constraints. So we encounter optimization problems in almost every engineering field.</p> <p>How can we solve such an optimization problem? That is the topic that will be addressed in this course. We will in particular discuss several classes of optimization problems and next focus on how to select most efficient numerical algorithms to solve a given optimization problem. The examples and case studies of this course are primarily oriented towards systems and control.</p> <p>More specifically, the following topics will be treated:</p> <ul style="list-style-type: none"> <li>* Linear Programming</li> <li>* Quadratic Programming</li> <li>* Nonlinear Optimization</li> <li>* Constraints in Nonlinear Optimization</li> <li>* Convex Optimization</li> <li>* Global Optimization</li> <li>* Matlab Optimization Toolbox</li> <li>* Multi-Objective Optimization</li> <li>* Integer Optimization</li> </ul> |   |
| <b>Study Goals</b>                      | <p>After successfully completing this course the students should be able to select the most efficient and best suited optimization algorithm for a given optimization problem.</p> <p>They should have insight into the basic operation of some of the most important and relevant optimization algorithms.</p> <p>They should be able to reduce the complexity of the problem using simplifications and/or reformulations so as to augment the efficiency of the solution approach.</p> <p>Finally, they should be able to solve the resulting optimization problem in practice.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Method</b>                 | lectures + 2 mandatory assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                       | <p>Attending the lectures is not mandatory, provided the recorded lectures and lecture materials are processed via self-study.</p> <p>Written exam (individual, fully closed-book, no calculators nor formulas sheets allowed, counts for 70% of the final marks) + 2 assignments (group assignments, assessed through written reports, count for 30% of the final marks).</p> <p>To successfully pass the course the final weighted average score for the exam and the assignments together should be at least 5.75.</p> <p>Important: Partial marks for the exam or the assignments do not carry over from one academic year to the next.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Permitted Materials during Tests</b> | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Enrolment / Application</b>          | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Department</b>                       | ME Department Delft Center for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact</b>                          | Questions are preferably asked during, before, or after the lectures, or via the Discussion forum on Brightspace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| SC42125                                | Model Predictive Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ing. S. Grammatico                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/0/4/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 3<br>4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | Model Predictive Control (MPC) is perhaps the most effective optimal control strategy for constrained dynamical systems. The basic concept of MPC is to exploit a dynamic model to forecast the system behavior, and optimize the forecast to determine the control input as the best decision at the current time. With emphasis on constrained linear systems, the course will present the theoretical fundamentals of MPC, such as Lyapunov stability, optimality and robustness, and its computational methods, such as quadratic programming.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <ul style="list-style-type: none"> <li>- Derive state prediction matrices from discrete-time linear models</li> <li>- Design the cost function, state and input constraints</li> <li>- Design MPC controllers with guaranteed recursive feasibility and asymptotic stability via appropriate terminal cost and constraint set</li> <li>- Formulate and solve constrained-linear-quadratic MPC problems via quadratic programming</li> <li>- Implement and simulate closed-loop systems controlled by MPC on Matlab</li> <li>- Design MPC controllers with integral action for reference tracking</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Method</b>                | Lectures, instructions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Assessment</b>                      | Final project and oral exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Tags</b>                            | Mathematics<br>Matlab<br>Optimalisation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Department</b>                      | ME Department Delft Center for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

| SC42130                                | Fault Diagnosis and Fault Tolerant Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R. Ferrari                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0.0.0.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Education Period</b>                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Exam Period</b>                     | 4<br>5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Expected prior knowledge</b>        | Control Systems Design, System Identification, Signal Processing, Statistics and Probability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Summary</b>                         | In critical systems such as transportation systems, power plants, distribution networks or life-support systems it is not enough to design a control system that meets the required performance levels. It is also of paramount importance to guarantee that such requirements, or a subset of them, are met also during non ideal conditions such as in the presence of physical failures or cyber attacks. Most of all, safety must be guaranteed at all times. This can be obtained by designing into the control system additional functionalities that will allow to detect the occurrence of faults or cyber attacks, and deploy adequate countermeasures, which may include the reconfiguration of the control system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Course Contents</b>                 | Faults and failures in dynamical systems; Cybersecurity and cyber attacks to Industrial Control Systems; Components and Services model; Availability and Reliability; FTA (Fault tree Analysis); FMEA (Failure Modes and Effects Analysis); Signal based methods for Diagnosis; Model based methods for Diagnosis; Residuals computation and evaluation; Learning of fault model; Fault Tolerant Control (Model Matching, Virtual Sensor and Actuator, Adaptive Control)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Study Goals</b>                     | <p>after the course the student should be able to:</p> <ol style="list-style-type: none"> <li>1) analyse the structure of a system and identify main components, the services provided by each and carry out a Fault Modes and Effects Analysis;</li> <li>2) analyse and model faults or cyber attacks that can occur in a given dynamical system;</li> <li>3) design an algorithm for detecting, isolate and identify them;</li> <li>4) design a policy for reconfiguring the control system in order to accommodate them.</li> <li>5) analyze the vulnerability of an industrial control system to cyber attacks and design prevention and detection algorithms</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Method</b>                | in person lectures, in-class exercises, homework                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>  | Recorded lectures, Slides, Example exercises with partial solutions, Suggested textbook:<br>Blanke, M., Kinnaert, M., Lunze, J., Staroswiecki, M., & Schröder, J. (2006). Diagnosis and fault-tolerant control (Vol. 691). Berlin: Springer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                      | <p>Assignments 30%, final exam (written) 70%.</p> <p>The grade for the assignments will be the average grade of all the assignments.</p> <p>To pass the course, the final grade must be a 5.75 or higher. The grade for the exam is not allowed to be lower than a 5.0.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Elective</b>                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Tags</b>                            | <p>Design</p> <p>Group work</p> <p>Mathematics</p> <p>Matlab</p> <p>Mechatronics</p> <p>Modelling</p> <p>Numeric Methods</p> <p>Process</p> <p>Signals and Systems</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Department</b>                      | ME Department Delft Center for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |

| SC42155                                | Modelling of Dynamical Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. R. Ferrari                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Dr. M. Mazo Espinosa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4.0.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>The course will cover the basics of modeling of both physical systems (in the form of ordinary differential equations), computing and scheduling systems (as finite state machines), and combinations thereof (as hybrid models).</p> <p>Physical systems modeling. The course will cover the basics of Lagrangian modelling, I.e. employing energy balances and the Euler-Lagrange equations to arrive to closed form state-space dynamical equations for system analysis and control design. Bond-graph modelling will be covered as a powerful tool for modelling engineering systems, especially when different physical domains are involved. The basic ideas of model reduction, a technique to reduce the complexity of dynamical models, may also be covered in the course. The fundamental differences and connections between continuous vs discrete time modelling will be discussed. Similarly, frequency domain modelling is also discussed with practical motivations for frequency vs time domain modelling.</p> <p>Finite State Models. A brief introduction to models traditionally employed for computation, e.g. finite state machines, automata, and petri-nets, will be provided. Particular attention will be given to the applicability of these models to provide high-level dynamic descriptions of discrete-event systems and for behaviour specifications of designs.</p> <p>Hybrid systems. The course will finalize with a short introduction to hybrid systems through the formalisms of: Hybrid automata and jump-flow systems, and presenting a brief number of interesting subclasses of hybrid systems, e.g. PWA systems, and timed -automata.</p> |   |
| <b>Study Goals</b>                     | <p>The purpose of the course is to introduce the students to basic concepts and results in physical modeling and the theory of nonlinear control systems.</p> <p>After successful completion of the course, the student is able to</p> <ul style="list-style-type: none"> <li>construct models of physical systems from the knowledge of the basic laws of physics (first principles)</li> <li>construct models of systems using the theory of bond graphs.</li> <li>construct models of systems using the Euler-Lagrange method (perform a model reduction for linear systems using a balanced realization)</li> <li>construct finite state models (automata, FSMs, Petri-Nets) of simple computing and scheduling systems</li> <li>analyze basic properties of finite state systems: blocking, determinism, and language expressed by a model.</li> <li>describe cyber-physical systems as hybrid dynamical models.</li> <li>select the appropriate modelling approach for a given problem.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Method</b>                | Lectures 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Literature and Study Materials</b>  | Lecture notes and additional reader                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Assessment</b>                      | <p>Assignments (35%) and a mini-project (65%). The grade for the assignments will be the average grade of all the assignments. To pass the course, the final grade must be a 5.75 or higher. The grade for the mini-project is not allowed to be lower than a 5.0.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Enrolment / Application</b>         | <p>Registration:</p> <p>To participate in this course, registration via MyTUDelft is not needed.</p> <p>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Tags</b>                            | <p>Embedded systems</p> <p>Group work</p> <p>Lineair Algebra</p> <p>Mathematics</p> <p>Matlab</p> <p>Mechatronics</p> <p>Modelling</p> <p>Numeric Methods</p> <p>Signals and Systems</p> <p>Stochastics</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Department</b>                      | ME Department Delft Center for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |



| SC42165                                | Machine Learning for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | A. Dabiri                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Instructor</b>                      | Dr.ing. R. van de Plas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Instructor</b>                      | Dr. L. Laurenti                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Expected prior knowledge</b>        | Linear algebra, basics of mathematical optimization, basics of probability theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Contents</b>                 | The course will lay the methodological groundwork for machine learning in signals, systems, and control contexts and corresponding application domains. The course covers three modules: supervised learning, unsupervised learning, and reinforcement learning. Throughout the course, students will delve into topics such as linear regression, logistic regression, classification, neural networks, Bayesian supervised learning, and learn about model assessment techniques. In addition, the course covers clustering and matrix factorization, as well as fundamentals of reinforcement learning and its connection with dynamic programming. Furthermore, students will gain insight into model-free RL control algorithms like Sarsa and Q-learning, and approximation methods in value space.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>After this course, the student should be able to</p> <ol style="list-style-type: none"> <li>1. Understand the theory of supervised learning</li> <li>2. Apply state-of-the-art supervised learning algorithms for control applications</li> <li>3. Understand basic unsupervised machine learning methods (e.g. clustering, principal component analysis)</li> <li>4. Understand the link between reinforcement learning and dynamic programming</li> <li>5. Apply RL algorithms like Q-learning or Sarsa for control applications</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Method</b>                | Lectures and computer exercises.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Computer Use</b>                    | Matlab , Python                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Literature and Study Materials</b>  | <p>Course material:<br/>Lecture notes are made available on Brightspace</p> <p>Textbook:</p> <p>Bishop, Christopher M., and Nasser M. Nasrabadi. Pattern recognition and machine learning. Vol. 4. No. 4. New York: springer, 2006.</p> <p>Sutton, R. S., Barto, A. G., Reinforcement Learning: An Introduction. The MIT Press, 2018.</p> <p>Bertsekas D., Lessons from AlphaZero for Optimal, Model Predictive, and Adaptive Control, Athena Scientific, 2022</p> <p>Li, S.E., Introduction to Reinforcement Learning. In: Reinforcement Learning for Sequential Decision and Optimal Control. Springer, Singapore, 2023.</p> <p>Hastie, T., Tibshirani, R., and Friedman, J., The Elements of Statistical Learning", 2nd Ed., Springer, New York, 2009.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | Written exam and assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Enrolment / Application</b>         | <p>Registration:<br/>To participate in this course, registration via MyTUDelft is not needed.<br/>More information regarding registration for courses, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses">https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses</a>.</p> <p>To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on <a href="https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams">https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams</a>.</p> <p>*in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, <a href="https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations">https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations</a>.</p> |   |
| <b>Department</b>                      | ME Department Delft Center for Systems and Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |

Dr. A.P. Afghari

|                   |                             |
|-------------------|-----------------------------|
| <b>Department</b> | Safety and Security Science |
| <b>Telephone</b>  | +31 15 27 83026             |
| <b>Room</b>       | 31.C1.020                   |

Dr.ir. D.J.N. Allaerts

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Wind Energy     |
| <b>Telephone</b>  | +31 15 27 81321 |

Dr. J. Alonso-Mora

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Learning & Autonomous Control  |
| <b>Telephone</b>  | +31 15 27 85489                |
| <b>Room</b>       | 34.F-2-100                     |

Prof.dr.ir. M. Bakker

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch    |
| <b>Department</b> | Surface and Groundwater Hydrology |
| <b>Telephone</b>  | +31 15 27 83714                   |
| <b>Room</b>       | 23.HG 4.92.1                      |

Prof.dr.ir. P. Bauer

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.          |
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 84654                         |
| <b>Room</b>       | 36.LB 03.600                            |

Dr.ir. R. van Bergem

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | Economie van Technologie en Innovatie |
| <b>Telephone</b>  | +31 15 27 84819                       |

Dr.ir. W.A.A.M. Bierbooms

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Wind Energy                    |
| <b>Telephone</b>  | +31 15 27 82097                |
| <b>Room</b>       | 62.5.21                        |

Dr.ir. L. van Biert

|                   |                                        |
|-------------------|----------------------------------------|
| <b>Department</b> | Ship Design, Production and Operations |
| <b>Telephone</b>  | +31 15 27 86479                        |
| <b>Room</b>       | 34.B-4-220                             |

Dr.ing. R.K. Bishnoi

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Computer Engineering |
| <b>Telephone</b>  | +31 15 27 86101      |
| <b>Room</b>       | 36.HB 09.140         |

Dr.ir. J.M. Bloemendal

|                   |                           |
|-------------------|---------------------------|
| <b>Department</b> | Water Systems Engineering |
| <b>Telephone</b>  | +31 15 27 89916           |
| <b>Room</b>       | 23.HG 4.82                |

Dipl.-Ing. Y. Blom

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 89733                    |

Prof.dr.ing. A.C. Boerstra

|                   |                                |
|-------------------|--------------------------------|
| <b>Department</b> | Environmental & Climate Design |
|-------------------|--------------------------------|

Dr. R.M.J. Bokel

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Bouwkunde                      |
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 88124                |
| <b>Room</b>       | 08.01WEST110                   |

Dr.ir. A.J.J. van den Boom

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Team Ton van den Boom          |
| <b>Telephone</b>  | +31 15 27 84052                |
| <b>Room</b>       | 34.C-3-180                     |

L. Botto

|                   |                              |
|-------------------|------------------------------|
| <b>Department</b> | Complexe Vloeistof Processen |
|-------------------|------------------------------|

Dr. W.G. Bouwman

|                   |                                               |
|-------------------|-----------------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen                |
| <b>Department</b> | RST/Neutron and Positron Methods in Materials |
| <b>Telephone</b>  | +31 15 27 85687                               |
| <b>Room</b>       | 50.02.01.240                                  |

Dr.ir. I. Bouwmans

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie en Industrie           |
| <b>Telephone</b>  | +31 15 27 84807                |
| <b>Room</b>       | 31.a3.240                      |

Dr. S.J. van Boxtel

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management                   |
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 81893                                  |
| <b>Room</b>       | 31.c0.090                                        |

Dr. S. Brancart

|                   |                        |
|-------------------|------------------------|
| <b>Department</b> | Structures & Materials |
| <b>Room</b>       | 08.01+.West.230        |

Dr.ir. W.P. Breugem

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Multi Phase Systems            |
| <b>Telephone</b>  | +31 15 27 88663                |
| <b>Room</b>       | 34.B-K-170                     |

Dr. B.M. van Breukelen

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch    |
| <b>Department</b> | Surface and Groundwater Hydrology |
| <b>Telephone</b>  | +31 15 27 85227                   |
| <b>Room</b>       | 23.HG 4.49                        |

Ir. S. Broersma

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Bouwkunde                      |
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 87926                |
| <b>Room</b>       | 08.01WEST130                   |

Prof.dr. D.F. Bruhn

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Reservoir Engineering          |
| <b>Telephone</b>  | +31 15 27 82875                |
| <b>Room</b>       | 23.HG 3.20                     |

Ir. B.J.E. de Bruin

|                   |                    |
|-------------------|--------------------|
| <b>Department</b> | Cognitive Robotics |
| <b>Department</b> | Cognitive Robotics |

Dr.ir. K. Bruninx

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Energie en Industrie |
| <b>Telephone</b>  | +31 15 27 86556      |

Dr. J.G. Buijnsters

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Micro and Nano Engineering     |
| <b>Telephone</b>  | +31 15 27 85396                |
| <b>Room</b>       | 34.G-1-440                     |

Dr. T.E. Burdyny

|                   |                                                   |
|-------------------|---------------------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen                    |
| <b>Department</b> | ChemE/Materials for Energy Conversion and Storage |

K. Bölat

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Intelligent Electrical Power Grids |
| <b>Telephone</b>  | +31 15 27 85483                    |

Dr. G.R. Chandra Mouli

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.          |
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 81898                         |
| <b>Room</b>       | 36.LB 03.670                            |

Dr.ir. E.J.L. Chappin

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie en Industrie           |
| <b>Telephone</b>  | +31 15 27 83410                |
| <b>Room</b>       | 31.A3.280                      |

Dr. A.F. Correljé

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management        |
| <b>Department</b> | Economie van Technologie en Innovatie |
| <b>Telephone</b>  | +31 15 27 87924                       |

Dr.ir. S. van Cranenburgh

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Transport en Logistiek         |
| <b>Telephone</b>  | +31 15 27 86957                |
| <b>Room</b>       | 31.A3.210                      |

Dr. J.L. Cremer

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Intelligent Electrical Power Grids |
| <b>Telephone</b>  | +31 15 27 86339                    |

Dr. L.L. Cutz IJchajchal

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Large Scale Energy Storage     |
| <b>Telephone</b>  | +31 15 27 85797                |
| <b>Room</b>       | 34b.K-1-140                    |

Dr. M. Cvetkovic

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Intelligent Electrical Power Grids |
| <b>Telephone</b>  | +31 15 27 87935                    |
| <b>Room</b>       | 36.LB 03.190                       |

---

A. Dabiri

---

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Team Azita Dabiri |
| <b>Telephone</b>  | +31 15 27 82259   |

---

Dr. A. Daniilidis

---

|                   |                       |
|-------------------|-----------------------|
| <b>Department</b> | Reservoir Engineering |
| <b>Telephone</b>  | +31 15 27 81553       |
| <b>Room</b>       | 23.HG 3.30            |

---

Dr. J.H.G. Dauwels

---

|                   |                           |
|-------------------|---------------------------|
| <b>Department</b> | Signal Processing Systems |
|-------------------|---------------------------|

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F. De Domenico

---

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Flight Performance and Propulsion |
| <b>Telephone</b>  | +31 15 27 84205                   |

---

Prof.dr.ir. B.H.K. De Schutter

---

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng       |
| <b>Department</b> | Delft Center for Systems and Control |
| <b>Telephone</b>  | +31 15 27 85113                      |

---

Dr.ir. D.A.M. De Tavernier

---

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Arts & Crafts   |
| <b>Telephone</b>  | +31 15 27 82606 |

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Wind Energy                    |
| <b>Telephone</b>  | +31 15 27 82606                |

---

Dr. R. Delfos

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Large Scale Energy Storage     |
| <b>Telephone</b>  | +31 15 27 82963                |
| <b>Room</b>       | 34b.K-0-270                    |

---

Dr. C. Della Santina

---

|                   |                               |
|-------------------|-------------------------------|
| <b>Department</b> | Learning & Autonomous Control |
| <b>Telephone</b>  | +31 15 27 81630               |
| <b>Room</b>       | 34.F-2-320                    |

---

H. Dialani

---

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | ESP LAB         |
| <b>Telephone</b>  | +31 15 27 84699 |

---

Dr.ir. A.C. Dieudonné

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Telephone</b>  | +31 15 27 86134                |
| <b>Room</b>       | 23.KG 00.490                   |

---

Dr.ir. N.H. van Dijk

---

|                   |                                                 |
|-------------------|-------------------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen                  |
| <b>Department</b> | RST/Fundamental Aspects of Materials and Energy |
| <b>Telephone</b>  | +31 15 27 86775                                 |

---

Prof.dr.ir. A.A.J.F. van den Dobbels

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Bouwkunde                      |
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 83563                |

---

---

Dr. J. Dong

---

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.          |
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 88115                         |
| <b>Room</b>       | 36.LB 03.630                            |

---

R.M. Ensing

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Intelligent Vehicles           |
| <b>Telephone</b>  | +31 15 27 89767                |
| <b>Room</b>       | 34.E-0-200                     |

---

Dr. H.B. Eral

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Complexe Vloeistof Processen   |
| <b>Telephone</b>  | +31 15 27 86715                |
| <b>Room</b>       | 34b.K-0-250                    |

---

T. Famprakis

---

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | RST/Storage of Electrochemical Energy |
|-------------------|---------------------------------------|

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|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | RST/Storage of Electrochemical Energy |
|-------------------|---------------------------------------|

---

Dr. R. Ferrari

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Team Riccardo Ferrari          |
| <b>Telephone</b>  | +31 15 27 83519                |
| <b>Room</b>       | 34.C-2-260                     |

---

A.R. Gammon

---

|                   |                                 |
|-------------------|---------------------------------|
| <b>Department</b> | Ethiek & Filosofie van Techniek |
| <b>Telephone</b>  | +31 15 27 81387                 |

---

Prof.dr. A. Gangoli Rao

---

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn    |
| <b>Department</b> | Flight Performance and Propulsion |
| <b>Telephone</b>  | +31 15 27 83833                   |
| <b>Room</b>       | 62.7.13                           |

---

Prof.dr. D. Gavrilă

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Intelligent Vehicles           |
| <b>Telephone</b>  | +31 15 27 88504                |
| <b>Room</b>       | 34.E-0-330                     |

---

Prof.dr. S. Geiger

---

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Applied Geology |
| <b>Telephone</b>  | +31 15 27 82203 |
| <b>Room</b>       | 23.HG 3.22      |

---

Dr. M. Ghaffarian Niasar

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform. |
| <b>Department</b> | High Voltage Technology Group  |
| <b>Telephone</b>  | +31 15 27 82764                |
| <b>Room</b>       | 36.LB 03.620                   |

---

Dr. A. Giga

---

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Delft Centre for Entrepreneurship |
| <b>Telephone</b>  | +31 15 27 89845                   |

---

Prof.dr.ir. E.L.V. Goetheer

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Energy Technology |
|-------------------|-------------------|

Prof.dr. I.M.F. Gordon

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 89288                    |

Dr.ir. C.M. van der Graaf

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Resource Engineering |
|-------------------|----------------------|

Dr.ing. S. Grammatico

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Team Sergio Grammatico         |
| <b>Telephone</b>  | +31 15 27 83593                |

Dr. J.B. van Grunsven

|                   |                                 |
|-------------------|---------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management  |
| <b>Department</b> | Ethiek & Filosofie van Techniek |
| <b>Telephone</b>  | +31 15 27 84715                 |

Prof.dr. H. Hajibeygi

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Reservoir Engineering          |
| <b>Telephone</b>  | +31 15 27 88482                |
| <b>Room</b>       | 23.HG 3.25                     |

P. Harish

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Delft Centre for Entrepreneurship |
|-------------------|-----------------------------------|

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Delft Centre for Entrepreneurship |
|-------------------|-----------------------------------|

Dr.ir. R.M. Hartkamp

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Complexe Vloeistof Processen   |
| <b>Telephone</b>  | +31 15 27 86674                |

Dr. L. Hartmann

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management    |
| <b>Department</b> | Delft Centre for Entrepreneurship |
| <b>Telephone</b>  | +31 15 27 88073                   |
| <b>Room</b>       | 31.C2.140                         |

J.W. Haverkort

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Large Scale Energy Storage     |
| <b>Telephone</b>  | +31 15 27 86651                |
| <b>Room</b>       | 34b.K-0-260                    |

Dr.ir. P.W. Heijnen

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie en Industrie           |
| <b>Telephone</b>  | +31 15 27 88072                |
| <b>Room</b>       | 31.a3.240                      |

Prof.dr.ir. J. Hellendoorn

|                   |                    |
|-------------------|--------------------|
| <b>Department</b> | Cognitive Robotics |
| <b>Telephone</b>  | +31 15 27 89007    |
| <b>Room</b>       | 34.F-2-440         |

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | College van Bestuur            |
| <b>Telephone</b>  | +31 15 27 89007                |

|             |            |
|-------------|------------|
| <b>Room</b> | 34.F-2-440 |
|-------------|------------|

Prof.dr.ir. S. Hickel

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Aerodynamics                   |
| <b>Telephone</b>  | +31 15 27 89570                |
| <b>Room</b>       | 64.HSL 0.32                    |

Dr.ir. O.A.C. Hoes

|                   |                                      |
|-------------------|--------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch       |
| <b>Department</b> | Water Systems Monitoring & Modelling |
| <b>Telephone</b>  | +31 15 27 84772                      |
| <b>Room</b>       | 23.HG 4.90                           |

Drs. G. Hoezen

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management                   |
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 82915                                  |
| <b>Room</b>       | 31.C0.110                                        |

Drs. R.J. Hoffmann

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 81549                                  |
| <b>Unit</b>       | Techniek, Bestuur & Management                   |
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 81549                                  |

Dr.ir. W.J. Hogendoorn

|                   |                              |
|-------------------|------------------------------|
| <b>Department</b> | Complexe Vloeistof Processen |
| <b>Telephone</b>  | +31 15 27 85711              |
| <b>Room</b>       | 34.F-1-490                   |
| <b>Department</b> | Multi Phase Systems          |
| <b>Telephone</b>  | +31 15 27 85711              |
| <b>Room</b>       | 34.F-1-490                   |

Dr.ir. M.F.M. Hoogreef

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn    |
| <b>Department</b> | Flight Performance and Propulsion |
| <b>Telephone</b>  | +31 15 27 85269                   |
| <b>Room</b>       | 62.7.20                           |

Dr. F.L. Hooimeijer

|                   |                                     |
|-------------------|-------------------------------------|
| <b>Unit</b>       | Bouwkunde                           |
| <b>Department</b> | Environmental Technology and Design |
| <b>Telephone</b>  | +31 15 27 89638                     |

Prof.dr. K. Hooman

|                   |                                |
|-------------------|--------------------------------|
| <b>Department</b> | Heat Transformation Technology |
|-------------------|--------------------------------|

Dr. A.J. Houtepen

|                   |                                 |
|-------------------|---------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen  |
| <b>Department</b> | ChemE/Opto-electronic Materials |
| <b>Telephone</b>  | +31 15 27 82157                 |
| <b>Room</b>       | 58.D1.140                       |

Dr.ir. P. Ibarra Gonzalez

|                   |                       |
|-------------------|-----------------------|
| <b>Department</b> | Energie en Industrie  |
| <b>Department</b> | Energie and Industrie |



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Prof.dr.ir. O. Isabella

---

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 81947                    |
| <b>Room</b>       | 36.LB 03.430                       |

---

Dr. L.C.M. Itard

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Bouwkunde                      |
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 86341                |
| <b>Room</b>       | 08.01WEST540                   |

---

Dr.ir. A. Jarquin Laguna

---

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch    |
| <b>Department</b> | Offshore and Dredging Engineering |
| <b>Telephone</b>  | +31 15 27 83297                   |

---

S.F. Johnson

---

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management                   |
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 82793                                  |

---

Prof.dr.ir. W. de Jong

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Large Scale Energy Storage     |
| <b>Telephone</b>  | +31 15 27 89476                |

---

A.I. Joseph Thaddeus

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 81533                |

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|                   |                                |
|-------------------|--------------------------------|
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 81533                |

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Dr. O. Kammouh

---

|                   |              |
|-------------------|--------------|
| <b>Department</b> | Systeemkunde |
|-------------------|--------------|

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Dr. L.M. Kamp

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie en Industrie           |
| <b>Telephone</b>  | +31 15 27 86488                |

---

Dr. E.M. Kelder

---

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen        |
| <b>Department</b> | RST/Storage of Electrochemical Energy |
| <b>Telephone</b>  | +31 15 27 83262                       |

---

G. Kelesidis

---

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Flight Performance and Propulsion |
|-------------------|-----------------------------------|

---

H. Khodaei

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|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management    |
| <b>Department</b> | Delft Centre for Entrepreneurship |
| <b>Telephone</b>  | +31 15 27 82759                   |
| <b>Room</b>       | 31.C2.130                         |

---

Prof.dr.ir. A.A. Kiss

---

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | ChemE/Process Systems Engineering |
| <b>Room</b>       | 58.D2.160                         |

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Dr.ir. R. Kleerebezem

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen |
| <b>Department</b> | BT/Milieubiotechnologie        |
| <b>Telephone</b>  | +31 15 27 81091                |

M. Kolagar Daronkola

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Delft Centre for Entrepreneurship |
|-------------------|-----------------------------------|

Dr. R. Kortlever

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Large Scale Energy Storage     |
| <b>Telephone</b>  | +31 15 27 86640                |
| <b>Room</b>       | 34b.K-1-150                    |

Dr. I. Langella

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Flight Performance and Propulsion |
| <b>Telephone</b>  | +31 15 27 89591                   |
| <b>Room</b>       | 62.7.19                           |

Drs. D.W. Laponder

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 89799                                  |

Dr. L. Laurenti

|                   |                    |
|-------------------|--------------------|
| <b>Department</b> | Team Luca Laurenti |
| <b>Telephone</b>  | +31 15 27 81822    |

Dr. A. Leki

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Intelligent Electrical Power Grids |
| <b>Telephone</b>  | +31 15 27 82461                    |
| <b>Room</b>       | 36.LB 03.210                       |

Drs. K.I. van der Linden

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management                   |
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 86384                                  |
| <b>Room</b>       | 31.c0.140                                        |

Dr. F. Lombardi

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Energie en Industrie |
| <b>Telephone</b>  | +31 15 27 81809      |

Dr.ir. M.C. Lugten

|                   |                                     |
|-------------------|-------------------------------------|
| <b>Department</b> | Environmental Technology and Design |
| <b>Department</b> | Environmental & Climate Design      |

Prof.dr.ir. Z. Lukszo

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie en Industrie           |
| <b>Telephone</b>  | +31 15 27 81147                |
| <b>Room</b>       | 31.B3.030                      |

Dr. A. Luna Navarro

|                   |                              |
|-------------------|------------------------------|
| <b>Department</b> | Building Design & Technology |
|-------------------|------------------------------|

Dr. G. Marangoni

|                   |                |
|-------------------|----------------|
| <b>Department</b> | Beleidsanalyse |
|-------------------|----------------|

|                  |                 |
|------------------|-----------------|
| <b>Telephone</b> | +31 15 27 81973 |
|------------------|-----------------|

## Dr. M. Mazo Espinosa

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Team Manuel Mazo Jr            |
| <b>Telephone</b>  | +31 15 27 88131                |
| <b>Room</b>       | 34.C-2-250                     |

## Dr. L. Mazzarella

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 88885                    |
| <b>Room</b>       | 36.LB 03.500                       |

## D. Mehta

|                   |                           |
|-------------------|---------------------------|
| <b>Department</b> | Water Systems Engineering |
| <b>Telephone</b>  | +31 15 27 83583           |

## Dr. A. Meijer

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Bouwkunde                      |
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 85658                |
| <b>Room</b>       | 08.01+.West.020                |

## Dr. D. Modesti

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Aerodynamics    |
| <b>Telephone</b>  | +31 15 27 83364 |

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|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Aerodynamics    |
| <b>Telephone</b>  | +31 15 27 83364 |

## Dr. M.A. Mosteiro Romero

|                   |                                |
|-------------------|--------------------------------|
| <b>Department</b> | Environmental & Climate Design |
|-------------------|--------------------------------|

## Dr. O. Moulτος

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Engineering Thermodynamics     |
| <b>Telephone</b>  | +31 15 27 81307                |
| <b>Room</b>       | 34b.K-0-250                    |

## N. Mouter

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Transport en Logistiek         |
| <b>Telephone</b>  | +31 15 27 89748                |
| <b>Room</b>       | 31.A3.060                      |

## Prof.dr. F.M. Mulder

|                   |                                                   |
|-------------------|---------------------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen                    |
| <b>Department</b> | ChemE/Materials for Energy Conversion and Storage |
| <b>Telephone</b>  | +31 15 27 85037                                   |
| <b>Room</b>       | 58.D1.320                                         |

## G.O. Ndubuisi

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | Economie van Technologie en Innovatie |
| <b>Telephone</b>  | +31 15 27 88895                       |

## Dr.ir. I. Nikolic

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Systeemkunde                   |
| <b>Telephone</b>  | +31 15 27 81135                |
| <b>Room</b>       | 31.B1.050                      |

Prof.dr.ir. H.J. Noorman

|                   |                           |
|-------------------|---------------------------|
| <b>Department</b> | BT/Bioprocess Engineering |
|-------------------|---------------------------|

Dr. F. Oliviero

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn    |
| <b>Department</b> | Flight Performance and Propulsion |
| <b>Telephone</b>  | +31 15 27 89045                   |

H.J.M. Olsthoorn

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform. |
| <b>Department</b> | ESP LAB                        |
| <b>Telephone</b>  | +31 15 27 86163                |
| <b>Room</b>       | 36.LB 02.620                   |

Prof.dr.ir. J.R. van Ommen

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen        |
| <b>Department</b> | ChemE/Product and Process Engineering |
| <b>Telephone</b>  | +31 15 27 82133                       |
| <b>Room</b>       | 58.F2.390                             |

Prof.dr. M. Overend

|                   |                        |
|-------------------|------------------------|
| <b>Department</b> | Structures & Materials |
|-------------------|------------------------|

Prof.dr. P. Palensky

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform. |
| <b>Department</b> | Electrical Sustainable Energy  |
| <b>Telephone</b>  | +31 15 27 88341                |
| <b>Room</b>       | 36.LB 03.230                   |

J. Palomo Jiménez

|                   |                             |
|-------------------|-----------------------------|
| <b>Department</b> | ChemE/Catalysis Engineering |
|-------------------|-----------------------------|

Prof.dr. R. Pecnik

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Energy Technology |
| <b>Telephone</b>  | +31 15 27 89153   |
| <b>Room</b>       | 34b.K-1-220       |

Dr.ir. J.W.R. Peeters

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Energy Technology              |
| <b>Telephone</b>  | +31 15 27 86628                |
| <b>Room</b>       | 34b.K-1-150                    |

Dr. S.J. Pfenninger-Lee

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Energie en Industrie |
| <b>Telephone</b>  | +31 15 27 86336      |
| <b>Room</b>       | 31.A3.200            |

Dr.ing. R. van de Plas

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Team Raf Van de Plas           |
| <b>Telephone</b>  | +31 15 27 84106                |
| <b>Room</b>       | 34.C-1-260                     |

Dr. A.M.H. Pluymakers

|                   |                                     |
|-------------------|-------------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch      |
| <b>Department</b> | Applied Geophysics and Petrophysics |
| <b>Telephone</b>  | +31 15 27 87599                     |
| <b>Room</b>       | 23.HG 3.30                          |

Prof.dr.ir. C. Poelma

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Process and Energy             |
| <b>Telephone</b>  | +31 15 27 82620                |
| <b>Room</b>       | 34b.K-0-120                    |

Drs. P.C. Post

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management                   |
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 83175                                  |
| <b>Room</b>       | 31.c0.040                                        |

Dr.ir. I.W.M. Pothof

|                   |                           |
|-------------------|---------------------------|
| <b>Department</b> | Water Systems Engineering |
|-------------------|---------------------------|

Dr.ir. M.J.B.M. Pourquie

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Fluid Mechanics                |
| <b>Telephone</b>  | +31 15 27 82997                |
| <b>Room</b>       | 34.F-1-600                     |

Prof.dr. A. Purushothaman Vellayani

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Energy Technology |
| <b>Telephone</b>  | +31 15 27 83550   |
| <b>Room</b>       | 34b.J-0-900       |

Dr. Z. Qin

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.          |
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 86584                         |
| <b>Room</b>       | 36.LB 03.640                            |

L. Qu

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Bouwkunde                         |
| <b>Department</b> | Ruimtelijke Planning en Strategie |
| <b>Telephone</b>  | +31 15 27 84079                   |

Dr.ir. J.N. Quist

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie en Industrie           |
| <b>Telephone</b>  | +31 15 27 85584                |

Dr. D. Ragni

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Wind Energy                    |
| <b>Telephone</b>  | +31 15 27 82053                |

M. Ramdin

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Engineering Thermodynamics     |
| <b>Room</b>       | 34b.K-1-290                    |

Dr.ir. L.M. Ramirez Elizondo

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.          |
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 81848                         |
| <b>Room</b>       | 36.LB 03.490                            |

M. Rana

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 84891                    |

## A. Rasooli

|                   |                                |
|-------------------|--------------------------------|
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 87383                |

## S. Renes

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | Economie van Technologie en Innovatie |
| <b>Telephone</b>  | +31 15 27 87566                       |
| <b>Room</b>       | 31.C3.090                             |

## S.A. Rivera Iunnissi

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 88176                         |

## Dr.ir. G. la Rocca

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn    |
| <b>Department</b> | Flight Performance and Propulsion |
| <b>Telephone</b>  | +31 15 27 85384                   |
| <b>Room</b>       | 62.7.21                           |

## Dr.ir. J.L. Rueda Torres

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Intelligent Electrical Power Grids |
| <b>Telephone</b>  | +31 15 27 86239                    |

## F.S. Saitta

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 89024                    |

## M. Sand

|                   |                                 |
|-------------------|---------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management  |
| <b>Department</b> | Ethiek & Filosofie van Techniek |
| <b>Telephone</b>  | +31 15 27 87885                 |
| <b>Room</b>       | 31.B4.250                       |

## Dr.ir. R. Santbergen

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 84425                    |
| <b>Room</b>       | 36.LB 03.500                       |

## Dr.ing. R. Schmehl

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Wind Energy                    |
| <b>Telephone</b>  | +31 15 27 85318                |
| <b>Room</b>       | 62.5.14                        |

## J. Schmutzler

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Delft Centre for Entrepreneurship |
| <b>Department</b> | Delft Centre for Entrepreneurship |

## Dr.ing. V.E. Scholten

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management    |
| <b>Department</b> | Delft Centre for Entrepreneurship |
| <b>Telephone</b>  | +31 15 27 89596                   |
| <b>Room</b>       | 31.C2.070                         |

## Dr. E. Schröder

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management        |
| <b>Department</b> | Economie van Technologie en Innovatie |
| <b>Telephone</b>  | +31 15 27 84273                       |
| <b>Room</b>       | 31.C3.130                             |

---

Dr. H. Schut

---

|                   |                                               |
|-------------------|-----------------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen                |
| <b>Department</b> | RST/Neutron and Positron Methods in Materials |
| <b>Telephone</b>  | +31 15 27 81961                               |

---

M.A. Sharifi Kolarijani

---

|                   |                              |
|-------------------|------------------------------|
| <b>Department</b> | Team Amin Sharifi Kolarijani |
| <b>Telephone</b>  | +31 15 27 85331              |
| <b>Room</b>       | 34.C-3-200                   |

---

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Team Peyman Mohajerin Esfahani |
| <b>Telephone</b>  | +31 15 27 85331                |
| <b>Room</b>       | 34.C-3-200                     |

---

W. Shi

---

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 88831                         |

---

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|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.          |
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 88831                         |

---

B. Shyrokau

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Intelligent Vehicles           |
| <b>Telephone</b>  | +31 15 27 84536                |
| <b>Room</b>       | 34.E-0-220                     |

---

Dr.ir. C.J. Simao Ferreira

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Wind Energy                    |
| <b>Telephone</b>  | +31 15 27 82073                |
| <b>Room</b>       | 62.5.22                        |

---

Dr. P. Simões Costa

---

|                   |                   |
|-------------------|-------------------|
| <b>Department</b> | Energy Technology |
|-------------------|-------------------|

---

Dr.ir. T. Sinnige

---

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn    |
| <b>Department</b> | Flight Performance and Propulsion |
| <b>Telephone</b>  | +31 15 27 89550                   |
| <b>Room</b>       | 62.7.08                           |

---

Prof.dr.ir. A.H.M. Smets

---

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 88739                    |
| <b>Room</b>       | 36.LB 03.420                       |

---

I.M. Splinter

---

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | ESP LAB         |
| <b>Telephone</b>  | +31 15 27 89190 |

---

Dr.ir. W.H. van der Spoel

---

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Bouwkunde                      |
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 83386                |

---

Ir. A.C.M. van der Stap

---

|             |                                |
|-------------|--------------------------------|
| <b>Unit</b> | Civiele Techniek & Geowetensch |
|-------------|--------------------------------|

---

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Offshore Engineering |
| <b>Room</b>       | 23.HG 3.48           |

#### Dr. A.I. Stefanov

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Intelligent Electrical Power Grids |
| <b>Telephone</b>  | +31 15 27 81528                    |
| <b>Room</b>       | 36.LB 03.180                       |

#### Dr. S.T.H. Storm

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management        |
| <b>Department</b> | Economie van Technologie en Innovatie |
| <b>Telephone</b>  | +31 15 27 83548                       |
| <b>Room</b>       | 31.c2.160                             |

#### Dr.ir. A. Straub

|                   |                                  |
|-------------------|----------------------------------|
| <b>Unit</b>       | Bouwkunde                        |
| <b>Department</b> | Design & Construction Management |
| <b>Telephone</b>  | +31 15 27 82769                  |

#### R. Stöllinger

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | Economie van Technologie en Innovatie |
| <b>Telephone</b>  | +31 15 27 88583                       |

#### Dr. R.A.C.M.M. van Swaaij

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 87259                    |
| <b>Room</b>       | 36.LB 03.410                       |

#### Drs. M.A. Swennen

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management                   |
| <b>Department</b> | Instituut voor Talen en Academische Vaardigheden |
| <b>Telephone</b>  | +31 15 27 82912                                  |
| <b>Room</b>       | 31.C0.160                                        |

#### Dr. F. Taruffi

|                   |             |
|-------------------|-------------|
| <b>Department</b> | Wind Energy |
|-------------------|-------------|

#### Prof.dr.ir. M.J. Tenpierik

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Bouwkunde                      |
| <b>Department</b> | Environmental & Climate Design |
| <b>Telephone</b>  | +31 15 27 84411                |
| <b>Room</b>       | 08.01WEST130                   |

#### Prof.dr. D.A. von Terzi

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Wind Energy     |
| <b>Telephone</b>  | +31 15 27 83379 |

#### Dr. B.P. Tighe

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Multi Phase Systems            |
| <b>Telephone</b>  | +31 15 27 81103                |
| <b>Room</b>       | 34b.K-1-250                    |

#### Dr. S.H. Tindemans

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Intelligent Electrical Power Grids |
| <b>Telephone</b>  | +31 15 27 84487                    |
| <b>Room</b>       | 36.LB 03.190                       |



G. Tran

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | Economie van Technologie en Innovatie |
| <b>Telephone</b>  | +31 15 27 89651                       |

Dr. M. Turrin

|                   |                      |
|-------------------|----------------------|
| <b>Unit</b>       | Bouwkunde            |
| <b>Department</b> | Digital Technologies |
| <b>Telephone</b>  | +31 15 27 82390      |
| <b>Room</b>       | 08.01.West.120       |

Prof.dr.ir. A. Urakawa

|                   |                             |
|-------------------|-----------------------------|
| <b>Department</b> | ChemE/Catalysis Engineering |
| <b>Telephone</b>  | +31 15 27 89402             |
| <b>Room</b>       | 58.E2.100                   |

Prof.ir. P.T.M. Vaessen

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform. |
| <b>Department</b> | High Voltage Technology Group  |
| <b>Telephone</b>  | +31 15 27 88833                |
| <b>Room</b>       | 36.LB 03.650                   |

Dr.ir. H. Vahedi

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>Department</b> | DC systems, Energy conversion & Storage |
| <b>Telephone</b>  | +31 15 27 89086                         |

Prof.dr. P.J. Vardon

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Telephone</b>  | +31 15 27 81456                |
| <b>Room</b>       | 23.KG 00.490                   |

A. Vasileiadis

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | RST/Storage of Electrochemical Energy |
| <b>Department</b> | RST/Storage of Electrochemical Energy |

Dr. P.P. Vergara Barrios

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Intelligent Electrical Power Grids |
| <b>Telephone</b>  | +31 15 27 85478                    |

D.A. Vermaas

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen |
| <b>Department</b> | ChemE/Transport Phenomena      |
| <b>Telephone</b>  | +31 15 27 89276                |
| <b>Room</b>       | 58.F2.210                      |

Prof.dr.ir. A.C. Viré

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Flow Physics and Technology    |
| <b>Telephone</b>  | +31 15 27 81385                |
| <b>Room</b>       | 62.5.19                        |

Dr.ir. R. Vismara

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 88902                    |
| <b>Room</b>       | 36.LB 03.460                       |

Prof.dr.ir. T.J.H. Vlugt

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Engineering Thermodynamics     |
| <b>Telephone</b>  | +31 15 27 87551                |

|             |             |
|-------------|-------------|
| <b>Room</b> | 34b.K-0-150 |
|-------------|-------------|

Dr. M.R. Vogt

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 87032                    |

Dr. D.V. Voskov

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Reservoir Engineering          |
| <b>Telephone</b>  | +31 15 27 87829                |
| <b>Room</b>       | 23.HG 3.28                     |

Prof.dr.ir. L.J. de Vries

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie en Industrie           |
| <b>Telephone</b>  | +31 15 27 81137                |
| <b>Room</b>       | 31.B3.030                      |

M. Wackers

Dr.ir. M. Wagemaker

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen        |
| <b>Department</b> | RST/Storage of Electrochemical Energy |
| <b>Telephone</b>  | +31 15 27 83800                       |
| <b>Room</b>       | 50.02.01.030                          |

Dr. C.N. van der Wal

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Systeemkunde    |
| <b>Telephone</b>  | +31 15 27 89941 |
| <b>Room</b>       | 31.B1.030       |

C. Wallmeier

|                   |                       |
|-------------------|-----------------------|
| <b>Department</b> | Reservoir Engineering |
|-------------------|-----------------------|

Dr. X. Wang

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Department</b> | RST/Storage of Electrochemical Energy |
|-------------------|---------------------------------------|

Prof.dr. S.J. Watson

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Wind Energy                    |
| <b>Telephone</b>  | +31 15 27 81896                |

Prof.dr. A.W. Weeber

|                   |                                    |
|-------------------|------------------------------------|
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 88873                    |
| <b>Room</b>       | 36.LB 03.250                       |

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 88873                    |
| <b>Room</b>       | 36.LB 03.250                       |

W. Yu

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Wind Energy     |
| <b>Telephone</b>  | +31 15 27 81419 |

Dr.ir. M.B. Zaayer

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Wind Energy                    |
| <b>Telephone</b>  | +31 15 27 86426                |
| <b>Room</b>       | 62.5.20                        |

Dr. D. Zappalá

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Wind Energy     |
| <b>Telephone</b>  | +31 15 27 81510 |
| <b>Room</b>       | 62.5.20         |

Dr.ir. H. Ziar

|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 81907                    |

Dr.ir. A.H. van Zuijlen

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Aerodynamics                   |
| <b>Telephone</b>  | +31 15 27 82046                |
| <b>Room</b>       | 64.HSL 0.39                    |

E. Çelik

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Department</b> | Delft Centre for Entrepreneurship |
|-------------------|-----------------------------------|